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Chapter 1

Introduction

HI!

The purpose of this course is to introduce some basic concepts and tools which are useful in the analysis and control of **dynamical systems**. The concept of a dynamical system is very general; it refers to anything which evolves with time. Every aerospace vehicle (aircraft, spacecraft, motorcycle) is a dynamical system. If one perturbs an aerospace vehicle from a steady state condition, the vehicle, in general, does not simply remain at the new condition, but, has a resulting time-varying behavior which may or may not result in the vehicle returning to its original condition. Other engineering examples of dynamical systems include automobiles, metal cutting machines such as lathes and milling machines, robots, chemical plants, electrical circuits, and communication networks. Even civil engineering structures such as bridges and skyscrapers are examples; think of a structure subject to strong winds or an earthquake. The concept of a dynamical system is not restricted to engineering; non-engineering examples include plants, animals, yourself, and the economy.

A system interacts with its environment via **inputs** and **outputs**. Inputs can be considered to be impressed on the system by its environment whereas outputs are exerted by the system on the environment. Inputs are usually divided into control inputs and disturbance inputs. In an aircraft, the deflection of a control surface such as an elevator would be considered a control input; a wind gust would be considered a disturbance input. An actuator is a physical device used for the implementation of a control input. Some examples of actuators are: the elevator on an aircraft; the throttle twistgrip on a motorcycle; a valve in a chemical plant.

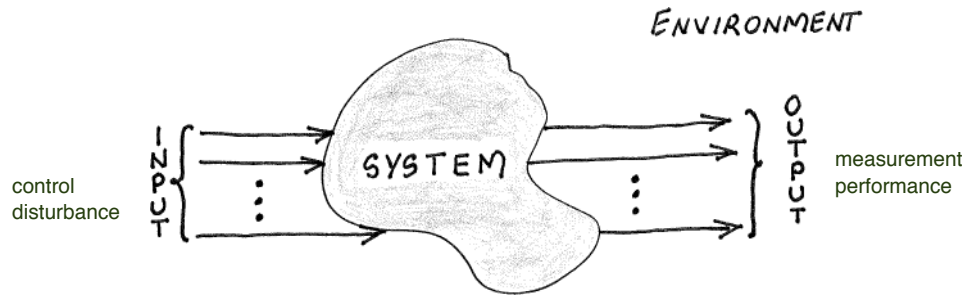


Figure 1.1: System, Inputs, Outputs

Outputs are usually divided into performance outputs and measured outputs. **Performance** outputs are those outputs whose **behavior or performance** you are interested in, for example, **heading and speed of an aircraft**. The **measured** outputs are the outputs you actually measure, for example, **speed of an aircraft**. Usually, all the performance outputs are measured. **Sensors** are the physical devices used to obtain the measured outputs. Some examples of sensors are: altimeter and airspeed sensor on an aircraft; a pressure gauge in a chemical plant.

A fundamental concept in describing the behavior of a dynamical system is the **state** of the system. A precise definition will be given later. However, a main desirable property of a system state is that the current state uniquely determines all future states, that is, if one knows the current state of the system and all future inputs then, one can predict the future state of the system.

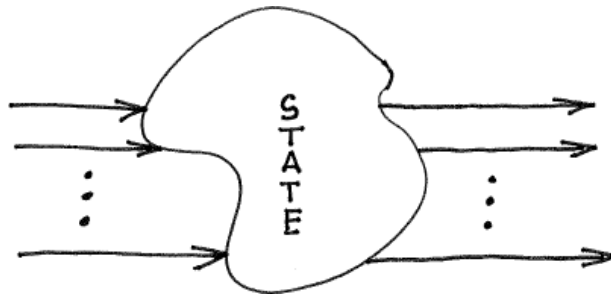


Figure 1.2: State

Feedback is a fundamental concept in system control. In applying a control input to a system the **controller** usually takes into account the behavior of the system; the controller bases its control inputs on the measured outputs of the **plant** (the system under control). Control based on feedback is called **closed loop control**. The term **open loop control** is usually used when one applies a control input which is pre-specified function of time; hence no feedback is involved.

The state space description of an input-output system usually involves a **time variable** and the following three sets of variables:

- **State variables:** $x_1, x_2, \dots, x_n$

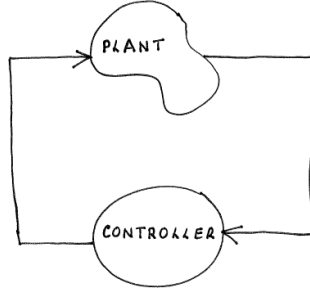


Figure 1.3: Feedback

- Input variables: $u_1, u_2, \dots, u_m$
- Output variables: $y_1, y_2, \dots, y_p$

Dynamical systems evolve with time. In a **continuous-time system**, the time-variable t can be any real number. For **discrete-time systems**, the time-variable k only takes integer values, that is, $\dots, -2, -1, 0, 1, 2, \dots$

For **continuous-time systems**, the description usually takes the following form

$$\begin{array}{l} \text{state eq.} \quad \begin{cases} \dot{x}_1 = F_1(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\ \dot{x}_2 = F_2(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\ \vdots \\ \dot{x}_n = F_n(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \end{cases} \end{array} \quad (1.1)$$

and

$$\begin{array}{l} \text{output eq} \quad \begin{cases} y_1 = H_1(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\ y_2 = H_2(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\ \vdots \\ y_p = H_p(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \end{cases} \end{array} \quad (1.2)$$

The first set of equations are called the **state equations** and the second set are called the **output equations**.

For **discrete-time systems**, the description usually takes the following form

$$\begin{array}{l} \begin{cases} x_1(k+1) = F_1(x_1(k), x_2(k), \dots, x_n(k), u_1(k), u_2(k), \dots, u_m(k)) \\ x_2(k+1) = F_2(x_1(k), x_2(k), \dots, x_n(k), u_1(k), u_2(k), \dots, u_m(k)) \\ \vdots \\ x_n(k+1) = F_n(x_1(k), x_2(k), \dots, x_n(k), u_1(k), u_2(k), \dots, u_m(k)) \end{cases} \end{array} \quad (1.3)$$

and

$$\begin{array}{l} \begin{cases} y_1 = H_1(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\ y_2 = H_2(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\ \vdots \\ y_p = H_p(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \end{cases} \end{array} \quad (1.4)$$

The first set of equations are called the **state equations** and the second set are called the **output equations**.

1.1 Course ingredients

Dynamical systems

Linear algebra

Aerospace/mechanical systems

MATLAB (including Simulink)

1.2 Some notation

Sets : $s \in \mathcal{S}$

$\mathbb{Z}$ represents the set of integers

$\mathbb{R}$ represents the set of real numbers

$\mathbb{C}$ represents the set of complex numbers

Functions: $f : \mathcal{S} \rightarrow \mathcal{T}$

1.3 MATLAB

Introduce yourself to MATLAB

```
%matlab  
>> lookfor  
>> help  
>> quit
```

Learn the representation, addition and multiplication of real and complex numbers

Chapter 2

State space representation of dynamical systems

In this section, we consider a bunch of simple physical systems and demonstrate how one can obtain state space mathematical models of these systems. We begin with some linear examples.

2.1 Linear examples

2.1.1 A first example

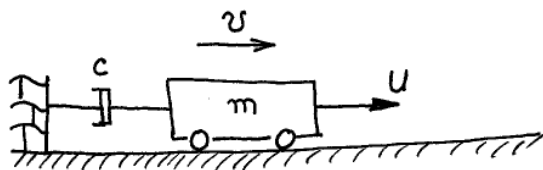


Figure 2.1: First example

Consider a small cart of mass m which is constrained to move along a horizontal line. The cart is subject to viscous friction which we model with a linear dashpot with coefficient c ; it is also subject to an input force which can be represented by a real scalar variable $u(t)$ where the real scalar variable t represents time. Suppose we are just interested in the velocity of the cart and not its position. Let the real scalar variable $v(t)$ represent the velocity of the cart at time t ; we will regard this as the output of the cart. Then, using Newton's second law, the motion of the cart can be described by the following first order ordinary differential equation (ODE):

$$m\dot{v}(t) = -cv(t) + u$$

Introducing the state $x := v$ results in the following state space description:

$$\begin{aligned}\dot{x} &= ax + bu \\ y &= x\end{aligned}$$

where $a := -c/m < 0$ and $b := 1/m$.

2.1.2 The unattached mass

Consider a small cart of mass m which is constrained to move without friction along a horizontal line; see Figure 2.2. It is also subject to an input force which can be represented by a real scalar variable $u(t)$. Here we are interested in the position of the cart which can be represented by $q(t)$, the horizontal displacement of the cart from a fixed point on its line of motion; we will regard $y = q$ as the output of the cart. Application of Newton's second

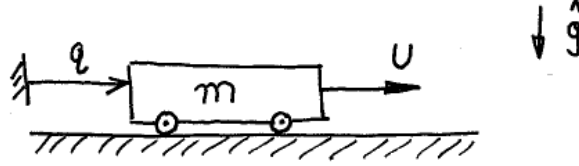


Figure 2.2: The unattached mass

law to the cart results in

$$m\ddot{q} = u$$

Introducing the state variables,

$$x_1 := q \quad \text{and} \quad x_2 := \dot{q},$$

yields the following state space description:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= u/m \\ y &= x_1 \end{aligned}$$

2.1.3 Spring-mass-damper

Consider a system which can be modelled as a simple mechanical system consisting of a body of mass m attached to a base via a linear spring of spring constant k and linear dashpot with damping coefficient c . The base is subject to an acceleration u which we will regard as an input to the system. As output y , we will consider the force transmitted to the mass from the spring-damper combination. Letting q be the deflection of the spring, the motion of the system can be described by

$$m(\ddot{q} + u) = -c\dot{q} - kq - mg$$

and $y = -kq - c\dot{q}$ where g is the gravitational acceleration constant. Introducing $x_1 := q$ and $x_2 := \dot{q}$ results in the following state space description:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -(k/m)x_1 - (c/m)x_2 - u - g \\ y &= -kx_1 - cx_2. \end{aligned}$$

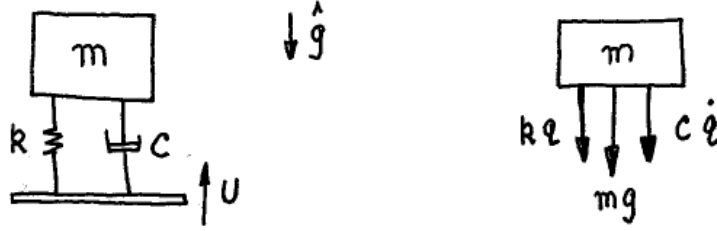


Figure 2.3: Spring-mass-damper with exciting base

2.1.4 A simple structure

Consider a structure consisting of two floors. The scalar variables q_1 and q_2 represent the lateral displacement of the floors from their nominal positions. Application of Newton's

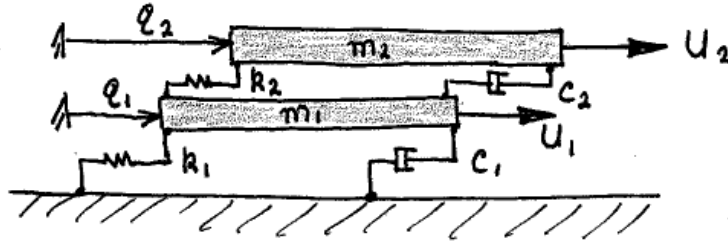


Figure 2.4: A simple structure

second law to each floor results in

$$\begin{aligned} m_1 \ddot{q}_1 + (c_1 + c_2) \dot{q}_1 + (k_1 + k_2) q_1 - c_2 \dot{q}_2 - k_2 q_2 &= u_1 \\ m_2 \ddot{q}_2 - c_2 \dot{q}_1 - k_2 q_1 + c_2 \dot{q}_2 + k_2 q_2 &= u_2 \end{aligned}$$

Here u_2 is a control input resulting from a force applied to the second floor and u_1 is a disturbance input resulting from a force applied to the first floor. We have not considered any outputs here.

Shortly, we will see how to obtain a state space description of this system.

2.2 Nonlinear examples

2.2.1 A first nonlinear system

Recall the first example, and suppose now that the friction force on the cart is due to Coulomb (or dry) friction with coefficient of friction $\mu > 0$. Then application of Newton's

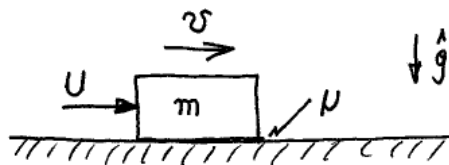


Figure 2.5: A first nonlinear system

second law to the cart results in

$$m\dot{v} = -\mu mg \operatorname{sgm}(v) + u$$

where the **signum function** sgm is defined by

$$\operatorname{sgm}(v) := \begin{cases} -1 & \text{if } v < 0 \\ 0 & \text{if } v = 0 \\ 1 & \text{if } v > 0 \end{cases}$$

and g is the gravitational acceleration constant of the planet on which the block resides.

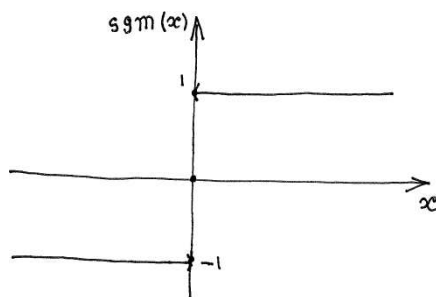


Figure 2.6: Signum function

With $x := v$ we obtain the following nonlinear state space description:

$$\begin{aligned} \dot{x} &= -\alpha \operatorname{sgm}(x) + bu \\ y &= x \end{aligned}$$

where $\alpha := \mu g$ and $b = 1/m$.

2.2.2 Pendulum

Consider a rigid body which is constrained to rotate about a horizontal axis which is perpendicular to the body. Here the input u is a torque applied to the pendulum and the output is the angle θ that the pendulum makes with a vertical line.

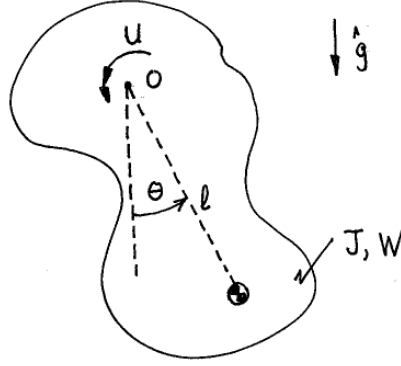


Figure 2.7: Simple pendulum

The motion of this system is governed by

$$J\ddot{\theta} + Wl \sin \theta = u$$

where $J > 0$ is the moment of inertia of the body about its axis of rotation, $W > 0$ is the weight of the body and l is the distance between the mass center of the body and the axis of rotation. Introducing state variables

$$x_1 := \theta \quad \text{and} \quad x_2 := \dot{\theta}$$

results in the following nonlinear state space description:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -a \sin x_1 + b_2 u \\ y &= x_1 \end{aligned}$$

where $a := Wl/J > 0$ and $b_2 = 1/J$.

2.2.3 Spacecraft attitude dynamics

Suppose we model as spacecraft as a rigid body and suppose that the only force acting on it is gravity acting at its mass center. The equations describing the rotational motion of a rigid body are given by Euler's equations of motion. If we choose the axes of a body-fixed

Figure 2.8: Spacecraft attitude dynamics

reference frame along the principal axes of inertia of the rigid body with origin at the center of mass, Euler's equations of motion take the simplified form

$$\begin{aligned} I_1 \dot{\omega}_1 &= (I_2 - I_3) \omega_2 \omega_3 \\ I_2 \dot{\omega}_2 &= (I_3 - I_1) \omega_3 \omega_1 \\ I_3 \dot{\omega}_3 &= (I_1 - I_2) \omega_1 \omega_2 \end{aligned}$$

where $\omega_1, \omega_2, \omega_3$ denote the components of the body angular velocity vector with respect to the body principal axes, and the positive scalars I_1, I_2, I_3 are the principal moments of inertia of the body with respect to its mass center. The above description is in state space form with state variables $\omega_1, \omega_2, \omega_3$.

2.2.4 Body in central force motion and orbit mechanics

Consider a body of mass m subject only to a central force whose line of action always passes through an inertially fixed point O . If the force is as illustrated in Figure 2.2.4, then application of Newton's Second Law yields

Figure 2.9: Body in central force motion and orbit mechanics

$$\begin{aligned} \ddot{r} - r\omega^2 + g(r) &= 0 \\ r\dot{\omega} + 2\dot{r}\omega &= 0 \end{aligned}$$

where r is the distance between the body and O while ω is the rotation rate of the line joining the body and O .

For the simplest situation in orbit mechanics (a “satellite” orbiting YFHB)

$$g(r) = \mu/r^2 \quad \mu = GM$$

where G is the universal constant of gravitation and M is the mass of YFHB.

2.2.5 Ballistics in drag

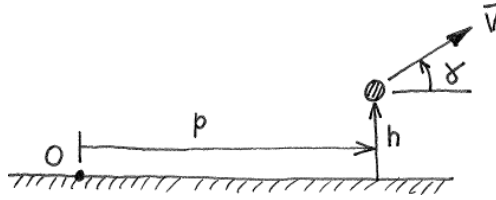


Figure 2.10: A cannonball: ballistics in drag

$$\begin{aligned} \dot{p} &= v \cos \gamma \\ \dot{h} &= v \sin \gamma \\ m\dot{v} &= -mg \sin \gamma - D(v) \\ mv\dot{\gamma} &= -mg \cos \gamma \end{aligned}$$

2.2.6 Double pendulum on cart

Consider the the double pendulum on a cart illustrated in Figure 2.11. The motion of this system can be described by the cart displacement y and the two pendulum angles θ_1 , θ_2 . The input u is a force applied to the cart. Application of your favorite laws of mechanics results in the following equations of motion:

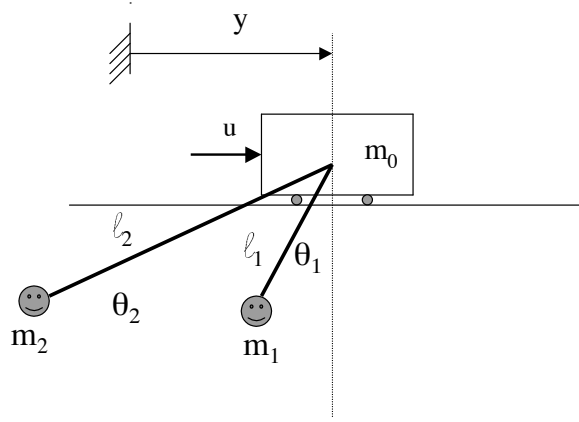


Figure 2.11: Double pendulum on cart

$ \begin{aligned} (m_0 + m_1 + m_2)\ddot{y} - m_1 l_1 \cos \theta_1 \ddot{\theta}_1 - m_2 l_2 \cos \theta_2 \ddot{\theta}_2 &+ m_1 l_1 \sin \theta_1 \dot{\theta}_1^2 + m_2 l_2 \sin \theta_2 \dot{\theta}_2^2 &= u \\ -m_1 l_1 \cos \theta_1 \ddot{y} + m_1 l_1^2 \ddot{\theta}_1 &+ m_1 l_1 g \sin \theta_1 &= 0 \\ -m_2 l_2 \cos \theta_2 \ddot{y} + m_2 l_2^2 \ddot{\theta}_2 &+ m_2 l_2 g \sin \theta_2 &= 0 \end{aligned} $

2.2.7 Two-link robotic manipulator

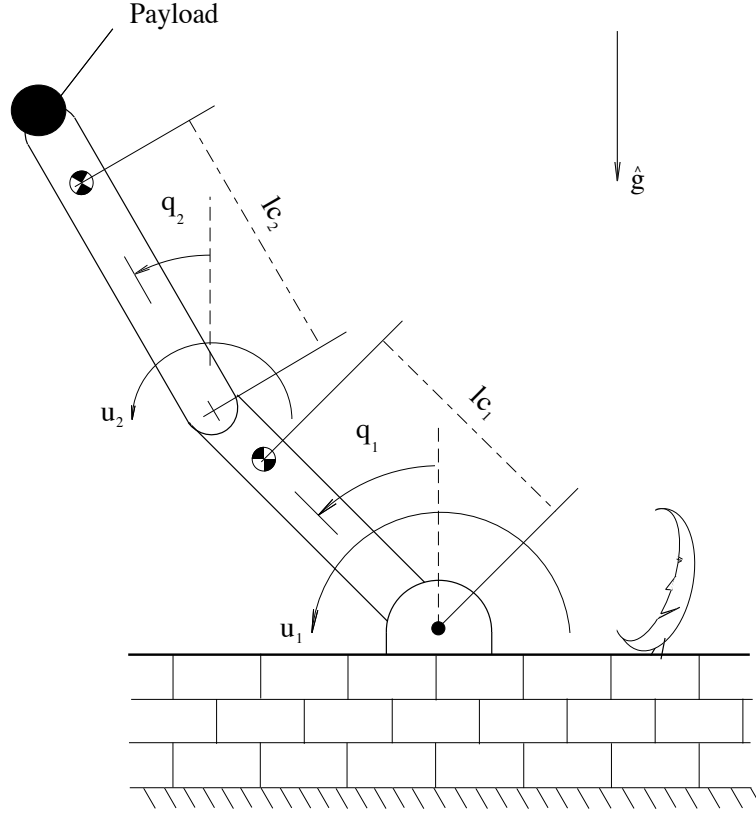


Figure 2.12: A simplified model of a two link manipulator

The coordinates q_1 and q_2 denote the angular location of the first and second links relative to the local vertical, respectively. The second link includes a payload located at its end. The masses of the first and the second links are m_1 and m_2 , respectively. The moments of inertia of the first and the second links about their centers of mass are I_1 and I_2 , respectively. The locations of the center of mass of links one and two are determined by lc_1 and lc_2 , respectively; l_1 is the length of link 1. The equations of motion for the two arms are described by:

$$\begin{aligned} m_{11}\ddot{q}_1 + m_{12}\cos(q_1 - q_2)\ddot{q}_2 + c_1\sin(q_1 - q_2)\dot{q}_2^2 + g_1\sin(q_1) &= u_1 \\ m_{21}\cos(q_1 - q_2)\ddot{q}_1 + m_{22}\ddot{q}_2 + c_2\sin(q_1 - q_2)\dot{q}_1^2 + g_2\sin(q_2) &= u_2 \end{aligned}$$

where

$$\begin{aligned} m_{11} &= I_1 + m_1lc_1^2 + m_2l_1^2, & m_{12} &= m_{21} = m_2l_1lc_2, & m_{22} &= I_2 + m_2lc_2^2 \\ g_1 &= -(m_1lc_1 + m_2l_1)g, & g_2 &= -m_2lc_2g \\ c_1 &= m_2l_1lc_2, & c_2 &= -m_2l_1lc_2 \end{aligned}$$

2.3 Discrete-time examples

All the examples considered so far are continuous-time examples. Now we consider discrete-time examples; here the time variable k is an integer, that is, $\dots, -2, -1, 0, 1, 2, \dots$. Sometimes, a discrete-time system results from the **discretization** of a continuous-time system (see first example) and sometimes it arises naturally (see second example).

2.3.1 The discrete unattached mass

Here we consider a discretization of the unattached mass of Section 2.1.2 described by

$$m\ddot{q} = u.$$

Suppose that the input to this system is given by the following discrete-time signal:

$$u(t) = u_d(k) \quad \text{for } kT \leq t < (k+1)T$$

where k is an integer, that is, $k = \dots, -2, -1, 0, 1, 2, \dots$ and $T > 0$ is some specified time interval; we call it the **sampling time**. Thus the input to the system is constant over each sampling interval $[kT, (k+1)T)$.

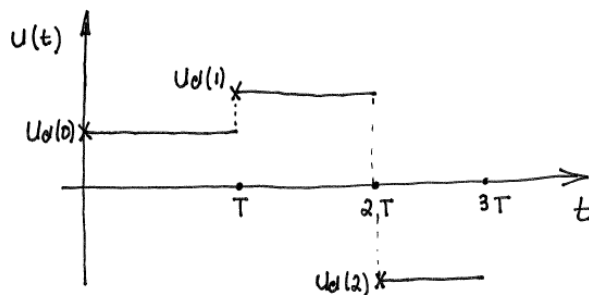


Figure 2.13: Zero order hold

Suppose we **sample** the state of the system at integer multiples of the sampling time and let

$$x_1(k) = q(kT) \quad x_2(k) := \dot{q}(kT)$$

for every integer k . Note that, for any k and any t satisfying $kT \leq t < (k+1)T$ we have

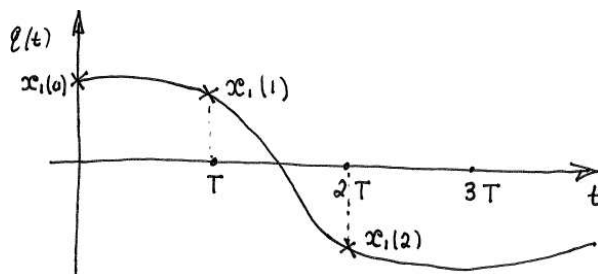


Figure 2.14: Sampling

$$\dot{q}(t) = \dot{q}(kT) + \int_{kT}^t \ddot{q}(\tau) d\tau = x_2(k) + \int_{kT}^t \frac{1}{m} u(k) dt = x_2(k) + \frac{t-kT}{m} u(k) .$$

Hence,

$$x_2(k+1) = \dot{q}((k+1)T) = x_2(k) + \frac{T}{m} u(k)$$

and

$$\begin{aligned} x_1(k+1) &= q((k+1)T) = q(kT) + \int_{kT}^{(k+1)T} \dot{q}(t) dt = x_1(k) + \int_{kT}^{(k+1)T} x_2(k) + \frac{t-kT}{m} u(k) dt \\ &= x_1(k) + T x_2(k) + \frac{T^2}{2m} u(k) . \end{aligned}$$

Thus, the sampled system is described by the two first order linear difference equations:

$$\boxed{\begin{array}{lcl} x_1(k+1) & = & x_1(k) + T x_2(k) + \frac{T^2}{2m} u(k) \\ x_2(k+1) & = & x_2(k) + \frac{T}{m} u(k) \end{array}}$$

and the output equation:

$$\boxed{y(k) = x_1(k)}$$

This is a discrete-time state space description.

2.3.2 Additive increase multiplicative decrease (AIMD) algorithm for resource allocation

Consider a resource such as a communications router which is serving n users and has a maximum capacity of C . Ideally, we would like the resource to be fully utilized at all times.

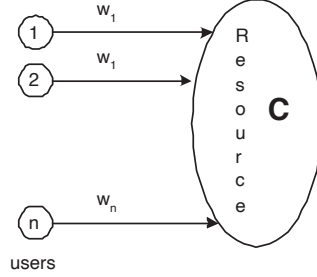


Figure 2.15: Users and resource

Each user determines how much capacity it will request without knowledge of the usage of the other users. How does each user determine its use of the resource? One approach is to use an **additive increase multiplicative decrease (AIMD)** algorithm. In this approach, each user increases its use of the resource linearly with time (additive increase phase) until it is notified that the resource has reached maximum capacity. It then (instantaneously) reduces its usage to a fraction of its usage at notification (multiplicative decrease phase). It again starts to increase its usage linearly with time.

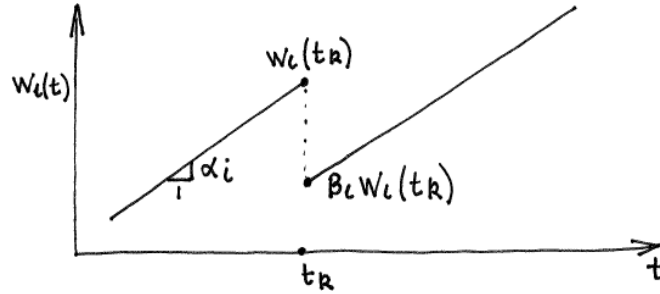


Figure 2.16: AIMD

To describe AIMD, let $w_i(t) \geq 0$ represent the i -th users share of the resource capacity at time t . Each user increases its share linearly with time until the resource reaches maximum capacity C . When maximum capacity is reached, we refer to this as a **capacity event**. Suppose we number the capacity events consecutively with index $k = 1, 2, \dots$ and let t_k denote the time at which the k -th capacity event occurs. Then.

$$w_1(t_k) + w_2(t_k) + \dots + w_n(t_k) = C.$$

Suppose that immediately after a capacity event, each user i decreases its share of the resource to β_i times its share at congestion where

$$0 \leq \beta_i < 1.$$

This is the **multiplicative decrease** phase of the algorithm. If t_k^+ represents a time immediately after the k -th congestion event then,

$$w_i(t_k^+) = \beta_i w_i(t_k)$$

Following the multiplicative decrease of its share after a capacity event, each user i increases its share linearly in time at a rate $\alpha_i > 0$ until congestion occurs again, that is,

$$w_i(t) = \beta_i w_i(t_k) + \alpha_i(t - t_k)$$

for $t_k < t \leq t_{k+1}$. This is the **additive increase** phase of the algorithm.

Let T_k be the time between the k -th capacity event and capacity event $k + 1$. Then $T_k = t_{k+1} - t_k$ and

$$w_i(t_{k+1}) = \beta_i w_i(t_k) + \alpha_i T_k$$

Abusing notation and letting $w_i(k) = w_i(t_k)$, the AIMD algorithm is described by

$$w_i(k+1) = \beta_i w_i(k) + \alpha_i T_k \quad \text{for } i = 1, \dots, n \quad (2.1)$$

where the inter-capacity time T_k is determined by the capacity condition

$$w_1(k+1) + \dots + w_n(k+1) = C$$

Using this, (2.1) and

$$w_1(k) + \dots + w_n(k) = C$$

results in

$$(1 - \beta_1)w_1(k) + \dots + (1 - \beta_n)w_n(k) = (\alpha_1 + \dots + \alpha_n)T_k$$

Solving the above equation for T_k and substitution into (2.1) results in

$$\boxed{w_i(k+1) = \beta_i w_i(k) + \gamma_i(1 - \beta_1)w_1(k) + \dots + \gamma_i(1 - \beta_n)w_n(k)} \quad (2.2)$$

for $i = 1, \dots, n$. where

$$\gamma_i := \frac{\alpha_i}{\alpha_1 + \dots + \alpha_n}.$$

Note that

$$\gamma_1 + \dots + \gamma_n = 1. \quad (2.3)$$

Not also that the above system is described by n first order difference equations.

2.4 General representation

2.4.1 Continuous-time

With the exception of the discrete-time examples, all of the preceding systems can be described by a bunch of first order ordinary differential equations of the form

$$\begin{array}{lcl} \dot{x}_1 & = & F_1(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\ \dot{x}_2 & = & F_2(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\ & \vdots & \\ \dot{x}_n & = & F_n(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \end{array}$$

Such a description of a dynamical system is called a **state space description**; the real scalar variables, $x_i(t)$ are called the **state variables**; the real scalar variables, $u_i(t)$ are called the **input variables** and the real scalar variable t is called the **time variable**. If the system has outputs, they are described by

$$\begin{array}{lcl} y_1 & = & H_1(x_1, \dots, x_n, u_1, \dots, u_m) \\ y_2 & = & H_2(x_1, \dots, x_n, u_1, \dots, u_m) \\ & \vdots & \\ y_p & = & H_p(x_1, \dots, x_n, u_1, \dots, u_m) \end{array}$$

where the real scalar variables, $y_i(t)$ are called the **output variables**

When a system has no inputs or the inputs are fixed at some constant values, the system is described by

$$\begin{array}{lcl} \dot{x}_1 & = & f_1(x_1, x_2, \dots, x_n) \\ \dot{x}_2 & = & f_2(x_1, x_2, \dots, x_n) \\ & \vdots & \\ \dot{x}_n & = & f_n(x_1, x_2, \dots, x_n) \end{array}$$

Limitations of the above description.

Systems with delays.

Systems described by partial differential equations.

Higher order ODE descriptions

Single equation. (Recall the spring-mass-damper system.) Consider a dynamical system described by a single n^{th} -order differential equation of the form

$$\boxed{F(q, \dot{q}, \dots, q^{(n)}, u) = 0}$$

where $q(t)$ is a real scalar and

$$q^{(n)} := \frac{d^n q}{dt^n}.$$

To obtain an equivalent state space description, we proceed as follows.

(a) First solve for the highest order derivative $q^{(n)}$ of q as a function of $q, \dot{q}, \dots, q^{(n-1)}$ and u to obtain something like:

$$q^{(n)} = a(q, \dot{q}, \dots, q^{(n-1)}, u).$$

If this is not possible, then we cannot obtain a state space description.

(b) Now introduce state variables,

$$\begin{aligned} x_1 &:= q \\ x_2 &:= \dot{q} \\ &\vdots \\ x_n &:= q^{(n-1)} \end{aligned}$$

to obtain the following state space description:

$$\boxed{\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ &\vdots \\ \dot{x}_{n-1} &= x_n \\ \dot{x}_n &= a(x_1, x_2, \dots, x_n, u) \end{aligned}}$$

Multiple equations. (Recall the simple structure and the pendulum cart system.) Consider a dynamical system described by N scalar differential equations in N scalar variables:

$$\boxed{\begin{array}{rcl} F_1(q_1, \dot{q}_1, \dots, q_1^{(n_1)}, q_2, \dot{q}_2, \dots, q_2^{(n_2)}, \dots, q_N, \dot{q}_N, \dots, q_N^{(n_N)}, u_1, u_2, \dots, u_m) & = & 0 \\ F_2(q_1, \dot{q}_1, \dots, q_1^{(n_1)}, q_2, \dot{q}_2, \dots, q_2^{(n_2)}, \dots, q_N, \dot{q}_N, \dots, q_N^{(n_N)}, u_1, u_2, \dots, u_m) & = & 0 \\ & \vdots & \\ F_N(q_1, \dot{q}_1, \dots, q_1^{(n_1)}, q_2, \dot{q}_2, \dots, q_2^{(n_2)}, \dots, q_N, \dot{q}_N, \dots, q_N^{(n_N)}, u_1, u_2, \dots, u_m) & = & 0 \end{array}}$$

where $t, q_1(t), q_2(t), \dots, q_N(t)$ are real scalars. Note that $q_i^{(n_i)}$ is the highest order derivative of q_i which appears in the above equations. To obtain an equivalent state space description, we proceed as follows.

(a) Solve for the highest order derivatives, $q_1^{(n_1)}, q_2^{(n_2)}, \dots, q_N^{(n_N)}$ of $q_1, \dots, q_N$ which appear in the differential equations to obtain something like:

$$\begin{aligned} q_1^{(n_1)} &= a_1(q_1, \dot{q}_1, \dots, q_1^{(n_1-1)}, q_2, \dot{q}_2, \dots, q_2^{(n_2-1)}, \dots, q_N, \dot{q}_N, \dots, q_N^{(n_N-1)}, u_1, u_2, \dots, u_m) \\ q_2^{(n_2)} &= a_2(q_1, \dot{q}_1, \dots, q_1^{(n_1-1)}, q_2, \dot{q}_2, \dots, q_2^{(n_2-1)}, \dots, q_N, \dot{q}_N, \dots, q_N^{(n_N-1)}, u_1, u_2, \dots, u_m) \\ &\vdots \\ q_N^{(n_N)} &= a_N(q_1, \dot{q}_1, \dots, q_1^{(n_1-1)}, q_2, \dot{q}_2, \dots, q_2^{(n_2-1)}, \dots, q_N, \dot{q}_N, \dots, q_N^{(n_N-1)}, u_1, u_2, \dots, u_m) \end{aligned}$$

If one cannot uniquely solve for the above derivatives, then one cannot obtain a unique state space description

(b) As state variables, consider each q_i variable and its derivatives up to but not including its highest order derivative. One way to do this is as follows. Let

$$\begin{array}{llll} x_1 := q_1 & x_2 := \dot{q}_1 & \dots & x_{n_1} := q_1^{(n_1-1)} \\ x_{n_1+1} := q_2, & x_{n_1+2} := \dot{q}_2 & \dots & x_{n_1+n_2} := q_2^{(n_2-1)} \\ & & \vdots & \\ x_{n_1+\dots+n_{N-1}+1} := q_N & x_{n_1+\dots+n_{N-1}+2} := \dot{q}_N & \dots & x_n := q_N^{(n_N-1)} \end{array}$$

where

$$n := n_1 + n_2 + \dots + n_N$$

to obtain

$$\begin{aligned}
\dot{x}_1 &= x_2 \\
\dot{x}_2 &= x_3 \\
&\vdots \\
\dot{x}_{n_1-1} &= x_{n_1} \\
\dot{x}_{n_1} &= a_1(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\
\dot{x}_{n_1+1} &= x_{n_1+2} \\
&\vdots \\
\dot{x}_{n_1+n_2} &= a_2(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\
&\vdots \\
\dot{x}_n &= a_N(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m)
\end{aligned}$$

Example 1

$$\begin{aligned}
\ddot{q}_1 + \dot{q}_2 + 2q_1 &= 0 \\
-\ddot{q}_1 + \dot{q}_1 + \dot{q}_2 + 4q_2 &= 0
\end{aligned}$$

The highest order derivatives of q_1 and q_2 appearing in these equations are $\ddot{q}_1$ and $\dot{q}_2$, respectively. Solving for $\ddot{q}_1$ and $\dot{q}_2$, we obtain

$$\begin{aligned}
\ddot{q}_1 &= -q_1 + \frac{1}{2}\dot{q}_1 + 2q_2 \\
\dot{q}_2 &= -q_1 - \frac{1}{2}\dot{q}_1 - 2q_2.
\end{aligned}$$

Introducing state variables $x_1 := q_1$, $x_2 := \dot{q}_1$, and $x_3 = q_2$ we obtain the following state space description:

$$\begin{aligned}
\dot{x}_1 &= x_2 \\
\dot{x}_2 &= -x_1 + \frac{1}{2}x_2 + 2x_3 \\
\dot{x}_3 &= -x_1 - \frac{1}{2}x_2 - 2x_3
\end{aligned}$$

2.4.2 Discrete-time

Recall the discrete unattached mass and the AIMD algorithm of Section 2.3.1. The general discrete-time system is described by a bunch of first order difference equations of the form:

$$\begin{aligned}
x_1(k+1) &= F_1(x_1(k), x_2(k), \dots, x_n(k), u_1(k), u_2(k), \dots, u_m(k)) \\
x_2(k+1) &= F_2(x_1(k), x_2(k), \dots, x_n(k), u_1(k), u_2(k), \dots, u_m(k)) \\
&\vdots \\
x_n(k+1) &= F_n(x_1(k), x_2(k), \dots, x_n(k), u_1(k), u_2(k), \dots, u_m(k))
\end{aligned}$$

The real scalar variables, $x_i(k)$, $i = 1, 2, \dots, n$ are called the **state variables**; the real scalar variables, $u_i(k)$, $i = 1, 2, \dots, m$ are called the **input variables** and the integer variable k is called the **time variable**. If the system has outputs, they are described by

$$\begin{array}{lcl} y_1 & = & H_1(x_1, \dots, x_n, u_1, \dots, u_m) \\ y_2 & = & H_2(x_1, \dots, x_n, u_1, \dots, u_m) \\ & \vdots & \\ y_p & = & H_p(x_1, \dots, x_n, u_1, \dots, u_m) \end{array}$$

where the real scalar variables, $y_i(k)$, $i = 1, 2, \dots, p$ are called the **output variables**

When a system has no inputs or the inputs are fixed at some constant values, the system is described by

$$\begin{array}{lcl} x_1(k+1) & = & f_1(x_1(k), x_2(k), \dots, x_n(k)) \\ x_2(k+1) & = & f_2(x_1(k), x_2(k), \dots, x_n(k)) \\ & \vdots & \\ x_n(k+1) & = & f_n(x_1(k), x_2(k), \dots, x_n(k)) \end{array}$$

Higher order difference equation descriptions

One converts higher order difference equation descriptions into state space descriptions in a similar manner to that of converting higher order ODE descriptions into state space descriptions; see Example 2.3.1.

2.4.3 Exercises

Exercise 1 Consider a system described by a single n^{th} -order linear differential equation of the form

$$q^{(n)} + a_{n-1}q^{(n-1)} + \dots a_1\dot{q} + a_0q = 0$$

where $q(t) \in \mathbb{R}$ and $q^{(n)} := \frac{d^n q}{dt^n}$. By appropriate definition of state variables, obtain a first order state space description of this system.

Exercise 2 By appropriate definition of state variables, obtain a first order state space description of the following systems.

- (i) The ‘simple structure’
- (ii) The ‘body in central force motion’
- (iii)

$$\begin{aligned}\ddot{q}_1 + q_1 + 2\dot{q}_2 &= 0 \\ \ddot{q}_1 + \dot{q}_2 + q_2 &= 0\end{aligned}$$

where q_1, q_2 are real scalars.

Exercise 3 By appropriate definition of state variables, obtain a first order state space description of the following systems where q_1 and q_2 are real scalars.

- (i)

$$\begin{aligned}2\ddot{q}_1 + \ddot{q}_2 + \sin q_1 &= 0 \\ \ddot{q}_1 + 2\ddot{q}_2 + \sin q_2 &= 0\end{aligned}$$

- (ii)

$$\begin{aligned}\ddot{q}_1 + \dot{q}_2 + q_1^3 &= 0 \\ \dot{q}_1 + \dot{q}_2 + q_2^3 &= 0\end{aligned}$$

Exercise 4 By appropriate definition of state variables, obtain a first order state space description of the following systems where q_1, q_2 are real scalars.

- (i)

$$\begin{aligned}2\ddot{q}_1 + 3\ddot{q}_2 + \dot{q}_1 - q_2 &= 0 \\ \ddot{q}_1 + 2\ddot{q}_2 - \dot{q}_2 + q_1 &= 0\end{aligned}$$

(ii)

$$\begin{aligned}\ddot{q}_2 + 2\dot{q}_1 + q_2 - q_1 &= 0 \\ \ddot{q}_2 + \dot{q}_1 - q_2 + q_1 &= 0\end{aligned}$$

Exercise 5 By appropriate definition of state variables, obtain a first order state space description of the following systems where q_1, q_2 are real scalars.

(i)

$$\begin{aligned}3\ddot{q}_1 - \ddot{q}_2 + 2\dot{q}_1 + 4q_2 &= 0 \\ -\ddot{q}_1 + 3\ddot{q}_2 + 2\dot{q}_1 - 4q_2 &= 0\end{aligned}$$

(ii)

$$\begin{aligned}3\ddot{q}_1 - \dot{q}_2 + 4q_2 - 4q_1 &= 0 \\ -\ddot{q}_1 + 3\dot{q}_2 - 4q_2 + 4q_1 &= 0\end{aligned}$$

Exercise 6 Obtain a state-space description of the following single-input single-output system with input u and output y .

$$\begin{aligned}\ddot{q}_1 + \dot{q}_2 + q_1 &= 0 \\ \ddot{q}_1 - \dot{q}_2 + q_2 &= u \\ y &= \ddot{q}_1\end{aligned}$$

Exercise 7 Consider the second order difference equation

$$q(k+2) = q(k+1) + q(k)$$

Obtain the solution corresponding to

$$q(0) = 1, \quad q(1) = 1$$

and $k = 0, 1, \dots, 10$. Obtain a state space description of this system.

Exercise 8 Obtain a state space representation of the following system:

$$q(k+n) + a_{n-1}q(k+n-1) + \dots + a_1q(k+1) + a_0q(k) = 0$$

where $q(k) \in \mathbb{R}$.

2.5 Vectors

When dealing with systems containing many variables, the introduction of vectors for system description can considerably simplify system analysis and control design

2.5.1 Vector spaces and $\mathbb{R}^n$

A **scalar** is a real or a complex number. The symbols $\mathbb{R}$ and $\mathbb{C}$ represent the set of real and complex numbers, respectively. In this section, all the definitions and results are given for real scalars. However, they also hold for complex scalars; to get the results for complex scalars, simply replace ‘real’ with ‘complex’ and $\mathbb{R}$ with $\mathbb{C}$.

Consider any positive integer n . A **real n -vector** x is an **ordered n -tuple** of real numbers, $x_1, x_2, \dots, x_n$. This is usually written as $x = (x_1, x_2, \dots, x_n)$ or

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{or} \quad x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

The real numbers $x_1, x_2, \dots, x_n$ are called the **scalar components** of x and x_i is called the **i -th component**. The symbol $\mathbb{R}^n$ represents the set of ordered n -tuples of real numbers.

Addition

The addition of any two real n -vectors x and y yields another real n -vector $x + y$ which is defined by:

$$x + y = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} := \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

Zero element of $\mathbb{R}^n$:

$$0 := \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Note that we are using the same symbol, 0, for a zero scalar and a zero vector.

The negative of a vector:

$$-x := \begin{pmatrix} -x_1 \\ -x_2 \\ \vdots \\ -x_n \end{pmatrix}$$

Properties of addition

(a) (Commutative). For each pair x, y in $\mathbb{R}^n$,

$$x + y = y + x$$

(b) (Associative). For each x, y, z in $\mathbb{R}^n$,

$$(x + y) + z = x + (y + z)$$

(c) There is an element 0 in $\mathbb{R}^n$ such that for every x in $\mathbb{R}^n$,

$$x + 0 = x$$

(d) For each x in $\mathbb{R}^n$, there is an element $-x$ in $\mathbb{R}^n$ such that

$$x + (-x) = 0$$

Scalar multiplication.

The multiplication of an element x of $\mathbb{R}^n$ by a real scalar α yields an element of $\mathbb{R}^n$ and is defined by:

$$\alpha x = \alpha \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} := \begin{pmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{pmatrix}$$

Properties of scalar multiplication

(a) For each scalar α and pair x, y in $\mathbb{R}^n$

$$\alpha(x + y) = \alpha x + \alpha y$$

(b) For each pair of scalars α, β and x in $\mathbb{R}^n$,

$$(\alpha + \beta)x = \alpha x + \beta x$$

(c) For each pair of scalars α, β , and x in $\mathbb{R}^n$,

$$\alpha(\beta x) = (\alpha\beta)x$$

(d) For each x in $\mathbb{R}^n$,

$$1x = x$$

Vector space

Consider *any* set $\mathcal{V}$ equipped with an addition operation and a scalar multiplication operation. Suppose the addition operation assigns to each pair of elements x, y in $\mathcal{V}$ a unique element $x + y$ in $\mathcal{V}$ and it satisfies the above four properties of addition (with $\mathbb{R}^n$ replaced by $\mathcal{V}$). Suppose the scalar multiplication operation assigns to each scalar α and element x in $\mathcal{V}$ a unique element αx in $\mathcal{V}$ and it satisfies the above four properties of scalar multiplication (with $\mathbb{R}^n$ replaced by $\mathcal{V}$). Then this set (along with its addition and scalar multiplication) is called a **vector space**. Thus $\mathbb{R}^n$ equipped with its definitions of addition and scalar multiplication is a specific example of a vector space. We shall meet other examples of vector spaces later. An element x of a vector space is called a **vector**. A vector space with real (complex) scalars is called a **real (complex) vector space**.

As a more abstract example of a vector space, let $\mathcal{V}$ be the set of continuous real-valued functions which are defined on the interval $[0, \infty)$; thus, an element x of $\mathcal{V}$ is a function which maps $[0, \infty)$ into $\mathbb{R}$, that is, $x : [0, \infty) \rightarrow \mathbb{R}$. Addition and scalar multiplication are defined as follows. Suppose x and y are any two elements of $\mathcal{V}$ and α is any scalar, then

$$\begin{aligned}(x + y)(t) &= x(t) + y(t) \\ (\alpha x)(t) &= \alpha x(t)\end{aligned}$$

Subtraction in a vector space is defined by:

$$x - y := x + (-y)$$

Hence in $\mathbb{R}^n$,

$$x - y = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} - \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 - y_1 \\ x_2 - y_2 \\ \vdots \\ x_n - y_n \end{pmatrix}$$

2.5.2 $\mathbb{R}^2$ and pictures

An element of $\mathbb{R}^2$ can be represented in a plane by a point or a directed line segment.

2.5.3 Derivatives

Suppose $x(\cdot)$ is a function of a real variable t where $x(t)$ is an n -vector. Then

$$\dot{x} := \frac{dx}{dt} := \begin{pmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \\ \vdots \\ \frac{dx_n}{dt} \end{pmatrix} = \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{pmatrix}$$

2.5.4 MATLAB

Representation of real and complex vectors

Addition and subtraction of vectors

Multiplication of a vector by a scalar

2.6 Vector representation of dynamical systems

Recall the general descriptions (continuous and discrete) of dynamical systems given in Section 2.4. We define the **state (vector)** x as the vector with components, $x_1, x_2, \dots, x_n$, that is,

$$x := \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

If the system has inputs, we define the **input (vector)** u as the vector with components, $u_1, u_2, \dots, u_m$, that is,

$$u := \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{pmatrix}.$$

If the system has outputs, we define the **output (vector)** y as the vector with components, $y_1, y_2, \dots, y_p$, that is,

$$y := \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{pmatrix}.$$

We introduce the vector valued functions F and H defined by

$$F(x, u) := \begin{pmatrix} F_1(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\ F_2(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\ \vdots \\ F_n(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \end{pmatrix}$$

and

$$H(x, u) := \begin{pmatrix} H_1(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\ H_2(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\ \vdots \\ H_p(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \end{pmatrix}$$

respectively.

Continuous-time systems. The general representation of a continuous-time dynamical system can be compactly described by the following equations:

$$\boxed{\begin{array}{l} \dot{x} = F(x, u) \\ y = H(x, u) \end{array}} \quad (2.4)$$

where $x(t)$ is an n -vector, $u(t)$ is an m -vector, $y(t)$ is a p -vector and the real variable t is the time variable. The first equation above is a **first order vector differential equation** and is called the **state equation**. The second equation is called the **output equation**.

When a system has no inputs of the input vector vector is constant, it can be described by

$$\dot{x} = f(x).$$

A system described by the above equations is called **autonomous** or **time-invariant** because the right-hand sides of the equations do not depend explicitly on time t . For the first part of the course, we will only concern ourselves with these systems.

However, one can have a system containing time-varying parameters. In this case the system might be described by

$$\begin{aligned}\dot{x} &= F(t, x, u) \\ y &= H(t, x, u)\end{aligned}$$

that is, the right-hand sides of the differential equation and/or the output equation depend explicitly on time. Such a system is called **non-autonomous** or **time-varying**. We will look at them later.

Discrete-time systems. The general representation of a continuous-time dynamical system can be compactly described by the following equations:

$$\boxed{\begin{aligned}x(k+1) &= F(x(k), u(k)) \\ y(k) &= H(x(k), u(k))\end{aligned}} \quad (2.5)$$

where $x(k)$ is an n -vector, $u(k)$ is an m -vector, $y(k)$ is a p -vector and the integer variable k is the time variable. The first equation above is a **first order vector difference equation** and is called the **state equation**. The second equation is called the **output equation**. A system described by the above equations is called **autonomous** or **time-invariant** because the right-hand sides of the equations do not depend explicitly on time k .

However, one can have a system containing time-varying parameters. In this case the system might be described by

$$\begin{aligned}x(k+1) &= F(k, x(k), u(k)) \\ y(k) &= H(k, x(k), u(k))\end{aligned}$$

that is, the right-hand sides of the differential equation and/or the output equation depend explicitly on time. Such a system is called **non-autonomous** or **time-varying**.

2.7 Solutions and equilibrium states

2.7.1 Continuous-time

Consider a system described by

$$\dot{x} = f(x) \quad (2.6)$$

A **solution** or **motion** of this system is any continuous function $x(\cdot)$ satisfying $\dot{x}(t) = f(x(t))$ for all t .

An **equilibrium solution** is the simplest type of solution; *it is constant for all time*, that is, it satisfies

$$\boxed{x(t) \equiv x^e}$$

for some fixed state vector x^e . The state x^e is called an **equilibrium state**. Since an equilibrium solution must satisfy the above differential equation, all equilibrium states must satisfy the equilibrium condition:

$$\boxed{f(x^e) = 0}$$

or, in scalar terms,

$$\begin{aligned} f_1(x_1^e, x_2^e, \dots, x_n^e) &= 0 \\ f_2(x_1^e, x_2^e, \dots, x_n^e) &= 0 \\ &\vdots \\ f_n(x_1^e, x_2^e, \dots, x_n^e) &= 0 \end{aligned}$$

Conversely, if a state x^e satisfies the above equilibrium condition, then there is a solution satisfying $x(t) \equiv x^e$; hence x^e is an equilibrium state.

Example 2 Spring mass damper. With $u = 0$ this system is described by

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -(k/m)x_1 - (c/m)x_2 - g. \end{aligned}$$

Hence equilibrium states are given by:

$$\begin{aligned} x_2^e &= 0 \\ -(k/m)x_1^e - (c/m)x_2^e - g &= 0 \end{aligned}$$

Hence

$$x^e = \begin{pmatrix} -mg/k \\ 0 \end{pmatrix}$$

This system has a single equilibrium state.

Example 3 The unattached mass. With $u = 0$ this system is described by

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= 0. \end{aligned}$$

Hence

$$x^e = \begin{pmatrix} x_1^e \\ 0 \end{pmatrix}$$

where x_1^e is arbitrary. *This system has an infinite number of equilibrium states.*

Example 4 Pendulum. With $u = 0$, this system is described by

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -a \sin x_1.\end{aligned}$$

The equilibrium condition yields:

$$x_2^e = 0 \quad \text{and} \quad \sin(x_1^e) = 0.$$

Hence, all equilibrium states are of the form

$$x^e = \begin{pmatrix} m\pi \\ 0 \end{pmatrix}$$

where m is an arbitrary integer. Physically, there are *only two distinct equilibrium states*

$$x^e = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{and} \quad x^e = \begin{pmatrix} \pi \\ 0 \end{pmatrix}$$

Higher order ODEs

$$F(q, \dot{q}, \dots, q^{(n)}) = 0$$

An **equilibrium solution** is the simplest type of solution; *it is constant for all time*, that is, it satisfies

$$q(t) \equiv q^e$$

for some fixed scalar q^e . Clearly, y^e must satisfy

$$F(q^e, 0, \dots, 0) = 0 \tag{2.7}$$

For the state space description of this system introduced earlier, all equilibrium states are given by

$$x^e = \begin{pmatrix} q^e \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

where q^e solves (2.7).

Multiple higher order ODEs

Equilibrium solutions

$$q_i(t) \equiv q_i^e, \quad i = 1, 2, \dots, N$$

Hence

$$\begin{aligned}F_1(q_1^e, 0, \dots, q_2^e, 0, \dots, \dots, q_N^e, \dots, 0) &= 0 \\ F_2(q_1^e, 0, \dots, q_2^e, 0, \dots, \dots, q_N^e, \dots, 0) &= 0 \\ &\vdots \\ F_N(q_1^e, 0, \dots, q_2^e, 0, \dots, \dots, q_N^e, \dots, 0) &= 0\end{aligned} \tag{2.8}$$

For the state space description of this system introduced earlier, all equilibrium states are given by

$$x^e = \begin{pmatrix} q_1^e \\ 0 \\ \vdots \\ q_2^e \\ 0 \\ \vdots \\ \vdots \\ q_N^e \\ \vdots \\ 0 \end{pmatrix}$$

where $q_1^e, q_2^e, \dots, q_N^e$ solve (2.8).

Example 5 Central force motion in inverse square gravitational field

$$\begin{aligned} \ddot{r} - r\omega^2 + \mu/r^2 &= 0 \\ r\dot{\omega} + 2\dot{r}\omega &= 0 \end{aligned}$$

Equilibrium solutions

$$r(t) \equiv r^e, \quad \omega(t) \equiv \omega^e$$

Hence,

$$\dot{r}, \ddot{r}, \dot{\omega} = 0$$

This yields

$$\begin{aligned} -r^e(\omega^e)^2 + \mu/(r^e)^2 &= 0 \\ 0 &= 0 \end{aligned}$$

Thus there are infinite number of equilibrium solutions given by:

$$\omega^e = \pm \sqrt{\mu/(r^e)^3}$$

where r^e is arbitrary. Note that, for this state space description, an equilibrium state corresponds to a circular orbit.

Controlled equilibrium states

Consider now a system with inputs described by

$$\dot{x} = F(x, u) \tag{2.9}$$

Suppose the input is constant and equal to u^e , that is, $u(t) \equiv u^e$. Then the resulting system is described by

$$\dot{x}(t) = F(x(t), u^e)$$

The equilibrium states x^e of this system are given by $F(x^e, u^e) = 0$. This leads to the following definition.

A state x^e is a controlled equilibrium state of the system, $\dot{x} = F(x, u)$, if there is a constant input u^e such that

$$\boxed{F(x^e, u^e) = 0}$$

Example 6 (*Controlled pendulum*)

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\sin(x_1) + u\end{aligned}$$

Any state of the form

$$x^e = \begin{pmatrix} x_1^e \\ 0 \end{pmatrix}$$

is a controlled equilibrium state. The corresponding constant input is $u^e = \sin(x_1^e)$.

2.7.2 Discrete-time

Consider a discrete-time system described by

$$x(k+1) = f(x(k)). \quad (2.10)$$

A solution of (2.10) is a sequence

$$\{ x(0), x(1), x(2), \dots \}$$

which satisfies (2.5).

Example 7 Simple scalar nonlinear system

$$x(k+1) = x(k)^2$$

Sample solutions are:

$$\begin{aligned}(0, 0, 0, \dots) \\ (-1, 1, 1, \dots) \\ (2, 4, 16, \dots)\end{aligned}$$

Equilibrium solutions and states

$$x(k) \equiv x^e$$

$$f(x^e) = x^e$$

In scalar terms,

$$\begin{aligned}f_1(x_1^e, x_2^e, \dots, x_n^e) &= x_1^e \\ f_2(x_1^e, x_2^e, \dots, x_n^e) &= x_2^e \\ &\vdots \\ f_n(x_1^e, x_2^e, \dots, x_n^e) &= x_m^e\end{aligned}$$

Example 8 Simple scalar nonlinear system

$$x(k+1) = x(k)^2$$

All equilibrium states are given by:

$$(x^e)^2 = x^e$$

Solving yields

$$x^e = 0, 1$$

Example 9 The discrete unattached mass

$$\begin{aligned} x_1^e + T x_2^e &= x_1^e \\ x_2^e &= x_2^e \end{aligned}$$

$$x^e = \begin{pmatrix} x_1^e \\ 0 \end{pmatrix}$$

where x_1^e is arbitrary.

Example 10 (AIMD algorithm) Recalling (2.2) we see that an equilibrium share vector w must satisfy

$$w_i = \beta_i w_i + \gamma_i(1-\beta_1)w_1 + \cdots + \gamma_i(1-\beta_n)w_n \quad \text{for } i = 1, \dots, n. \quad (2.11)$$

Letting

$$\eta_i = \frac{(1-\beta_i)w_i}{\gamma_i}$$

above equations simplify to

$$\eta_i = \gamma_1 \eta_1 + \cdots + \gamma_n \eta_n \quad \text{for } i = 1, \dots, n.$$

Thus every solution must satisfy

$$\eta_i = K \quad \text{for } i = 1, \dots, n.$$

for some K . Since $\gamma_1 + \cdots + \gamma_n = 1$, K can be any real number. Hence

$$w_i = K \frac{\gamma_i}{1-\beta_i} \quad \text{for } i = 1, \dots, n.$$

Considering $w_1 + \cdots + w_n = C$, the constant K is uniquely given by

$$K = \frac{C}{\frac{\gamma_1}{1-\beta_1} + \cdots + \frac{\gamma_n}{1-\beta_n}}.$$

Controlled equilibrium states

Consider now a system with inputs described by

$$x(k+1) = F(x(k), u(k)) \quad (2.12)$$

Suppose the input is constant and equal to u^e , that is, $u(k) \equiv u^e$. Then the resulting system is described by

$$x(k+1) = F(x(k), u^e)$$

The equilibrium states x^e of this system are given by $x^e = F(x^e, u^e)$. This leads to the following definition.

A state x^e is a controlled equilibrium state of the system, $x(k+1) = F(x(k), u(k))$, if there is a constant input u^e such that

$$\boxed{F(x^e, u^e) = x^e}$$

2.7.3 Exercises

Exercise 9 Find all equilibrium states of the following systems

- (i) The first nonlinear system.
- (ii) The attitude dynamics system.
- (iii) The two link manipulator.

2.8 Numerical simulation

2.8.1 MATLAB

```
>> help ode23
```

ODE23 Solve differential equations, low order method.

ODE23 integrates a system of ordinary differential equations using 2nd and 3rd order Runge-Kutta formulas.

[T,Y] = ODE23('yprime', T0, Tfinal, Y0) integrates the system of ordinary differential equations described by the M-file YPRIME.M, over the interval T0 to Tfinal, with initial conditions Y0.

[T, Y] = ODE23(F, T0, Tfinal, Y0, TOL, 1) uses tolerance TOL and displays status while the integration proceeds.

INPUT:

F - String containing name of user-supplied problem description.

Call: yprime = fun(t,y) where F = 'fun'.

t - Time (scalar).

y - Solution column-vector.

yprime - Returned derivative column-vector; yprime(i) = dy(i)/dt.

t0 - Initial value of t.

tfinal- Final value of t.

y0 - Initial value column-vector.

tol - The desired accuracy. (Default: tol = 1.e-3).

trace - If nonzero, each step is printed. (Default: trace = 0).

OUTPUT:

T - Returned integration time points (column-vector).

Y - Returned solution, one solution column-vector per tout-value.

The result can be displayed by: plot(tout, yout).

See also ODE45, ODEDEMO.

```
>> help ode45
```

ODE45 Solve differential equations, higher order method.
ODE45 integrates a system of ordinary differential equations using 4th and 5th order Runge-Kutta formulas.
[T,Y] = ODE45('yprime', T0, Tfinal, Y0) integrates the system of ordinary differential equations described by the M-file YPRIME.M, over the interval T0 to Tfinal, with initial conditions Y0.
[T, Y] = ODE45(F, T0, Tfinal, Y0, TOL, 1) uses tolerance TOL and displays status while the integration proceeds.

INPUT:

F - String containing name of user-supplied problem description.
Call: yprime = fun(t,y) where F = 'fun'.
t - Time (scalar).
y - Solution column-vector.
yprime - Returned derivative column-vector; yprime(i) = dy(i)/dt.
t0 - Initial value of t.
tfinal - Final value of t.
y0 - Initial value column-vector.
tol - The desired accuracy. (Default: tol = 1.e-6).
trace - If nonzero, each step is printed. (Default: trace = 0).

OUTPUT:

T - Returned integration time points (column-vector).
Y - Returned solution, one solution column-vector per tout-value.

The result can be displayed by: plot(tout, yout).

See also ODE23, ODEDEMO.

2.8.2 Simulink

2.9 Exercises

Exercise 10 Obtain a state-space description of the following system.

$$\begin{aligned}\ddot{q}_1 + \dot{q}_2 + q_1 &= 0 \\ \ddot{q}_1 - \dot{q}_2 + q_2 &= 0\end{aligned}$$

Exercise 11 **Exercise 12** By appropriate definition of state variables, obtain a first order state space description of the following systems where q_1, q_2 are real scalars.

(i)

$$\begin{aligned}3\ddot{q}_1 - \ddot{q}_2 + 2\dot{q}_1 + 4q_2 &= 0 \\ -\ddot{q}_1 + 3\ddot{q}_2 + 2\dot{q}_1 - 4q_2 &= 0\end{aligned}$$

(ii)

$$\begin{aligned}3\ddot{q}_1 - \dot{q}_2 + 4q_2 - 4q_1 &= 0 \\ -\ddot{q}_1 + 3\dot{q}_2 - 4q_2 + 4q_1 &= 0\end{aligned}$$

Obtain a state-space description of the following single-input single-output system with input u and output y .

$$\begin{aligned}\ddot{q}_1 + \dot{q}_2 + q_1 &= 0 \\ \ddot{q}_1 - \dot{q}_2 + q_2 &= u \\ y &= \ddot{q}_1\end{aligned}$$

Obtain a state-space description of the following system.

$$\begin{aligned}\ddot{q}_1 + \dot{q}_2 + \sin q_1 &= u \\ \dot{q}_2 + q_1 + q_2 &= 0 \\ y &= q_1 + q_2\end{aligned}$$

Exercise 13 Recall the ‘two link manipulator’ example in the notes.

- (i) By appropriate definition of state variables, obtain a first order state space description of this system.
- (ii) Find all equilibrium states.
- (iii) Numerically simulate this system using MATLAB. Use the following data and initial conditions:

$$\begin{aligned}q_1(0) &= \frac{\pi}{2}; & \dot{q}_1(0) &= 0 \\ q_2(0) &= \frac{\pi}{4}; & \dot{q}_2(0) &= 0\end{aligned}$$

m_1	l_1	lc_1	I_1	m_2	l_2	lc_2	I_2	$m_{payload}$
kg	m	m	$kg.m^2$	kg	m	m	$kg.m^2$	kg
10	1	0.5	10/12	5	1	0.5	5/12	0

Exercise 14 Recall the two pendulum cart example in the notes.

- (a) Obtain all equilibrium configurations corresponding to $u = 0$.
- (b) Consider the following parameter sets

	m_0	m_1	m_2	l_1	l_2	g
P1	2	1	1	1	1	1
P2	2	1	1	1	0.99	1
P3	2	1	0.5	1	1	1
P4	2	1	1	1	0.5	1

and initial conditions,

	y	θ_1	θ_2	$\dot{y}$	$\dot{\theta}_1$	$\dot{\theta}_2$
IC1	0	-10°	10°	0	0	0
IC2	0	10°	10°	0	0	0
IC3	0	-90°	90°	0	0	0
IC4	0	-90.01°	90°	0	0	0
IC5	0	100°	100°	0	0	0
IC6	0	100.01°	100°	0	0	0
IC7	0	179.99°	0°	0	0	0

Simulate the system using the following combinations:

- $P1 :$ $IC1, IC2, IC3, IC4, IC5, IC6, IC7$
- $P2 :$ $IC1, IC2, IC3, IC5, IC7$
- $P3 :$ $IC1, IC2, IC5, IC7$
- $P4 :$ $IC1, IC2, IC3, IC4, IC5, IC6, IC7$

Chapter 3

Linear time-invariant (LTI) systems

To discuss linear systems, we need matrices. In this section we briefly review some properties of matrices. All the definitions and results of this section are stated for real scalars and matrices. However, they also hold for complex scalars and matrices; to obtain the results for the complex case, simply replace ‘real’ with ‘complex’ and $\mathbb{R}$ with $\mathbb{C}$.

3.1 Matrices

An $m \times n$ matrix A is an array of scalars consisting of m rows and n columns.

$$A = \underbrace{\left(\begin{array}{cccc} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{array} \right)}_{n \text{ columns}} \left. \vphantom{\begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix}} \right\} m \text{ rows}$$

The scalars a_{ij} are called the **elements** of A . If the scalars are real numbers, A is called a real matrix. The set of real $m \times n$ matrices is denoted by $\mathbb{R}^{m \times n}$.

Matrix addition: $A + B$

Multiplication of a matrix by a scalar: αA

Zero matrices: 0

Negative of the matrix A : $-A$

One can readily show that $\mathbb{R}^{m \times n}$ with the usual definitions of addition and scalar multiplication is a *vector space*.

Matrix multiplication: AB

Some properties:

$$(AB)C = A(BC) \quad (\text{associative})$$

$$\text{In general } AB \neq BA \quad (\text{non-commutative})$$

$$\begin{aligned} A(B+C) &= AB+AC \\ (B+C)A &= BA+CA \end{aligned}$$

$$\begin{aligned} A(\alpha B) &= \alpha AB \\ (\alpha A)B &= \alpha AB \end{aligned}$$

$n \times n$ identity matrix

$$I := \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} = \text{diag } (1, 1, \dots, 1)$$

$$AI = IA = A$$

Inverse of a square ($n = m$) matrix A : A^{-1}

$$AA^{-1} = A^{-1}A = I$$

Some properties:

$$\begin{aligned} (\alpha A)^{-1} &= \frac{1}{\alpha} A^{-1} \\ (AB)^{-1} &= B^{-1}A^{-1} \end{aligned}$$

Transpose of A : A^T

Some properties:

$$\begin{aligned} (A+B)^T &= A^T + B^T \\ (\alpha A)^T &= \alpha A^T \\ (AB)^T &= B^T A^T \end{aligned}$$

Determinant of a square matrix A : Associated with any square matrix A is a scalar called the **determinant** of A and is denoted by $\det A$.

Some properties:

$$\begin{aligned}\det(AB) &= \det(A) \det(B) \\ \det(A^T) &= \det(A)\end{aligned}$$

Note that, in general,

$$\begin{aligned}\det(A+B) &\neq \det(A) + \det(B) \\ \det(\alpha A) &\neq \alpha \det(A)\end{aligned}$$

The following fact provides us with a very useful property of determinants.

Fact: *A square matrix A is invertible if and only if it has a non-zero determinant.*

Exercise 15 Compute the determinant and inverse of the following matrix.

$$\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

Exercise 16 Prove that if A is invertible then,

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

MATLAB. *Representation of real and complex matrices*

Addition and multiplication of matrices

Multiplication of a matrix by a scalar

Matrix powers

Transpose of a matrix

Inverse of a matrix

```
>> help zeros
```

```
ZEROS All zeros.
```

```
ZEROS(N) is an N-by-N matrix of zeros.
```

```
ZEROS(M,N) or ZEROS([M,N]) is an M-by-N matrix of zeros.
```

```
ZEROS(SIZE(A)) is the same size as A and all zeros.
```

```
>> help eye
```

```
EYE Identity matrix.
```

```
EYE(N) is the N-by-N identity matrix.
```

```
EYE(M,N) or EYE([M,N]) is an M-by-N matrix with 1's on  
the diagonal and zeros elsewhere.
```

```
EYE(SIZE(A)) is the same size as A.
```

```
>> help det
```

```
DET      Determinant.  
DET(X) is the determinant of the square matrix X.
```

```
>> help inv
```

```
INV      Matrix inverse.  
INV(X) is the inverse of the square matrix X.  
A warning message is printed if X is badly scaled or  
nearly singular.
```

Exercise 17 Determine which of the following matrices are invertible. Obtain the inverses of those which are. Check your answers using MATLAB.

$$(a) \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad (b) \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad (c) \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad (d) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Exercise 18 Determine which of the following matrices are invertible. Using MATLAB, check your answers and determine the inverse of those that are invertible.

$$(a) \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \quad (b) \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \quad (c) \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Powers and polynomials of a matrix. Suppose A is a square matrix. We can define powers of a matrix as follows:

$$\begin{aligned} A^0 &:= I \\ A^1 &:= A \\ A^2 &:= AA \\ &\vdots \\ A^{k+1} &:= AA^k \end{aligned}$$

We can also define **polynomials of a matrix** as follows. If

$$p(s) = a_0 + a_1s + a_2s^2 + \dots + a_ms^m$$

where $a_0, \dots, a_m$ are scalars, then,

$$p(A) := a_0I + a_1A + a_2A^2 + \dots + a_mA^m$$

Exercise 19 Compute $p(A)$ for the following polynomial-matrix pairs. Check your answers using MATLAB.

(a)

$$p(s) = s^3 - 3s^2 + 3s - 1 \quad A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

(b)

$$p(s) = s^2 - 3s \quad A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

(c)

$$p(s) = s^3 + s^2 + s + 1 \quad A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

(d)

$$p(s) = s^3 - 2s \quad A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Linear maps. Suppose $\mathcal{V}, \mathcal{W}$ are two real vector spaces and $\mathcal{L}$ is a function from $\mathcal{V}$ to $\mathcal{W}$, that is, $\mathcal{L} : \mathcal{V} \rightarrow \mathcal{W}$.

DEFN. $\mathcal{L}$ is a linear map (function) if

$$\mathcal{L}(\alpha x + \beta y) = \alpha \mathcal{L}(x) + \beta \mathcal{L}(y)$$

for all pairs of vectors x, y in $\mathcal{V}$ and scalars α, β .

Example 11 Consider the map $\mathcal{L} : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by

$$\mathcal{L}(x) = \begin{pmatrix} 2x_2 \\ x_1 \\ x_1 + x_2 \end{pmatrix}$$

Example 12 Suppose A is a real $m \times n$ matrix. If we regard each $x \in \mathbb{R}^n$ as an $n \times 1$ matrix we can define a map $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ by

$$\mathcal{L}(x) = Ax$$

It can be readily shown that this is a linear map: Consider any pair of vectors x, y in $\mathbb{R}^n$ and any pair of scalars α, β . Then using the properties of matrix addition and multiplication we have:

$$\begin{aligned} \mathcal{L}(\alpha x + \beta y) &= A(\alpha x + \beta y) \\ &= A(\alpha x) + A(\beta y) \\ &= \alpha Ax + \beta Ay \\ &= \alpha \mathcal{L}(x) + \beta \mathcal{L}(y) \end{aligned}$$

- **Fact:** Every linear map $\mathcal{L}$ from $\mathbb{R}^n$ to $\mathbb{R}^m$ can be expressed uniquely as

$$\mathcal{L}(x) = Ax$$

where A is a real $m \times n$ matrix.

Example 13 As a more abstract example of a linear map, let $\mathcal{V}$ be the set of differentiable real-valued functions which are defined on the interval $[0, \infty)$. Let $\mathcal{L}$ be the map defined by

$$\mathcal{L}(x) = \dot{x}.$$

Then $\mathcal{L}$ is a linear map.

Exercise 20 Suppose A and B are fixed real $n \times n$ matrices and consider the map $\mathcal{L}$ from $\mathbb{R}^{n \times n}$ to $\mathbb{R}^{n \times n}$ given by

$$\mathcal{L}(P) = PA + BP$$

for any $n \times n$ matrix P . Show that $\mathcal{L}$ is a linear map.

Exercise 21 Suppose A and B are fixed $n \times n$ matrices and consider the map $\mathcal{L}$ from $\mathbb{R}^{n \times n}$ to $\mathbb{R}^{n \times n}$ given by

$$\mathcal{L}(P) = APB$$

for any $n \times n$ matrix P . Show that $\mathcal{L}$ is a linear map.

- If we define the addition of two maps $\mathcal{L}$ and $\mathcal{J}$ by

$$(\mathcal{L} + \mathcal{J})(x) = \mathcal{L}(x) + \mathcal{J}(x)$$

and we define scalar multiplication of a map $\mathcal{L}$ by a scalar α by

$$(\alpha\mathcal{L})(x) = \alpha\mathcal{L}(x)$$

then, the set of linear maps from $\mathcal{V}$ to $\mathcal{W}$ constitute a *vector space*.

3.2 Linear time-invariant systems

3.2.1 Continuous-time

A (finite-dimensional) continuous-time, linear time-invariant (LTI) input-output system can be described by

$$\begin{cases} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{cases}$$

where the n -vector $x(t)$ is the **state vector** at time t , the m -vector $u(t)$ is the **input vector** at time t , and the p -vector $y(t)$ is the **output vector** at time t . The $n \times n$ matrix A is sometimes called the **system matrix**. The matrices B , C , and D have dimensions $n \times m$, $p \times n$, and $p \times m$, respectively. Thus, an LTI system is completely specified by specifying the matrices A, B, C, D .

In scalar terms, this system is described by:

$$\begin{aligned} \dot{x}_1 &= a_{11}x_1 + \dots + a_{1n}x_n + b_{11}u_1 + \dots + b_{1m}u_m \\ &\vdots \\ \dot{x}_n &= a_{n1}x_1 + \dots + a_{nn}x_n + b_{n1}u_1 + \dots + b_{nm}u_m \end{aligned}$$

and

$$\begin{aligned} y_1 &= c_{11}x_1 + \dots + c_{1n}x_n + d_{11}u_1 + \dots + d_{1m}u_m \\ &\vdots \\ y_p &= c_{p1}x_1 + \dots + c_{pn}x_n + d_{p1}u_1 + \dots + d_{pm}u_m. \end{aligned}$$

Example 14 The unattached mass

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1/m \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad D = 0.$$

Example 15 Spring-mass-damper

$$A = \begin{pmatrix} 0 & 1 \\ -k/m & -c/m \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad C = \begin{pmatrix} -k & -c \end{pmatrix}, \quad D = 0.$$

Example 16 Axisymmetric spacecraft

Suppose we are interested in the attitude dynamics of an axisymmetric spacecraft. Recalling the equations which describe the rotational motion of a rigid body and choosing principal axes so that the 3-axis corresponds to an axis of symmetry, we have $I_1 = I_2$; hence, the rotational motion of an axisymmetric spacecraft can be described by

$$\begin{aligned} I_1 \dot{\omega}_1 &= (I_1 - I_3)\omega_2\omega_3 \\ I_1 \dot{\omega}_2 &= (I_3 - I_1)\omega_3\omega_1 \\ I_3 \dot{\omega}_3 &= 0 \end{aligned}$$

From the third equation above, ω_3 is constant. If we introduce the constant

$$\beta := (I_1 - I_3)\omega_3/I_1$$

and define state variables

$$x_1 := \omega_1 \quad x_2 := \omega_2$$

we obtain

$$\begin{aligned} \dot{x}_1 &= \beta x_2 \\ \dot{x}_2 &= -\beta x_1 \end{aligned}$$

This is a LTI system with

$$A = \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix}$$

Other systems. It should be mentioned that the results of this section and most of these notes only apply to **finite-dimensional systems**, that is, systems whose state space is finite-dimensional. As an example of a system not included here, consider a system with a **time delay** T described by

$$\begin{aligned} \dot{q}(t) &= A_0 q(t) + A_1 q(t-T) + Bu(t) \\ y(t) &= Cq(t) + Du(t) \end{aligned}$$

As another example, consider any system described by a partial differential equation, for example

$$\frac{\partial^2 q}{\partial t^2} + k^2 \frac{\partial^4 q}{\partial \eta^4} = 0.$$

3.2.2 Discrete-time

A discrete-time linear time-invariant (LTI) input-output system can be described by

$$\boxed{\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}}$$

where the n -vector $x(k)$ is the **state vector** at time k , the m -vector $u(k)$ is the **input vector** at time k , and the p -vector $y(k)$ is the **output vector** at time k . The $n \times n$ matrix A is sometimes called the **system matrix**. The matrices B , C , and D have dimensions $n \times m$, $p \times n$, and $p \times m$, respectively.

In scalar terms:

$$\begin{aligned} x_1(k+1) &= a_{11}x_1(k) + \dots + a_{1n}x_n(k) + b_{11}u_1(k) + \dots + b_{1m}u_m(k) \\ &\vdots \\ x_n(k+1) &= a_{n1}x_1(k) + \dots + a_{nn}x_n(k) + b_{n1}u_1(k) + \dots + b_{nm}u_m(k) \end{aligned}$$

and

$$\begin{aligned} y_1 &= c_{11}x_1 + \dots + c_{1n}x_n + d_{11}u_1 + \dots + d_{1m}u_m \\ &\vdots \\ y_p &= c_{p1}x_1 + \dots + c_{pn}x_n + d_{p1}u_1 + \dots + d_{pm}u_m \end{aligned}$$

Example 17 The discrete unattached mass

$$A = \begin{pmatrix} 1 & T \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} T^2/2m \\ T/m \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad D = 0.$$

Example 18 AIMD This is a system with no inputs and

$$A = \begin{pmatrix} \beta_1 + \gamma_1(1-\beta_1) & \gamma_1(1-\beta_2) & \dots & \gamma_1(1-\beta_n) \\ \gamma_2(1-\beta_1) & \beta_2 + \gamma_2(1-\beta_2) & \dots & \gamma_2(1-\beta_n) \\ \vdots & & \ddots & \vdots \\ \gamma_n(1-\beta_1) & \gamma_n(1-\beta_2) & \dots & \beta_n + \gamma_n(1-\beta_n) \end{pmatrix}$$

3.3 Linearization about an equilibrium solution

In this section we present systematic procedures for approximating a nonlinear system by a linear system. These procedures rely heavily on derivatives.

3.3.1 Derivative as a matrix

Consider an m -vector-valued function f of an n -vector variable x . Suppose that each of the partial derivatives, $\frac{\partial f_i}{\partial x_j}(x^*)$ exist and are continuous about some x^* .

The derivative of f (Jacobian of f) at x^* is the following $m \times n$ matrix:

$$Df(x^*) = \frac{\partial f}{\partial x}(x^*) := \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(x^*) & \frac{\partial f_1}{\partial x_2}(x^*) & \dots & \frac{\partial f_1}{\partial x_n}(x^*) \\ \frac{\partial f_2}{\partial x_1}(x^*) & \frac{\partial f_2}{\partial x_2}(x^*) & \dots & \frac{\partial f_2}{\partial x_n}(x^*) \\ \vdots & \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1}(x^*) & \frac{\partial f_m}{\partial x_2}(x^*) & \dots & \frac{\partial f_m}{\partial x_n}(x^*) \end{pmatrix}.$$

So,

$$Df(x^*)_{ij} = \frac{\partial f_i}{\partial x_j}(x^*),$$

that is, *the ij -th element of Df is the partial derivative of f_i with respect to x_j* . Sometimes Df is written as $\frac{\partial f}{\partial x}$.

Example 19 If

$$f(x) = \begin{pmatrix} x_1 x_2 \\ \cos(x_1) \\ e^{x_2} \end{pmatrix} .$$

then

$$Df(x) = \begin{pmatrix} x_2 & x_1 \\ -\sin(x_1) & 0 \\ 0 & e^{x_2} \end{pmatrix} .$$

Some properties:

$$\begin{aligned} D(f+g) &= Df + Dg \\ D(\alpha f) &= \alpha Df \quad (\alpha \text{ is a constant scalar}) \\ \text{(Constant function:)} \quad f(x) &\equiv c & Df(x) &= 0 \\ \text{(Linear function:)} \quad f(x) &= Ax & Df(x) &= A \end{aligned}$$

A fundamental property of the derivative. All linearization is based on the following fundamental property of the derivative. If x^* and x are any two vectors and x is ‘close’ to x^* , then

$$\boxed{f(x) \approx f(x^*) + \frac{\partial f}{\partial x}(x^*)(x - x^*)}$$

3.3.2 Linearization of continuous-time systems

Consider a continuous-time system described by

$$\begin{aligned} \dot{x} &= F(x, u) \\ y &= H(x, u) \end{aligned} \tag{3.1}$$

and suppose x^e is an equilibrium state corresponding to a constant input u^e ; hence,

$$F(x^e, u^e) = 0 .$$

Let y^e be the corresponding constant output, that is,

$$y^e = H(x^e, u^e) .$$

Sometimes we call the triple (x^e, u^e, y^e) a **trim condition** of the system.

Introducing the **perturbed state** δx , the **perturbed input** δu and the **perturbed output** δy defined by

$$\boxed{\delta x := x - x^e, \quad \delta u := u - u^e, \quad \delta y := y - y^e}$$

the system can also be described by

$$\begin{aligned} \delta \dot{x} &= F(x^e + \delta x, u^e + \delta u) \\ \delta y &= H(x^e + \delta x, u^e + \delta u) - H(x^e, u^e) \end{aligned}$$

Define now the following derivative matrices:

$$\frac{\partial F}{\partial x}(x^e, u^e) := \left(\frac{\partial F_i}{\partial x_j}(\ast) \right) = \begin{pmatrix} \frac{\partial F_1}{\partial x_1}(\ast) & \frac{\partial F_1}{\partial x_2}(\ast) & \dots & \frac{\partial F_1}{\partial x_n}(\ast) \\ \frac{\partial F_2}{\partial x_1}(\ast) & \frac{\partial F_2}{\partial x_2}(\ast) & \dots & \frac{\partial F_2}{\partial x_n}(\ast) \\ \vdots & \vdots & & \vdots \\ \frac{\partial F_n}{\partial x_1}(\ast) & \frac{\partial F_n}{\partial x_2}(\ast) & \dots & \frac{\partial F_n}{\partial x_n}(\ast) \end{pmatrix}$$

$$\frac{\partial F}{\partial u}(x^e, u^e) := \left(\frac{\partial F_i}{\partial u_j}(\ast) \right) = \begin{pmatrix} \frac{\partial F_1}{\partial u_1}(\ast) & \frac{\partial F_1}{\partial u_2}(\ast) & \dots & \frac{\partial F_1}{\partial u_m}(\ast) \\ \frac{\partial F_2}{\partial u_1}(\ast) & \frac{\partial F_2}{\partial u_2}(\ast) & \dots & \frac{\partial F_2}{\partial u_m}(\ast) \\ \vdots & \vdots & & \vdots \\ \frac{\partial F_n}{\partial u_1}(\ast) & \frac{\partial F_n}{\partial u_2}(\ast) & \dots & \frac{\partial F_n}{\partial u_m}(\ast) \end{pmatrix}$$

$$\frac{\partial H}{\partial x}(x^e, u^e) := \left(\frac{\partial H_i}{\partial x_j}(\ast) \right) = \begin{pmatrix} \frac{\partial H_1}{\partial x_1}(\ast) & \frac{\partial H_1}{\partial x_2}(\ast) & \dots & \frac{\partial H_1}{\partial x_n}(\ast) \\ \frac{\partial H_2}{\partial x_1}(\ast) & \frac{\partial H_2}{\partial x_2}(\ast) & \dots & \frac{\partial H_2}{\partial x_n}(\ast) \\ \vdots & \vdots & & \vdots \\ \frac{\partial H_p}{\partial x_1}(\ast) & \frac{\partial H_p}{\partial x_2}(\ast) & \dots & \frac{\partial H_p}{\partial x_n}(\ast) \end{pmatrix}$$

$$\frac{\partial H}{\partial u}(x^e, u^e) := \left(\frac{\partial H_i}{\partial u_j}(\ast) \right) = \begin{pmatrix} \frac{\partial H_1}{\partial u_1}(\ast) & \frac{\partial H_1}{\partial u_2}(\ast) & \dots & \frac{\partial H_1}{\partial u_m}(\ast) \\ \frac{\partial H_2}{\partial u_1}(\ast) & \frac{\partial H_2}{\partial u_2}(\ast) & \dots & \frac{\partial H_2}{\partial u_m}(\ast) \\ \vdots & \vdots & & \vdots \\ \frac{\partial H_p}{\partial u_1}(\ast) & \frac{\partial H_p}{\partial u_2}(\ast) & \dots & \frac{\partial H_p}{\partial u_m}(\ast) \end{pmatrix}$$

where $\ast = (x^e, u^e)$. When x is ‘close’ to x^e and u is ‘close’ to u^e , that is, when δx and δu are ‘small’,

$$\begin{aligned} F(x^e + \delta x, u^e + \delta u) &\approx F(x^e, u^e) + \frac{\partial F}{\partial x}(x^e, u^e)\delta x + \frac{\partial F}{\partial u}(x^e, u^e)\delta u \\ &= \frac{\partial F}{\partial x}(x^e, u^e)\delta x + \frac{\partial F}{\partial u}(x^e, u^e)\delta u \end{aligned}$$

and

$$H(x^e + \delta x, u^e + \delta u) \approx H(x^e, u^e) + \frac{\partial H}{\partial x}(x^e, u^e) \delta x + \frac{\partial H}{\partial u}(x^e, u^e) \delta u$$

Hence

$$\begin{aligned} \delta \dot{x} &\approx \frac{\partial F}{\partial x}(x^e, u^e) \delta x + \frac{\partial F}{\partial u}(x^e, u^e) \delta u \\ \delta y &\approx \frac{\partial H}{\partial x}(x^e, u^e) \delta x + \frac{\partial H}{\partial u}(x^e, u^e) \delta u \end{aligned}$$

This leads to the following definition:

The linearization of system (3.1) about (x^e, u^e) :

$$\boxed{\begin{aligned} \delta \dot{x} &= A \delta x + B \delta u \\ \delta y &= C \delta x + D \delta u \end{aligned}} \quad (3.2)$$

where

$$A = \frac{\partial F}{\partial x}(x^e, u^e), \quad B = \frac{\partial F}{\partial u}(x^e, u^e), \quad C = \frac{\partial H}{\partial x}(x^e, u^e), \quad D = \frac{\partial H}{\partial u}(x^e, u^e).$$

Example 20 (Simple pendulum.) The linearization of this system about any (x^e, u^e) is given by (3.2) with

$$A = \begin{pmatrix} 0 & 1 \\ -\cos(x_1^e) & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 0 \end{pmatrix} \quad D = 0$$

For eqm. state

$$x^e = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

we obtain

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

For eqm. state

$$x^e = \begin{pmatrix} \pi \\ 0 \end{pmatrix}$$

we obtain

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

3.3.3 Implicit linearization

This is the way to go when starting from a higher order ODE description.

It is usually easier to first linearize and then obtain a state space description than vice-versa.

For a single equation,

$$F(q, \dot{q}, \dots, q^{(n)}, u) = 0$$

linearization about an equilibrium solution $q(t) \equiv q^e$ corresponding to constant input $u(t) \equiv u^e$ is defined by:

$$\frac{\partial F}{\partial q}(\ast)\delta q + \frac{\partial F}{\partial \dot{q}}(\ast)\delta \dot{q} + \dots + \frac{\partial F}{\partial q^{(n)}}(\ast)\delta q^{(n)} + \frac{\partial F}{\partial u}(\ast)\delta u = 0$$

where $(\ast) = (q^e, 0, 0, \dots, 0)$, $\delta q = q - q^e$ and $\delta u = u - u^e$.

For multiple equations, see next example.

Example 21 Orbit mechanics

$$\begin{aligned} \ddot{r} - r\omega^2 + \mu/r^2 &= 0 \\ r\dot{\omega} + 2\dot{r}\omega &= 0 \end{aligned}$$

Linearization about equilibrium solutions corresponding to constant values r^e, ω^e of r and ω , respectively, results in:

$$\begin{aligned} \delta\ddot{r} - (\omega^e)^2\delta r - (2r^e\omega^e)\delta\omega - (2\mu/(r^e)^3)\delta r &= 0 \\ \dot{\omega}^e\delta r + r^e\delta\dot{\omega} + 2\omega^e\delta\dot{r} + 2\dot{r}^e\delta\omega &= 0 \end{aligned}$$

Using the relationship

$$(\omega^e)^2 = \mu/(r^e)^3$$

we obtain

$$\begin{aligned} \delta\ddot{r} - 3(\omega^e)^2\delta r - (2r^e\omega^e)\delta\omega &= 0 \\ r^e\delta\dot{\omega} + 2\omega^e\delta\dot{r} &= 0 \end{aligned}$$

Introducing state variables:

$$x_1 := \delta r, \quad x_2 := \delta\dot{r} \quad x_3 := \delta\omega$$

we obtain a LTI system with system matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 3(\omega^e)^2 & 0 & 2r^e\omega^e \\ 0 & -2\omega^e/r^e & 0 \end{pmatrix}$$

Example: vehicle lateral dynamics

Consider a vehicle moving in a horizontal plane. We can describe the motion of this vehicle with the velocity $\bar{v}$ of its center of gravity (CG) and its yaw rate $\dot{\psi}$. The velocity of the mass center can be described by its magnitude v (the speed of the CG) and the vehicle sideslip angle β which is the angle between $\bar{v}$ and the vehicle. We will use these three variables to describe the state of the vehicle. As inputs we consider the steering angle δ of the front wheels and the braking and acceleration forces F_{fx} and F_{rx} on the front and rear wheels of the vehicle.

Application of Newton's second law to the vehicle mass center results in

$$m\dot{v} = F_{rx} \cos \beta + F_{ry} \sin \beta + F_{fx} \cos(\beta - \delta) + F_{fy} \sin(\beta - \delta) \quad (3.3a)$$

$$mv(\dot{\beta} + \dot{\psi}) = -F_{rx} \sin \beta + F_{fy} \cos \beta - F_{fx} \sin(\beta - \delta) + F_{ry} \cos(\beta - \delta) \quad (3.3b)$$

where F_{fx}, F_{rx} are the longitudinal forces on the front and rear wheels due to braking and acceleration; F_{fy}, F_{ry} are the lateral forces on the front and rear wheels. Considerations of angular momentum yield

$$J_z \ddot{\psi} = a(F_{fy} \cos \delta + F_{fx} \sin \delta) - bF_{ry} \quad (3.4)$$

where J_z is the moment of inertia of the vehicle about the vertical line through its CG, and a, b are the distances between the mass center and the front and rear of the vehicle.

The lateral force F_{ry} depends on the rear sideslip angle α_r which is the angle between the rear tires and the velocity of the rear of the vehicle. The lateral force F_{fy} depends on the front sideslip angle α_f which is the angle between the front tires and the velocity of the front of the vehicle. Kinematical considerations yield the following relationships:

$$\tan \alpha_r = \tan \beta - \frac{b\dot{\psi}}{v \cos \beta} \quad (3.5a)$$

$$\tan(\alpha_f + \delta) = \tan \beta + \frac{a\dot{\psi}}{v \cos \beta} \quad (3.5b)$$

Assuming the lateral forces are zero for zero sideslip angles, and considering $F_{rx} = F_{fx} = \delta = 0$, the vehicle has an equilibrium with $\beta = \dot{\psi} = \alpha_r = \alpha_f = 0$ and v an arbitrary constant.

Linearization of the nonlinear equations about these equilibrium conditions results in

$$\begin{aligned} \delta\dot{v} &= a \\ mv(\dot{\beta} + \dot{\psi}) &= -C_{\alpha f}\alpha_f - C_{\alpha r}\alpha_r \\ J_z \ddot{\psi} &= -aC_{\alpha f}\alpha_f + bC_{\alpha r}\alpha_r \\ \alpha_r &= \beta - \frac{b\dot{\psi}}{v} \\ \alpha_f &= \beta + \frac{a\dot{\psi}}{v} - \delta \end{aligned}$$

where $a = (F_{rx} + F_{fx})/m$ and

$$C_{\alpha f} = -\frac{dF_{yf}}{d\alpha_f} \quad \text{and} \quad C_{\alpha r} = -\frac{dF_{yr}}{d\alpha_r}$$

are the front and rear “axle stiffnesses”.

Notice that the dynamics of δv can be decoupled from that of $(\beta, \dot{\psi})$. With $x = (\beta, \dot{\psi})$ and $u = \delta$, the lateral vehicle dynamics can be described by $\dot{x} = Ax + Bu$ where

$$A = \begin{pmatrix} \frac{-(C_{\alpha f} + C_{\alpha r})}{mv} & \frac{bC_{\alpha r} - aC_{\alpha f}}{mv^2} - 1 \\ \frac{bC_{\alpha r} - aC_{\alpha f}}{J_z} & -\frac{a^2C_{\alpha f} + b^2C_{\alpha r}}{J_z v} \end{pmatrix}, \quad B = \begin{pmatrix} \frac{C_{\alpha f}}{mv} \\ \frac{aC_{\alpha f}}{J_z} \end{pmatrix} \quad (3.6)$$

3.3.4 Linearization of discrete-time systems

Consider a discrete-time system described by

$$\begin{aligned}x(k+1) &= F(x(k), u(k)) \\ y(k) &= H(x(k), u(k))\end{aligned}\tag{3.7}$$

and suppose x^e, u^e, y^e is a trim condition, that is,

$$\begin{aligned}F(x^e, u^e) &= x^e \\ H(x^e, u^e) &= y^e\end{aligned}$$

The linearization of system (3.7) about (x^e, u^e) :

$$\boxed{\begin{aligned}\delta x(k+1) &= A \delta x(k) + B \delta u(k) \\ \delta y(k) &= C \delta x(k) + D \delta u(k)\end{aligned}}\tag{3.8}$$

where

$$A = \frac{\partial F}{\partial x}(x^e, u^e), \quad B = \frac{\partial F}{\partial u}(x^e, u^e), \quad C = \frac{\partial H}{\partial x}(x^e, u^e), \quad D = \frac{\partial H}{\partial u}(x^e, u^e).$$

Example 22 Simple scalar nonlinear system

$$x(k+1) = x(k)^2$$

We have $x^e = 0, 1$. For $x^e = 0$:

$$\delta x(k+1) = 0$$

For $x^e = 1$:

$$\delta x(k+1) = 2\delta x(k)$$

Exercises

Exercise 22 Consider the two-degree-of-freedom spring-mass-damper system described by

$$\begin{aligned} m\ddot{q}_1 + C(\dot{q}_1 - \dot{q}_2) + K(q_1 - q_2) + c\dot{q}_1 + kq_1 &= 0 \\ m\ddot{q}_2 + C(\dot{q}_2 - \dot{q}_1) + K(q_2 - q_1) + c\dot{q}_2 + kq_2 &= 0 \end{aligned}$$

Obtain an A -matrix for a state space description of this system.

Exercise 23 Linearize each of the following systems about the zero equilibrium state.

(i)

$$\begin{aligned} \dot{x}_1 &= (1 + x_1^2)x_2 \\ \dot{x}_2 &= -x_1^3 \end{aligned}$$

(ii)

$$\begin{aligned} \dot{x}_1 &= \sin x_2 \\ \dot{x}_2 &= (\cos x_1)x_3 \\ \dot{x}_3 &= e^{x_1}x_2 \end{aligned}$$

Exercise 24

(a) Obtain all equilibrium states of the following system:

$$\begin{aligned} \dot{x}_1 &= 2x_2(1 - x_1) - x_1 \\ \dot{x}_2 &= 3x_1(1 - x_2) - x_2 \end{aligned}$$

(b) Linearize the above system about the zero equilibrium state.

Exercise 25 For each of the following systems, linearize about each equilibrium solution and obtain the system A -matrix for a state space representation of these linearized systems.

(a)

$$\ddot{q} + (q^2 - 1)\dot{q} + q = 0.$$

where $q(t)$ is a scalar.

(b)

$$\ddot{q} + \dot{q} + q - q^3 = 0$$

where $q(t)$ is a scalar.

(c)

$$\begin{aligned} (M + m)\ddot{q} + ml\ddot{\theta}\cos\theta - ml\dot{\theta}^2\sin\theta + kq &= 0 \\ ml\ddot{q}\cos\theta + ml^2\ddot{\theta} + mgl\sin\theta &= 0 \end{aligned}$$

where $q(t)$ and $\theta(t)$ are scalars.

(d)

$$\ddot{q} + 0.5\dot{q}|\dot{q}| + q = 0.$$

where $q(t)$ is a scalar.

Exercise 26

(a) Obtain all equilibrium solutions of the following system:

$$\begin{aligned} (\cos q_1)\ddot{q}_1 + (\sin q_1)\ddot{q}_2 &+ (\sin q_2)\dot{q}_1^2 + \sin q_2 = 0 \\ -(\sin q_1)\ddot{q}_1 + (\cos q_1)\ddot{q}_2 &+ (\cos q_2)\dot{q}_2^2 + \sin q_1 = 0 \end{aligned}$$

(b) Linearize the above system about its zero solution.

Exercise 27 (Simple pendulum in drag)

Figure 3.1: Pendulum in drag

Recall the simple pendulum in drag whose motion is described by

$$ml\ddot{\theta} + \kappa V(l\dot{\theta} - w \sin \theta) + mg \sin \theta = 0$$

where

$$V = \sqrt{l^2\dot{\theta}^2 + w^2 - 2lw \sin(\theta)\dot{\theta}} \quad \text{with} \quad \kappa = \frac{\rho S C_D}{2}$$

and

g is the gravitational acceleration constant of the earth,

S is the reference area associated with the ball,

C_D is the drag coefficient of the ball,

ρ is the air density.

(a) Obtain the equilibrium values θ^e of θ .

(b) Linearize the system about θ^e .

(c) Obtain an expression for the A matrix for a state space representation of the linearized system.

(d) Compare the behavior of the nonlinear system with that of the linearized system for the following parameters.

$$\begin{aligned} l = 0.1 \text{ m} & \quad m = 10 \text{ grams} & g = 9.81 \text{ m/s}^2 \\ C_D = 0.2 & \quad S = .01 \text{ m}^2 & \rho = 0.3809 \text{ kg/m}^3 \end{aligned}$$

Consider the following cases:

θ^e	w (m/s)
0	0
0	5
0	15
0	20
180°	20

Illustrate your results with time histories of $\delta\theta$ in degrees. Comment on your results.

Chapter 4

Transfer functions

4.1 ← Transfer functions ←

4.1.1 The Laplace transform

A state space description of an input-output system can be regarded as a time-domain description. For linear time-invariant systems, transfer functions provide another way of describing input-output systems; this description is sometimes called a frequency domain description.

To discuss transfer functions of continuous-time systems, we need the **Laplace transform**. Suppose f is any function of the continuous-time variable t . The Laplace transform of f , which we will denote by $\hat{f}$, is a function of a complex variable s and is defined by

$$\hat{f}(s) = \mathcal{L}(f)(s) = \int_0^{\infty} e^{-st} f(t) dt. \quad (4.1)$$

Note that $f(t)$ could be a scalar, a vector or even a matrix.

Example 23 Consider $f(t) = e^{at}$ where a is any scalar. Then,

$$\hat{f}(s) = \int_0^{\infty} e^{-st} e^{at} dt = \int_0^{\infty} e^{(a-s)t} dt = \left. \frac{e^{(a-s)t}}{a-s} \right|_0^{\infty}$$

When, $\Re(s) > a$, we have $\Re(a-s) < 0$ and $\lim_{t \rightarrow \infty} e^{(a-s)t} = 0$. Hence,

$$\mathcal{L}(e^{at}) = \frac{1}{s-a} \quad \text{for} \quad \Re(s) > a.$$

Note that the Laplace transform is a linear operator, that is, for any two functions f and g

$$\mathcal{L}(f+g) = \mathcal{L}(f) + \mathcal{L}(g).$$

Also, if M is any constant matrix of appropriate dimensions:

$$\mathcal{L}(Mf) = M\mathcal{L}(f).$$

We will need the following very useful relationship:

$$\boxed{\mathcal{L}(\dot{f})(s) = s\hat{f}(s) - f(0)} \quad (4.2)$$

The following example illustrates the use of the above relationship.

Example 24 Consider $f(t) = e^{at}$ where a is any scalar. Since f satisfies

$$\dot{f} = af$$

and $f(0) = 1$, we obtain that

$$s\hat{f}(s) - 1 = a\hat{f}(s).$$

Hence, $\hat{f}(s) = 1/(s - a)$.

4.1.2 Transfer functions

Consider now the LTI system

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t) \end{aligned}$$

along with the initial condition

$$x(0) = x_0.$$

Taking the Laplace transform of the state equation and using relationship (4.2) results in

$$s\hat{x}(s) - x_0 = A\hat{x}(s) + B\hat{u}(s)$$

which can be rewritten as

$$(sI - A)\hat{x}(s) = x_0 + B\hat{u}(s)$$

where $\hat{x}$ and $\hat{u}$ are the Laplace transforms of x and u , respectively. When the matrix $sI - A$ is invertible, the above equation may be uniquely solved for $\hat{x}$ yielding

$$\hat{x}(s) = (sI - A)^{-1}x_0 + (sI - A)^{-1}B\hat{u}(s).$$

Taking the Laplace transform of the above output equation yields

$$\hat{y}(s) = C\hat{x}(s) + D\hat{u}(s)$$

where $\hat{y}$ is the Laplace transform of y . Using the expression for $\hat{x}$ now results in

$$\hat{y}(s) = C(sI - A)^{-1}x_0 + \hat{G}(s)\hat{u}(s)$$

where the transfer function (matrix) $\hat{G}$ is defined by

$$\boxed{\hat{G}(s) = C(sI - A)^{-1}B + D}$$

Zero initial conditions response. Suppose that $x(0) = 0$. Then, the relationship between the Laplace transform of the input $\hat{u}$ and the Laplace transform of the output $\hat{y}$ is given by the simple relationship

$$\boxed{\hat{y} = \hat{G}\hat{u}}$$

Graphically, this can be represented by

$$\hat{y} \longleftarrow \boxed{\hat{G}} \longleftarrow \hat{u}$$

For a SISO system, we obtain that, for $\hat{u} \neq 0$,

$$\hat{G} = \frac{\hat{y}}{\hat{u}}.$$

Thus, for a SISO system the transfer function is the ratio of the Laplace transform of the output to the Laplace transform of the input when all initial conditions are zero.

Example 25

$$\begin{aligned}\dot{x}(t) &= -x(t) + u(t) \\ y(t) &= x(t)\end{aligned}$$

Here $A = -1$, $B = 1$, $C = 1$ and $D = 0$; hence

$$\hat{G}(s) = C(sI - A)^{-1} + D = \boxed{\frac{1}{s + 1}}$$

Example 26 Unattached mass

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= u \\ y &= x_1\end{aligned}$$

Here

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad C = (1 \ 0) \quad D = 0$$

Hence

$$\begin{aligned}\hat{G}(s) &= C(sI - A)^{-1}B \\ &= \boxed{\frac{1}{s^2}}\end{aligned}$$

Exercise 28 Show that the transfer function of

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\alpha_0 x_1 - \alpha_1 x_2 + u \\ y &= \beta_0 x_1 + \beta_1 x_2\end{aligned}$$

is given by

$$\hat{G}(s) = \frac{\beta_1 s + \beta_0}{s^2 + \alpha_1 s + \alpha_0}$$

Exercise 29 Show that the transfer function of

$$\begin{aligned}\dot{x}_1 &= -\alpha_0 x_2 + \beta_0 u \\ \dot{x}_2 &= x_1 - \alpha_1 x_2 + \beta_1 u \\ y &= x_2\end{aligned}$$

is given by

$$\hat{G}(s) = \frac{\beta_1 s + \beta_0}{s^2 + \alpha_1 s + \alpha_0}$$

The next example illustrates the following observation which is useful for hand calculation of transfer functions.

It is usually easier to compute the transfer function of a system by taking the Laplace transform of the equations describing the system and solving for $\hat{y}$ in terms of $\hat{u}$ to obtain $\hat{G}$.

This is particularly true if the original description is in terms of higher order ODEs. It does not require one to obtain a state space description.

Example 27 MIMO (Multi-input multi-output) B&B Consider the system illustrated in Figure 4.1 which has two inputs and two outputs. Its motion can be described by

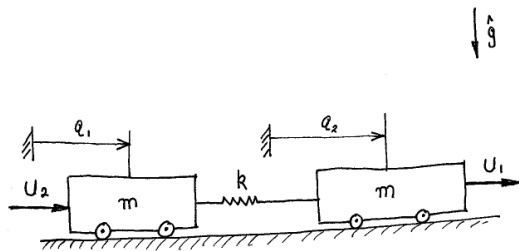


Figure 4.1: B&B

$$\begin{aligned}m\ddot{q}_1 &= k(q_2 - q_1) + u_1 \\ m\ddot{q}_2 &= -k(q_2 - q_1) + u_2 \\ y_1 &= q_1 \\ y_2 &= q_2\end{aligned}$$

With state variable, $x_1 = q_1, x_2 = q_2, x_3 = \dot{q}_1$ and $x_4 = \dot{q}_2$, one can obtain a state space description with

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -k/m & k/m & 0 & 0 \\ k/m & -k/m & 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1/m & 0 \\ 0 & 1/m \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad D = 0$$

Hence

$$\begin{aligned} \hat{G}(s) &= C(sI - A)^{-1}B + D \\ &= \begin{pmatrix} \frac{ms^2 + k}{ms^2(ms^2 + 2k)} & \frac{k}{ms^2(ms^2 + 2k)} \\ \frac{k}{ms^2(ms^2 + 2k)} & \frac{ms^2 + k}{ms^2(ms^2 + 2k)} \end{pmatrix} \end{aligned}$$

Note that the above calculation requires one to invert the 4×4 matrix $sI - A$.

• We can also compute $\hat{G}(s)$ as follows. Taking the Laplace transform of the original second order differential equations with zero initial conditions, we get

$$\begin{aligned} ms^2\hat{q}_1 &= -k(\hat{q}_1 - \hat{q}_2) + \hat{u}_1 \\ ms^2\hat{q}_2 &= k(\hat{q}_1 - \hat{q}_2) + \hat{u}_2 \\ \hat{y}_1 &= \hat{q}_1 \\ \hat{y}_2 &= \hat{q}_2 \end{aligned}$$

hence,

$$\begin{pmatrix} ms^2 + k & -k \\ -k & ms^2 + k \end{pmatrix} \begin{pmatrix} \hat{q}_1 \\ \hat{q}_2 \end{pmatrix} = \begin{pmatrix} \hat{u}_1 \\ \hat{u}_2 \end{pmatrix}$$

Solving, yields

$$\begin{aligned} \hat{y} = \begin{pmatrix} \hat{q}_1 \\ \hat{q}_2 \end{pmatrix} &= \frac{1}{\Delta(s)} \begin{pmatrix} ms^2 + k & k \\ k & ms^2 + k \end{pmatrix} \begin{pmatrix} \hat{u}_1 \\ \hat{u}_2 \end{pmatrix} \\ \Delta(s) &= ms^2(ms^2 + 2k) \end{aligned}$$

So,

$$\hat{G}(s) = \begin{pmatrix} \frac{ms^2 + k}{ms^2(ms^2 + 2k)} & \frac{k}{ms^2(ms^2 + 2k)} \\ \frac{k}{ms^2(ms^2 + 2k)} & \frac{ms^2 + k}{ms^2(ms^2 + 2k)} \end{pmatrix}$$

Using this second method, the matrix to be inverted was only 2×2 .

MATLAB

```
>> help ss2tf
```

```
SS2TF  State-space to transfer function conversion.  
[NUM,DEN] = SS2TF(A,B,C,D,iu)  calculates the transfer function:
```

$$H(s) = \frac{\text{NUM}(s)}{\text{DEN}(s)} = C(sI-A)^{-1} B + D$$

of the system:

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}$$

from the *iu*'th input. Vector DEN contains the coefficients of the denominator in descending powers of *s*. The numerator coefficients are returned in matrix NUM with as many rows as there are outputs *y*.

See also `ltimodels`, `ss` and `tf`.

Rational transfer functions. A scalar valued function of a complex variable is **rational** if it can be expressed as the ratio of two polynomials. It is **proper** if the order of the numerator polynomial is less than or equal to the order of the denominator polynomial. It is **strictly proper** if the order of the numerator polynomial is strictly less than that of the denominator polynomial. A matrix valued function of a complex variable is rational, proper or strictly proper if each element of the matrix function is respectively rational, proper or strictly proper.

Example 28

$$\begin{aligned}\hat{G}(s) &= \frac{s}{s^2+1} && \text{rational, strictly proper} \\ \hat{G}(s) &= \frac{s^2}{s^2+1} && \text{rational, proper} \\ \hat{G}(s) &= \frac{s^3}{s^2+1} && \text{rational, not proper} \\ \hat{G}(s) &= \frac{e^{-s}}{s^2+1} && \text{not rational}\end{aligned}$$

We now demonstrate the following result:

The transfer function of a finite-dimensional LTI system is always rational and proper.

The above result follows from the fact that every element $\hat{G}_{ij}$ of the transfer function of a finite dimensional LTI system can be expressed as

$$\hat{G}_{ij} = \frac{N_{ij}}{\Delta} \quad (4.3)$$

where $\Delta(s) = \det(sI - A)$ is called the **characteristic polynomial** of A . It is a monic polynomial of order n . A monic polynomial is one whose coefficient of its highest order term is one. Thus

$$\det(sI - A) = s^n + \alpha_{n-1}s^{n-1} + \cdots + \alpha_1s + \alpha_0$$

for some scalars $\alpha_0, \cdots, \alpha_{n-1}$. The order of the polynomial $\hat{N}_{ij}$ is at most n . It is less than n if D is zero.

To demonstrate the above fact, recall that the inverse of an invertible $n \times n$ matrix M can be expressed as $M^{-1} = \frac{1}{\det(M)} \text{adj}(M)$ where each element of the matrix $\text{adj}(M)$ is equal to $\pm$ the determinant of an $(n-1) \times (n-1)$ submatrix of M . Hence $(sI - A)^{-1}$ can be expressed as

$$(sI - A)^{-1} = \frac{1}{\det(sI - A)} \text{adj}(sI - A)$$

where each element of the matrix $\text{adj}(sI - A)$ is equal to $\pm$ the determinant of an $(n-1) \times (n-1)$ submatrix of $sI - A$. Hence, if A is $n \times n$, then each element of $\text{adj}(sI - A)$ is a polynomial in s whose degree is at most $n-1$. Also, $\det(sI - A)$ is a polynomial of order n .

From the above form of $(sI - A)^{-1}$ it follows that if

$$\hat{G}(s) = C(sI - A)^{-1}B$$

where A is $n \times n$, then

$$\hat{G}(s) = \frac{1}{\Delta(s)} N(s)$$

where Δ is the n -th order characteristic polynomial of A , that is, $\Delta(s) = \det(sI - A)$ and each element N_{ij} of N is a polynomial of order less than or equal to $n-1$. So,

$$\hat{G}_{ij} = \frac{N_{ij}}{\Delta}$$

that is, each element $\hat{G}_{ij}$ of $\hat{G}$ is a rational function (ratio of two polynomials) and is strictly proper (order of numerator strictly less than order of denominator). When each element of $\hat{G}$ is rational and strictly proper, we say that $\hat{G}$ is rational and strictly proper.

If

$$\hat{G}(s) = C(sI - A)^{-1}B + D$$

then

$$\boxed{\hat{G}(s) = \frac{1}{\Delta(s)} N(s), \quad \Delta(s) = \det(sI - A)} \quad (4.4)$$

where each element N_{ij} of N is a polynomial of order less than or equal to n . So, each element of $\hat{G}$ is a rational function and is **proper** (order of numerator less than or equal to

order of denominator). When each element of $\hat{G}$ is rational and proper, we say that $\hat{G}$ is rational and proper.

So, we see that if $\hat{G}$ is the transfer function of a finite-dimensional LTI system then, $\hat{G}$ is rational and proper. Is the converse true? That is, is every proper rational $\hat{G}$ the transfer function of some finite-dimensional system and if so, what is(are) the system(s). This is answered in the next section.

Invariance of transfer function under state transformations. The transfer function of a linear time-invariant system describes the input output behavior of the system when all initial conditions are zero. Hence, it should be independent of the state variables used in a state space description of the system. We now demonstrate this explicitly.

Suppose $\hat{G}$ is the transfer function of a system described by

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}$$

Then $\hat{G}(s) = C(sI - A)^{-1}B + D$. Consider any state transformation

$$x = T\xi$$

where T is square and invertible. Then, using the new state, the system can also be described by

$$\begin{aligned}\dot{\xi} &= \bar{A}\xi + \bar{B}u \\ y &= \bar{C}\xi + \bar{D}u\end{aligned}$$

where

$$\begin{aligned}\bar{A} &= T^{-1}AT & \bar{B} &= T^{-1}B \\ \bar{C} &= CT & \bar{D} &= D\end{aligned}$$

It can be readily verified that

$$\bar{C}(sI - \bar{A})^{-1}\bar{B} + \bar{D} = C(sI - A)^{-1}B + D$$

that is, the transfer function matrix is unaffected by a state transformation.

General input-output systems. Consider *any linear time-invariant* input-output system described by the convolution integral

$$y(t) = \int_0^t G(t - \tau)u(\tau) d\tau$$

where G is Laplace transformable. Then, using properties of the Laplace transform, one can show that

$$\hat{y} = \hat{G}\hat{u}$$

where $\hat{G}$ is the Laplace transform of G .

Example 29

$$\begin{aligned}\dot{x}(t) &= -x(t) + u(t-1) \\ y(t) &= x(t)\end{aligned}$$

Taking Laplace transform with $x(0) = 0$ yields

$$\begin{aligned}s\hat{x}(s) &= -\hat{x}(s) + e^{-s}\hat{u}(s) \\ \hat{y}(s) &= \hat{x}(s)\end{aligned}$$

Hence $\hat{y}(s) = \hat{G}(s)\hat{u}(s)$ with

$$\hat{G}(s) = \frac{e^{-s}}{s+1}$$

Note that this transfer function is not rational.

Exercise 30 What is the transfer function of the following system?

$$\begin{aligned}\dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= u(t-2) \\ y(t) &= x_1(t)\end{aligned}$$

Exercise 31 Obtain the transfer function of the following system

$$\begin{aligned}\dot{x}(t) &= -x(t) + 2x(t-h) + u(t) \\ y(t) &= x(t)\end{aligned}$$

4.1.3 Poles and zeros

Poles. Suppose $\hat{G}$ is a matrix valued function of a complex variable. A complex number λ is defined to be a **pole** of $\hat{G}$ if for some element $\hat{G}_{ij}$ of $\hat{G}$,

$$\lim_{s \rightarrow \lambda} \hat{G}_{ij}(s) = \infty.$$

Example 30

$\hat{G}(s)$	poles
$\frac{1}{s^2+3s+2}$	-1, -2
$\frac{s+1}{s^2+3s+2}$	-2
$\begin{pmatrix} \frac{1}{s} & \frac{e^{-s}}{s+1} \\ \frac{s^2}{s+1} & \frac{s+1}{s+2} \end{pmatrix}$	0, -1, -2

When $\hat{G}(s) = C(sI - A)^{-1}B + D$ it should be clear from (4.4) that if λ is a pole of $\hat{G}$ then λ must be a root of the characteristic polynomial of A ; we shall see later that this means λ is an eigenvalue of A . However, as the following example illustrates, the converse is not always true.

Example 31 If,

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad C = \begin{pmatrix} -1 & 1 \end{pmatrix} \quad D = 0,$$

then

$$\det(sI - A) = s^2 - 1$$

and

$$\hat{G}(s) = C(sI - A)^{-1}B = \frac{s-1}{s^2-1} = \frac{1}{s+1}.$$

Here Δ has roots -1 and 1 whereas $\hat{G}$ only has a single pole at -1 .

We say that a complex number λ is a **pole of order m** of a transfer function $\hat{G}$ if λ is a pole of $\hat{G}$ and m is the smallest integer such that λ is not a pole of

$$(s - \lambda)^m \hat{G}.$$

Example 32 The transfer function

$$\hat{G}(s) = \frac{s+1}{(s-3)(s+4)^2}$$

has two poles: 3 and -4 . The order of the pole at 3 is 1 whereas the order of the pole at -4 is 2 .

Suppose $\lambda_1, \lambda_2, \dots, \lambda_l$ are the poles of a rational transfer function $\hat{G}$. Then, it should be clear that we can express $\hat{G}$ as

$$\hat{G}(s) = \frac{1}{d(s)} N(s) \tag{4.5}$$

where

$$d(s) = (s - \lambda_1)^{m_1} (s - \lambda_2)^{m_2} \dots (s - \lambda_l)^{m_l},$$

with m_i being the order of the pole λ_i , and N is a matrix with polynomial elements.

Zeroes. Here, we only define zeroes for scalar-input scalar-output transfer functions. The definition of zeroes for MIMO systems is much more complicated. We say that a complex number η is a **zero** of a scalar-input scalar-output transfer function $\hat{G}$ if

$$\hat{G}(\eta) = 0. \tag{4.6}$$

4.1.4 Discrete-time

The transfer function matrix for

$$\begin{aligned}x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k)\end{aligned}$$

is defined by

$$\hat{G}(z) = C(zI - A)^{-1}B + D$$

where $z \in \mathbb{C}$. This is the z -transform of the impulse response matrix G .

4.1.5 Exercises

Exercise 32 (BB in Laundromat.) Consider

$$\begin{aligned}m\ddot{\phi}_1 - m\Omega^2\phi_1 + \frac{k}{2}(\phi_1 - \phi_2) &= b_1u \\ m\ddot{\phi}_2 - m\Omega^2\phi_2 - \frac{k}{2}(\phi_1 - \phi_2) &= b_2u \\ y &= c_1\phi_1 + c_2\phi_2\end{aligned}$$

For each of the following cases, obtain the system transfer function and its poles; consider $\omega := \sqrt{k/m} > \Omega$.

(a)

$$\begin{aligned}b_1 &= 1 & b_2 &= 0 \\ c_1 &= 1 & c_2 &= 0\end{aligned}$$

(b) (Self excited.)

$$\begin{aligned}b_1 &= 1 & b_2 &= -1 \\ c_1 &= 1 & c_2 &= 0\end{aligned}$$

(c) (Mass center measurement.)

$$\begin{aligned}b_1 &= 1 & b_2 &= 0 \\ c_1 &= 1 & c_2 &= 1\end{aligned}$$

(d) (Self excited and mass center measurement.)

$$\begin{aligned}b_1 &= 1 & b_2 &= -1 \\ c_1 &= 1 & c_2 &= 1\end{aligned}$$

Comment on your results.

4.2 Some transfer function properties*

4.2.1 Power series expansion and Markov parameters

Suppose $\hat{G}$ is a transfer function for (A, B, C, D) , that is $\hat{G}(s) = C(sI - A)^{-1}B + D$. Then we will show that, for sufficiently large s , we have the following power series expansion:

$$\hat{G}(s) = D + \sum_{k=1}^{\infty} \frac{1}{s^k} CA^{k-1}B = D + \frac{1}{s}CB + \frac{1}{s^2}CAB + \frac{1}{s^3}CA^2B + \cdots.$$

To see this recall that, for sufficiently large s ,

$$(sI - A)^{-1} = \frac{1}{s}I + \frac{1}{s^2}A + \frac{1}{s^3}A^2 + \cdots.$$

The result follows from this. The coefficient matrices

$$D, \quad CB, \quad CAB, \quad CA^2B, \quad \dots$$

are called the Markov parameters of $\hat{G}$.

4.2.2 More properties

Recall that a transfer function $\hat{G}$ for a finite dimensional system must be rational and proper, that is, it can be expressed as

$$\hat{G} = \frac{1}{d}N$$

where d is a scalar valued polynomial and N is a matrix of polynomials whose degrees are not greater than that of d . Without loss of generality, we consider d to be **monic**, that is the coefficient of the term of highest power is one.. Then we can express d and N as

$$\begin{aligned} d(s) &= s^n + d_{n-1}s^{n-1} + \cdots + d_1s + d_0 \\ N(s) &= s^n N_n + s^{n-1}N_{n-1} + \cdots + sN_1 + N_0 \end{aligned}$$

where $d_0, \dots, d_{n-1}$ are scalars and $N_0, \dots, N_n$ are constant matrices whose dimensions are the same as those of $\hat{G}(s)$. We now show that $\hat{G} = \frac{1}{d}N$ is a transfer function for a system (A, B, C, D) if and only if d satisfies

$$CA^k d(A)B = 0 \quad \text{for } k = 0, 1, 2, \dots. \quad (4.7)$$

and

$$\begin{aligned} N_n &= D \\ N_{n-1} &= CB + d_{n-1}D \\ N_{n-2} &= CAB + d_{n-1}CB + d_{n-2}D \\ &\vdots \\ N_0 &= CA^{n-1}B + d_{n-1}CA^{n-2}B + \cdots + d_1CB + d_0D \end{aligned} \quad (4.8)$$

To obtain the above result, we first use the power series expansion for $\hat{G}$ to obtain

$$\frac{1}{d(s)}N(s) = D + \frac{1}{s}CB + \frac{1}{s^2}CAB + \frac{1}{s^3}CA^2B + \dots$$

Multiplying both sides by $d(s)$ yields

$$\begin{aligned} & s^n N_n + s^{n-1} N_{n-1} + \dots + s N_1 + N_0 \\ &= [s^n + d_{n-1}s^{n-1} + \dots + d_1 s + d_0] \left[D + \frac{1}{s}CB + \frac{1}{s^2}CAB + \frac{1}{s^3}CA^2B + \dots \right] \end{aligned}$$

Since the coefficients of like powers of s on both sides of the equation must be equal, this equation holds if and only if

$$\begin{aligned} N_n &= D \\ N_{n-1} &= CB + d_{n-1}D \\ N_{n-2} &= CAB + d_{n-1}CB + d_{n-2}D \\ &\vdots \\ N_0 &= CA^{n-1}B + d_{n-1}CA^{n-2}B + \dots + d_1 CB + d_0 D \end{aligned}$$

and

$$\begin{aligned} 0 &= CA^n B + d_{n-1}CA^{n-1}B + \dots + d_0 CB \\ 0 &= CA^{n+1}B + d_{n-1}CA^n B + \dots + d_0 CAB \\ &\vdots \\ 0 &= CA^{n+k}B + d_{n-1}CA^{n+k-1}B + \dots + d_0 CA^k B \\ &\vdots \end{aligned}$$

Now note that the first set of equations above is the same as (4.8) while the second set is equivalent to (4.7) ■

Suppose $\hat{G}$ is a transfer function for (A, B, C, D) and d is *any* polynomial for which $d(A) = 0$. It follows immediately that (4.8) holds. Hence we can express $\hat{G}$ as $\hat{G} = \frac{1}{d}N$ where the coefficients of N are computed from (4.7).

A proof of the Cayley Hamilton Theorem. Consider any square matrix A and let $\hat{G}(s) = (sI - A)^{-1}$. The function $\hat{G}$ is the transfer function for the system with $B = I$, $C = I$ and $D = 0$. Hence $\hat{G}$ can be expressed as $\hat{G} = \frac{1}{d}N$ where d is the characteristic polynomial of A . Considering $k = 1$ in (4.7), it now follows that $d(A) = 0$.

4.3 State space realization of transfer functions

Here we consider the following type of problem: Given a matrix valued function $\hat{G}$ of a complex variable s , find constant matrices A, B, C, D (or determine that none exist) so that

$$\hat{G}(s) = C(sI - A)^{-1}B + D. \quad (4.9)$$

When the above holds, we say that (A, B, C, D) is a finite-dimensional **state space realization** of $\hat{G}$. The following result provides conditions under which such realizations exist.

Theorem 1 *Suppose $\hat{G}$ is a matrix valued function of a complex variable. Then $\hat{G}$ has a finite-dimensional state space realization if and only if $\hat{G}$ is rational and proper.*

PROOF: We have already seen the necessity of $\hat{G}$ being proper rational. We prove sufficiency in the next two sections by demonstrating specific realizations. ■

Minimal realizations. It should be clear from Section 4.1 that a given transfer function has an infinite number of state space realizations. One might expect that the dimension of every state space realization is the same. This is not even true as the next example illustrates. We say that a realization is a **minimal realization** if its dimension is less than or equal to that of any other realization. Clearly, all minimal realizations have the same dimension. Later on we will characterize minimal state space realizations.

Example 33 Consider a SISO system of dimension two described by

$$A = \begin{pmatrix} 1 & -3 \\ -3 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 0 \end{pmatrix} \quad D = 0$$

Here

$$\hat{G}(s) = C(sI - A)^{-1}B + D = \frac{s - 4}{s^2 - 2s - 8} = \frac{1}{s + 2}$$

This also has the following one dimensional realization

$$A = -2 \quad B = 1 \quad C = 1 \quad D = 0$$

Note that this system is stable, whereas the original system is unstable.

4.4 Realization of SISO transfer functions

4.4.1 Controllable canonical form realization

Suppose that $\hat{G}$ is a proper rational scalar function of a complex variable s . If $\hat{G} \neq 0$, then for some positive integer n , the function $\hat{G}$ can be expressed as

$$\boxed{\hat{G}(s) = \frac{\beta_{n-1}s^{n-1} + \cdots + \beta_1s + \beta_0}{s^n + \alpha_{n-1}s^{n-1} + \cdots + \alpha_1s + \alpha_0} + d} \quad (4.10)$$

This is illustrated in the following example.

Example 34 Consider

$$G(s) = \frac{4s + 8}{-2s + 10}$$

This is a proper rational function. Also $4s + 8 = -2(-2s + 10) + 28$ which results in

$$\begin{aligned} G(s) &= \frac{-2(-2s + 10) + 28}{-2s + 10} = -2 + \frac{28}{-2s + 10} \\ &= \frac{-14}{s - 5} - 2. \end{aligned}$$

When $\hat{G}$ is expressed in the form shown in (4.10) we will show that

$$\hat{G}(s) = C(sI - A)^{-1}B + D$$

where

$$\boxed{\begin{aligned} A &= \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ -\alpha_0 & -\alpha_1 & -\alpha_2 & \cdots & -\alpha_{n-2} & -\alpha_{n-1} \end{pmatrix} & B &= \begin{pmatrix} 0 \\ 0 \\ \vdots \\ \vdots \\ 0 \\ 1 \end{pmatrix} \\ C &= (\beta_0 \quad \beta_1 \quad \beta_2 \quad \cdots \quad \beta_{n-2} \quad \beta_{n-1}) & D &= d \end{aligned}} \quad (4.11)$$

It then follows that one can obtain a state space realization of any proper rational scalar SISO transfer function.

To demonstrate the above claim, let Δ be the denominator polynomial in the above expression for $\hat{G}$, that is,

$$\Delta(s) := \alpha_0 + \alpha_1s + \cdots + s^n$$

and define

$$v(s) := \begin{pmatrix} 1 \\ s \\ \vdots \\ s^{n-1} \end{pmatrix}.$$

Then,

$$(sI - A)v(s) = \begin{pmatrix} s & -1 & 0 & \dots & 0 & 0 \\ 0 & s & -1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & & \vdots \\ \vdots & \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & s & -1 \\ \alpha_0 & \alpha_1 & \alpha_2 & \dots & \alpha_{n-2} & s + \alpha_{n-1} \end{pmatrix} \begin{pmatrix} 1 \\ s \\ \vdots \\ s^{n-2} \\ s^{n-1} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ \vdots \\ 0 \\ \Delta(s) \end{pmatrix} = \Delta(s)B$$

For any value of s for which $\delta(s)^{-1}$ is nonzero, the matrix $sI - A$ is invertible, and we can multiply both sides of the above equation by $(sI - A)^{-1}$ and $\delta(s)^{-1}$ to obtain

$$(sI - A)^{-1}B = \frac{1}{\Delta(s)}v(s);$$

hence

$$\begin{aligned} C(sI - A)^{-1}B &= \frac{1}{\Delta(s)} \begin{pmatrix} \beta_0 & \beta_1 & \dots & \beta_{n-1} \end{pmatrix} \begin{pmatrix} 1 \\ s \\ \vdots \\ s^{n-1} \end{pmatrix} \\ &= \frac{\beta_0 + \beta_1 s + \dots + \beta_{n-1} s^{n-1}}{\alpha_0 + \alpha_1 s + \dots + \alpha_{n-1} s^{n-1} + s^n} \end{aligned}$$

So,

$$\hat{G}(s) = C(sI - A)^{-1}B + D$$

■

We will see later why this realization is called controllable.

4.4.2 Observable canonical form realization

Suppose $\hat{G}(s)$ is a proper rational scalar function of a complex variable s . If $\hat{G} \neq 0$, then, for some positive integer n ,

$$\hat{G}(s) = \frac{\beta_{n-1}s^{n-1} + \dots + \beta_1 s + \beta_0}{s^n + \alpha_{n-1}s^{n-1} + \dots + \alpha_1 s + \alpha_0} + d$$

We will show that

$$\hat{G}(s) = C(sI - A)^{-1}B + D$$

where

$$A = \begin{pmatrix} 0 & 0 & \dots & 0 & -\alpha_0 \\ 1 & 0 & \dots & 0 & -\alpha_1 \\ 0 & 1 & \ddots & 0 & -\alpha_2 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & -\alpha_{n-1} \end{pmatrix} \quad B = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \vdots \\ \beta_{n-1} \end{pmatrix}$$

$$C = (0 \ 0 \ \dots \ 0 \ 1) \quad D = d$$

- Noting that $C(sI - A)^{-1}B + D$ is a scalar and using the results of the previous section, we have

$$C(sI - A)^{-1}B + D = B^T(sI - A^T)^{-1}C^T + D = \hat{G}(s)$$

■

We will see later why this realization is called observable.

MATLAB

>> help tf2ss

TF2SS Transfer function to state-space conversion.
 [A,B,C,D] = TF2SS(NUM,DEN) calculates the state-space representation:

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned}$$

of the system:

$$H(s) = \frac{\text{NUM}(s)}{\text{DEN}(s)}$$

from a single input. Vector DEN must contain the coefficients of the denominator in descending powers of s. Matrix NUM must contain the numerator coefficients with as many rows as there are outputs y. The A,B,C,D matrices are returned in controller canonical form. This calculation also works for discrete systems. To avoid confusion when using this function with discrete systems, always use a numerator polynomial that has been padded with zeros to make it the same length as the denominator. See the User's guide for more details.

The commands `tf` and `ss` are very useful here.

Example 35 Consider

$$\hat{G}(s) = \frac{s^2}{s^2 + 1}$$

This can be written as

$$\hat{G}(s) = \frac{-1}{s^2 + 1} + 1$$

hence a state space representation is given by

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad C = \begin{pmatrix} -1 & 0 \end{pmatrix} \quad D = 1$$

or,

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1 + u \\ y &= -x_1 + u \end{aligned}$$

Matlab time:

```
>> num = [1 0 0]
>> den = [1 0 1]
>> [a,b,c,d]=tf2ss(num,den)
```

```
a =
    0    -1
    1     0
```

```
b =
    1
    0
```

```
c =
    0    -1
```

```
d =
    1
```

4.5 Realization of MIMO systems

Suppose $\hat{G}$ is an $p \times m$ proper rational transfer function. Because $\hat{G}$ is a proper rational function it has a representation of the form

$$\hat{G}(s) = \frac{1}{d(s)}N(s) + D \quad (4.12)$$

where D is a constant $p \times m$ matrix, N is a $p \times m$ matrix of polynomials and d is a scalar valued monic polynomial.

4.5.1 Controllable realizations

Sometimes it is more convenient to express $\hat{G}$ as

$$\hat{G}(s) = N(s)\Delta(s)^{-1} + D \quad (4.13)$$

where Δ is a $m \times m$ matrix of polynomials and N is a $p \times m$ matrix of polynomials. Let us consider this case. Note that if $\hat{G}$ is expressed as in (4.12) then, $\hat{G}$ can be expressed as in (4.13) with $\Delta(s) = d(s)I$ where I is the $m \times m$ identity matrix.

Without loss of generality, we can express N and Δ as

$$\begin{aligned} N(s) &= s^{l-1}N_{l-1} + s^{l-2}N_{l-2} + \cdots + sN_1 + N_0 \\ \Delta(s) &= s^l I + s^{l-1}D_{l-1} + s^{l-2}D_{l-2} + \cdots + sD_1 + D_0 \end{aligned} \quad (4.14)$$

Here, $N_0, \dots, N_{l-1}$ are constant $p \times m$ matrices while $D_0, \dots, D_{l-1}$ are constant $m \times m$ matrices. With

$$n = ml,$$

let A, B and C be the following matrices of dimensions $n \times n$, $n \times m$ and $p \times n$, respectively,

$$A = \begin{pmatrix} 0 & I & 0 & \cdots & 0 & 0 \\ 0 & 0 & I & \cdots & 0 & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & I \\ -D_0 & -D_1 & -D_2 & \cdots & -D_{l-2} & -D_{l-1} \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ \vdots \\ 0 \\ I \end{pmatrix}, \quad (4.15a)$$

$$C = \begin{pmatrix} N_0 & N_1 & N_2 & \cdots & N_{l-2} & N_{l-1} \end{pmatrix}. \quad (4.15b)$$

Note that in the above matrices, I and 0 represent the $m \times m$ identity and zero matrices, respectively. We claim that (A, B, C, D) is a realization of $\hat{G}$. This is proven at the end of this section.

Note that, when

$$\hat{G}(s) = \frac{1}{d(s)}N(s) + D \quad \text{and} \quad d(s) = \alpha_0 + \alpha_1 s + \cdots + \alpha_{l-1} s^{l-1} + s^l \quad (4.16)$$

we have

$$\Delta(s) = d(s)I$$

and

$$A = \begin{pmatrix} 0 & I & 0 & \cdots & 0 & 0 \\ 0 & 0 & I & \cdots & 0 & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & I \\ -\alpha_0 I & -\alpha_1 I & -\alpha_2 I & \cdots & -\alpha_{l-2} I & -\alpha_{l-1} I \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ \vdots \\ 0 \\ I \end{pmatrix}, \quad (4.17a)$$

$$C = \begin{pmatrix} N_0 & N_1 & N_2 & \cdots & N_{l-2} & N_{l-1} \end{pmatrix}. \quad (4.17b)$$

Example 36 Consider

$$\hat{G}(s) = \begin{pmatrix} \frac{3}{s-2} \\ \frac{s+4}{s^2+2s+3} \end{pmatrix}$$

The lowest common denominator (LCD) of the two elements of $\hat{G}(s)$ is

$$(s-2)(s^2+2s+3) = s^3 - s - 6 =: d(s)$$

and $\hat{G} = \frac{1}{d}N$ where

$$N(s) = \begin{pmatrix} 3s^2+6s+9 \\ s^2+2s-8 \end{pmatrix} = s^2 \begin{pmatrix} 3 \\ 1 \end{pmatrix} + s \begin{pmatrix} 6 \\ 2 \end{pmatrix} + \begin{pmatrix} 9 \\ -8 \end{pmatrix}$$

Hence, a controllable realization of $\hat{G}$ is given by

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & 1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad C = \begin{pmatrix} 9 & 6 & 3 \\ -8 & 2 & 1 \end{pmatrix} \quad D = 0$$

Example 37 Consider the strictly proper rational function,

$$\hat{G}(s) = \begin{pmatrix} \frac{1}{s+1} & \frac{1}{s^2+3s+2} \\ 0 & \frac{1}{s+2} \end{pmatrix}.$$

Here $\hat{G} = \frac{1}{d}N$ where

$$d(s) = s^2 + 3s + 2$$

and

$$N(s) = \begin{pmatrix} s+2 & 1 \\ 0 & s+1 \end{pmatrix} = s \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

Hence a realization of $\hat{G}$ is given by

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & 0 & -3 & 0 \\ 0 & -2 & 0 & -3 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 2 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Here we show that (A, B, C, D) as defined above is a realization of $\hat{G}$. Let V be the $n \times m$ matrix of polynomials defined by

$$V(s) = \begin{pmatrix} I \\ sI \\ \vdots \\ s^{l-1}I \end{pmatrix}.$$

where I represents the $m \times m$ identity matrix. Then we have

$$\begin{aligned}
(sI - A)V(s) &= \begin{pmatrix} sI & -I & 0 & \dots & 0 & 0 \\ 0 & sI & -I & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & & \vdots \\ \vdots & \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & sI & -I \\ D_0 & D_1 & D_2 & \dots & D_{l-2} & sI + D_{l-1} \end{pmatrix} \begin{pmatrix} I \\ sI \\ \vdots \\ \vdots \\ s^{l-2}I \\ s^{l-1}I \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ \vdots \\ 0 \\ \Delta(s) \end{pmatrix} \\
&= B\Delta(s)
\end{aligned}$$

Hence, whenever $\Delta(s)$ is invertible,

$$(sI - A)^{-1}B = V(s)\Delta(s)^{-1}$$

Now note that

$$CV(s) = \begin{pmatrix} N_0 & N_1 & \dots & N_{l-1} \end{pmatrix} \begin{pmatrix} I \\ sI \\ \vdots \\ \vdots \\ s^{l-2}I \\ s^{l-1}I \end{pmatrix} = N(s).$$

Hence

$$C(sI - A)^{-1}B = CV(s)\Delta(s)^{-1} = N(s)\Delta(s)^{-1}$$

and

$$\hat{G}(s) = C(sI - A)^{-1}B + D.$$

■

4.5.2 Observable realizations

Sometimes it is more convenient to express $\hat{G}$ as

$$\hat{G}(s) = \Delta(s)^{-1}N(s) + D \tag{4.18}$$

where Δ is a $p \times p$ matrix of polynomials. Note that if $\hat{G}$ is expressed as in (4.12) then, $\hat{G}$ can be expressed as in (4.18) with $\Delta(s) = d(s)I$ where I is the $p \times p$ identity matrix.

Without loss of generality, we can express N and Δ as

$$\begin{aligned}
N(s) &= s^{l-1}N_{l-1} + s^{l-2}N_{l-2} + \dots + sN_1 + N_0 \\
\Delta(s) &= s^lI + s^{l-1}D_{l-1} + s^{l-2}D_{l-2} + \dots + sD_1 + D_0
\end{aligned} \tag{4.19}$$

Here, $N_0, \dots, N_{l-1}$ are constant $p \times m$ matrices while $D_0, \dots, D_{l-1}$ are constant $p \times p$ matrices. With

$$n = pl$$

let A , B and C be the following matrices of dimensions $n \times n$, $n \times m$ and $p \times n$, respectively,

$$A = \begin{pmatrix} -D_{l-1} & I & 0 & \dots & 0 & 0 \\ -D_{l-2} & 0 & I & \dots & 0 & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ -D_1 & 0 & 0 & \dots & 0 & I \\ -D_0 & 0 & 0 & \dots & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} N_{n-1} \\ N_{n-2} \\ \vdots \\ \vdots \\ N_1 \\ N_0 \end{pmatrix}, \quad (4.20a)$$

$$C = \begin{pmatrix} I & 0 & 0 & \dots & 0 & 0 \end{pmatrix}. \quad (4.20b)$$

Note that in the above matrices, I and 0 represent the $p \times p$ identity and zero matrices, respectively. We claim that (A, B, C, D) is a realization of $\hat{G}$. This can be shown as follows.

Let V be the $p \times n$ matrix of polynomials defined by

$$V(s) = \begin{pmatrix} s^{l-1}I & s^{l-2}I & \dots & sI & I \end{pmatrix}.$$

where I represents the $p \times p$ identity matrix. Then we have

$$\begin{aligned} V(s)(sI - A) &= \begin{pmatrix} s^{l-1}I & s^{l-2}I & \dots & \dots & sI & I \end{pmatrix} \begin{pmatrix} sI + D_{l-1} & -I & 0 & \dots & 0 & 0 \\ D_{l-2} & sI & -I & \dots & 0 & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ D_1 & 0 & 0 & \dots & 0 & -I \\ D_0 & 0 & 0 & \dots & 0 & sI \end{pmatrix} \\ &= \begin{pmatrix} \Delta(s) & 0 & \dots & \dots & 0 & 0 \end{pmatrix} \\ &= \Delta(s)C \end{aligned}$$

Hence, whenever s is not an eigenvalue of A ,

$$C(sI - A)^{-1} = \Delta(s)^{-1}V(s)$$

Now note that

$$V(s)B = \begin{pmatrix} s^{l-1}I & s^{l-2}I & \dots & sI & I \end{pmatrix} \begin{pmatrix} N_{l-1} \\ N_{l-2} \\ \vdots \\ N_1 \\ N_0 \end{pmatrix} = N(s).$$

Hence

$$C(sI - A)^{-1}B = \Delta(s)^{-1}V(s)B = \Delta(s)^{-1}N(s)$$

and

$$\hat{G}(s) = C(sI - A)^{-1}B + D.$$

■

4.5.3 Alternative realizations*

Suppose $\hat{G}$ is an $p \times 1$ transfer matrix function and

$$\hat{G} = \begin{pmatrix} \hat{g}_1 \\ \vdots \\ \hat{g}_p \end{pmatrix}$$

that is $\hat{g}_i$ is the i -th element of $\hat{G}$. Choosing a monic polynomial Δ as the least common denominator polynomial of $\hat{g}_1, \dots, \hat{g}_p$, we can write each $\hat{g}_i$ as

$$\hat{g}_i(s) = \frac{n_i(s)}{\Delta(s)} + d_i$$

where

$$\Delta(s) = s^n + \alpha_{n-1}s^{n-1} + \dots + \alpha_1s + \alpha_0$$

and

$$n_i(s) = \beta_{n-1}^i s^{n-1} + \dots + \beta_1^i s + \beta_0^i$$

Letting

$$A = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ -\alpha_0 & -\alpha_1 & -\alpha_2 & \dots & -\alpha_{n-2} & -\alpha_{n-1} \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

$$C = \begin{pmatrix} \beta_0^1 & \beta_1^1 & \beta_2^1 & \dots & \beta_{n-2}^1 & \beta_{n-1}^1 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ \beta_0^p & \beta_1^p & \beta_2^p & \dots & \beta_{n-2}^p & \beta_{n-1}^p \end{pmatrix} \quad D = \begin{pmatrix} d_1 \\ \vdots \\ d_p \end{pmatrix}$$

we obtain

$$C(sI - A)^{-1}B + D = \hat{G}(s)$$

Example 38 Consider

$$\hat{G}(s) = \begin{pmatrix} \frac{3}{s-2} \\ \frac{s+4}{s^2+2s+3} \end{pmatrix}$$

The lowest common denominator (LCD) of the two elements of $\hat{G}(s)$ is

$$(s-2)(s^2+2s+3) = s^3 - s - 6 =: \Delta(s)$$

and $\hat{G}(s)$ can be expressed as

$$\hat{G}(s) = \begin{pmatrix} \frac{3s^2+6s+9}{\Delta(s)} \\ \frac{s^2+2s-8}{\Delta(s)} \end{pmatrix}$$

Hence, a controllable realization of $\hat{G}$ is given by

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & 1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad C = \begin{pmatrix} 9 & 6 & 3 \\ -8 & 2 & 1 \end{pmatrix}$$

MIMO systems

Suppose $\hat{G}$ is an $p \times m$ transfer matrix function and

$$\hat{G} = (\hat{G}_1 \quad \dots \quad \hat{G}_m)$$

that is $\hat{G}_i$ is the i -th column of $\hat{G}$. Using the results of the previous section, for $i = 1, \dots, m$, we can obtain matrices A_i, B_i, C_i, D_i such that

$$\hat{G}_i(s) = C_i(sI - A_i)^{-1}B_i + D_i$$

Let

$$A = \begin{pmatrix} A_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & A_m \end{pmatrix} \quad B = \begin{pmatrix} B_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & B_m \end{pmatrix}$$

$$C = (C_1 \quad \dots \quad C_m) \quad D = (D_1 \quad \dots \quad D_m)$$

Then

$$C(sI - A)^{-1}B + D = \hat{G}(s)$$

- Nonminimal realization, in general.

Example 39

$$\hat{G}(s) = \begin{pmatrix} \frac{1}{s} & \frac{1}{s^2} \\ 0 & \frac{1}{s} \end{pmatrix}$$

The first and second columns of $\hat{G}$ have realizations

$$A_1 = 0 \quad B_1 = 1 \quad C_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad D_1 = 0$$

and

$$A_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad B_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad C_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad D_2 = 0$$

respectively. Hence a realization of $\hat{G}$ is given by

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Note that this transfer function also has the following lower dimensional realization:

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

4.6 State space realizations of systems described by higher order differential equations

Consider a system described by

$$A_l y^{(l)} + A_{l-1} y^{(l-1)} + \cdots + A_0 y = B_l u^{(l)} + B_{l-1} u^{(l-1)} + \cdots + B_0 u$$

where the input $u(t)$ is an m -vector, the output $y(t)$ is a p -vector, $l \geq 1$, and the matrix A_l is invertible. How do we put such a system into state space form? If any of the matrices $B_1, \dots, B_l$ are non-zero, we cannot use our previous approach. If we use the previous approach, we will obtain differential equations which depend on some of the derivatives of u .

To obtain a state representation in the general case, we first obtain the transfer function of the system and then obtain a state space realization of the transfer function. By taking the Laplace transform of the above equation and considering zero initial values for $y, \dot{y}, \dots, y^{(l-1)}$ and $u, \dot{u}, \dots, u^{(l-1)}$, we obtain

$$[s^l A_l + s^{l-1} A_{l-1} + \cdots + A_0] \hat{y}(s) = [s^l B_l + s^{l-1} B_{l-1} + \cdots + B_0] \hat{u}(s)$$

Hence $\hat{y}(s) = \hat{G}(s) \hat{u}(s)$ where

$$\hat{G}(s) = [s^l A_l + s^{l-1} A_{l-1} + \cdots + A_0]^{-1} [s^l B_l + s^{l-1} B_{l-1} + \cdots + B_0]$$

To obtain a state space description of the system under consideration, we obtain a state-space realization of $\hat{G}$, that is we obtain matrices A, B, C, D such that $\hat{G}(s) = C(sI - A)^{-1}B + D$.

Exercises

Exercise 33 Obtain a state space realization of the transfer function,

$$\hat{G}(s) = \frac{s^2 + 3s + 2}{s^2 + 5s + 6}.$$

Is your realization minimal?

Exercise 34 Obtain a state space realization of the transfer function,

$$\hat{G}(s) = \begin{pmatrix} 1 & \frac{1}{s-1} \\ \frac{1}{s+1} & \frac{1}{s^2-1} \end{pmatrix}.$$

Exercise 35 Obtain a state space realization of the following transfer function.

$$\hat{G}(s) = \begin{pmatrix} \frac{s+2}{s+1} & \frac{1}{s+3} \\ \frac{5}{s+1} & \frac{5s+1}{s+2} \end{pmatrix}$$

Exercise 36 Obtain a state space realization of the transfer function,

$$\hat{G}(s) = \begin{pmatrix} \frac{1}{s-1} & \frac{1}{s-1} \\ \frac{1}{s+1} & -\frac{1}{s+1} \end{pmatrix}.$$

Exercise 37 Obtain a state space realization of the transfer function,

$$\hat{G}(s) = \begin{pmatrix} \frac{s}{s-1} & \frac{1}{s+1} \end{pmatrix}.$$

Is your realization minimal?

Exercise 38 Obtain a state space representation of the following transfer function.

$$\hat{G}(s) = \begin{pmatrix} \frac{s^2+1}{s^2-1} & \frac{2}{s-1} \end{pmatrix}$$

Exercise 39 Obtain a state space representation of the following input-output system.

$$2\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 2y = 6\frac{d^2u}{dt^2} + 10\frac{du}{dt} + 8u$$

Exercise 40 (a) Obtain a state space realization of the following single-input single-output system.

$$\ddot{y} - 3\dot{y} - 4y = \ddot{u} - 2\dot{u} - 8u$$

(b) Is your realization minimal?

Exercise 41 Obtain a state space realization of the following input-output system.

$$\begin{aligned} \dot{y}_1 - y_1 + y_2 &= \dot{u}_1 + u_1 \\ \ddot{y}_2 + y_2 - y_1 &= u_2 \end{aligned}$$

Exercise 42 Obtain a state space realization of the following input-output system.

$$\begin{aligned}\dot{y}_1 &= \dot{u}_1 + u_2 \\ \dot{y}_2 &= u_1 + \dot{u}_2\end{aligned}$$

Exercise 43 Obtain a state space realization of the following input-output system.

$$\begin{aligned}\dot{y}_1 + y_2 &= \dot{u}_2 + u_1 \\ \dot{y}_2 + y_1 &= \dot{u}_1 + u_2\end{aligned}$$

Exercise 44 Consider the single-input single-output system described by

$$\begin{aligned}\dot{x}_1 &= -2x_2 + u \\ \dot{x}_2 &= -2x_1 + u \\ y &= x_1 - x_2\end{aligned}$$

(a) Compute e^{At} for the matrix,

$$A = \begin{pmatrix} 0 & -2 \\ -2 & 0 \end{pmatrix}.$$

(b) Obtain the impulse response of the system.

(c) What is the transfer function of this system?

Exercise 45 Consider the single-input single-output system described by

$$\begin{aligned}\dot{x}_1 &= -3x_1 + x_2 + w \\ \dot{x}_2 &= x_1 - 3x_2 + w \\ y &= x_1 + x_2\end{aligned}$$

(a) Compute e^{At} for the matrix,

$$A = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}$$

(b) Obtain the impulse response of the system.

(c) What is the transfer function of this system?

(d) What are the poles and zeros of the transfer function?

Chapter 5

Some linear algebra

All the definitions and results of this section are stated for real scalars. However, they also hold for complex scalars; to get the results for complex scalars, simply replace ‘real’ with ‘complex’ and $\mathbb{R}$ with $\mathbb{C}$.

5.1 Linear equations

Many problems in linear algebra can be reduced to that of solving a bunch of linear equations in a bunch of unknowns. Consider the following m linear scalar equations in n scalar unknowns $x_1, x_2, \dots, x_n$:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned} \tag{5.1}$$

Given the scalar coefficients a_{ij} and b_i , a basic problem is that of determining whether or not the above system of linear equations has a solution for $x_1, x_2, \dots, x_n$ and, if a solution exists, determining all solutions.

A systematic approach to solving the above problem is to carry out a sequence of **elementary operations** on the above equations to reduce them to a much simpler system of equations which have the same solutions as the original system. There are three elementary operations:

- 1) *Interchanging two equations.*
- 2) *Multiplication of an equation by a nonzero scalar.*
- 3) *Addition of a multiple of one equation to another different equation.*

Each of the above operations is reversible and does not change the set of solutions.

The above approach is usually implemented by expressing the above equations in matrix form and carrying out the above elementary operations via **elementary matrix row operations**.

Introducing

$$b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}, \quad A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \quad \text{and} \quad x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix},$$

the above system of scalar equations can be written in matrix form as:

$$Ax = b \tag{5.2}$$

where x is an n -vector, b is an m -vector and A is an $m \times n$ matrix.

Elementary row operations. The above elementary operations on the original scalar equations are respectively equivalent to the following **elementary row operations** on the **augmented matrix**:

$$(A \ b)$$

associated with the scalar equations.

- 1) *Interchanging two rows.*
- 2) *Multiplication of a row by a nonzero scalar.*
- 3) *Addition of a multiple of one row to another different row.*

Any matrix can be transformed into its unique corresponding reduced row echelon form by carrying out a sequence of the above elementary row operations.

The reduced row echelon form of a matrix. Consider any matrix M . For each non-zero row, the **leading element** of that row is the first non-zero element of the row. The matrix M is said to be in **reduced row echelon form** if its structure satisfies the following four conditions.

- 1) *All zero rows come after the nonzero rows.*
- 2) *All leading elements are equal to one.*
- 3) *The leading elements have a staircase pattern.*
- 4) *If a column contains a leading element of some row, then all other elements of the column are zero.*

This is illustrated with the following matrix.

$$M = \begin{pmatrix} 0 & \cdots & 0 & \boxed{1} & * & \cdots & 0 & * & \cdots & 0 & \cdots & * \\ 0 & \cdots & 0 & 0 & 0 & \cdots & \boxed{1} & * & \cdots & 0 & \cdots & * \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & \cdots & \boxed{1} & \cdots & * \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & \cdots & 0 \end{pmatrix} \tag{5.3}$$

So, the matrices

$$\begin{pmatrix} 0 & \boxed{1} & 0 & 0 & 2 \\ 0 & 0 & \boxed{1} & 0 & 3 \\ 0 & 0 & 0 & \boxed{1} & 4 \end{pmatrix} \quad \begin{pmatrix} \boxed{1} & 2 & 0 & 0 & 2 \\ 0 & 0 & \boxed{1} & 0 & 3 \\ 0 & 0 & 0 & \boxed{1} & 4 \end{pmatrix} \quad \begin{pmatrix} \boxed{1} & 0 & 0 & 2 & 0 \\ 0 & 0 & \boxed{1} & 3 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

are in reduced row echelon form, whereas the following matrices are not in reduced row echelon form.

$$\begin{pmatrix} \boxed{1} & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \boxed{1} & 4 \end{pmatrix} \quad \begin{pmatrix} \boxed{1} & 0 & 0 & 3 \\ 0 & \boxed{2} & 0 & 0 \\ 0 & 0 & \boxed{1} & 4 \end{pmatrix} \quad \begin{pmatrix} 0 & \boxed{1} & 2 & 3 \\ \boxed{1} & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} \boxed{1} & 2 & 0 & 0 \\ 0 & \boxed{1} & 2 & 0 \\ 0 & 0 & 0 & \boxed{1} \end{pmatrix}$$

The above matrices respectively violate conditions 1, 2, 3, 4 above.

Sometimes the leading elements of a reduced row echelon matrix are called **pivots**.

Fact 1 *Every matrix can be transformed into reduced row echelon form by a sequence of elementary row operations. Although the sequence of row operations is not unique, the resulting row echelon form matrix is unique.*

We will denote the row echelon form of a matrix M by $\text{rref}(M)$. The following example illustrates how to obtain the reduced row echelon form of a matrix.

Example 40

$$M = \begin{pmatrix} 0 & 2 & 2 & 4 & -2 \\ 1 & 2 & 3 & 4 & -1 \\ 2 & 0 & 2 & 0 & 2 \end{pmatrix}$$

We reduce M to its reduced row echelon form as follows:

$$\begin{pmatrix} 0 & 2 & 2 & 4 & -2 \\ 1 & 2 & 3 & 4 & -1 \\ 2 & 0 & 2 & 0 & 2 \end{pmatrix}$$

Interchange Rows 1 and 2.

$$\begin{pmatrix} 1 & 2 & 3 & 4 & -1 \\ 0 & 2 & 2 & 4 & -2 \\ 2 & 0 & 2 & 0 & 2 \end{pmatrix}$$

Add multiples of row 1 to all succeeding rows.

$$\begin{pmatrix} 1 & 2 & 3 & 4 & -1 \\ 0 & 2 & 2 & 4 & -2 \\ 0 & -4 & -4 & -8 & 4 \end{pmatrix}$$

Multiply row 2 by 1/2.

$$\begin{pmatrix} 1 & 2 & 3 & 4 & -1 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & -4 & -4 & -8 & 4 \end{pmatrix}$$

Add multiples of row 2 to all succeeding rows.

$$\begin{pmatrix} 1 & 2 & 3 & 4 & -1 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Add multiples of row 2 to all preceeding rows.

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Gauss-Jordan procedure for solving systems of linear equations. Recall the problem of solving the system of scalar linear equations (5.1) or their vector/matrix counterpart (5.2). The Gauss-Jordan procedure is a systematic method for solving such equations. One simply uses elementary row operations to transform the augmented matrix $[A \ b]$ into its reduced row echelon form which we denote by $[\hat{A} \ \hat{b}]$. Since $[\hat{A} \ \hat{b}]$ is obtained from $[A \ b]$ by elementary row operations, the set of vectors x which satisfy $Ax = b$ is exactly the same as the set which satisfy $\hat{A}x = \hat{b}$. However, since $[\hat{A} \ \hat{b}]$ is in reduced row echelon form, it can be simply determined whether or not the equation $\hat{A}x = \hat{b}$ has a solution for x , and, if a solution exists, to determine all solutions.

Example 41 Determine whether or not the following system of linear equations has a solution. If a solution exists, determine whether or not it is unique; if not unique, obtain an expression for all solutions and give two solutions.

$$\begin{aligned} y - 2z &= 4 \\ x - y &= 1 \\ 3x - 3y + 4z &= -9 \\ -2y + 10z &= -26 \end{aligned}$$

Here the augmented matrix is given by

$$(A \ b) = \begin{pmatrix} 0 & 1 & -2 & 4 \\ 1 & -1 & 0 & 1 \\ 3 & -3 & 4 & -9 \\ 0 & -2 & 10 & -26 \end{pmatrix}$$

Carrying our elementary row operations, we obtain

$$\begin{pmatrix} 0 & 1 & -2 & 4 \\ 1 & -1 & 0 & 1 \\ 3 & -3 & 4 & -9 \\ 0 & -2 & 10 & -26 \end{pmatrix} \rightarrow \begin{pmatrix} \boxed{1} & -1 & 0 & 1 \\ 0 & 1 & -2 & 4 \\ 3 & -3 & 4 & -9 \\ 0 & -2 & 10 & -26 \end{pmatrix} \rightarrow \begin{pmatrix} \boxed{1} & -1 & 0 & 1 \\ 0 & \boxed{1} & -2 & 4 \\ 0 & 0 & 4 & -12 \\ 0 & -2 & 10 & -26 \end{pmatrix} \rightarrow \begin{pmatrix} \boxed{1} & -1 & 0 & 1 \\ 0 & \boxed{1} & -2 & 4 \\ 0 & 0 & 4 & -12 \\ 0 & 0 & 6 & -18 \end{pmatrix} \\ \rightarrow \begin{pmatrix} \boxed{1} & -1 & 0 & 1 \\ 0 & \boxed{1} & -2 & 4 \\ 0 & 0 & \boxed{1} & -3 \\ 0 & 0 & 6 & -18 \end{pmatrix} \rightarrow \begin{pmatrix} \boxed{1} & -1 & 0 & 1 \\ 0 & \boxed{1} & -2 & 4 \\ 0 & 0 & \boxed{1} & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} \boxed{1} & -1 & 0 & 1 \\ 0 & \boxed{1} & 0 & -2 \\ 0 & 0 & \boxed{1} & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} \boxed{1} & 0 & 0 & -1 \\ 0 & \boxed{1} & 0 & -2 \\ 0 & 0 & \boxed{1} & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The reduced row echelon form of $[A \ b]$ (the last matrix above) tells us that the original equations are equivalent to:

$$\begin{aligned} x &= -1 \\ y &= -2 \\ z &= -3 \end{aligned}$$

Thus, the original equations have a unique solution given by $x = -1$, $y = -2$ and $z = -3$.

From the above procedure we see that there are three possibilities when it comes to solving linear equations:

- 1) A unique solution. (All columns except the last have a leading element.)
- 2) No solution. (The last column has a leading element, that is, there is a row in which all elements are zero except the last element.)
- 3) Infinite number of solutions. (Case 2 does not occur and there is a column in $\hat{A}$ which does not have a leading element.)

MATLAB.

help rref

rref Reduced row echelon form.

$R = \text{rref}(A)$ produces the reduced row echelon form of A .

$[R, \text{jb}] = \text{rref}(A)$ also returns a vector, jb , so that:

$r = \text{length}(\text{jb})$ is this algorithm's idea of the rank of A ,

$x(\text{jb})$ are the bound variables in a linear system, $Ax = b$,

$A(:, \text{jb})$ is a basis for the range of A ,

$R(1:r, \text{jb})$ is the r -by- r identity matrix.

$[R, \text{jb}] = \text{rref}(A, \text{TOL})$ uses the given tolerance in the rank tests.

Roundoff errors may cause this algorithm to compute a different value for the rank than `RANK`, `ORTH` and `NULL`.

Class support for input A :

float: double, single

See also `rank`, `orth`, `null`, `qr`, `svd`.

Reference page for `rref`

Other functions named `rref`

Exercises

Exercise 46 Determine whether or not the following system of linear equations has a solution. If a solution exists, determine whether or not it is unique, and if not unique, obtain an expression for all solutions.

$$\begin{array}{rrrrrrcl} x_1 & - & x_2 & + & 2x_3 & + & x_4 & = & 5 \\ x_1 & - & x_2 & + & x_3 & & & = & 3 \\ -2x_1 & + & 2x_2 & & & + & 2x_4 & = & -2 \\ 2x_1 & - & 2x_2 & - & x_3 & - & 3x_4 & = & 0 \end{array}$$

Exercise 47 Determine whether or not the following system of linear equations has a solution. If a solution exists, determine whether or not it is unique; if not unique, obtain an expression for all solutions and given two solutions.

$$\begin{array}{rrrrcl} x_1 & - & 2x_2 & - & 4x_3 & = & -1 \\ 2x_1 & - & 3x_2 & - & 5x_3 & = & -1 \\ x_1 & & & + & 2x_3 & = & 1 \\ x_1 & + & x_2 & + & 5x_3 & = & 2 \end{array}$$

Exercise 48 Determine whether or not the following system of linear equations has a solution. If a solution exists, determine whether or not it is unique; if not unique, obtain an expression for all solutions and give two solutions.

$$\begin{array}{rrrrrrrr} x_1 & - & 2x_2 & & x_3 & - & 2x_4 & = & -1 \\ -3x_1 & + & 7x_2 & - & 3x_3 & + & 7x_4 & = & 4 \\ 2x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & 2 \end{array}$$

Exercise 49 Determine whether or not the following system of linear equations has a solution. If a solution exists, determine whether or not it is unique; if not unique, obtain an expression for all solutions and give two solutions.

$$\begin{array}{rrrrrrrr} & & x_2 & & & + & x_4 & = & 2 \\ 2x_1 & + & 3x_2 & + & 2x_3 & + & 3x_4 & = & 8 \\ x_1 & + & x_2 & + & x_3 & + & x_4 & = & 3 \end{array}$$

Exercise 50 Determine whether or not the following system of linear equations has a solution. If a solution exists, determine whether or not it is unique; if not unique, obtain an expression for all solutions and given two solutions.

$$\begin{array}{rrrrrr} x_1 & - & 2x_2 & - & 4x_3 & = & -1 \\ 2x_1 & - & 3x_2 & - & 5x_3 & = & -1 \\ x_1 & & & + & 2x_3 & = & 1 \\ x_1 & + & x_2 & + & 5x_3 & = & 2 \end{array}$$

Exercise 51 Determine whether or not the following system of linear equations has a solution. If a solution exists, determine whether or not it is unique; if not unique, obtain an expression for all solutions and given two solutions.

$$\begin{array}{rrrr} y - 2z & = & 4 \\ x - y & = & 1 \\ 3x - 3y + 4z & = & -9 \\ -2y + 10z & = & -26 \end{array}$$

Exercise 52 We consider here systems of linear equations of the form

$$Ax = b$$

with $x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$

For each of the following cases, we present the reduced row echelon form of the augmented matrix $[Ab]$. Determine whether or not the corresponding system of linear equations has a solution. If a solution exists, determine whether or not it is unique; if not unique, obtain an expression for all solutions and given two solutions.

$$\begin{array}{llll} \text{(a)} \begin{pmatrix} 1 & 5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \text{(b)} \begin{pmatrix} 1 & -2 & 0 & 6 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix} & \text{(c)} \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} & \text{(d)} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

5.2 Subspaces

Recall that a vector space is a set equipped with two operations, vector addition and multiplication by a scalar, and these operations must satisfy certain properties. The following concept is a generalization of the notion of a line or a plane which passes through the origin.

*A non-empty subset $\mathcal{S}$ of a vector space is a **subspace** if for every pair of vectors x, y in $\mathcal{S}$ and every pair of scalars α, β , the vector $\alpha x + \beta y$ is contained in $\mathcal{S}$.*

In other words, a subset of a vector space is a subspace if it is closed under the operations of addition and multiplication by a scalar. Note that a subspace can be considered a vector space (Why?). An immediate consequence of the above definition is that a subspace must contain the zero vector; to see this consider α and β to be zero.

Example 42 Consider the set of real 2-vectors x which satisfy

$$x_1 - x_2 = 0$$

This set is a subspace of $\mathbb{R}^2$.

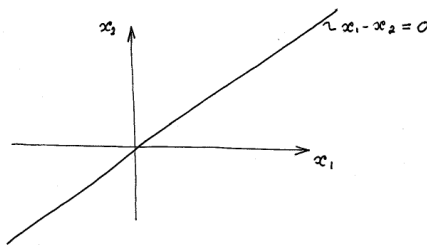


Figure 5.1: A simple subspace

Note that the set of vectors satisfying

$$x_1 - x_2 - 1 = 0$$

is not a subspace; it does not contain the zero vector.

Example 43 Consider any real $m \times n$ matrix A and consider the set of real n -vectors x which satisfy

$$Ax = 0$$

This set is a subspace of $\mathbb{R}^n$.

Example 44 The set of real matrices of the form

$$\begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}$$

is a subspace of the space of real 2×2 matrices.

Example 45 As a more abstract example of a subspace space, let $\mathcal{V}$ be the set of continuous real-valued functions which are defined on the interval $[0, \infty)$ and let addition and scalar multiplication be as defined earlier. Suppose $\mathcal{W}$ be the subset of $\mathcal{V}$ which consists of the differentiable functions $\mathcal{V}$. Then $\mathcal{W}$ is a subspace of $\mathcal{V}$.

5.3 Basis

To describe a subspace, we use a **basis**, which is a minimal set of vectors with the property that every vector in the subspace can be expressed in terms of the vectors in the basis. To discuss bases, we need the concepts of the **span** of a bunch of vectors and the concept of linear independence.

5.3.1 Span (not Spam)

Consider a bunch $\mathcal{X}$ of vectors. A **linear combination** of these vectors is a vector which can be expressed as

$$\alpha_1 x^1 + \alpha_2 x^2 + \dots + \alpha_m x^m$$

where each x^i is in the bunch and each α_i is a scalar.

*The **span** of a bunch $\mathcal{X}$ of vectors is the set of all vectors of the form*

$$\alpha_1 x^1 + \alpha_2 x^2 + \dots + \alpha_m x^m$$

where each x^i is in the bunch and each α_i is a scalar; it is denoted by $\text{span } \mathcal{X}$.

In other words, the span of a bunch of vectors is the set of all linear combinations of vectors from the bunch.

Exercise 53 Show that $\text{span } \mathcal{X}$ is a subspace.

Example 46

$$\text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} = \mathbb{R}^2$$

To see this, note that any element x of $\mathbb{R}^2$ can be expressed as

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Example 47

$$\text{span} \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\} = \mathbb{R}^{2 \times 2}$$

To see this, note that any element A of $\mathbb{R}^{2 \times 2}$ can be expressed as:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = a_{11} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + a_{12} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + a_{21} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + a_{22} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Fact 2 Consider a bunch $\mathcal{X}$ of vectors. The following ‘experiences’ do not affect span.

- (a) Multiplication of one of the vectors by a nonzero scalar.
- (b) Replacing a vector by the sum of itself and a scalar multiple of another (different) vector in the bunch.
- (c) Removal of a vector which is a linear combination of other vectors in the bunch.
- (d) Inclusion of a vector which is in the span of the original bunch.

Span and rref. One can use rref to solve the following problem.

Given a bunch of n -vectors $x^1, x^2, \dots, x^m$ and another n -vector b , determine whether or not b is in the span of the bunch.

To solve this problem, consider any linear combination $\alpha_1 x^1 + \alpha_2 x^2 + \dots + \alpha_m x^m$ of the vectors $x^1, x^2, \dots, x^m$. Let A be the matrix whose columns are the vectors $x^1, x^2, \dots, x^m$, that is,

$$X = \begin{pmatrix} x^1 & x^2 & \cdots & x^m \end{pmatrix}$$

and let α be the m -vector whose components are $\alpha_1, \dots, \alpha_m$, that is,

$$\alpha = \begin{pmatrix} \alpha_1 & \alpha_2 & \cdots & \alpha_m \end{pmatrix}^T.$$

Then the above linear combination can be expressed as

$$\alpha_1 x^1 + \alpha_2 x^2 + \dots + \alpha_m x^m = \begin{pmatrix} x^1 & x^2 & \cdots & x^m \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_m \end{pmatrix} = X\alpha.$$

Thus, b is in the span of the bunch if and only if the equation

$$X\alpha = b$$

has a solution for α . One can determine whether or not a solution exists by considering $\hat{M} = \text{rref}(M)$ where

$$M = [X \ b] = \begin{pmatrix} x^1 & x^2 & \cdots & x^m & b \end{pmatrix}.$$

Example 48 Suppose we wish to determine whether or not the vector $(1, 1, -2)$ is a linear combination of the vectors

$$\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}.$$

Since

$$\text{rref} \left(\begin{pmatrix} 2 & 0 & -1 & 1 \\ 1 & 3 & 1 & 1 \\ 1 & 1 & 0 & -2 \end{pmatrix} \right) = \begin{pmatrix} 1 & 0 & -0.5 & 0 \\ 0 & 1 & 0.5 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

it follows that the last vector is not a linear combination of the other three.

5.3.2 Linear independence

A bunch of vectors, $x^1, x^2, \dots, x^m$ is linearly dependent if there exist scalars $\alpha_1, \alpha_2, \dots, \alpha_m$, not all zero, such that

$$\alpha_1 x^1 + \alpha_2 x^2 + \dots + \alpha_m x^m = 0 \tag{5.4}$$

The bunch is linearly independent if it is not linearly dependent, that is, the bunch is linearly independent if and only if (5.4) implies that

$$\alpha_1, \alpha_2, \dots, \alpha_m = 0$$

If the bunch contains only a single vector, then it is linearly independent if and only if that vector is not the zero vector. If a bunch consists of more than one vector, then it is linearly dependent if and only if one of the vectors can be expressed as a linear combination of the preceding vectors in the bunch, that is for some j ,

$$x^j = \beta_1 x^1 + \beta_2 x^2 + \dots + \beta_{j-1} x^{j-1}$$

for some scalars β_i . (Why?)

Example 49 The following bunches of vectors are linearly dependent.

(i)

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

(ii)

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

(iii)

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 7 \\ 6 \\ 5 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$$

Why?

(iv)

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

Example 50 The following bunches of vectors are linearly independent.

(a)

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

(b)

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

(c)

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 7 \\ 6 \\ 5 \end{pmatrix}$$

(d)

$$\begin{pmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 494 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 5 \\ 6 \\ 4 \\ 74 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 203 \\ 11 \end{pmatrix}$$

To see this:

$$\alpha_1 \begin{pmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 494 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 \\ 0 \\ 5 \\ 6 \\ 4 \\ 74 \end{pmatrix} + \alpha_3 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 203 \\ 11 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

implies

$$\begin{aligned} \alpha_1 &= 0 \\ 2\alpha_1 + 5\alpha_2 &= 0 \\ 4\alpha_1 + 4\alpha_2 + 203\alpha_3 &= 0 \end{aligned}$$

The first equation implies $\alpha_1 = 0$; combining this with the second equation yields $\alpha_2 = 0$; combining these results with the last equation produces $\alpha_3 = 0$. So, no nontrivial linear combination of this bunch equals zero; hence the bunch is linearly independent.

(e)

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Exercise 54 Useful generalization of example 50 (d)

Consider a bunch of m vectors in $\mathbb{R}^n$ ($m \leq n$) which have the following structure.

$$\begin{pmatrix} 0 \\ \vdots \\ 0 \\ x_{n_1}^1 \\ \vdots \\ \vdots \\ \vdots \end{pmatrix}, \quad \begin{pmatrix} 0 \\ \vdots \\ \vdots \\ 0 \\ x_{n_2}^2 \\ \vdots \\ \vdots \end{pmatrix}, \quad \dots, \quad \begin{pmatrix} 0 \\ \vdots \\ \vdots \\ \vdots \\ 0 \\ x_{n_m}^m \\ \vdots \\ \vdots \end{pmatrix}$$

$$x_{n_1}^1, x_{n_2}^2, \dots, x_{n_m}^m \neq 0$$

In words, the first $n_i - 1$ components of the i -th vector are zero; its n_i component is nonzero; and $n_1 < n_2 < \dots < n_m$.

Prove that this bunch is linearly independent.

- Consider a bunch

$$x^1, x^2, \dots, x^m$$

of vectors. It can readily be shown that:

Fact 3 *The following ‘experiences’ do not affect linear independence status.*

- (a) *Multiplication of one of the vectors by a nonzero scalar.*
- (b) *Replacing a vector by the sum of itself and a scalar multiple of another (different) vector in the bunch.*
- (c) *Removal of a vector from the bunch.*
- (d) *Inclusion of a vector which is not in the span of the original bunch.*

Using rref to determine linear independence. Suppose we want to determine whether or not a bunch of n -vectors, $x^1, x^2, \dots, x^m$, are linearly independent. We can achieve this using rref. To see this, consider any linear combination $\alpha_1 x^1 + \alpha_2 x^2 + \dots + \alpha_m x^m$ of the vectors $x^1, x^2, \dots, x^m$. Let A be the $n \times m$ matrix whose columns are the vectors $x^1, x^2, \dots, x^m$, that is,

$$X = \begin{pmatrix} x^1 & x^2 & \cdots & x^m \end{pmatrix},$$

and let α be the m -vector whose components are $\alpha_1, \dots, \alpha_m$, that is,

$$\alpha = [\alpha_1 \quad \alpha_2 \quad \cdots \quad \alpha_m]^T.$$

Then the above linear combination of vectors can be expressed as

$$\alpha_1 x^1 + \alpha_2 x^2 + \dots + \alpha_m x^m = \begin{pmatrix} x^1 & x^2 & \cdots & x^m \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_m \end{pmatrix} = X\alpha.$$

Thus the linear combination is zero if and only if $X\alpha = 0$. Letting $\hat{X}$ be the reduced row echelon form of X , it follows that α satisfies $X\alpha = 0$ if and only if $\hat{X}\alpha = 0$. So, the columns of X are linearly independent if and only if the columns of $\hat{X}$ are linearly independent. Moreover, *a column of X is a linear combination of the preceding columns of X if and only if the corresponding column of $\hat{X}$ is the same linear combination of the preceding columns of $\hat{X}$.* From the above observations, we can obtain the following results where $\hat{X}$ is the reduced row echelon form of X .

- (a) *The vectors are linear independent if and only if every column of $\hat{X}$ contains a leading element, that is, the number of leading elements of $\hat{X}$ is the same as the number of vectors.*
- (b) *If $\hat{X}$ has a zero column, then the corresponding x -vector is zero; hence the vectors are linearly dependent.*

(c) If a non-zero column of $\hat{X}$ does not contain a leading element, then the corresponding vector can be written as a linear combination of the preceding vectors, hence the vectors are linearly dependent.

Example 51 Show that the vectors,

$$\begin{pmatrix} 1 \\ 1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 3 \\ -2 \end{pmatrix},$$

are linear dependent.

Solution:

$$\text{rref} \left(\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 3 \\ -1 & 0 & -2 \end{pmatrix} \right) = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The vectors are linearly dependent. The third vector is the twice the first vector minus the second vector.

Example 52 Are the vectors

$$\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix},$$

linearly independent?

Solution:

$$\text{rref} \left(\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 0 & -1 \end{pmatrix} \right) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The vectors are linearly independent.

5.3.3 Basis

A (non-zero) space (or subspace) contains an infinite number of vectors; however, if the space is finite-dimensional (to be defined shortly), it can be described by a finite number of vectors. A basis for a space is a minimal set of vectors which describe the space. Every vector in the space can be written uniquely as a linear combination of basis vectors.

A basis for a vector space is a bunch of linearly independent vectors whose span is the space.

A vector space is called **finite dimensional** if it has a finite basis; otherwise it is **infinite dimensional**.

Some facts. For a finite dimensional vector space $\mathcal{V}$, it can be shown that the number of vectors in every basis is the same; this number is called the **dimension** of the space and is denoted $\dim \mathcal{V}$. We refer to a vector space with dimension n as an **n -dimensional space**. In an n -dimensional space, every bunch of more than n vectors space must be linearly dependent, and every bunch of n linearly independent vectors must be a basis.

Example 53 The following vectors constitute a basis for $\mathbb{R}^2$ (Why?) :

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Hence $\mathbb{R}^2$ is a 2-dimensional space (Big surprise!).

Example 54 The following vectors constitute a basis for $\mathbb{R}^{2 \times 2}$ (Why?) :

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Hence $\mathbb{R}^{2 \times 2}$ is a 4-dimensional space.

Example 55 The following vectors constitute a basis for the subspace of example 44.

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Exercise 55 Give an example of a basis for $\mathbb{R}^3$ and for $\mathbb{R}^{2 \times 3}$. What are the dimensions of these spaces?

Coordinates. If $e^1, e^2, \dots, e^n$ is a basis for an n -dimensional vector space, then every vector x in the space has a *unique* representation of the form:

$$x = \xi_1 e^1 + \xi_2 e^2 + \dots + \xi_n e^n$$

The scalars $\xi_1, \xi_2, \dots, \xi_n$ are called the **coordinates** of x wrt the basis $e^1, e^2, \dots, e^n$. Thus there is a one-to-one correspondence between any real n -dimensional vector space and $\mathbb{R}^n$.

$$x \begin{matrix} \xrightarrow{\quad} \\ \xleftarrow{\quad} \end{matrix} \begin{pmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_n \end{pmatrix}$$

Any linear map from a real n -dimensional $\mathcal{V}$ space to a real m -dimensional space $\mathcal{W}$ can be ‘uniquely’ represented by a real $m \times n$ matrix; that is, there is a one-to-one correspondence between the set of linear maps from $\mathcal{V}$ to $\mathcal{W}$ and $\mathbb{R}^{m \times n}$. This is achieved by choosing bases for $\mathcal{V}$ and $\mathcal{W}$.

5.4 Range and null space of a matrix

5.4.1 Null space

Consider any real $m \times n$ matrix A .

The null space of A or the kernel of A denoted by $\mathcal{N}(A)$, is the set of real n -vectors x which satisfy

$$Ax = 0$$

that is,

$$\mathcal{N}(A) := \{x \in \mathbb{R}^n : Ax = 0\}$$

- $\mathcal{N}(A)$ is a subspace of $\mathbb{R}^n$. (**Exercise**)
- The nullity of A is the dimension of its null space, that is,

$$\text{nullity } A := \dim \mathcal{N}(A)$$

Determining the null space of a matrix A simply amounts to solving the homogeneous equation $Ax = 0$. This we can do by obtaining $\text{rref}(A)$.

Example 56 A null space calculation

Let's compute a basis for the null space of the following matrix.

$$A = \begin{pmatrix} 0 & 2 & 2 & 4 & -2 \\ 1 & 2 & 3 & 4 & -1 \\ 2 & 0 & 2 & 0 & 2 \end{pmatrix}$$

We saw in Example 40 that the reduced row echelon form of A is given by

$$\hat{A} = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Hence, the vectors in the null space of A are those vectors x which satisfy $\hat{A}x = 0$, that is,

$$\begin{aligned} x_1 + x_3 + x_5 &= 0 \\ x_2 + x_3 + 2x_4 - x_5 &= 0 \end{aligned}$$

these equations are equivalent to

$$\begin{aligned} x_1 &= -x_3 - x_5 \\ x_2 &= -x_3 - 2x_4 + x_5 \end{aligned}$$

Hence, we can consider x_3, x_4, x_5 arbitrary and for any given x_3, x_4, x_5 , the remaining components x_1 and x_2 are uniquely determined; hence, all vectors in the null space must be of the form

$$\begin{pmatrix} -x_3 - x_5 \\ -x_3 - 2x_4 + x_5 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = x_3 \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 0 \\ -2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + x_5 \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

where x_3, x_4, x_5 are arbitrary. Thus the null space of A is the subspace spanned by the three vectors:

$$\begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ -2 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Since the three vectors are linearly independent, they form a basis for the null space of A ; hence the nullity of A is 3.

Example 57 Consider any non-zero (angular velocity) matrix of the form:

$$A = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}$$

The null space of A is the set of vectors x which satisfy:

$$\begin{array}{rclcl} & -\omega_3 x_2 & + & \omega_2 x_3 & = & 0 \\ \omega_3 x_1 & & & - & \omega_1 x_3 & = & 0 \\ -\omega_2 x_1 & + & \omega_1 x_2 & & = & 0 \end{array}$$

Every solution to these equations is a scalar multiple of

$$x = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}$$

Hence, the null space of A is the 1-dimensional subspace of $\mathbb{R}^3$ spanned by this vector.

- The set of equilibrium states of a continuous LTI system $\dot{x} = Ax$ is the null space of A .
- The set of equilibrium states of a discrete LTI system $x(k+1) = Ax(k)$ is the null space of $A - I$.

5.4.2 Range

Consider any real $m \times n$ matrix A .

The **range** of A , denoted by $\mathcal{R}(A)$, is the set of m -vectors of the form Ax where x is an n -vector, that is,

$$\mathcal{R}(A) := \{Ax : x \in \mathbb{R}^n\}$$

Thus, the range of A is the set of vectors b for which the equation $Ax = b$ has a least one solution for x .

- $\mathcal{R}(A)$ is a subspace of $\mathbb{R}^m$. (**Exercise**)

The **rank** of A is the dimension of its range space, that is,

$$\text{rank } A := \dim \mathcal{R}(A)$$

Since

$$A = \begin{pmatrix} a_1 & a_2 & \dots & a_n \end{pmatrix}$$

where $a_1, a_2, \dots, a_n$ are the columns of A , it should be clear that

$$Ax = \begin{pmatrix} a_1 & a_2 & \dots & a_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1 a_1 + x_2 a_2 + \dots + x_n a_n$$

Hence, the range of a matrix is the set of all linear combinations of its columns. Thus the rank of A equals the maximum number of linearly independent columns of A .

From Fact 2 it should be clear that if a matrix $\tilde{A}$ is obtained from a matrix A by a sequence of the following operations then A and $\tilde{A}$ have the same range.

- Interchanging two columns.
- Multiplication of one of the columns by a nonzero scalar.
- Replacing a column by the sum of itself and a scalar multiple of another (different) column in the bunch.
- Removal of a column which is a linear combination of other columns in the bunch.
- Inclusion of a column which is a linear combination of other columns in the bunch.

Calculation of range basis and rank with rref

Here, we present two ways to calculate a basis for the range of a matrix using rref.

Method one: $\text{rref}(A)$ A basis for the range of A is given by the columns of A corresponding to the columns of $\text{rref}(A)$ which contain leading elements. Thus the rank of A is the number of leading elements in $\text{rref}(A)$. This method yields a basis whose members are columns from A .

Method two: $\text{rref}(A^T)$ Let $\hat{B}$ be the reduced row echelon form of A^T , that is

$$\hat{B} = \text{rref}(A^T).$$

Then the nonzero columns of $\hat{B}^T$ form a basis for the range of A . Thus the rank of A is the number of leading elements in $\text{rref}(A^T)$.

From the above considerations, it follows that, for any matrix A ,

$$\text{rank } A^T = \text{rank } A$$

hence,

$$\begin{aligned} \text{rank } A &= \text{maximum number of linearly independent columns of } A \\ &= \text{maximum number of linearly independent rows of } A \end{aligned}$$

Example 58 (A range calculation) Let's compute a basis for the range of the matrix,

$$A = \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 0 & 2 & 0 \end{pmatrix}$$

Using Method one, we compute

$$\text{rref}(A) = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Since the leading elements of $\text{rref}(A)$ occur in columns 1 and 2, a basis for the range of A is given by columns 1 and 2 of A , that is

$$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}.$$

Also, the rank of A is two.

Using Method two, we compute

$$\text{rref}(A^T) = \begin{pmatrix} 1 & 0 & -4 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Hence,

$$\text{rref}(A^T)^T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -4 & 2 & 0 & 0 \end{pmatrix}$$

Since columns 1 and 2 are the non-zero columns of $\text{rref}(A^T)^T$, a basis for the range of A is given by columns 1 and 2 of $\text{rref}(A^T)^T$, that is

$$\begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}.$$

Again we obtain that the rank of A is two.

Fact 4 *If A is an $m \times n$ matrix,*

$$\boxed{\dim \mathcal{R}(A) + \dim \mathcal{N}(A) = n}$$

Fact 5 *If A and B are two matrices which have the same numbers of columns, then*

$$\mathcal{N}(A) = \mathcal{N}(B) \quad \text{if and only if} \quad \mathcal{R}(A^T) = \mathcal{R}(B^T)$$

Exercise 56 Picturize the range and the null space of the following matrices.

$$\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$$

Exercise 57

(a) Find a basis for the null space of the matrix,

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{pmatrix}.$$

(b) What is the nullity of A ?

Check your answer using MATLAB.

Exercise 58

(a) Find a basis for the range of the matrix,

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{pmatrix}.$$

(b) What is the rank of A ?

Check your answer using MATLAB.

Exercise 59 Obtain a basis for the range of each of the following matrices. Also, what is the rank and nullity of each matrix?

$$\begin{pmatrix} 4 & 1 & -1 \\ 3 & 2 & -3 \\ 1 & 3 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 1 & 2 \\ 3 & 4 & 5 & 0 & 0 \end{pmatrix}$$

Check your answers using MATLAB.

Exercise 60 Obtain a basis for the null space of each of the matrices in exercise 59. Check your answers using MATLAB

5.4.3 Solving linear equations revisited

Recall the problem of solving the following m scalar equations for n scalar unknowns $x_1, x_2, \dots, x_n$:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

These equations can be written in matrix form as:

$$Ax = b$$

where x is $n \times 1$, b is $m \times 1$ and A is $m \times n$.

- This matrix equation has at least one solution for x if and only if b is in the range of A . This is equivalent to

$$\text{rank}(A \ b) = \text{rank } A$$

The matrix equation has at least one solution for *every* b if and only if the range of A is $\mathbb{R}^m$, that is, the rank of A is m , or equivalently, the nullity of A is $n - m$; hence $n \geq m$.

- If x^* is any particular solution of $Ax = b$, then all other solutions have the form

$$x^* + e$$

where e is in the null space of A .

To see this:

Hence, a solution is *unique* if and only if the nullity of A is zero or equivalently, the rank of A is n ; hence $n \leq m$.

- It follows from the above that A is invertible if and only if its rank is m and its nullity is 0; since $\text{rank } A + \text{nullity } A = n$, we must have $m = n$, that is A must be square.

MATLAB A solution (if one exists) to the equation $Ax = b$ can be solved by

`x = A\b`

`>> help orth`

ORTH Orthogonalization.

`Q = orth(A)` is an orthonormal basis for the range of `A`.

`Q'*Q = I`, the columns of `Q` span the same space as the columns of `A` and the number of columns of `Q` is the rank of `A`.

See also `SVD`, `ORTH`, `RANK`.

`>> help rank`

RANK Number of linearly independent rows or columns.

`K = RANK(X)` is the number of singular values of `X` that are larger than `MAX(SIZE(X)) * NORM(X) * EPS`.

`K = RANK(X,tol)` is the number of singular values of `X` that are larger than `tol`.

`>> help null`

NULL Null space.

`Z = null(A)` is an orthonormal basis for the null space of `A`.

`Z'*Z = I`, `A*Z` has negligible elements, and the number of columns of `Z` is the nullity of `A`.

See also `SVD`, `ORTH`, `RANK`.

Exercise 61 Using MATLAB, compute the rank of the matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & \delta \end{pmatrix}$$

for

$$\delta = 1, 10^{-3}, 10^{-10}, 10^{-15}, 10^{-16}$$

Any comments?

Range and null space of linear maps*

Suppose $\mathcal{A}$ is a linear map from an n -dimensional space $\mathcal{V}$ to an m -dimensional space $\mathcal{W}$. Then the above definitions and results for matrices can be generalized as follows.

Range of $\mathcal{A}$

$$\mathcal{R}(\mathcal{A}) := \{\mathcal{A}x : x \in \mathcal{V}\}$$

- $\mathcal{R}(\mathcal{A})$ is a subspace of $\mathcal{W}$.

$$\text{rank } \mathcal{A} := \dim \mathcal{R}(\mathcal{A})$$

Null space of $\mathcal{A}$

$$\mathcal{N}(\mathcal{A}) := \{x \in \mathcal{V} : \mathcal{A}x = 0\}$$

- $\mathcal{N}(\mathcal{A})$ is a subspace of $\mathcal{V}$.

$$\text{nullity of } \mathcal{A} := \dim \mathcal{N}(\mathcal{A})$$

Fact 6

$$\dim \mathcal{R}(\mathcal{A}) + \dim \mathcal{N}(\mathcal{A}) = n$$

- Suppose $\mathcal{W} = \mathcal{V}$. Then $\mathcal{A}$ is invertible if and only if $\text{rank } \mathcal{A} = n$.

Suppose b is given vector in $\mathcal{W}$ and consider the equation.

$$\mathcal{A}x = b$$

This equation has a solution for x if and only if b is in the range of $\mathcal{A}$. If x^* is any particular solution of $\mathcal{A}x = b$, then all other solutions have the form

$$x^* + e$$

where e is in the null space of $\mathcal{A}$

5.5 Coordinate transformations

Suppose we have n scalar variables $x_1, x_2, \dots, x_n$ and we implicitly define new scalar variables $\xi_1, \xi_2, \dots, \xi_n$ by

$$\boxed{x = U\xi}$$

where U is an $n \times n$ invertible matrix. Then, ξ is explicitly given by

$$\xi = U^{-1}x$$

We can obtain a geometric interpretation of this change of variables as follows. First, observe that

$$\underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_x = x_1 \underbrace{\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{e^1} + x_2 \underbrace{\begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}}_{e^2} + \dots + x_n \underbrace{\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}}_{e^n}$$

that is the scalars $x_1, x_2, \dots, x_n$ are the coordinates of the vector x wrt the *standard basis* $e^1, e^2, \dots, e^n$, or, x is the coordinate vector of itself wrt the standard basis. Suppose

$$U = \begin{pmatrix} u^1 & \dots & u^j & \dots & u^n \end{pmatrix}$$

that is, u^j is the j -th column of U . Since U is invertible, its columns $u^1, u^2, \dots, u^n$ are linearly independent; hence they form a basis for $\mathbb{R}^n$. Since $x = U\xi$ can be written as

$$x = \xi_1 u^1 + \xi_2 u^2 + \dots + \xi_n u^n$$

we see that $\xi_1, \xi_2, \dots, \xi_n$ are the coordinates of x wrt to the new basis $u^1, u^2, \dots, u^n$ and the vector ξ is the coordinate vector of the vector x wrt this new basis. So $x = U\xi$ defines a *coordinate transformation*.

5.5.1 Coordinate transformations and linear maps

Suppose we have a linear map from $\mathbb{R}^n$ into $\mathbb{R}^m$ defined by

$$y = Ax$$

where A is an $m \times n$ matrix. Suppose we introduce coordinate transformations

$$x = U\xi, \quad y = V\eta$$

where U and V are invertible matrices. Then the map is described by

$$\eta = \Lambda\xi$$

where

$$\Lambda = V^{-1}AU$$

Recalling that the columns $u^1, u^2, \dots, u^n$ of U form a basis for $\mathbb{R}^n$, and the columns $v^1, v^2, \dots, v^m$ of V form a basis for $\mathbb{R}^m$, we have now the following very useful result.

• **Useful result:** Suppose

$$Au^j = \alpha_{1j}v^1 + \alpha_{2j}v^2 + \dots + \alpha_{mj}v^m$$

Then the matrix $\Lambda = V^{-1}AU$ is uniquely given by

$$\Lambda_{ij} = \alpha_{ij}$$

that is Λ_{ij} is the i -th coordinate of the vector Au^j wrt the basis $v^1, v^2, \dots, v^m$, or, the j -th column of Λ is the coordinate vector of Au^j wrt the basis $v^1, v^2, \dots, v^m$.

Example 59 Suppose A is a 2×2 matrix and there are two linearly independent vectors u^1 and u^2 which satisfy:

$$Au^1 = 2u^1 \quad \text{and} \quad Au^2 = u^1 + 2u^2$$

Letting $U = \begin{pmatrix} u^1 & u^2 \end{pmatrix}$, we can use the above useful result to obtain that

$$U^{-1}AU = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

Example 60 Suppose A is a 3×2 matrix while $U = \begin{pmatrix} u^1 & u^2 \end{pmatrix}$ and $V = \begin{pmatrix} v^1 & v^2 & v^3 \end{pmatrix}$ are invertible 3×3 and 2×2 matrices, respectively, which satisfy

$$Au^1 = v_1 + 2v_3 \quad Au^2 = 2v_2 + v_3$$

Then, applying the above useful result, we obtain

$$V^{-1}AU = \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 2 & 1 \end{pmatrix}$$

Proof of useful result. Premultiply $\Lambda = V^{-1}AU$ by V to get:

$$V\Lambda = AU$$

On the right we have

$$AU = A \begin{pmatrix} u^1 & \dots & u^j & \dots & u^n \end{pmatrix} = \begin{pmatrix} Au^1 & \dots & Au^j & \dots & Au^n \end{pmatrix}$$

And on the left,

$$\begin{aligned}
V\Lambda &= \begin{pmatrix} v^1 & v^2 & \dots & \dots & v^m \end{pmatrix} \begin{pmatrix} \Lambda_{11} & \dots & \Lambda_{1j} & \dots & \Lambda_{1n} \\ \Lambda_{21} & \dots & \Lambda_{2j} & \dots & \Lambda_{2n} \\ \vdots & & \vdots & & \vdots \\ \vdots & & \vdots & & \vdots \\ \Lambda_{m1} & \dots & \Lambda_{mj} & \dots & \Lambda_{mn} \end{pmatrix} \\
&= \begin{pmatrix} \Lambda_{11}v^1 + \dots + \Lambda_{m1}v^m : & \dots & \underbrace{\Lambda_{1j}v^1 + \dots + \Lambda_{mj}v^m}_{j\text{-th column}} : & \dots & \Lambda_{1n}v^1 + \dots + \Lambda_{mn}v^m \end{pmatrix}
\end{aligned}$$

Comparing the expressions for the j -th columns of AU and $V\Lambda$ yields

$$\Lambda_{1j}v^1 + \dots + \Lambda_{mj}v^m = Au^j = \alpha_{1j}v^1 + \dots + \alpha_{mj}v^m$$

Since the $\{v^1, \dots, v^m\}$ is a basis we must have $\Lambda_{ij} = \alpha_{ij}$. ■

5.5.2 The structure of a linear map*

Consider again a linear map from $\mathbb{R}^n$ into $\mathbb{R}^m$ defined by

$$y = Ax$$

where A is an $m \times n$ matrix. Suppose the rank of A is r . Then the dimension of the null space of A is $n - r$. Choose any basis

$$\{v^1, \dots, v^r\}$$

for the range of A and extend it to a basis

$$\{v^1, v^2, \dots, v^m\}$$

for $\mathbb{R}^m$. Choose any basis

$$\{u^{r+1}, \dots, u^n\}$$

for the null space of A and extend it to a basis

$$\{u^1, u^2, \dots, u^n\}$$

for $\mathbb{R}^n$.

If we introduce the coordinate transformations

$$y = V\eta, \quad x = U\xi$$

where

$$V = [v^1 \dots v^m], \quad U = [u^1 \dots u^n]$$

the original linear map is described by

$$\eta = \Lambda \xi$$

where

$$\Lambda = V^{-1}AU.$$

Recall the useful result on coordinate transformations. Since the vectors $u^j, j = r + 1, \dots, n$ are in the null space of A we have

$$Au^j = 0 \quad \text{for } j = r + 1, \dots, n$$

Hence, the last r columns of Λ are zero. For $j = 1, \dots, r$ the vector Au^j is in the range of A . Hence it can be expressed as a linear combination of the vectors $v^1, \dots, v^r$. Consequently, the last $m - r$ rows of Λ are zero and Λ has the following structure:

$$\Lambda = \begin{pmatrix} \Lambda_{11} & 0 \\ 0 & 0 \end{pmatrix}$$

where $\Lambda_{11} \in \mathbb{R}^{r \times r}$. Since the rank of A equals rank Λ which equals rank Λ_{11} , the matrix Λ_{11} must be invertible.

If we decompose ξ and η as

$$\xi = \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix}, \quad \eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$$

where $\xi_1 \in \mathbb{R}^r, \xi_2 \in \mathbb{R}^{n-r}, \eta_1 \in \mathbb{R}^r, \eta_2 \in \mathbb{R}^{m-r}$, then the linear map is described by

$$\begin{aligned} \eta_1 &= \Lambda_{11}\xi_1 \\ \eta_2 &= 0 \end{aligned}$$

where Λ_{11} is invertible.

Also, we have shown that any matrix any $m \times n$ matrix A can be decomposed as

$$\boxed{A = V \begin{pmatrix} \Lambda_{11} & 0 \\ 0 & 0 \end{pmatrix} U^{-1}}$$

where U, V , and Λ are square invertible matrices of dimensions $n \times n, m \times m$, and $r \times r$, respectively, and r is the rank of A .

Chapter 6

Behavior of (LTI) systems : I

6.1 Initial stuff

6.1.1 Continuous-time

Here we look at the behavior of linear time-invariant systems described by

$$\boxed{\dot{x} = Ax} \tag{6.1}$$

where the state $x(t)$ is a complex n -vector and A is a complex $n \times n$ matrix. (The reason for complexifying things will be seen shortly.) Sometimes A is called the **system matrix**.

Solutions. By a solution of system (6.1) we mean any continuous function $x(\cdot)$ which is defined on some time interval $[t_0, \infty)$ and which satisfies (6.1). Sometimes we are interested in solutions which satisfy an **initial condition** of the form

$$\boxed{x(t_0) = x_0} \tag{6.2}$$

where t_0 is called the **initial time** and x_0 is called the **initial state**. As a consequence of the linearity of the system, one can prove the following result:

For each initial condition, system (6.1) has a unique solution.

Shortly, we will study the nature of these solutions. Since the system is time-invariant, knowledge of the solutions for zero initial time, that is $t_0 = 0$, yields knowledge of the solutions for any other initial time. To see this suppose $\tilde{x}(\cdot)$ is the solution with $\tilde{x}(0) = x_0$. Then, the solution corresponding to $x(t_0) = x_0$ is given by

$$x(t) = \tilde{x}(t - t_0).$$

Since the system is linear, one can readily show that *solutions depends linearly on the initial state*; more specifically, there is a matrix valued function Φ (called the **state transition matrix**) such that at each time t the solution due to $x(0) = x_0$ is uniquely given by

$$x(t) = \Phi(t)x_0$$

Later, we develop explicit expressions for Φ .

6.1.2 Discrete-time

Here we look at the behavior of discrete-time linear time-invariant systems described by

$$\boxed{x(k+1) = Ax(k)}$$

The state $x(k)$ is a complex n -vector and A is a complex $n \times n$ matrix. Sometimes A is called the **system matrix**.

Solutions: Consider any *initial condition*:

$$\boxed{x(k_0) = x_0} \tag{6.3}$$

where $k_0 = \dots, -1, 0, 1, \dots$ is called the **initial time** and the n -vector x_0 is called the **initial state**. Since the system is time-invariant we can, wlog (without loss of generality), consider only zero initial time, that is, $k_0 = 0$. For suppose $\tilde{x}(\cdot)$ is a solution with $\tilde{x}(0) = x_0$. Then

$$x(k) = \tilde{x}(k - k_0)$$

is a solution corresponding to (6.3).

For $k \geq k_0 = 0$, we have existence and uniqueness of solutions; specifically,

$$\boxed{x(k) = A^k x_0}$$

where $x(0) = x_0$.

Proof: (Example of a proof by induction) Suppose there is a $k^* \geq 0$ for which the above holds, that is,

$$x(k^*) = A^{k^*} x_0$$

Then

$$\begin{aligned} x(k^*+1) &= Ax(k^*) \\ &= AA^{k^*} x_0 \\ &= A^{(k^*+1)} x_0 \end{aligned}$$

So, if it holds for k^* it holds for $k^* + 1$. Since, $x(0) = A^0 x_0$, it holds for $k^* = 0$. It now follows by induction that it holds for all $k^* \geq 0$. ■

Solution depends linearly on initial state; specifically,

$$x(k) = \Phi(k)x(0), \quad \text{for } k \geq 0$$

where $\Phi(k) = A^k$. Hence, for any $k \geq k_0$

$$\boxed{x(k) = A^{(k-k_0)} x(k_0)} \tag{6.4}$$

6.2 Scalar systems

6.2.1 Continuous-time

Consider a scalar system described by

$$\dot{x} = ax$$

where $x(t)$ and a are (real or complex) scalars. All solutions are given by:

$$x(t) = e^{at}x_0; \quad x_0 := x(0)$$

Real a . So,

- (a) If $a < 0$ you decay with increasing age.
- (b) If $a > 0$ you grow with increasing age.
- (c) If $a = 0$ you stay the same; every state is an equilibrium state.

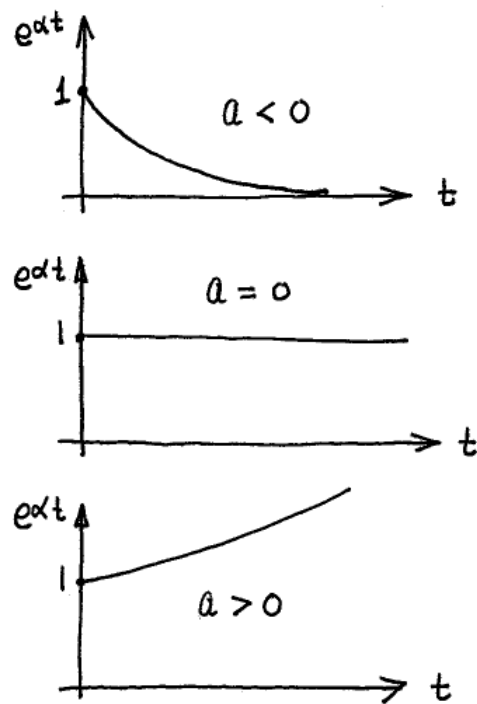


Figure 6.1: The three types of modes for real scalar systems: $\alpha = a$

Example 61 (A first example.) Here

$$a = -c/m < 0$$

Thus, all solutions are of the form $e^{-(c/m)t}x_0$; hence they decay exponentially.

Complex a . Suppose a is a genuine complex number (unreal); that is,

$$a = \alpha + j\omega$$

where α and ω are real numbers. We call α the real part of a and denote this by $\alpha = \Re(a)$. Similarly, we call ω the imaginary part of a and denote this by $\omega = \Im(a)$.

We now show that the growth behavior of e^{at} depends only on the real part of a . To this end, we first note that

$$e^{at} = e^{(\alpha+j\omega)t} = e^{(\alpha t + j\omega t)} = e^{\alpha t} e^{j\omega t},$$

and, hence, $|e^{at}| = |e^{\alpha t}| |e^{j\omega t}|$. Since $e^{\alpha t}$ is positive, $|e^{\alpha t}| = e^{\alpha t}$. Since,

$$e^{j\omega t} = \cos \omega t + j \sin \omega t,$$

we obtain that $|e^{j\omega t}| = 1$. Hence,

$$\boxed{|e^{at}| = e^{\alpha t} \quad \text{where} \quad \alpha = \Re(a)}$$

Thu, $|x(t)| = |e^{at}x_0| = |e^{at}||x_0| = e^{\alpha t}|x_0|$. So,

- (a) If $\Re(a) < 0$ you decay with increasing age.
- (b) If $\Re(a) > 0$ you grow with increasing age.
- (c) If $\Re(a) = 0$ your magnitude remains constant.

6.2.2 First order complex and second order real.

Suppose a is a genuine complex number (unreal); that is,

$$a = \alpha + j\omega \quad \text{where} \quad \alpha = \Re(a), \quad \omega = \Im(a)$$

Since

$$x = \xi_1 + j\xi_2 \quad \text{where} \quad \xi_1 = \Re(x), \quad \xi_2 = \Im(x)$$

Then

$$\begin{aligned} \dot{\xi}_1 &= \alpha \xi_1 - \omega \xi_2 \\ \dot{\xi}_2 &= \omega \xi_1 + \alpha \xi_2 \end{aligned} = \underbrace{\begin{pmatrix} \alpha & -\omega \\ \omega & \alpha \end{pmatrix}}_{\text{axisymmetric spacecraft}} \xi$$

Exercise 62 Verify the last set of equations.

Also,

$$\begin{aligned}\xi_1(t) &= e^{\alpha t} \cos(\omega t) \xi_{10} - e^{\alpha t} \sin(\omega t) \xi_{20} \\ \xi_2(t) &= e^{\alpha t} \sin(\omega t) \xi_{10} + e^{\alpha t} \cos(\omega t) \xi_{20}\end{aligned} = \begin{pmatrix} e^{\alpha t} \cos(\omega t) & -e^{\alpha t} \sin(\omega t) \\ e^{\alpha t} \sin(\omega t) & e^{\alpha t} \cos(\omega t) \end{pmatrix} \xi_0$$

where $\xi_{10} = \xi_1(0)$, $\xi_{20} = \xi_2(0)$

Exercise 63 Verify the last set of equations. Hint: $e^{j\omega t} = \cos \omega t + j \sin \omega t$

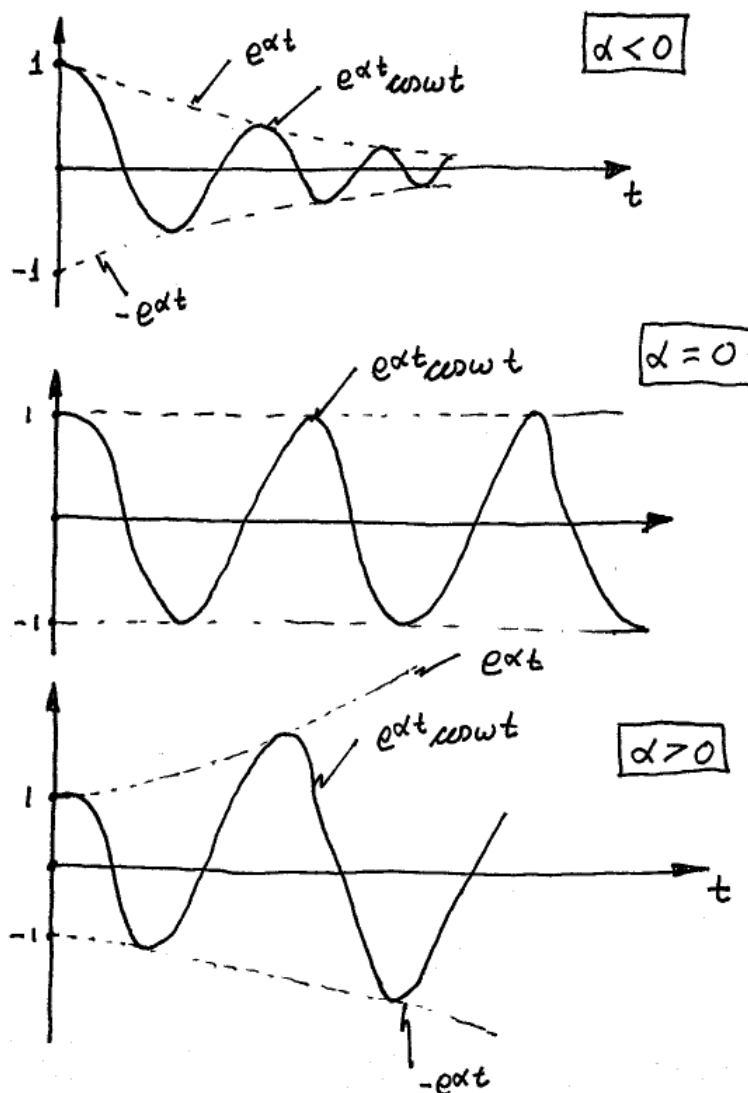


Figure 6.2: The three complex modes

6.2.3 DT

$$\boxed{x(k+1) = ax(k)}$$

where $x(k)$ and a are scalars. All solutions are given by:

$$\boxed{x(k) = a^k x_0 ; \quad x_0 := x(0)}$$

Hence, $|x(k)| = |a^k x_0| = |a|^k |x_0|$. So,

- (a) If $|a| < 1$ you decay with increasing age.
- (b) If $|a| > 1$ you grow with increasing age.
- (c) If $|a| = 1$ your magnitude remains the same.
- (d) If $a = 1$ you stay the same; every state is an equilibrium state.

6.3 Eigenvalues and eigenvectors

Shortly, we shall see that the behavior of a linear time invariant system is completely determined by the eigenvalues and eigenvectors of its system matrix.

Suppose A is a complex $n \times n$ matrix. (The use of the word complex also includes real. To indicate that something is complex but not real we say genuine complex.)

DEFN. A complex number λ is an **eigenvalue** of A if there is a nonzero vector v such that

$$\boxed{Av = \lambda v}$$

The nonzero vector v is called an **eigenvector** of A corresponding to λ .

So, an eigenvector of A is a nonzero complex n -vector with the property that there is a complex number λ such that the above relationship holds or, equivalently,

$$\boxed{(\lambda I - A)v = 0} \tag{6.5}$$

The existence of a non-zero vector v such that the above relation holds is equivalent to $\lambda I - A$ being singular; this in turn is equivalent to

$$\boxed{\det(\lambda I - A) = 0}$$

- The **characteristic polynomial** of A :

$$\text{charpoly}(A) := \det(sI - A)$$

Note that $\det(sI - A)$ is an n -th order **monic polynomial**, that is, it has the form

$$\det(sI - A) = a_0 + a_1 s + \dots + s^n$$

- We conclude that a complex number λ is an eigenvalue of A if and only if it is a root of the n th order characteristic polynomial of A . Hence, A has at least one eigenvalue and at most n distinct eigenvalues. Suppose A has l **distinct** eigenvalues, $\lambda_1, \lambda_2, \dots, \lambda_l$; since these are the distinct roots of the characteristic polynomial of A , we must have

$$\det(sI - A) = \prod_{i=1}^l (s - \lambda_i)^{m_i}$$

- The integer m_i is called the **algebraic multiplicity** of λ_i
- It should be clear that if v^1 and v^2 are any two eigenvectors corresponding to the same eigenvalue λ and ξ_1 and ξ_2 are any two numbers then (provided it is nonzero) $\xi_1 v^1 + \xi_2 v^2$ is also an eigenvector for λ . The set of eigenvectors corresponding to λ along with the zero vector is called the **eigenspace** of A associated with λ . This is simply the null space of $\lambda I - A$
- The **geometric multiplicity** of λ is the nullity of $\lambda I - A$, that is, it is the maximum number of linearly independent eigenvectors associated with λ .
- It follows that A is invertible if and only if all its eigenvalues are nonzero.

Example 62 (An eigenvalue-eigenvector calculation)

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

The characteristic polynomial of A is given by

$$\begin{aligned} \det(sI - A) &= \det \begin{pmatrix} s-3 & 1 \\ 1 & s-3 \end{pmatrix} \\ &= (s-3)(s-3) - 1 = s^2 - 6s + 8 \\ &= (s-2)(s-4) \end{aligned}$$

The roots of this polynomial yield *two distinct real eigenvalues*:

$$\lambda_1 = 2, \quad \lambda_2 = 4$$

To compute eigenvectors for λ_1 , we use $(\lambda_1 I - A)v = 0$ to obtain

$$\begin{aligned} -v_1 + v_2 &= 0 \\ v_1 - v_2 &= 0 \end{aligned}$$

which is equivalent to

$$v_1 - v_2 = 0$$

So, only one linearly independent eigenvector for λ_1 ; let's take

$$v = v^1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

In a similar fashion, λ_2 has one linearly independent eigenvector; we take

$$v^2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

To check the above calculations, note that:

$$Av^1 = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \lambda_1 v^1$$

Similarly for λ_2 .

Example 63 (Another eigenvalue-eigenvector calculation)

$$A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

The characteristic polynomial of A is given by

$$\begin{aligned} \det(sI - A) &= \det \begin{pmatrix} s-1 & -1 \\ 1 & s-1 \end{pmatrix} = (s-1)(s-1) + 1 \\ &= s^2 - 2s + 2 \end{aligned}$$

Computing the roots of this polynomial yields *two distinct complex eigenvalues*:

$$\lambda_1 = 1 + j, \quad \lambda_2 = 1 - j$$

To compute eigenvectors for λ_1 , we use $(\lambda_1 I - A)v = 0$ to obtain

$$\begin{aligned} jv_1 - v_2 &= 0 \\ v_1 + jv_2 &= 0 \end{aligned}$$

which is equivalent to

$$-jv_1 + v_2 = 0$$

So, only one linearly independent eigenvector; let's take

$$v = v^1 = \begin{pmatrix} 1 \\ j \end{pmatrix}$$

(We could also take $\begin{pmatrix} j & 1 \end{pmatrix}^T$.) In a similar fashion, λ_2 has one linearly independent eigenvector; we take

$$v^2 = \begin{pmatrix} 1 \\ -j \end{pmatrix}$$

To check the above calculations, note that

$$Av^1 = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ j \end{pmatrix} = \begin{pmatrix} 1+j \\ -1+j \end{pmatrix} = (1+j) \begin{pmatrix} 1 \\ j \end{pmatrix} = \lambda_1 v^1$$

Similarly for λ_2 .

Example 64

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Exercise 64 Compute expressions for eigenvalues and linearly independent eigenvectors of

$$A = \begin{pmatrix} \alpha & \omega \\ -\omega & \alpha \end{pmatrix} \quad \alpha, \omega \neq 0$$

6.3.1 Real A

Recall Examples 62 and 63. They illustrate the following facts.

- If A is real, its eigenvalues and eigenvectors occur in complex conjugate pairs.

To see this, suppose λ is an eigenvalue of A with eigenvector v . Then

$$Av = \lambda v$$

Taking complex conjugate of both sides of this equation yields

$$\bar{A}\bar{v} = \bar{\lambda}\bar{v}$$

Since A is real, $\bar{A} = A$; hence

$$A\bar{v} = \bar{\lambda}\bar{v}$$

that is, $\bar{\lambda}$ is an eigenvalue of A with eigenvector $\bar{v}$ ■

- Real eigenvectors for real eigenvalues.
- Genuine complex eigenvectors for (genuine) complex eigenvalues.
- The real and imaginary parts of a complex eigenvector are linearly independent.

Example 65 Consider any non-zero angular velocity matrix of the form:

$$A = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}$$

$$\text{charpoly}(A) = s(s^2 + \Omega^2), \quad \Omega := \sqrt{\omega_1^2 + \omega_2^2 + \omega_3^2}$$

Hence, A has one real eigenvalue 0 and two complex conjugate eigenvalues, $j\Omega, -j\Omega$.

6.3.2 MATLAB

```
>> help eig
```

EIG Eigenvalues and eigenvectors.
EIG(X) is a vector containing the eigenvalues of a square matrix X.

[V,D] = EIG(X) produces a diagonal matrix D of eigenvalues and a full matrix V whose columns are the corresponding eigenvectors so that $X*V = V*D$.

[V,D] = EIG(X,'nobalance') performs the computation with balancing disabled, which sometimes gives more accurate results for certain problems with unusual scaling.

```
>> help poly
```

POLY Characteristic polynomial.
If A is an N by N matrix, POLY(A) is a row vector with N+1 elements which are the coefficients of the characteristic polynomial, $\text{DET}(\text{lambd}\text{a}*\text{EYE}(\text{A}) - \text{A})$.
If V is a vector, POLY(V) is a vector whose elements are the coefficients of the polynomial whose roots are the elements of V . For vectors, ROOTS and POLY are inverse functions of each other, up to ordering, scaling, and roundoff error.
ROOTS(POLY(1:20)) generates Wilkinson's famous example.

```
>> help roots
```

ROOTS Find polynomial roots.
ROOTS(C) computes the roots of the polynomial whose coefficients are the elements of the vector C. If C has N+1 components, the polynomial is $C(1)*X^N + \dots + C(N)*X + C(N+1)$.

6.3.3 Companion matrices

These matrices will be useful later in control design. We met these matrices when constructing state space descriptions for systems described by a single higher order linear differential equation.

Exercise 65 Compute the characteristic polynomial of

$$\begin{pmatrix} 0 & 1 \\ -a_0 & -a_1 \end{pmatrix}$$

Hence, write down the characteristic polynomials for the A matrices associated with the unattached mass and the damped linear oscillator.

Exercise 66 Compute the characteristic polynomial of

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{pmatrix}$$

Fact 7 *The characteristic polynomial of*

$$\begin{pmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-2} & -a_{n-1} \end{pmatrix}$$

is given by

$$p(s) = a_0 + a_1 s + \dots + a_{n-1} s^{n-1} + s^n$$

Hence, given any bunch of n complex numbers, there is a companion matrix whose eigenvalues are exactly these numbers.

Exercise 67 What is the real 2×2 companion matrix with eigenvalues $1 + j$, $1 - j$?

Exercise 68 What is the companion matrix whose eigenvalues are -1 , -2 , and -3 ?

6.4 Behavior of continuous-time systems

6.4.1 System significance of eigenvectors and eigenvalues

In this section, we demonstrate that the eigenvalues and eigenvectors of a square matrix A play a fundamental role in the behavior of the linear system described by

$$\dot{x} = Ax. \quad (6.6)$$

We first demonstrate the following result.

The system $\dot{x} = Ax$ has a non-zero solution of the form

$$\boxed{x(t) = e^{\lambda t}v} \quad (6.7)$$

if and only if λ is an eigenvalue of A and v is a corresponding eigenvector.

To demonstrate the above claim, note that if $x(t) = e^{\lambda t}v$, then

$$\dot{x} = e^{\lambda t}\lambda v \quad \text{and} \quad Ax(t) = Ae^{\lambda t}v = e^{\lambda t}Av.$$

Hence, $x(t) = e^{\lambda t}v$ is a non-zero solution of $\dot{x} = Ax$ if and only if v is nonzero and

$$e^{\lambda t}\lambda v = e^{\lambda t}Av.$$

Since $e^{\lambda t}$ is nonzero, the above condition is equivalent to

$$Av = \lambda v.$$

Since v is non-zero, this means λ is an eigenvalue of A and v is a corresponding eigenvector. ■

A solution of the form $e^{\lambda t}v$ is a very special type of solution and is sometimes called a **mode** of the system. Note that if $x(t) = e^{\lambda t}v$ is a solution, then $x(0) = v$, that is, v is the initial value of x . Hence, we can make the following statement.

If v is an eigenvector of A , then the solution to $\dot{x} = Ax$ with initial condition $x(0) = v$ is given by $x(t) = e^{\lambda t}v$ where λ is the eigenvalue corresponding to v .

If v is an eigenvector then, $e^{\lambda t}v$ is also an eigenvector for each t . Also, considering the magnitude of $e^{\lambda t}v$, we obtain

$$\|e^{\lambda t}v\| = |e^{\lambda t}| \|v\| = e^{\alpha t} \|v\|$$

where α is the real part of λ . So,

- (a) Once an eigenvector, always an eigenvector!
- (b) If $\Re(\lambda) < 0$ you decay with increasing age.
- (c) If $\Re(\lambda) > 0$ you grow with increasing age.

- (d) If $\Re(\lambda) = 0$ your magnitude remains constant.
- (e) If $\lambda = 0$ you stay put; every eigenvector corresponding to $\lambda = 0$ is an equilibrium state.

Example 66 The matrix A of example 62 had 2 as an eigenvalue with corresponding eigenvector $\begin{bmatrix} 1 & 1 \end{bmatrix}^T$. Hence with initial state,

$$x_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

the system $\dot{x} = Ax$ has the following solution:

$$x(t) = \begin{pmatrix} e^{2t} \\ e^{2t} \end{pmatrix}$$

What is the solution for

$$x_0 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} ?$$

Exercise 69 Prove the following statement. *Suppose that $v^1, \dots, v^m$ are eigenvectors of A with corresponding eigenvalues, $\lambda_1, \dots, \lambda_m$. Then the solution of (6.6) with initial state*

$$x_0 = \xi_{10}v^1 + \dots + \xi_{m0}v^m,$$

where $\xi_{10}, \dots, \xi_{m0}$ are scalars, is given by

$$x(t) = \xi_{10}e^{\lambda_1 t}v^1 + \dots + \xi_{m0}e^{\lambda_m t}v^m.$$

Behaviour in the v subspace. Suppose v is an eigenvector of A with corresponding eigenvalue λ and consider any initial state of the form $x_0 = \xi_0 v$. Thus, the initial state x_0 is in the 1-dimensional subspace spanned by v . We have shown that the solution corresponding to $x(0) = x_0$ has the form $x(t) = \xi(t)v$; hence $x(t)$ remains in this subspace. Forgetting about the actual solution, we note that $\xi v = \dot{x} = Ax = A(\xi v) = \lambda \xi v$. Hence

$$\dot{\xi} = \lambda \xi \qquad \xi(0) = \xi_0$$

In other words, the behavior of the system in the 1-dimensional subspace spanned by v is that of a scalar system.

Some state space pictures

Real systems and complex eigenvalues. Suppose we are interested in the behavior of a ‘real’ system with real state x and real system matrix A and suppose λ is a (genuine) complex eigenvalue of A , that is,

$$Av = \lambda v$$

for a nonzero v . Since A is real and λ is complex, the eigenvector v is complex. The complex system can start with a complex initial state but the ‘real’ system cannot. So is the above result useful? It is, just wait.

Since A is real, $\bar{\lambda}$ is also an eigenvalue of A with corresponding eigenvector $\bar{v}$; hence the (complex) solution corresponding to $x(0) = \bar{v}$ is given by

$$e^{\bar{\lambda}t}\bar{v} = \overline{e^{\lambda t}v}$$

Consider now the real vector

$$u := \Re(v) = \frac{1}{2}(v + \bar{v})$$

as an initial state. Since the solution depends linearly on initial state, the solution for this real initial state is

$$\frac{1}{2}(e^{\lambda t}v + \overline{e^{\lambda t}v}) = \Re(e^{\lambda t}v)$$

Suppose we let

$$\alpha = \Re(\lambda) \quad \omega = \Im(\lambda) \quad w = \Im(v).$$

Then

$$\lambda = \alpha + j\omega \quad \text{and} \quad v = u + jw.$$

Hence,

$$\begin{aligned} e^{\lambda t}v &= e^{\alpha t}(\cos \omega t + j \sin \omega t)(u + jw) \\ &= [e^{\alpha t} \cos(\omega t) u - e^{\alpha t} \sin(\omega t) w] + j(e^{\alpha t} \sin(\omega t) u + e^{\alpha t} \cos(\omega t) w) \end{aligned}$$

Hence, the solution due to $x(0) = u$ is

$$e^{\alpha t} \cos(\omega t) u - e^{\alpha t} \sin(\omega t) w$$

Similarly, the solution due to initial state w is $\Im(e^{\lambda t}v)$, that is,

$$e^{\alpha t} \sin(\omega t) u + e^{\alpha t} \cos(\omega t) w$$

- So, if the initial state is of the form

$$x(0) = \xi_{10}u + \xi_{20}w$$

where ξ_{10}, ξ_{20} are any two real scalars (that is, the system starts in the real subspace spanned by the real span of u, w), the resulting solution is given by

$$x(t) = \xi_1(t)u + \xi_2(t)w$$

where

$$\begin{aligned}\xi_1(t) &= e^{\alpha t} \cos(\omega t) \xi_{10} + e^{\alpha t} \sin(\omega t) \xi_{20} \\ \xi_2(t) &= -e^{\alpha t} \sin(\omega t) \xi_{10} + e^{\alpha t} \cos(\omega t) \xi_{20}\end{aligned} = \begin{pmatrix} e^{\alpha t} \cos(\omega t) & e^{\alpha t} \sin(\omega t) \\ -e^{\alpha t} \sin(\omega t) & e^{\alpha t} \cos(\omega t) \end{pmatrix} \xi_0$$

Note that, if we let $A_0 = (\xi_{10}^2 + \xi_{20}^2)^{\frac{1}{2}}$ and let ϕ_0 be the unique number in the interval $[0, 2\pi)$ defined by

$$\sin(\phi_0) = \xi_{10}/A_0 \quad \text{and} \quad \cos(\phi_0) = \xi_{20}/A_0$$

then

$$\begin{aligned}\xi_1(t) &= A_0 e^{\alpha t} \sin(\omega t + \phi_0) \\ \xi_2(t) &= A_0 e^{\alpha t} \cos(\omega t + \phi_0)\end{aligned}$$

and

$$x(t) = A_0 e^{\alpha t} (\sin(\omega t + \phi_0)u + \cos(\omega t + \phi_0)w)$$

So, we conclude that for real systems:

- (a) If the initial state is in the real 2D subspace spanned by the real and imaginary parts of an complex eigenvector, then so is the resulting solution.
- (b) If $\Re(\lambda) < 0$ you decay with increasing age.
- (c) If $\Re(\lambda) > 0$ you grow with increasing age.
- (d) If $\Re(\lambda) = 0$ your magnitude remains constant and if $\lambda \neq 0$ you oscillate forever.
- (e) If $\lambda = 0$ you stay put; every eigenvector corresponding to $\lambda = 0$ is an equilibrium state.

Example 67 Recall example 63. Considering

$$v = v^1 = \begin{pmatrix} 1 \\ j \end{pmatrix}$$

we have

$$u = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad w = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

So,

$$x = x_1 u + x_2 w$$

Hence, all solutions are given by

$$\begin{aligned}x_1(t) &= e^{\alpha t} \cos(\omega t) x_{10} + e^{\alpha t} \sin(\omega t) x_{20} \\ x_2(t) &= -e^{\alpha t} \sin(\omega t) x_{10} + e^{\alpha t} \cos(\omega t) x_{20}\end{aligned} = \begin{pmatrix} e^{\alpha t} \cos(\omega t) & e^{\alpha t} \sin(\omega t) \\ -e^{\alpha t} \sin(\omega t) & e^{\alpha t} \cos(\omega t) \end{pmatrix} x_0$$

Behavior in the $u-w$ subspace. We have shown that if the state starts in the 2-dimensional subspace spanned by u, w , (let's call it the $u-w$ subspace) it remains there, that is, if $x_0 = \xi_{10}u + \xi_{20}w$ then $x(t) = \xi_1(t)u + \xi_2(t)w$. Forgetting about the actual solution, we now show that motion in this subspace is described by:

$$\begin{cases} \dot{\xi}_1 = \alpha \xi_1 + \omega \xi_2 \\ \dot{\xi}_2 = -\omega \xi_1 + \alpha \xi_2 \end{cases} = \begin{pmatrix} \alpha & \omega \\ -\omega & \alpha \end{pmatrix} \xi$$

To see this, we first equate real and imaginary parts of

$$A(u + jw) = (\alpha + j\omega)(u + jw)$$

to obtain

$$\begin{aligned} Au &= \alpha u - \omega w \\ Aw &= \omega u + \alpha w \end{aligned}$$

Now

$$\begin{aligned} \dot{\xi}_1 u + \dot{\xi}_2 w &= \dot{x} \\ &= Ax \\ &= A(\xi_1 u + \xi_2 w) \\ &= \xi_1 Au + \xi_2 Aw \\ &= (\alpha \xi_1 + \omega \xi_2)u + (-\omega \xi_1 + \alpha \xi_2)w \end{aligned}$$

The desired result now follows by equating coefficients of u and w on both sides of the equation.

Example 68 (Beavis and Butthead) Consider $u_1 = u_2 = 0$ in the system in Figure 6.3.

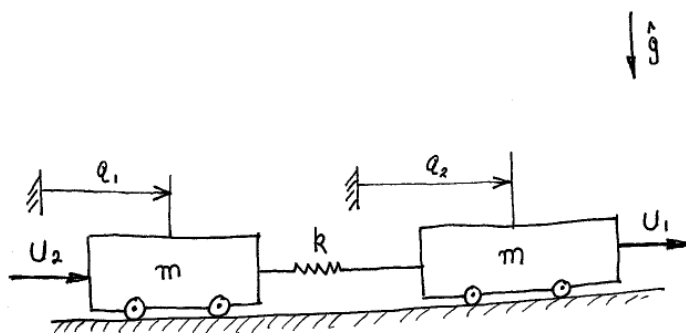


Figure 6.3: Beavis & Butthead

$$\begin{aligned} m\ddot{q}_1 &= k(q_2 - q_1) \\ m\ddot{q}_2 &= -k(q_2 - q_1) \end{aligned}$$

Introducing state variables

$$x_1 := q_1, \quad x_2 := q_2, \quad x_3 := \dot{q}_1, \quad x_4 := \dot{q}_2,$$

this system is described by $\dot{x} = Ax$ where A is the block partitioned matrix given by

$$A = \begin{pmatrix} 0 & I \\ A_{21} & 0 \end{pmatrix}, \quad A_{21} = k/m \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The characteristic polynomial of A is given by

$$p(s) = s^2(s^2 + \omega^2), \quad \omega = \sqrt{2k/m}$$

Hence, A has three eigenvalues

$$\lambda_1 = j\omega, \quad \lambda_2 = -j\omega, \quad \lambda_3 = 0.$$

Corresponding to each eigenvalue λ_i , there is at most one linearly independent eigenvector v^i , for example,

$$v^1 = \begin{pmatrix} 1 \\ -1 \\ j\omega \\ -j\omega \end{pmatrix}, \quad v^2 = \overline{v^1}, \quad v^3 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

Consider now the subspace $\mathcal{S}^1$ spanned by the real and imaginary parts of v^1 :

$$\mathcal{S}^1 = \text{span} \left\{ \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ \omega \\ -\omega \end{pmatrix} \right\} = \text{span} \left\{ \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix} \right\}.$$

A little reflection reveals that states in this subspace correspond to those states for which $q_2 = -q_1$ and $\dot{q}_2 = -\dot{q}_1$. A state trajectory which remains in this subspace is characterized by the condition that

$$q_2(t) \equiv -q_1(t).$$

We have seen that whenever a motion originates in this subspace it remains there; also all motions in this subspace are purely oscillatory and consist of terms of the form $\cos(\omega t)$ and $\sin(\omega t)$.

Consider now the subspace $\mathcal{S}^3$ spanned by the eigenvector v^3 . Every state in this subspace has the form

$$\begin{pmatrix} q_1 \\ q_1 \\ 0 \\ 0 \end{pmatrix},$$

that is, q_1 is arbitrary, $q_2 = q_1$, and $\dot{q}_1 = \dot{q}_2 = 0$. Since the eigenvalue corresponding to this eigenspace is zero, every state in this subspace is an equilibrium state.

6.4.2 Solutions for nondefective matrices

Recall that the algebraic multiplicity of an eigenvalue of a matrix is the number of times that the eigenvalue is repeated as a root of the characteristic polynomial of the matrix. Also the geometric multiplicity of an eigenvalue is the dimension of its eigenspace, or equivalently, the maximum number of linearly independent eigenvectors associated with that eigenvalue. It can be proven that the geometric multiplicity of an eigenvalue is always less than or equal to its algebraic multiplicity. An eigenvalue whose geometric multiplicity equals its algebraic multiplicity is called a **nondefective eigenvalue**. Otherwise, it is called **defective**. If an eigenvalue is not repeated, that is, it is not a repeated root of the matrix characteristic polynomial, then it has algebraic multiplicity one. Since its geometric multiplicity must be greater than zero and less than or equal to its algebraic multiplicity, it follows that its geometric multiplicity equals its algebraic multiplicity; hence the eigenvalue is nondefective.

A matrix with the property that all its eigenvalues are nondefective is called a **nondefective matrix**. Otherwise, it is called **defective**. When is a matrix A nondefective? Some examples are:

- (a) A has n *distinct* eigenvalues where A is $n \times n$.
- (b) A is **hermitian** that is, $A^* = A$.
- (c) A is **unitary**, that is, $A'A = I$.

Here A' is the complex conjugate transpose of A .

We have the following result.

Fact 8 *An $n \times n$ matrix has n linearly independent eigenvectors if and only if A is nondefective.*

Example 69 The following matrix is a simple example of a defective matrix.

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Example 70 The following matrix is another example of a defective matrix.

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -\omega^2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -\omega^2 & 0 \end{pmatrix}$$

where ω is any real number.

Suppose A is a nondefective matrix and consider any initial condition: $x(0) = x_0$. Since A is nondefective, it has a basis of eigenvectors $v^1, \dots, v^n$. Hence, there are unique scalars $\xi_{10}, \xi_{10}, \dots, \xi_{n0}$ such that x_0 can be expressed as

$$x_0 = \xi_{10}v^1 + \xi_{20}v^2 + \dots + \xi_{n0}v^n.$$

From this it follows that the solution due to $x(0) = x_0$ is given by

$$\boxed{x(t) = e^{\lambda_1 t} \xi_{10} v^1 + e^{\lambda_2 t} \xi_{20} v^2 + \dots + e^{\lambda_n t} \xi_{n0} v^n} \quad (6.8)$$

Note that each term $e^{\lambda_i t} \xi_{i0} v^i$ on the right-hand-side of the above equation is a mode. This results in the following fundamental observation.

Every motion (solution) of a linear time-invariant system with nondefective A matrix is simply a sum of modes.

6.4.3 Defective matrices and generalized eigenvectors

Consider a square matrix A . Then we have the following fact.

Fact 9 *An eigenvalue λ for A is defective if and only if there is a nonzero vector g such that*

$$(A - \lambda I)^2 g = 0 \quad \text{but} \quad (A - \lambda I)g \neq 0.$$

The vector g is called a **generalized eigenvector** for A .

Example 71

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Consider $\lambda = 0$. Then

$$g = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

is a generalized eigenvector of A .

We now demonstrate the consequence of a system having a defective eigenvalue. So, suppose λ is a defective eigenvalue for a square matrix A . Let g be any corresponding generalized eigenvector for λ . We now show that

$$\boxed{x(t) = e^{\lambda t} g + t e^{\lambda t} (A - \lambda I)g}$$

is a solution of $\dot{x} = Ax$. To see this, differentiate x to obtain

$$\begin{aligned} \dot{x}(t) &= e^{\lambda t} \lambda g + e^{\lambda t} (A - \lambda I)g + t e^{\lambda t} \lambda (A - \lambda I)g \\ &= e^{\lambda t} A g + t e^{\lambda t} \lambda (A - \lambda I)g \end{aligned}$$

Since $(A - \lambda I)^2 g = 0$, we have

$$A(A - \lambda I)g = \lambda(A - \lambda I)g.$$

Hence

$$\begin{aligned} \dot{x}(t) &= e^{\lambda t} A g + t e^{\lambda t} A(A - \lambda I)g = A(e^{\lambda t} g + t e^{\lambda t} (A - \lambda I)g) \\ &= Ax(t). \end{aligned}$$

Consider the function given by $te^{\lambda t}$ and note that

$$|te^{\lambda t}| = |t|e^{\alpha t}$$

where α is the real part of λ . So, if α is negative, the magnitude of the above function decays exponentially. If α is positive, the magnitude of the above function grows exponentially. If α is zero, the magnitude of the above function is given by $|t|$ which is also unbounded. From this discussion, we see that *the defectiveness of an eigenvalue is only significant when the eigenvalue lies on the imaginary axis of the complex plane, that is when it has zero real part.*

Example 72 (Resonance) Consider a simple spring-mass system, of mass m and spring constant k , subject to a sinusoidal force $w(t) = A \sin(\omega t + \phi)$ where $\omega = \sqrt{k/m}$ is the unforced natural frequency of the spring mass system and A and ϕ are arbitrary constants. Letting q be the displacement of the mass from its unforced equilibrium position, the motion of this system can be described by

$$m\ddot{q} + kq = w.$$

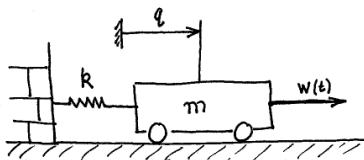


Figure 6.4: Resonance

Since $w(t) = A \sin(\omega t + \phi)$ it satisfies the following differential equation:

$$\ddot{w} + \omega^2 w = 0.$$

If we introduce state variables,

$$x_1 = q, \quad x_2 = \dot{q}, \quad x_3 = w, \quad x_4 = \dot{w}$$

the sinusoidally forced spring-mass system can be described by $\dot{x} = Ax$ where

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -\omega^2 & 0 & 1/m & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -\omega^2 & 0 \end{pmatrix}.$$

This matrix has characteristic polynomial

$$p(s) = (s^2 + \omega^2)^2.$$

Hence A has two distinct imaginary eigenvalues $\lambda_1 = j\omega$ and $\lambda_2 = -j\omega$, both of which are repeated twice as roots of the characteristic polynomial. All eigenvalues for λ_1 and λ_2 are of the form cv^1 and cv^2 , respectively, where

$$v^1 = \begin{pmatrix} 1 \\ j\omega \\ 0 \\ 0 \end{pmatrix} \quad \text{and} \quad v^2 = \begin{pmatrix} 1 \\ -j\omega \\ 0 \\ 0 \end{pmatrix}.$$

Since these eigenvalues have algebraic multiplicity two and geometric multiplicity one, they are defective. A generalized eigenvector corresponding to λ_1 and λ_2 is given by

$$g^1 = \begin{pmatrix} 1/2k \\ 0 \\ 1 \\ j\omega \end{pmatrix} \quad \text{and} \quad g^2 = \begin{pmatrix} 1/2k \\ 0 \\ 1 \\ -j\omega \end{pmatrix},$$

respectively. Hence this system has solutions with terms of the form $t \sin(\omega t)$.

Exercises

Exercise 70 (Beavis and Butthead at the laundromat)

$$m\ddot{q}_1 - m\Omega^2 q_1 + \frac{k}{2}(q_1 - q_2 - 2l_0) = 0$$

$$m\ddot{q}_2 - m\Omega^2 q_2 - \frac{k}{2}(q_1 - q_2 - 2l_0) = 0$$

with

$$\omega := \sqrt{k/m} > \Omega$$

Figure 6.5: At the laundromat

- (a) Obtain the equilibrium values q_1^e, q_2^e of q_1, q_2 .

(b) With

$$\begin{aligned}x_1 &= \delta q_1 := q_1 - q_1^e \\x_2 &= \delta q_2 := q_2 - q_2^e \\x_3 &= \dot{q}_1 \\x_4 &= \dot{q}_2\end{aligned}$$

Obtain a state space description of the form $\dot{x} = Ax$.

- (c) Obtain expressions for the eigenvalues of A . Write down expressions for the complex Jordan form and the real Jordan form of A ?
- (d) For the rest of the exercise, consider $\omega^2 = 2\Omega^2$. Obtain expressions for linearly independent eigenvectors of A .
- (e) For the rest of the exercise, let $\Omega = 1$. Numerically simulate with initial conditions

$$\delta q_1(0) = 1, \quad \delta q_2(0) = -1, \quad \dot{q}_1(0) = 0, \quad \dot{q}_2(0) = 0$$

On the same graph, plot $\delta q_1(t)$ and $-\delta q_2(t)$ vs t . Why is $\delta q_2(t) \equiv -\delta q_1(t)$? Plot $\delta q_2(t)$ vs $\delta q_1(t)$.

- (f) Numerically simulate with initial conditions

$$\delta q_1(0) = 1, \quad \delta q_2(0) = 1, \quad \dot{q}_1(0) = -1, \quad \dot{q}_2(0) = -1$$

On the same graph, plot $\delta q_1(t)$ and $\delta q_2(t)$ vs t . Why is $\delta q_2(t) \equiv \delta q_1(t)$? Why does the solution decay to zero? Plot $\delta q_2(t)$ vs $\delta q_1(t)$.

- (g) Numerically simulate with initial conditions

$$\delta q_1(0) = 1, \quad \delta q_2(0) = 1, \quad \dot{q}_1(0) = 1, \quad \dot{q}_2(0) = 1$$

On the same graph, plot $\delta q_1(t)$ and $\delta q_2(t)$ vs t . Why is $\delta q_2(t) \equiv \delta q_1(t)$? Is this solution bounded? Plot $\delta q_2(t)$ vs $\delta q_1(t)$.

- (h) Numerically simulate with initial conditions

$$\delta q_1(0) = 1, \quad \delta q_2(0) = -1, \quad \dot{q}_1(0) = j, \quad \dot{q}_2(0) = -j$$

Compute $\|x(t)\|$ and show (by plotting $\|x\|$ vs t) that it is constant. Why is it constant?

Exercise 71 Consider the delay differential equation described by

$$\dot{x}(t) = A_1 x(t) + A_2 x(t-1)$$

where $t \in \mathbb{R}$ and $x(t) \in \mathbb{C}^n$. Show that if a complex number λ satisfies

$$\det(\lambda I - A_1 - e^{-\lambda} A_2) = 0$$

then the delay differential equation has a solution of the form

$$e^{\lambda t} v$$

where $v \in \mathbb{C}^n$ is nonzero.

Exercise 72 Consider a system described by

$$M\ddot{q} + C\dot{q} + Kq = 0$$

where $q(t)$ is an N -vector and M , C and K are square matrices. Suppose λ is a complex number which satisfies

$$\det(\lambda^2 M + \lambda C + K) = 0.$$

Show that the above system has a solution of the form

$$q(t) = e^{\lambda t} v$$

where v is a constant N -vector.

6.5 Behavior of discrete-time systems

6.5.1 System significance of eigenvectors and eigenvalues

Consider now a discrete time system described by

$$x(k+1) = Ax(k) \quad (6.9)$$

We now demonstrate the following result:

The discrete-time system (6.9) has a solution of the form

$$\boxed{x(k) = \lambda^k v} \quad (6.10)$$

if and only if λ is an eigenvalue of A and v is a corresponding eigenvector.

To prove this, we first suppose that λ is an eigenvalue of A with v as a corresponding eigenvector. We need to show that $x(k) = \lambda^k v$ is a solution of (6.9). Since $Av = \lambda v$, it follows that $x(k+1) = \lambda^{(k+1)}v = \lambda^k \lambda v = \lambda^k Av = A(\lambda^k v) = Ax(k)$, that is $x(k+1) = Ax(k)$; hence $x(\cdot)$ is a solution of (6.9).

Now suppose that $x(k) = \lambda^k v$ is a solution of (6.9). Then $\lambda v = x(1) = Ax(0) = Av$, that is, $Av = \lambda v$; hence λ is an eigenvalue of A and v is a corresponding eigenvector. ■

A solution of the form $\lambda^k v$ is a very special type of solution and is sometimes called a **mode** of the system. Note that if $x(k) = \lambda^k v$ is a solution, then $x(0) = v$, that is, v is the initial value of x . Hence, we can make the following statement.

If v is an eigenvector of A , then the solution to $x(k+1) = Ax(k)$ with initial condition $x(0) = v$ is given by $x(k) = \lambda^k v$ where λ is the eigenvalue corresponding to v .

If v is an eigenvector then, $\lambda^k v$ is also an eigenvector for each k . Also, considering the magnitude of $\lambda^k v$, we obtain

$$\|\lambda^k v\| = |\lambda^k| \|v\| = |\lambda|^k \|v\|$$

So,

- (a) Once an eigenvector, always an eigenvector!
 - (b) If $|\lambda| < 1$ you decay with increasing age.
 - (c) If $|\lambda| > 1$ you grow with increasing age.
 - (d) If $|\lambda| = 1$ and $\lambda \neq 1$ you oscillate forever.
- If $\lambda = 1$ and $x(0) = v$, then $x(k) = v$ for all k , that is v is an equilibrium state for the system.
 - If $\lambda = 0$ and $x(0) = v$, then $x(k) = 0$ for all $k > 0$, that is *the state of the system goes to zero in one step.*

6.5.2 All solutions for nondefective matrices

Recall that an eigenvalue whose geometric multiplicity equals its algebraic multiplicity is called a nondefective eigenvalue. Otherwise, it is considered defective. A matrix with the property that all its eigenvalues are nondefective is called a nondefective matrix. Otherwise, it is considered defective. Recall also that an $n \times n$ matrix has n linearly independent eigenvectors if and only if A is nondefective.

Example 73 The following matrix is a simple example of a defective matrix.

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Suppose A is a nondefective matrix and consider any initial condition: $x(0) = x_0$. Since A is nondefective, it has a basis of eigenvectors $v^1, \dots, v^n$. Hence,

$$x_0 = \xi_{10}v^1 + \xi_{20}v^2 + \dots + \xi_{n0}v^n$$

From this it follows that

$$\boxed{x(k) = \lambda_1^k \xi_{10} v^1 + \lambda_2^k \xi_{20} v^2 + \dots + \lambda_n^k \xi_{n0} v^n} \quad (6.11)$$

Note that each term $\lambda_i^k \xi_{i0} v^i$ on the right-hand-side of the above equation is a mode. This results in the following fundamental observation.

Every motion (solution) of a discrete-time linear time-invariant system with nondefective A matrix is simply a sum of modes.

6.5.3 Some solutions for defective matrices

Consider a square matrix A and recall that an eigenvalue λ for A is defective if and only if there is a nonzero vector g such that

$$(A - \lambda I)^2 g = 0 \quad \text{but} \quad (A - \lambda I)g \neq 0.$$

The vector g is called a generalized eigenvector for A .

Example 74

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Consider $\lambda = 0$. Then

$$g = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

is a generalized eigenvector of A .

We now demonstrate the consequence of a system having a defective eigenvalue. So, suppose λ is a defective eigenvalue for a square matrix A . Let g be any corresponding generalized eigenvector for λ . We now show that

$$\boxed{x(k) = \lambda^k g + k\lambda^{(k-1)}(A - \lambda I)g}$$

is a solution of $x(k+1) = Ax(k)$. To see this, we first note that, since $(A - \lambda I)^2 g = 0$, we have

$$A(A - \lambda I)g = \lambda(A - \lambda I)g.$$

Hence

$$\begin{aligned} Ax(k) &= \lambda^k Ag + k\lambda^{(k-1)} A(A - \lambda I)g \\ &= \lambda^k Ag + k\lambda^k (A - \lambda I)g. \end{aligned}$$

Now note that

$$\begin{aligned} x(k+1) &= \lambda^{(k+1)} g + (k+1)\lambda^k (A - \lambda I)g \\ &= \lambda^{(k+1)} g + k\lambda^k (A - \lambda I)g + \lambda^k Ag - \lambda^{(k+1)} g \\ &= \lambda^k Ag + k\lambda^k (A - \lambda I)g. \end{aligned}$$

Hence $x(k+1) = Ax(k)$ ■

Consider the function given by $k\lambda^{k-1}$ and note that

$$|k\lambda^{k-1}| = |k||\lambda|^{k-1}.$$

So, if $|\lambda| < 1$, the magnitude of the above function decays exponentially. If $|\lambda| > 1$, the magnitude of the above function grows exponentially. If $|\lambda| = 1$, the magnitude of the above function is given by $|k|$ which is also unbounded. From this discussion, we see that *the defectiveness of an eigenvalue is only significant when the eigenvalue lies on the unit circle in the complex plane, that is when its magnitude is one.*

6.6 Similarity transformations

Motivating example. Consider

$$\begin{aligned}\dot{x}_1 &= 3x_1 - x_2 \\ \dot{x}_2 &= -x_1 + 3x_2\end{aligned}$$

Suppose we define new states:

$$\begin{aligned}\xi_1 &= (x_1 + x_2)/2 \\ \xi_2 &= (x_1 - x_2)/2\end{aligned}$$

Then,

$$\begin{aligned}\dot{\xi}_1 &= 2\xi_1 \\ \dot{\xi}_2 &= 4\xi_2\end{aligned}$$

That looks better!

Where did new states come from? First note that

$$\begin{aligned}x_1 &= \xi_1 + \xi_2 \\ x_2 &= \xi_1 - \xi_2\end{aligned}$$

and we have the following geometric interpretation. Since,

$$\begin{aligned}x &= x_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ &= \xi_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \xi_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix}\end{aligned}$$

the scalars x_1, x_2 are the coordinates of the vector x wrt the usual basis for $\mathbb{R}^2$; the scalars ξ_1, ξ_2 are the coordinates of x wrt the basis

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Where did these new basis vectors come from? These new basis vectors are eigenvectors of the matrix

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

and the system is described by $\dot{x} = Ax$.

Coordinate transformations. Suppose we have n scalar variables $x_1, x_2, \dots, x_n$ and we implicitly define new scalar variables $\xi_1, \xi_2, \dots, \xi_n$ by

$$\boxed{x = T\xi} \quad \text{where} \quad x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad \text{and} \quad \xi = \begin{pmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_n \end{pmatrix}$$

and T is an $n \times n$ invertible matrix. Then, ξ is explicitly given by

$$\xi = T^{-1}x$$

We can obtain a geometric interpretation of this change of variables as follows. First, observe that

$$\underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_x = x_1 \underbrace{\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{e^1} + x_2 \underbrace{\begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}}_{e^2} + \dots + x_n \underbrace{\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}}_{e^n},$$

that is the scalars $x_1, x_2, \dots, x_n$ are the coordinates of the vector x wrt the standard basis $e^1, e^2, \dots, e^n$, or, x is the coordinate vector of itself wrt the standard basis. Suppose

$$T = \begin{pmatrix} t^1 & \dots & t^j & \dots & t^n \end{pmatrix}$$

that is, t^j is the j -th column of T . Since T is invertible, its columns, $t^1, t^2, \dots, t^n$, form a basis for $\mathbb{C}^n$. Also, $x = T\xi$ can be written as

$$x = \xi_1 t^1 + \xi_2 t^2 + \dots + \xi_n t^n$$

From this we see that $\xi_1, \xi_2, \dots, \xi_n$ are the coordinates of x wrt to the new basis and the vector ξ is the coordinate vector of the vector x wrt this new basis. So $x = T\xi$ defines a coordinate transformation.

Similarity transformations. Consider now a CT (DT) system described by

$$\dot{x} = Ax \quad (x(k+1) = Ax(k))$$

and suppose we define new state variables by $x = T\xi$ where T is nonsingular; then the behavior of ξ is governed by

$$\dot{\xi} = \Lambda \xi \quad (\xi(k+1) = \Lambda \xi(k))$$

where

$$\boxed{\Lambda = T^{-1}AT}$$

A square matrix A is said to be **similar** to another square matrix Λ if there exists a nonsingular matrix T such that $\Lambda = T^{-1}AT$.

Recalling that the columns $t^1, t^2, \dots, t^n$ of T form a basis, we have now the following very useful result.

Useful result. Suppose

$$At^j = \alpha_{1j}t^1 + \alpha_{2j}t^2 + \dots + \alpha_{nj}t^n$$

Then the matrix $\Lambda = T^{-1}AT$ is uniquely given by

$$\Lambda_{ij} = \alpha_{ij}$$

that is Λ_{ij} is the i -th coordinate of the vector At^j wrt the basis $t^1, t^2, \dots, t^n$. Thus, *the j -th column of the matrix Λ is the coordinate vector of At^j wrt the basis $t^1, t^2, \dots, t^n$.*

PROOF. Premultiply $\Lambda = T^{-1}AT$ by T to get:

$$T\Lambda = AT$$

On the right we have

$$\begin{aligned} AT &= A \begin{pmatrix} t^1 & \dots & t^j & \dots & t^n \end{pmatrix} \\ &= \begin{pmatrix} At^1 & \dots & At^j & \dots & At^n \end{pmatrix} \end{aligned}$$

And on the left,

$$\begin{aligned} T\Lambda &= \begin{pmatrix} t^1 & t^2 & \dots & \dots & t^n \end{pmatrix} \begin{pmatrix} \Lambda_{11} & \dots & \Lambda_{1j} & \dots & \Lambda_{1n} \\ \Lambda_{21} & \dots & \Lambda_{2j} & \dots & \Lambda_{2n} \\ \vdots & & \vdots & & \vdots \\ \vdots & & \vdots & & \vdots \\ \Lambda_{n1} & \dots & \Lambda_{nj} & \dots & \Lambda_{nn} \end{pmatrix} \\ &= \begin{pmatrix} \Lambda_{11}t^1 + \dots + \Lambda_{n1}t^n \vdots & \dots & \underbrace{\Lambda_{1j}t^1 + \dots + \Lambda_{nj}t^n}_{j\text{-th column}} \vdots & \dots & \Lambda_{1n}t^1 + \dots + \Lambda_{nn}t^n \end{pmatrix} \end{aligned}$$

Comparing the expressions for the j -th columns of AT and $T\Lambda$ yields

$$\begin{aligned} \Lambda_{1j}t^1 + \dots + \Lambda_{nj}t^n &= At^j \\ &= \alpha_{1j}t^1 + \dots + \alpha_{nj}t^n \end{aligned}$$

Since the b_i 's form a basis we must have $\Lambda_{ij} = \alpha_{ij}$. ■

• Note that

$$A = T\Lambda T^{-1}$$

Example 75 Recall example 62. Note that the eigenvectors v^1 and v^2 are linearly independent; hence the matrix

$$T := (v^1 \ v^2) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

is invertible. Since

$$\begin{aligned} Av^1 &= 2v^1 \\ Av^2 &= 4v^2 \end{aligned}$$

we can use the above result and immediately write down (without computing T^{-1})

$$\Lambda = T^{-1}AT = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$$

Explicitly compute T^{-1} and then $T^{-1}AT$ to check.

Example 76 Recalling example 63 we can follow the procedure of the last example and define

$$T := \begin{pmatrix} 1 & 1 \\ j & -j \end{pmatrix}$$

to obtain (without computing T^{-1})

$$\Lambda = T^{-1}AT = \begin{pmatrix} 1+j & 0 \\ 0 & 1-j \end{pmatrix}$$

You may now compute T^{-1} and check this.

Example 77 Suppose A is a 4×4 matrix and v^1, v^2, v^3, v^4 are four linearly independent vectors which satisfy:

$$Av^1 = v^2, \quad Av^2 = v^3, \quad Av^3 = v^4, \quad Av^4 = 3v^1 - v^2 + 6v^3$$

If we let V be the matrix whose columns are v^1, v^2, v^3, v^4 , then V is invertible and we can immediately write down that

$$V^{-1}AV = \begin{pmatrix} 0 & 0 & 0 & 3 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Example 78 B&B

Some properties:

Exercise 73 Prove the following statements:

- (a) If A is similar to B , then B is similar to A
- (b) If A is similar to B and B is similar to C , then A is similar to C .

Exercise 74 Show that if A is similar to Λ then the following hold:

(a) $\det A = \det \Lambda$

(b) $\text{charpoly } A = \text{charpoly } \Lambda$ and, hence, A and Λ have the same eigenvalues with the same algebraic multiplicities.

Exercise 75 Suppose A is a 3×3 matrix, b is a 3-vector,

$$A^3 + 2A^2 + A + 1 = 0$$

and the matrix

$$T = \begin{pmatrix} b & Ab & A^2b \end{pmatrix}$$

is invertible. What is the matrix $\Lambda := T^{-1}AT$? Specify each element of Λ explicitly.

6.7 Hotel California

6.7.1 Invariant subspaces

Suppose A is a square matrix. We have already met several subspaces associated with A , namely, its range, null space, and eigenspaces. These subspaces are all examples of invariant subspaces for A

We say that a subspace $\mathcal{S}$ is an invariant subspace for A if for each vector x in $\mathcal{S}$, the vector Ax is also in $\mathcal{S}$.

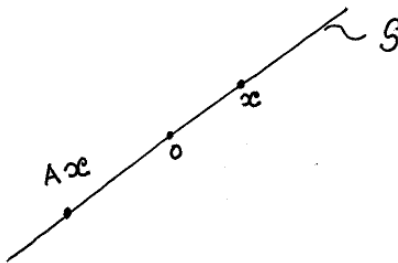


Figure 6.6: Invariant subspace

The significance of invariant subspaces for LTI systems lies in the following result which is proven later.

If the state of a linear-time-invariant system starts in an invariant subspace of the system matrix, it never leaves the subspace.

A coordinate transformation. Suppose $\mathcal{S}$ is an invariant subspace for a square matrix A . If $\mathcal{S}$ has dimension m , let

$$t^1, t^2, \dots, t^m$$

be a basis for $\mathcal{S}$. If A is $n \times n$, extend the above basis to a basis

$$t^1, \dots, t^m, t^{m+1}, \dots, t^n$$

for $\mathbb{C}^n$ (This can always be done.) Now introduce the coordinate transformation

$$x = T\xi$$

where

$$T := \begin{pmatrix} t^1 & t^2 & \dots & t^n \end{pmatrix}$$

Letting

$$\xi = \begin{pmatrix} \zeta \\ \eta \end{pmatrix},$$

where ζ is an m -vector and η is an $(n-m)$ -vector, we obtain

$$x = \zeta_1 t^1 + \dots + \zeta_m t^m + \eta_1 t^{m+1} + \dots + \eta_{n-m} t^n$$

So

$$x \text{ is in } \mathcal{S} \quad \text{if and only if} \quad \eta = 0$$

Let

$$\Lambda := T^{-1}AT.$$

Using our ‘useful result on similarity transformations’ and the fact that $\mathcal{S}$ is invariant for A , it follows that Λ must have the following structure:

$$\Lambda = \begin{pmatrix} \Lambda_{11} & \Lambda_{21} \\ 0 & \Lambda_{22} \end{pmatrix}$$

System significance of invariant subspaces. Consider a system described by

$$\dot{x} = Ax$$

and suppose $\mathcal{S}$ is an invariant subspace for A . Introduce the above coordinate transformation to obtain

$$\dot{\xi} = \Lambda\xi$$

or

$$\begin{aligned} \dot{\zeta} &= \Lambda_{11}\zeta + \Lambda_{12}\eta \\ \dot{\eta} &= \Lambda_{22}\eta \end{aligned}$$

Looking at the differential equation $\dot{\eta} = \Lambda_{22}\eta$, we see that if $\eta(0) = 0$, then $\eta(t) \equiv 0$; hence, if $x(0)$ is in $\mathcal{S}$ then $x(t)$ is always in $\mathcal{S}$. In other words, if you are in $\mathcal{S}$, you can check out any time, but you can never leave.....

6.7.2 More on invariant subspaces*

Suppose $\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_q$ are q invariant subspaces of a square matrix A with the property that there is basis

$$t^1, \dots, t^n$$

where

$$t^1, \dots, t^{n_1}$$

is a basis for $\mathcal{S}_1$,

$$t^{n_1+1}, \dots, t^{n_1+n_2}$$

is a basis for $\mathcal{S}_2$,

$$t^{n_1+\dots+n_{q-1}+1}, \dots, t^n$$

is a basis for $\mathcal{S}_q$,

$$\dot{x} = Ax$$

Coordinate transformation

$$x = T\xi$$

where

$$T = \begin{pmatrix} t^1 & t^2 & \dots & t^n \end{pmatrix}$$

that is, t^i is the i -th column of T . Consider

$$\Lambda = T^{-1}AT$$

Claim:

$$\Lambda = \text{blockdiag} (\Lambda_1, \dots, \Lambda_q)$$

6.8 Diagonalizable systems

6.8.1 Diagonalizable matrices

We say that a square matrix A is **diagonalizable** if it is similar to a diagonal matrix, that is, there is an invertible matrix T and a diagonal matrix Λ such that

$$\Lambda = T^{-1}AT.$$

In this section, we show that A is diagonalizable if and only if it has the following property: For each distinct eigenvalue of A ,

$$\boxed{\text{geometric multiplicity} = \text{algebraic multiplicity}}$$

Recall that the geometric multiplicity of an eigenvalue is the dimension of its eigenspace, or equivalently, the maximum number of linearly independent eigenvectors associated with that eigenvalue. An eigenvalue whose geometric multiplicity equals its algebraic multiplicity is called a **nondefective eigenvalue**. Otherwise, it is considered **defective**. A matrix with the property that all its eigenvalues are nondefective is called a **nondefective matrix**. Otherwise, it is considered defective. When is a matrix A nondefective? Some examples are:

- (a) A has n *distinct* eigenvalues where A is $n \times n$.
- (b) A is *hermitian* that is, $A^* = A$.

We first need the following result.

Fact 10 Suppose that geometric multiplicity equals algebraic multiplicity for each distinct eigenvalue of a square matrix A . Then A has a basis of eigenvectors:

$$v^1, v^2, \dots, v^n$$

Example 79

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Theorem 2 Suppose that for each distinct eigenvalue of A , its geometric multiplicity is the same as its algebraic multiplicity. Then A is similar to a diagonal matrix Λ whose diagonal elements are the eigenvalues of A . (If an eigenvalue has multiplicity m , it appears m times on the diagonal of Λ)

PROOF. Let

$$T := \begin{pmatrix} v^1 & v^2 & \dots & v^n \end{pmatrix}$$

where $v^1, \dots, v^n$ is a basis of eigenvectors for A . Let λ_i be the eigenvalue corresponding to v^i . Then,

$$Av^i = \lambda_i v^i$$

Defining

$$\Lambda := T^{-1}AT$$

it follows from the ‘useful result on similarity transformations’ that

$$\begin{aligned}\Lambda &= \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) \\ &:= \underbrace{\begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}}_{\text{Complex Jordan form of } A}\end{aligned}$$

So A is similar to the diagonal matrix Λ . ■

Example 80

$$A = \begin{pmatrix} 4 & -2 & -2 \\ 0 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix}$$

The characteristic polynomial of A is

$$p(s) = \det(sI - A) = (s - 4)[(s - 2)(s - 2) - 4] = s(s - 4)^2$$

So A has eigenvalues,

$$\lambda_1 = 0, \quad \lambda_2 = 4$$

with algebraic multiplicities 1 and 2.

The equation $(A - \lambda I)v = 0$ yields:

For $\lambda_1 = 0$,

$$\text{rref}(A - \lambda_1 I) = \text{rref} \begin{pmatrix} 4 & -2 & -2 \\ 0 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix} = \text{rref} \begin{pmatrix} 4 & 0 & -4 \\ 0 & 2 & -2 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

Hence, λ_1 has geometric multiplicity 1 and its eigenspace is spanned by

$$v^1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

For $\lambda_2 = 4$,

$$\text{rref}(A - \lambda_2 I) = \text{rref} \begin{pmatrix} 0 & -2 & -2 \\ 0 & -2 & -2 \\ 0 & -2 & -2 \end{pmatrix} = \text{rref} \begin{pmatrix} 0 & -2 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Hence, λ_2 has geometric multiplicity 2 and its eigenspace is spanned by

$$v^2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \text{and} \quad v^3 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

Since the geometric multiplicity of each eigenvalue is equal to its algebraic multiplicity, A is diagonalizable and its Jordan form is

$$\Lambda = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

A quick check with MATLAB yields $\Lambda = T^{-1}AT$ where $T = (v^1 v^2 v^3)$.

6.8.2 Diagonalizable continuous-time systems

Consider a continuous-time system

$$\dot{x} = Ax$$

and suppose A is an $n \times n$ diagonalizable matrix, that is, there is a nonsingular matrix T such that

$$\Lambda = T^{-1}AT$$

is diagonal. Hence, the diagonal elements of Λ are the eigenvalues (including possible multiplicities) $\lambda_1, \lambda_2, \dots, \lambda_n$ of A . Introducing the state transformation,

$$x = T\xi$$

we have $\xi = T^{-1}x$ and the system behavior is now equivalently described by

$$\dot{\xi} = \Lambda\xi.$$

This yields

$$\boxed{\begin{array}{rcl} \dot{\xi}_1 & = & \lambda_1 \xi_1 \\ \dot{\xi}_2 & = & \lambda_2 \xi_2 \\ & \vdots & \\ \dot{\xi}_n & = & \lambda_n \xi_n \end{array}} \quad (6.12)$$

that is, n decoupled first order systems.

Hence, for $i = 1, 2, \dots, n$,

$$\xi_i(t) = e^{\lambda_i t} \xi_{i0} \quad (6.13)$$

where $\xi_{i0} = \xi_i(0)$. In vector form, we obtain

$$\xi(t) = \begin{pmatrix} e^{\lambda_1 t} & 0 & \dots & 0 \\ 0 & e^{\lambda_2 t} & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & e^{\lambda_n t} \end{pmatrix} \xi_0 \quad (6.14)$$

where $\xi_0 = (\xi_{10} \ \xi_{20} \ \cdots \ \xi_{n0})^T = \xi(0)$. Since $x(t) = T\xi(t)$, we have

$$x(t) = \xi_1(t)v^1 + \xi_2(t)v^2 + \cdots + \xi_n(t)v^n$$

where $v^1, v^2, \dots, v^n$ are the columns of T . It now follows from (6.13) that the solution for $x(t)$ is given by

$$\boxed{x(t) = e^{\lambda_1 t} \xi_{10} v^1 + e^{\lambda_2 t} \xi_{20} v^2 + \cdots + e^{\lambda_n t} \xi_{n0} v^n} \quad (6.15)$$

Note that each term $e^{\lambda_i t} \xi_{i0} v^i$ on the right-hand-side of the above equation is a mode. This results in the following fundamental observation.

Every motion (solution) of a diagonalizable linear time-invariant system is simply a sum of modes.

Suppose x_0 is the initial value of x at $t = 0$, that is, $x(0) = x_0$. Then, the corresponding initial value $\xi_0 = \xi(0)$ of ξ is given by

$$\xi_0 = T^{-1}x_0.$$

State transition matrix. Since $x(t) = T\xi(t)$ and $\xi_0 = T^{-1}x_0$, it follows from (6.14) that

$$x(t) = \Phi(t)x_0 \quad (6.16)$$

where the state transition matrix $\Phi(\cdot)$ is given by

$$\boxed{\Phi(t) := T \begin{pmatrix} e^{\lambda_1 t} & 0 & \cdots & 0 \\ 0 & e^{\lambda_2 t} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & e^{\lambda_n t} \end{pmatrix} T^{-1}} \quad (6.17)$$

Example 81 Consider a CT system with A from example 62, that is,

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

Here,

$$T = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

and $\lambda_1 = 2, \lambda_2 = 4$.

Hence,

$$\begin{aligned} \Phi(t) &= T \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{4t} \end{pmatrix} T^{-1} \\ &= \frac{1}{2} \begin{pmatrix} e^{2t} + e^{4t} & e^{2t} - e^{4t} \\ e^{2t} - e^{4t} & e^{2t} + e^{4t} \end{pmatrix} \end{aligned}$$

and all solutions are given by

$$\begin{aligned}x_1(t) &= \frac{1}{2}(e^{2t} + e^{4t})x_{10} + \frac{1}{2}(e^{2t} - e^{4t})x_{20} \\x_2(t) &= \frac{1}{2}(e^{2t} - e^{4t})x_{10} + \frac{1}{2}(e^{2t} + e^{4t})x_{20}\end{aligned}$$

Exercise 76 Compute the state transition matrix for the CT system corresponding to the A matrix in example 80.

6.8.3 Diagonalizable DT systems

The story here is basically the same as that for CT. Consider a DT system

$$x(k+1) = Ax(k)$$

and suppose A is diagonalizable, that is, there is a nonsingular matrix T such that

$$\Lambda = T^{-1}AT$$

is diagonal and the diagonal elements of Λ are the eigenvalues $\lambda_1, \dots, \lambda_n$ of A .

Introducing the state transformation,

$$x = T\xi$$

we have $\xi = T^{-1}x$ and its behavior is described by:

$$\xi(k+1) = \Lambda\xi(k)$$

Hence,

$$\begin{aligned}\xi_1(k+1) &= \lambda_1\xi_1(k) \\ \xi_2(k+1) &= \lambda_2\xi_2(k) \\ &\vdots \\ \xi_n(k+1) &= \lambda_n\xi_n(k)\end{aligned}$$

If

$$x(0) = x_0$$

then

$$\xi(0) = \xi_0 = T^{-1}x_0$$

So,

$$\xi_i(k) = \lambda_i^k \xi_{i0}, \quad i = 1, 2, \dots, n$$

or,

$$\xi(k) = \begin{pmatrix} \lambda_1^k & 0 & \dots & 0 \\ 0 & \lambda_2^k & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n^k \end{pmatrix} \xi_0$$

Since

$$x = \xi_1 v^1 + \xi_2 v^2 + \dots + \xi_n v^n$$

where $v^1, v^2, \dots, v^n$ are the columns of T , the solution for $x(k)$ is given by

$$\boxed{x(k) = \lambda_1^k \xi_{10} v^1 + \lambda_2^k \xi_{20} v^2 + \dots + \lambda_n^k \xi_{n0} v^n}$$

Or,

$$x(k) = \Phi(k) x_0$$

where the *DT state transition matrix* $\Phi(\cdot)$ is given by

$$\Phi(k) := T \begin{pmatrix} \lambda_1^k & 0 & \dots & 0 \\ 0 & \lambda_2^k & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n^k \end{pmatrix} T^{-1}$$

Example 82 Consider a DT system with A from example 62, that is,

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

Here,

$$T = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

and $\lambda_1 = 2, \lambda_2 = 4$.

Hence,

$$\begin{aligned} \Phi(k) &= T \begin{pmatrix} 2^k & 0 \\ 0 & 4^k \end{pmatrix} T^{-1} \\ &= \frac{1}{2} \begin{pmatrix} 2^k + 4^k & 2^k - 4^k \\ 2^k - 4^k & 2^k + 4^k \end{pmatrix} \end{aligned}$$

and all solutions are given by

$$\begin{aligned} x_1(t) &= (2^k + 4^k)x_{10} + (2^k - 4^k)x_{20} \\ x_2(t) &= (2^k - 4^k)x_{10} + (2^k + 4^k)x_{20} \end{aligned}$$

6.9 Real Jordan form for real diagonalizable systems*

Suppose A is real and has a complex eigenvalue

$$\lambda = \alpha + j\omega$$

where α, ω are real. Then any corresponding eigenvector v is complex and can be expressed as

$$v = u + jw$$

where u, w are real and are linearly independent over the real scalars.

Since A is real, $\bar{\lambda}$ is also an eigenvalue of A with eigenvector $\bar{v}$. So we can replace each v and $\bar{v}$ in the basis of eigenvectors with the corresponding u and w to obtain a new basis; then we define a coordinate transformation where the columns of T are these new basis vectors.

The relationship,

$$Av = \lambda v$$

and A real implies

$$\begin{aligned} Au &= \alpha u - \omega w \\ Aw &= \omega u + \alpha w \end{aligned}$$

Hence, using the ‘useful result on similarity transformations’, we can choose T so that $\Lambda = T^{-1}AT$ is given by

$$\begin{aligned} \Lambda &= \text{block-diag} \left(\lambda_1, \dots, \lambda_l, \begin{pmatrix} \alpha_1 & \omega_1 \\ -\omega_1 & \alpha_1 \end{pmatrix}, \dots, \begin{pmatrix} \alpha_m & \omega_m \\ -\omega_m & \alpha_m \end{pmatrix} \right) \\ &= \underbrace{\begin{pmatrix} \lambda_1 & & & & \\ & \ddots & & & \\ & & \lambda_l & & \\ & & & \begin{pmatrix} \alpha_1 & \omega_1 \\ -\omega_1 & \alpha_1 \end{pmatrix} & \\ & & & \ddots & \\ & & & & \begin{pmatrix} \alpha_m & \omega_m \\ -\omega_m & \alpha_m \end{pmatrix} \end{pmatrix}}_{\text{Real Jordan form of } A} \end{aligned}$$

where $\lambda_1, \dots, \lambda_l$ are real eigenvalues of A and $\alpha_1 + j\omega_1, \dots, \alpha_m + j\omega_m$ are genuine complex eigenvalues.

Example 83 Recall example 63.

Example 84

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

By inspection (recall companion matrices),

$$\det(sI - A) = s^4 - 1$$

So eigenvalues are:

$$\lambda_1 = 1 \quad \lambda_2 = -1 \quad \lambda_3 = j \quad \lambda_4 = -j$$

Since there are $n = 4$ distinct eigenvalues, the complex Jordan form is:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & j & 0 \\ 0 & 0 & 0 & -j \end{pmatrix}$$

The real Jordan form is:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

6.9.1 CT systems

The solutions of

$$\dot{x} = Ax, \quad x(0) = x_0$$

are given by

$$x(t) = \Phi(t)x_0$$

where the state transition matrix $\Phi(\cdot)$ is given by

$$\Phi(t) = T\hat{\Phi}(t)T^{-1}$$

with $\hat{\Phi}(t)$ being the block diagonal matrix:

$$\text{block-diag} \left(e^{\lambda_1 t}, \dots, e^{\lambda_n t}, \begin{pmatrix} e^{\alpha_1 t} \cos(\omega_1 t) & e^{\alpha_1 t} \sin(\omega_1 t) \\ -e^{\alpha_1 t} \sin(\omega_1 t) & e^{\alpha_1 t} \cos(\omega_1 t) \end{pmatrix}, \dots, \begin{pmatrix} e^{\alpha_m t} \cos(\omega_m t) & e^{\alpha_m t} \sin(\omega_m t) \\ -e^{\alpha_m t} \sin(\omega_m t) & e^{\alpha_m t} \cos(\omega_m t) \end{pmatrix} \right)$$

6.10 Diagonalizable second order systems

Only two cases for real Jordan form

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \quad \begin{pmatrix} \alpha & \omega \\ -\omega & \alpha \end{pmatrix}$$

where all the above scalars are real.

6.10.1 State plane portraits for CT systems

Real eigenvalues

Genuine complex eigenvalues

6.11 Exercises

Exercise 77 Compute the eigenvalues and eigenvectors of the matrix,

$$A = \begin{pmatrix} 2 & -3 & 0 \\ 2 & -3 & 0 \\ 3 & -5 & 1 \end{pmatrix}$$

Exercise 78 Compute the eigenvalues and eigenvectors of the matrix,

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ -1 & 2 & 3 \end{pmatrix}$$

Exercise 79 Compute the eigenvalues and eigenvectors of the following matrix.

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

Exercise 80 Compute expressions for eigenvalues and linearly independent eigenvectors of

$$A = \begin{pmatrix} \alpha & \omega \\ -\omega & \alpha \end{pmatrix} \quad \alpha, \omega \neq 0$$

Exercise 81 Find expressions for the eigenvalues and linearly independent eigenvectors of matrices of the form:

$$\begin{pmatrix} \beta & \gamma \\ \gamma & \beta \end{pmatrix}$$

where γ and β are real with $\gamma \geq 0$. Obtain state plane portraits for

$$\begin{array}{ll} \beta = 0 & \gamma = 1 \\ \beta = -3 & \gamma = 2 \\ \beta = 3 & \gamma = 2 \end{array}$$

Exercise 82 Consider an LTI system described by $\dot{x} = Ax$.

(a) Suppose that the vectors

$$\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

are eigenvectors of A corresponding to eigenvalues -2 and 3 , respectively. What is the response $x(t)$ of the system to the initial condition

$$x(0) = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} ?$$

(b) Suppose A is a real matrix and the vector

$$\begin{pmatrix} 1 \\ j \\ -1 \\ -j \end{pmatrix}$$

is an eigenvector of A corresponding to the eigenvalue $2 + 3i$. What is the response $x(t)$ of the system to the initial condition

$$x(0) = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} ?$$

Exercise 83 Suppose A is a square matrix and the vectors

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

are eigenvectors of A corresponding to eigenvalues -1 and 1 , respectively. What is the response $x(t)$ of the system $\dot{x} = Ax$ to the following initial conditions.

$$(a) \quad x(0) = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} \quad (b) \quad x(0) = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} \quad (c) \quad x(0) = \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix} ?$$

(d) Suppose A is a real matrix and the vector

$$\begin{pmatrix} 1 + 2j \\ 1 - j \\ 3 \end{pmatrix}$$

is an eigenvector of A corresponding to the eigenvalue $2 - 3j$. What is the response $x(t)$ of the system $\dot{x} = Ax$ to the initial condition

$$x(0) = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} ?$$

Exercise 84 Suppose A is a real square matrix and the vectors

$$\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 2 \\ 4 \\ 8 \end{pmatrix} \quad \begin{pmatrix} 1 \\ j \\ -1 \\ -j \end{pmatrix}$$

are eigenvectors of A corresponding to eigenvalues -1 and 2 and j , respectively. What is the response $x(t)$ of the system $\dot{x} = Ax$ to the following initial conditions.

$$(a) \quad x(0) = \begin{pmatrix} 2 \\ -2 \\ 2 \\ -2 \end{pmatrix} \quad (b) \quad x(0) = \begin{pmatrix} -1 \\ -2 \\ -4 \\ -8 \end{pmatrix} \quad (c) \quad x(0) = \begin{pmatrix} 0 \\ 3 \\ 3 \\ 9 \end{pmatrix} \quad (d) \quad x(0) = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix} ?$$

Exercise 85 Suppose A is a real square matrix with characteristic polynomial

$$p(s) = (s + 2)(s^2 + 4)^2$$

Suppose that all eigenvectors of A are nonzero multiples of

$$\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1+j \\ -j \\ 0 \\ 1 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 0 \\ 1-j \\ j \\ 0 \\ 1 \end{pmatrix}$$

What is the solution to $\dot{x} = Ax$ with the following initial conditions.

$$(a) \quad x(0) = \begin{pmatrix} 2 \\ 4 \\ 6 \\ 8 \\ 0 \end{pmatrix}, \quad (b) \quad x(0) = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad (c) \quad x(0) = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

(d) All all solutions of the system $\dot{x} = Ax$ bounded? Justify your answer.

Exercise 86 Consider a discrete-time LTI system described by $x(k+1) = Ax(k)$.

(a) Suppose that the vectors

$$\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

are eigenvectors of A corresponding to eigenvalues -2 and 3 , respectively. What is the response $x(k)$ of the system to the initial condition

$$x(0) = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} ?$$

(b) Suppose A is a real matrix and the vector

$$\begin{pmatrix} 1 \\ j \\ -1 \\ -j \end{pmatrix}$$

is an eigenvector of A corresponding to the eigenvalue $2 + 3i$. What is the response $x(k)$ of the system to the initial condition

$$x(0) = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} ?$$

Exercise 87 Determine whether or not the following matrix is nondefective.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

Exercise 88 In vibration analysis, one usually encounters systems described by

$$M\ddot{q} + Kq = 0$$

where $q(t)$ is an N -vector and M and K are square matrices. Suppose ω is a real number which satisfies

$$\det(\omega^2 M - K) = 0.$$

Show that the above system has a solution of the form

$$q(t) = \sin(\omega t)v$$

where v is a constant N -vector.

Exercise 89 Determine whether or not the matrix,

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & 3 \end{pmatrix},$$

is diagonalizable, that is, determine whether or not there exists an invertible matrix T such that $T^{-1}AT$ is diagonal. If A is diagonalizable, determine T .

Problem 1 Recall that a square matrix is said to be diagonalizable if there exists an invertible matrix T so that $T^{-1}AT$ is diagonal. Determine whether or not each of the following matrices are diagonalizable. If not diagonalizable, give a reason. If diagonalizable, obtain T .

(a)

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

(b)

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}$$

(c)

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Chapter 7

$$e^{At}$$

Recall that all solutions to the scalar system $\dot{x} = ax$ satisfy $x(t) = e^{a(t-t_0)}x(t_0)$ for all t and t_0 . Recall also the power series expansion for e^{at} :

$$e^{at} = 1 + at + \frac{1}{2!}(at)^2 + \cdots + \frac{1}{k!}(at)^k + \cdots$$

The main result of this chapter is that all solutions of $\dot{x} = Ax$ satisfy

$$\boxed{x(t) = e^{A(t-t_0)}x(t_0)} \quad (7.1)$$

for all t and t_0 where the **matrix exponential** e^{At} is given by the convergent series

$$\boxed{e^{At} = 1 + At + \frac{1}{2!}(At)^2 + \cdots + \frac{1}{k!}(At)^k + \cdots} \quad (7.2)$$

e^{At} is referred to as the **state transition matrix** of the system.

7.1 State transition matrix

7.1.1 Continuous-time

Here we justify the existence of the state transition matrix and establish some of its properties.

Consider the initial value problem specified by

$$\dot{x} = Ax \quad \text{and} \quad x(t_0) = x_0. \quad (7.3)$$

It can be shown that for each initial time t_0 and each initial state x_0 there is a unique solution $x(\cdot)$ to the above initial value problem and this solution is defined for all time t . Let $\tilde{\phi}(t; t_0, x_0)$ be the value of this solution at t , that is, for any t_0 and x_0 , the solution to (7.3) is given by

$$x(t) = \tilde{\phi}(t; t_0, x_0).$$

Since the system $\dot{x} = Ax$ is **time-invariant**, we must have

$$\tilde{\phi}(t; t_0, x_0) = \phi(t - t_0; x_0)$$

for some function ϕ . Since the system under consideration is **linear**, one can show that, for any t and t_0 , the solution at time t depends linearly on the initial state x_0 at time t_0 ; hence there exists a matrix $\Phi(t-t_0)$ such that

$$\phi(t-t_0, x_0) = \Phi(t-t_0)x_0.$$

We call the matrix valued function $\Phi(\cdot)$ the **state transition matrix** for the system $\dot{x} = Ax$. So, we have concluded that there is a matrix valued function $\Phi(\cdot)$ such that for any t_0 and x_0 , the solution $x(\cdot)$ to the above initial value problem is given by

$$\boxed{x(t) = \Phi(t-t_0)x_0}$$

We now show that Φ must satisfy the following **matrix differential equation** and initial condition:

$$\boxed{\begin{array}{lcl} \dot{\Phi} & = & A\Phi \\ \Phi(0) & = & I \end{array}}$$

To demonstrate this, consider any initial state x_0 and let $x(\cdot)$ be the solution to (7.3) with $t_0 = 0$. Then $x(t) = \Phi(t)x_0$. Considering $t = 0$, we note that

$$x_0 = x(0) = \Phi(0)x_0;$$

that is $\Phi(0)x_0 = x_0$. Since the above holds for any x_0 , we must have $\Phi(0) = I$.

We now note that for all t ,

$$\dot{\Phi}(t)x_0 = \dot{x}(t) = Ax(t) = A\Phi(t)x_0;$$

that is $\dot{\Phi}(t)x_0 = A\Phi(t)x_0$. Since the above must hold for any x_0 , we obtain that $\dot{\Phi}(t) = A\Phi(t)$.

Note that the above two conditions on Φ uniquely specify Φ (Why?) The purpose of this section is to examine Φ and look at ways of evaluating it.

7.1.2 Discrete-time

$$x(k+1) = Ax(k) \quad \text{and} \quad x(k_0) = x_0$$

We have

$$\boxed{x(k) = \Phi(k - k_0)x_0}$$

The matrix valued function $\Phi(\cdot)$ is called the **state transition matrix** and is given by

$$\boxed{\Phi(k) = A^k}$$

Note that, if A is singular, $\Phi(k)$ is not defined for $k < 0$. That was easy; now for CT.

7.2 Polynomials of a square matrix

Before discussing the exponential of a matrix, let's first look at polynomials of a matrix.

7.2.1 Polynomials of a matrix

We define a polynomial of a square matrix as follows. Consider any polynomial p of a scalar variable s given by:

$$p(s) = a_0 + a_1s + a_2s^2 + \dots + a_ms^m$$

where $a_0, \dots, a_m$ are scalars. If A is a square matrix, we define $p(A)$ as follows:

$$p(A) := a_0I + a_1A + a_2A^2 + \dots + a_mA^m,$$

that is, we replace s^k with A^k . (Recall that $s^0 = 1$; hence $A^0 := I$)

The following properties can be readily deduced from the above definition.

- (a) The matrices A and $p(A)$ commute, that is,

$$Ap(A) = p(A)A$$

- (b) Consider any nonsingular matrix T , and suppose $A = T\Lambda T^{-1}$. Then

$$p(A) = Tp(\Lambda)T^{-1}$$

- (c) Suppose A is diagonal, that is,

$$A = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix};$$

then

$$p(A) = \begin{pmatrix} p(\lambda_1) & 0 & \cdots & 0 \\ 0 & p(\lambda_2) & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & p(\lambda_n) \end{pmatrix}$$

- (d) Suppose A is diagonalizable, that is, for some nonsingular T ,

$$A = T \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} T^{-1}$$

Then

$$p(A) = T \begin{pmatrix} p(\lambda_1) & 0 & \cdots & 0 \\ 0 & p(\lambda_2) & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & p(\lambda_n) \end{pmatrix} T^{-1}$$

- (e) If λ is an eigenvalue of A with eigenvector v , then $p(\lambda)$ is an eigenvalue of $p(A)$ with eigenvector v .

Properties (b)-(d) are useful for computing $p(A)$.

Example 85 For

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

we found that

$$T = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

results in

$$\Lambda = T^{-1}AT = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$$

Hence, $A = T\Lambda T^{-1}$ and

$$\begin{aligned} p(A) &= Tp(\Lambda)T^{-1} \\ &= T \begin{pmatrix} p(2) & 0 \\ 0 & p(4) \end{pmatrix} T^{-1} \\ &= \frac{1}{2} \begin{pmatrix} p(2) + p(4) & p(2) - p(4) \\ p(2) - p(4) & p(2) + p(4) \end{pmatrix} \end{aligned}$$

Suppose p is the characteristic polynomial of A . Since 2, 4 are the eigenvalues of A , we have $p(2) = p(4) = 0$; hence

$$p(A) = 0$$

7.2.2 Cayley-Hamilton Theorem

The following result is a fundamental result in linear algebra and is very useful in systems and control.

Theorem 3 *If p is the characteristic polynomial of a square matrix A , then*

$$\boxed{p(A) = 0}$$

PROOF. The proof is easy for a diagonalizable matrix A . Recall that such a matrix is similar to a diagonal matrix whose diagonal elements $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A ; hence $p(\lambda_1) = \dots = p(\lambda_n) = 0$. Now use property (d) above. We will leave the proof of the general nondiagonalizable case for another day. ■

Suppose the characteristic polynomial of A is

$$\det(sI - A) = a_0 + a_1s + \dots + a_{n-1}s^{n-1} + s^n$$

Then the Cayley Hamilton theorem states that

$$a_0I + a_1A + \dots + a_{n-1}A^{n-1} + A^n = 0$$

hence,

$$A^n = -a_0I - a_1A - \dots - a_{n-1}A^{n-1}$$

From this one can readily show that for any $m \geq n$, A^m can be expressed as a linear combination of $I, A, \dots, A^{n-1}$. Hence, we have the following conclusion:

Any polynomial of an $n \times n$ matrix A can be expressed as a linear combination of $I, A, \dots, A^{n-1}$.

Example 86 Consider a discrete-time system described by $x(k+1) = Ax(k)$ and suppose that all the eigenvalues of A are zero. Then the characteristic polynomial of A is given by $p(s) = s^n$ where A is $n \times n$. It now follows from the Cayley Hamilton theorem that

$$A^n = 0$$

Since all solutions of the system satisfy $x(k) = A^k x(0)$ for $k = 0, 1, 2, \dots$, it follows that

$$x(k) = 0 \quad \text{for } k \geq n.$$

Thus, *all solutions go to zero in at most n steps.*

Example 87 To illustrate the result above, consider the companion matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

The characteristic polynomial of A is $p(s) = s^3$; hence $A^3 = 0$.

Minimal polynomial of A

The minimal polynomial (minpoly) of a square matrix A is the monic polynomial q of lowest order for which $q(A) = 0$.

The order of minpoly q is always less than or equal to the order of charpoly p ; its roots are precisely the eigenvalues of A and $p(s) = g(s)q(s)$.

Example 88 The matrix

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

has charpoly $p(s) = s^2$ and minpoly $q(s) = s$. The matrix

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

has charpoly $p(s) = s^2$ and minimal poly $q(s) = s^2$.

- It follows from property (d) above that for a *diagonalizable matrix* A , its minimal polynomial is

$$\prod_{i=1}^l (s - \lambda_i)$$

where λ_i are the *distinct* eigenvalues of A .

Exercise 90 Without doing any matrix multiplications, compute A^4 for

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 564 & 1 & 0 & 0 \end{pmatrix}$$

Justify your answer.

7.3 Functions of a square matrix

Consider a complex valued function f of a complex variable λ . Suppose f is analytic in some open disk $\{\lambda \in \mathbb{C} : |\lambda| < \rho\}$ of radius $\rho > 0$. Then, provided $|\lambda| < \rho$, $f(\lambda)$ can be expressed as the sum of a convergent power series; specifically

$$f(\lambda) = \sum_{k=0}^{\infty} a_k \lambda^k = a_0 + a_1 \lambda + a_2 \lambda^2 + \dots$$

where

$$a_k = \frac{1}{k!} f^{(k)}(0) = \frac{1}{k!} \frac{d^k f}{d\lambda^k}(0)$$

Suppose A is a square complex matrix and $|\lambda_i| < \rho$ for every eigenvalue λ_i of A . Then, it can be shown that the power series $\{\sum_{k=0}^n a_k A^k\}_{n=0}^{\infty}$ is convergent. We define $f(A)$ to be the sum of this power series, that is,

$$f(A) := \sum_{k=0}^{\infty} a_k A^k = a_0 I + a_1 A + a_2 A^2 + \dots$$

Example 89 Consider

$$f(\lambda) = \frac{1}{1 - \lambda}$$

This is analytic in the open unit disk ($\rho = 1$) and

$$f(\lambda) = \sum_{k=0}^{\infty} \lambda^k = 1 + \lambda + \lambda^2 + \dots$$

Consider any matrix A with eigenvalues in the open unit disk. Then the series $\{\sum_{k=0}^n A^k\}_{n=0}^{\infty}$ is convergent and we let

$$f(A) := \sum_{k=0}^{\infty} A^k$$

Multiplication of both sides of the above equality by $I - A$ yields

$$(I - A)f(A) = \sum_{k=0}^{\infty} (A^k - A^{k+1}) = I;$$

hence the matrix $I - A$ is invertible with inverse $f(A)$ and we obtain the following power series expansion for $(I - A)^{-1}$:

$$(I - A)^{-1} = \sum_{k=0}^{\infty} A^k = I + A + A^2 + \dots$$

Power series expansion for $(sI - A)^{-1}$. We present here an expansion which will be useful later on. Consider any square matrix A and any complex number s which satisfies:

$$|s| > |\lambda_i| \quad \text{for every eigenvalue } \lambda_i \text{ of } A.$$

Then $sI - A$ is invertible and $(sI - A)^{-1} = \frac{1}{s}(I - \frac{1}{s}A)^{-1}$. Also, the eigenvalues of $\frac{1}{s}A$ are in the open unit disk; hence

$$(I - \frac{1}{s}A)^{-1} = \sum_{k=0}^{\infty} \frac{1}{s^k} A^k = I + \frac{1}{s}A + \frac{1}{s^2}A^2 + \dots$$

and

$$(sI - A)^{-1} = \sum_{k=0}^{\infty} \frac{1}{s^{k+1}} A^k = \frac{1}{s}I + \frac{1}{s^2}A + \frac{1}{s^3}A^2 + \dots$$

Properties of $f(A)$. The following properties can be readily deduced from the above definition of $f(A)$.

(a) The matrices A and $f(A)$ commute, that is,

$$Af(A) = f(A)A.$$

(b) Consider any nonsingular matrix T and suppose $A = T\Lambda T^{-1}$. Then

$$f(A) = Tf(\Lambda)T^{-1}.$$

(c) Suppose A is diagonal, that is,

$$A = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

then

$$f(A) = \begin{pmatrix} f(\lambda_1) & 0 & \cdots & 0 \\ 0 & f(\lambda_2) & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & f(\lambda_n) \end{pmatrix}$$

(d) Suppose A is diagonalizable, that is, for some nonsingular T ,

$$A = T \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} T^{-1}$$

Then

$$f(A) = T \begin{pmatrix} f(\lambda_1) & 0 & \cdots & 0 \\ 0 & f(\lambda_2) & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & f(\lambda_n) \end{pmatrix} T^{-1}$$

(e) If λ is an eigenvalue of A with eigenvector v , then $f(\lambda)$ is an eigenvalue of $f(A)$ with eigenvector v .

Properties (b)-(d) are useful for calculating $f(A)$. When A is diagonalizable, one simply has to evaluate the scalars $f(\lambda_1), \dots, f(\lambda_n)$ and then compute $T \operatorname{diag}(f(\lambda_1), \dots, f(\lambda_n)) T^{-1}$.

Recall that every power of A can be written as a linear combination of $I, A, \dots, A^{n-1}$. Hence we conclude that:

Any function of an $n \times n$ matrix A is a linear combination of $I, A, \dots, A^{n-1}$

Example 90 For

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

we found that

$$T = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

results in

$$\Lambda = T^{-1}AT = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$$

Hence, $A = T\Lambda T^{-1}$ and

$$\begin{aligned} f(A) &= Tf(\Lambda)T^{-1} \\ &= T \begin{pmatrix} f(2) & 0 \\ 0 & f(4) \end{pmatrix} T^{-1} \\ &= \frac{1}{2} \begin{pmatrix} f(2) + f(4) & f(2) - f(4) \\ f(2) - f(4) & f(2) + f(4) \end{pmatrix} \end{aligned}$$

Exercise 91 Compute an explicit expression (see last example) for $f(A)$ where

$$A = \begin{pmatrix} 9 & -2 \\ -2 & 6 \end{pmatrix}$$

7.3.1 The matrix exponential: e^A

Recall the exponential function:

$$f(\lambda) = e^\lambda$$

This function is analytic in the whole complex plane (that is $\rho = \infty$) and

$$\frac{d^k e^\lambda}{d\lambda^k} = e^\lambda$$

for $k = 1, 2, \dots$; hence for any complex number λ ,

$$e^\lambda = \sum_{k=0}^{\infty} \frac{1}{k!} \lambda^k = 1 + \lambda + \frac{1}{2} \lambda^2 + \frac{1}{3!} \lambda^3 + \dots$$

Thus the exponential of any square matrix A is defined by:

$$e^A := \sum_{k=0}^{\infty} \frac{1}{k!} A^k = I + A + \frac{1}{2!} A^2 + \dots$$

and this power series converges for every square matrix A . Some properties:

(a) For any zero matrix 0 ,

$$e^0 = I$$

(b) Suppose t is a scalar variable.

$$\frac{de^{At}}{dt} = Ae^{At} = e^{At}A$$

The following results also hold:

(i)

$$e^{A(t_1+t_2)} = e^{At_1} e^{At_2}$$
$$e^{-A} = (e^A)^{-1}$$

(ii) If λ is an eigenvalue of A with eigenvector v , then e^λ is an eigenvalue of e^A with eigenvector v . So, e^A has no eigenvalue at 0 and hence:

(iii) e^A is nonsingular.

(iv) The following result also holds

$$e^{A+B} = e^A e^B \quad \text{if } A \text{ and } B \text{ commute}$$

7.3.2 Other matrix functions

$$\cos(A)$$

$$\sin(A)$$

7.4 The state transition matrix: e^{At}

Using the above definitions and abusing notation (e^{At} instead of e^{tA}),

$$e^{At} := \sum_{k=0}^{\infty} \frac{1}{k!} (tA)^k = I + tA + \frac{1}{2!} (tA)^2 + \frac{1}{3!} (tA)^3 + \dots$$

Letting

$$\Phi(t) := e^{At}$$

we have

$$\begin{aligned}\Phi(0) &= I \\ \dot{\Phi} &= A\Phi\end{aligned}$$

Hence, the solution of

$$\dot{x} = Ax \quad x(0) = x_0$$

is given by

$$x(t) = e^{At}x_0$$

7.5 Computation of e^{At}

7.5.1 MATLAB

```
>> help expm
```

```
EXPM    Matrix exponential.  
EXPM(X) is the matrix exponential of X.  EXPM is computed using  
a scaling and squaring algorithm with a Pade approximation.  
Although it is not computed this way, if X has a full set  
of eigenvectors V with corresponding eigenvalues D, then  
[V,D] = EIG(X) and EXPM(X) = V*diag(exp(diag(D)))/V.  
See EXPM1, EXPM2 and EXPM3 for alternative methods.
```

```
EXP(X) (that's without the M) does it element-by-element.
```

7.5.2 Numerical simulation

Let ϕ_i be the i -th column of the state transition matrix $\Phi(t) = e^{At}$; thus

$$\Phi(t) = \begin{pmatrix} \phi_1(t) & \phi_2(t) & \cdots & \phi_n(t) \end{pmatrix}.$$

Let x be the unique solution to $\dot{x} = Ax$ and $x(0) = x_0$. Since $x(t) = \Phi(t)x_0$ we have

$$x(t) = x_{01}\phi_1(t) + \cdots + x_{0n}\phi_n$$

where x_{0i} is the i -th component of x_0 . Hence ϕ_i , is the solution to

$$\dot{x} = Ax \quad x(0) = e^i$$

where e^i is all zeros except for its i -th component which is 1.

7.5.3 Jordan form

$$\boxed{e^{At} = T e^{\Lambda t} T^{-1}}$$

where $\Lambda = T^{-1}AT$ and Λ is the Jordan form of A . We have already seen this for diagonalizable systems. For diagonalizable systems Λ is a diagonal matrix whose diagonal elements are the eigenvalues of A and the columns of T are eigenvectors of A ; the eigenvector in the i -th column of T corresponds to the eigenvalue in the ii position of Λ .

7.5.4 Laplace style

Recall that the Laplace transform of the scalar function e^{at} is $(s - a)^{-1}$. Here, we generalize this result and show that for any square matrix A ,

$$\boxed{\mathcal{L}(e^{At}) = (sI - A)^{-1}} \tag{7.4}$$

where $\mathcal{L}$ denotes Laplace transform.

To demonstrate the above result, let $\Phi(t) := e^{At}$ and recall that

$$\dot{\Phi} = A\Phi \quad \text{and} \quad \Phi(0) = I.$$

Taking the Laplace transform of the above differential equation yields

$$s\hat{\Phi}(s) - \Phi(0) = A\hat{\Phi}(s).$$

Using the initial condition and rearranging the above equation results in

$$(sI - A)\hat{\Phi}(s) = I.$$

This yields the desired result that

$$\mathcal{L}(e^{At}) = \hat{\Phi}(s) = (sI - A)^{-1}.$$

■

It follows from (7.4) that

$$\boxed{e^{At} = \mathcal{L}^{-1}((sI - A)^{-1})} \tag{7.5}$$

This identity is useful for calculating e^{At} ; see next example.

Example 91 Recall

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

So,

$$(sI - A)^{-1} = \begin{pmatrix} \frac{s-3}{(s-2)(s-4)} & \frac{-1}{(s-2)(s-4)} \\ \frac{-1}{(s-2)(s-4)} & \frac{s-3}{(s-2)(s-4)} \end{pmatrix}$$

Since

$$\frac{s-3}{(s-2)(s-4)} = \frac{1}{2} \left(\frac{1}{s-2} + \frac{1}{s-4} \right)$$

$$\frac{-1}{(s-2)(s-4)} = \frac{1}{2} \left(\frac{1}{s-2} - \frac{1}{s-4} \right)$$

and

$$\mathcal{L}^{-1}\left(\frac{1}{s-2}\right) = e^{2t} \quad \mathcal{L}^{-1}\left(\frac{1}{s-4}\right) = e^{4t}$$

we have,

$$\begin{aligned} e^{At} &= \mathcal{L}^{-1}((sI - A)^{-1}) \\ &= \frac{1}{2} \begin{pmatrix} e^{2t} + e^{4t} & e^{2t} - e^{4t} \\ e^{2t} - e^{4t} & e^{2t} + e^{4t} \end{pmatrix} \end{aligned}$$

Exercise 92 Compute e^{At} at $t = \ln(2)$ for

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

using *all* methods of this section.

An explicit formula for $(sI - A)^{-1}$

Recall that

$$\boxed{(sI - A)^{-1} = \frac{1}{d(s)} N(s)} \quad (7.6)$$

where d , the characteristic polynomial of A , can be expressed as

$$d(s) = s^n + a_{n-1}s^{n-1} + \cdots + a_1s + a_0$$

and N , an $n \times n$ matrix of polynomials of order less than n , can be expressed as

$$N(s) = s^{n-1}N_{n-1} + s^{n-2}N_{n-2} + \cdots + sN_1 + N_0$$

Thus, to compute $(sI - A)^{-1}$, we just need to compute the scalars, $a_{n-1}, \dots, a_0$ and the matrices $N_{n-1}, \dots, N_0$.

Assume that the scalars $a_{n-1}, \dots, a_0$ have been computed. Then

$$N(s) = d(s)(sI - A)^{-1}$$

Recall that for s sufficiently large,

$$(sI - A)^{-1} = \frac{1}{s}I + \frac{1}{s^2}A + \frac{1}{s^3}A^2 + \cdots$$

Hence,

$$s^{n-1}N_{n-1} + s^{n-2}N_{n-2} + \cdots + sN_1 + N_0 = (s^n + a_{n-1}s^{n-1} + \cdots + a_1s + a_0) \left(\frac{1}{s}I + \frac{1}{s^2}A + \frac{1}{s^3}A^2 + \cdots \right)$$

Equating coefficients of powers of s , we obtain the following expressions for $N_{n-1}, \dots, N_0$:

$$\begin{aligned} s^{n-1} : \quad & N_{n-1} = I \\ s^{n-2} : \quad & N_{n-2} = A + a_{n-1}I \\ s^{n-3} : \quad & N_{n-3} = A^2 + a_{n-1}A + a_{n-2}I \\ & \vdots \\ s^1 : \quad & N_1 = A^{n-2} + a_{n-1}A^{n-3} + a_{n-2}A^{n-4} + \cdots + a_2I \\ s^0 : \quad & N_0 = A^{n-1} + a_{n-1}A^{n-2} + a_{n-2}A^{n-3} + \cdots + a_2A + a_1I \end{aligned}$$

Note that, considering the coefficients of s^{-1} we obtain:

$$s^{-1} : \quad 0 = A^n + a_{n-1}A^{n-1} + a_{n-2}A^{n-2} + \cdots + a_1A + a_0I$$

We have just proven the Cayley Hamilton Theorem for a general square matrix.

7.6 Sampled-data systems

Consider a continuous-time LTI system described by

$$\dot{x}_c(t) = A_c x_c(t).$$

Suppose we only look at the behavior of this system at discrete instants of time, namely

$$\dots, -2T, T, 0, T, 2T, \dots$$

where $T > 0$ is called the **sampling time**.

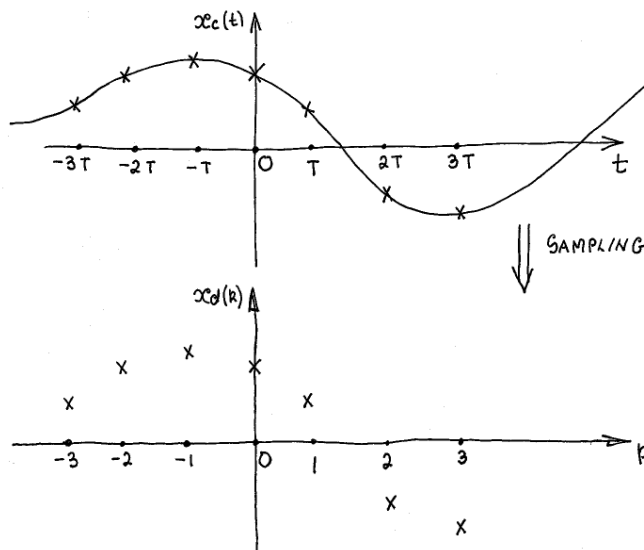


Figure 7.1: Sampling

If we introduce the discrete time state defined

$$x_d(k) = x_c(kT) \quad k = \dots, -2, -1, 0, 1, 2, \dots,$$

then,

$$\begin{aligned} x_d(k+1) &= x_c((k+1)T) \\ &= e^{A_c T} x_c(kT) \\ &= e^{A_c T} x_d(k) \end{aligned}$$

Hence, this **sampled data system** is described by the discrete-time system

$$x_d(k+1) = A_d x_d(k)$$

where

$$\boxed{A_d = e^{A_c T}}$$

The eigenvalues of A_d are

$$e^{\lambda_1 T}, \dots, e^{\lambda_l T}$$

where $\lambda_1, \dots, \lambda_l$ are the eigenvalues of A_c . Note that,

$$\begin{aligned} |e^{\lambda_i T}| < 1 & \quad \text{iff} \quad \Re(\lambda_i) < 0 \\ |e^{\lambda_i T}| = 1 & \quad \text{iff} \quad \Re(\lambda_i) = 0 \\ |e^{\lambda_i T}| > 1 & \quad \text{iff} \quad \Re(\lambda_i) > 0 \end{aligned}$$

$$e^{\lambda_i T} = 1 \quad \text{iff} \quad \lambda_i = 0$$

It should be clear that *all eigenvalues of A_d are nonzero, that is A_d is invertible*.

Note that if we approximate $e^{A_c T}$ by the first two terms in its power series, that is,

$$e^{A_c T} \approx I + A_c T$$

then,

$$A_d \approx I + A_c T.$$

7.7 Exercises

Exercise 93 Suppose A is a 3×3 matrix, b is a 3-vector and

$$\det(sI - A) = s^3 + 2s^2 + s + 1$$

(a) Express $A^3 b$ in terms of $b, Ab, A^2 b$.

(b) Suppose that the matrix

$$T = \begin{pmatrix} b & Ab & A^2 b \end{pmatrix}$$

is invertible. What is the matrix $\Lambda := T^{-1}AT$? Specify each element of Λ explicitly.

Exercise 94 Suppose A is a 3×3 matrix and

$$\det(sI - A) = s^3 + 2s^2 + s + 1$$

(a) Express A^3 in terms of I, A, A^2 .

(b) Express A^5 in terms of I, A, A^2 .

(c) Express A^{-1} in terms of I, A, A^2

Exercise 95 Suppose A is a 3×3 matrix with eigenvalues $1, 2\pi j$ and $-2\pi j$. What are the eigenvalues of e^A .

Exercise 96 Consider the differential equation

$$\dot{x} = Ax \tag{7.7}$$

where A is a square matrix. Show that if A has $j2\pi$ as an eigenvalue, then there is a nonzero initial state x_0 such that (7.7) has a solution x which satisfies $x(1) = x(0) = x_0$.

Exercise 97 Suppose A is a square matrix.

(a) Obtain expressions for

$$\cos(At) \quad \text{and} \quad \sin(At).$$

(b) Show that

$$\cos(At) = (e^{jAt} + e^{-jAt})/2 \quad \text{and} \quad \sin(At) = (e^{jAt} - e^{-jAt})/2j$$

(c) Show that

$$\frac{d \cos(At)}{dt} = -A \sin(At) \quad \text{and} \quad \frac{d \sin(At)}{dt} = A \cos(At).$$

Exercise 98 Compute e^{At} for the matrix

$$A = \begin{pmatrix} 3 & -5 \\ -5 & 3 \end{pmatrix}$$

Exercise 99 Compute e^{At} for the matrix,

$$A = \begin{pmatrix} -11 & 20 \\ -6 & 11 \end{pmatrix}.$$

Exercise 100 Compute e^{At} for the matrix

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Exercise 101 Suppose A is a matrix with four distinct eigenvalues: $-1, +1 - j, j$. Show that $A^4 = I$.

Exercise 102 Suppose

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

What is $A^{20} + 4A^4$?

Exercise 103 Suppose A is a 3×3 matrix with eigenvalues $1, 2\pi j$ and $-2\pi j$. What are the eigenvalues of e^A .

Exercise 104 Compute e^{At} for the matrix,

$$A = \begin{pmatrix} -3 & 2 \\ -4 & 3 \end{pmatrix}.$$

Exercise 105 Compute e^{At} for the matrix

$$A = \begin{pmatrix} 3 & -5 \\ -5 & 3 \end{pmatrix}$$

Exercise 106 Compute e^{At} at $t = \ln(2)$ for

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

using *all* four methods mentioned in the notes.

Chapter 8

Behavior of (LTI) systems : II*

8.1 Introduction

Recall that if the system matrix of an LTI system is diagonalizable, then every component of every solution can be expressed as a linear combination of terms of the form $e^{\lambda t}$. Consider now the unattached mass

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= 0\end{aligned}$$

With initial state $x(0) = (0, 1)$, this system has solution $x(t) = (t, 1)$. The term t cannot be expressed as a linear combination of exponential functions. So, the system matrix

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

is not diagonalizable; hence it does not have $n = 2$ linearly independent eigenvectors. To explicitly see this, note that this matrix has characteristic polynomial

$$p(s) = s^2$$

hence it has a single eigenvalue at 0. All eigenvectors are of the form

$$v = \begin{pmatrix} x_1 \\ 0 \end{pmatrix},$$

that is, this matrix only has one linearly independent eigenvector.

If an $n \times n$ matrix is not diagonalizable, it cannot have n distinct eigenvalues; hence at least one eigenvalue must be a repeated root of the characteristic polynomial of the matrix. This is very difficult to detect numerically.

8.2 Jordan blocks

For any complex number λ and any positive integer n , $J(\lambda, n)$ is a **Jordan block** of order n ; it is an $n \times n$ matrix with every diagonal element equal to λ , every element of the superdiagonal equal to 1 and every other element equal to 0, that is, it has the following structure:

$$J(\lambda, n) = \begin{pmatrix} \lambda & 1 & 0 & \dots & 0 & 0 \\ 0 & \lambda & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & & \vdots & \vdots \\ \vdots & \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & \lambda & 1 \\ 0 & 0 & 0 & \dots & 0 & \lambda \end{pmatrix}.$$

Example 92

$$\underbrace{\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}}$$

The unattached mass (discrete and otherwise)

$$\begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix} \quad \begin{pmatrix} \lambda & 1 & 0 & 0 \\ 0 & \lambda & 1 & 0 \\ 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & \lambda \end{pmatrix}$$

Example 93 (The Disturbed unattached mass)

$$\ddot{q} = w$$

The forcing term w is a constant disturbance input. Introducing state variables

$$x_1 := q \quad x_2 := \dot{q} \quad x_3 = w$$

we obtain

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= 0 \end{aligned}$$

This is an LTI system with system matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

The characteristic polynomial of $J(\lambda, n)$ is given by

$$p(s) = (s - \lambda)^n.$$

To see this, note that

$$J(\lambda, n) - \lambda I = J(0, n) = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & & \vdots & \vdots \\ \vdots & \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & 0 & \dots & 0 & 0 \end{pmatrix}$$

i.e., the matrix $J(\lambda, n) - \lambda I$ is a companion matrix whose last row is zero; hence its characteristic polynomial is given by

$$\det(sI - (J(\lambda, n) - \lambda I)) = s^n$$

We now have

$$\begin{aligned} \det(sI - J(\lambda, n)) &= \det((s - \lambda)I - (J(\lambda, n) - \lambda I)) \\ &= (s - \lambda)^n \end{aligned}$$

It can readily be shown that all eigenvectors of $J(\lambda, n)$ are of the form

$$v = \begin{pmatrix} v_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Hence λ has algebraic multiplicity n and geometric multiplicity 1.

Fact 11 *If $\Lambda = J(\lambda, n)$, then*

$$f(\Lambda) = \begin{pmatrix} f(\lambda) & f'(\lambda) & f''(\lambda)/2! & \dots & f^{(n-2)}(\lambda)/(n-2)! & f^{(n-1)}(\lambda)/(n-1)! \\ 0 & f(\lambda) & f'(\lambda) & \dots & f^{(n-3)}(\lambda)/(n-3)! & f^{(n-2)}(\lambda)/(n-2)! \\ \vdots & \vdots & \ddots & & \vdots & \vdots \\ \vdots & \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & f(\lambda) & f'(\lambda) \\ 0 & 0 & 0 & \dots & 0 & f(\lambda) \end{pmatrix}$$

Hence,

$$e^{\Lambda t} = \begin{pmatrix} e^{\lambda t} & te^{\lambda t} & t^2 e^{\lambda t}/2! & \dots & t^{n-2} e^{\lambda t}/(n-2)! & t^{n-1} e^{\lambda t}/(n-1)! \\ 0 & e^{\lambda t} & te^{\lambda t} & \dots & t^{n-3} e^{\lambda t}/(n-3)! & t^{n-2} e^{\lambda t}/(n-2)! \\ \vdots & \vdots & \ddots & & \vdots & \vdots \\ \vdots & \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & e^{\lambda t} & te^{\lambda t} \\ 0 & 0 & 0 & \dots & 0 & e^{\lambda t} \end{pmatrix}$$

What is $(sI - \Lambda)^{-1}$?

8.3 Generalized eigenvectors

8.3.1 Generalized eigenvectors

Recall that a vector v is an eigenvector of a square matrix A iff it satisfies

$$\begin{aligned}v &\neq 0 \\Av &= \lambda v\end{aligned}$$

for some complex number λ . Also, λ is called the eigenvalue of A corresponding to v . We can rewrite the above two equations as:

$$\begin{aligned}(A - \lambda I)^0 v &\neq 0 \\(A - \lambda I)^1 v &= 0\end{aligned}$$

We now introduce the following definition for any positive integer $k = 1, 2, \dots$.

DEFN. A non-zero vector $g^k \in \mathbb{C}^n$ is a **generalized eigenvector** of order k of A if it satisfies

$\begin{aligned}(A - \lambda I)^{k-1} g^k &\neq 0 \\(A - \lambda I)^k g^k &= 0\end{aligned}$
--

for some complex number λ .

It should be clear that an eigenvector is a generalized eigenvector of order 1. Also, if g^k is a generalized eigenvector of order k then

$$v = (A - \lambda I)^{k-1} g^k$$

is an eigenvector of A with eigenvalue λ .

Example 94

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Here $\lambda = 0$ is the only eigenvalue. For

$$g^1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

we have $Ag^1 = 0$. For

$$g^2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

we have

$$\begin{aligned}Ag^2 &= g^1 \neq 0 \\A^2 g^2 &= Ag^1 = 0\end{aligned}$$

hence g^2 is a generalized eigenvector of order 2.

If A is a Jordan block, then

$$e^k := \begin{pmatrix} 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \leftarrow k\text{-th element}$$

is a generalized eigenvector of order k .

8.3.2 Chain of generalized eigenvectors

It follows from the above definition that if g^k is a generalized eigenvector of order $k > 1$ then,

$$g^{k-1} := (A - \lambda I)g^k \neq 0$$

is a generalized eigenvector of order $k - 1$. To see this:

$$\begin{aligned} (A - \lambda I)^{k-1}g^{k-1} &= (A - \lambda I)^k g^k = 0 \\ (A - \lambda I)^{k-2}g^{k-1} &= (A - \lambda I)^{k-1}g^k \neq 0 \end{aligned}$$

Consider now any generalized eigenvector g^k of order $k > 1$, and recursively define the sequence

$$g^1, g^2, \dots, g^k$$

of k vectors by

$$\begin{aligned} g^{k-1} &:= (A - \lambda I)g^k \\ g^{k-2} &:= (A - \lambda I)g^{k-1} = (A - \lambda I)^2 g^k \\ &\vdots \\ g^{j-1} &:= (A - \lambda I)g^j = (A - \lambda I)^{k-j} g^k \\ &\vdots \\ g^1 &:= (A - \lambda I)g^2 = (A - \lambda I)^{k-1} g^k \end{aligned}$$

Then, g^j is a generalized eigenvector of order j . This is called a **chain of k generalized eigenvectors**.

If A is a Jordan block of length n then, the usual basis for $\mathbb{C}^n$,

$$e^1, e^2, \dots, e^n$$

is a chain of generalized eigenvectors.

Exercise 107 (Optional) Letting

$$\mathcal{G} := \{g^1, g^2, \dots, g^k\}$$

Show that $\mathcal{G}$ is linearly independent.

8.3.3 System significance of generalized eigenvectors

- If g^k is a generalized eigenvector of order k of A , then the solution of

$$\dot{x} = Ax ; \quad x(0) = g^k$$

is given by

$$\begin{aligned} x(t) &= \sum_{j=0}^{k-1} \frac{t^j e^{\lambda t}}{j!} (A - \lambda I)^j g^k \\ &= e^{\lambda t} g^k + t e^{\lambda t} (A - \lambda I) g^k + \dots + \frac{t^{k-1} e^{\lambda t}}{(k-1)!} (A - \lambda I)^{k-1} g^k \end{aligned}$$

Proof. Clearly $x(0) = g^k$. Differentiating the above expression for $x(t)$ yields

$$\begin{aligned} \dot{x}(t) &= \lambda \sum_{j=0}^{k-1} \frac{t^j e^{\lambda t}}{j!} (A - \lambda I)^j g^k + \sum_{j=1}^{k-1} \frac{t^{j-1} e^{\lambda t}}{(j-1)!} (A - \lambda I)^j g^k \\ &= \lambda x(t) + (A - \lambda I) \sum_{j=0}^{k-2} \frac{t^j e^{\lambda t}}{j!} (A - \lambda I)^j g^k \end{aligned}$$

Since $(A - \lambda I)^k g^k = 0$, the second term on the rhs of the previous equation equals $(A - \lambda I)x(t)$; hence

$$\begin{aligned} \dot{x}(t) &= \lambda x(t) + (A - \lambda I)x(t) \\ &= Ax(t) \end{aligned} \quad \blacksquare$$

- Note that the above solution may be written as

$$x(t) = e^{\lambda t} g^k + t e^{\lambda t} g^{k-1} + \dots + \frac{t^{k-1} e^{\lambda t}}{(k-1)!} g^1$$

where $g^j = (A - \lambda I)^{k-j} g^k$ is a generalized eigenvector of order j .

- So, if the initial state is a generalized eigenvector of order k , the resulting solution is a generalized eigenvector of order k and contains terms of the form:

$$\boxed{e^{\lambda t}, t e^{\lambda t}, \dots, t^{k-1} e^{\lambda t}}$$

So for $\boxed{k > 1}$, we conclude that:

- Once an generalized eigenvector of order k , always an generalized eigenvector order k !
- If $\Re(\lambda) < 0$ you decay with increasing age.
- If $\Re(\lambda) \geq 0$ you grow with increasing age.

Note that when the initial state is an eigenvector and $\Re(\lambda) = 0$ the solution does not grow, its magnitude remains constant.

8.4 The Jordan form of a square matrix

8.4.1 Special case

We consider first the special case in which the $n \times n$ matrix A has only a single eigenvalue λ and this eigenvalue has geometric multiplicity one.

Hence the algebraic multiplicity of λ is n and the charpoly of A is

$$p(s) = (s - \lambda)^n$$

Complex Jordan form of A . Let $g^1 = v$ where v is any eigenvector of A , i.e., $g^1 \neq 0$ and

$$Ag^1 = \lambda g^1$$

Recursively solve (solutions exist)

$$\begin{array}{rcl} (A - \lambda I)g^2 & = & g^1 \\ (A - \lambda I)g^3 & = & g^2 \\ & \vdots & \\ (A - \lambda I)g^n & = & g^{n-1} \end{array}$$

to obtain a chain of n generalized eigenvectors: $g^1, g^2, \dots, g^n$. This chain is a basis for $\mathbb{C}^n$. Also,

$$\begin{array}{rcl} Ag^1 & = & \lambda g^1 \\ Ag^2 & = & g^1 + \lambda g^2 \\ Ag^3 & = & g^2 + \lambda g^3 \\ & \vdots & \\ Ag^n & = & g^{n-1} + \lambda g^n \end{array}$$

Introduce similarity transformation:

$$T := (g^1 \ g^2 \ \dots \ g^n)$$

Letting

$$\Lambda := T^{-1}AT$$

we obtain (using useful result on similarity transformations)

$$\Lambda = J(\lambda, n) := \overbrace{\begin{pmatrix} \lambda & 1 & 0 & \dots & 0 & 0 \\ 0 & \lambda & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & & \vdots & \vdots \\ \vdots & \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & \lambda & 1 \\ 0 & 0 & 0 & \dots & 0 & \lambda \end{pmatrix}}^{\text{Complex Jordan form of } A}$$

Jordan block of order n

and

$$A = T\Lambda T^{-1}$$

Example 95

$$A = \begin{pmatrix} -3.5 & -0.5 \\ 0.5 & -4.5 \end{pmatrix}$$

Characteristic polynomial of A :

$$\det(sI - A) = (s + 4)^2.$$

So, A has a single eigenvalue $\lambda = -4$. Solving $(A - \lambda I)v = 0$ yields one linearly independent eigenvector

$$v = g^1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Now use

$$(A - \lambda I)g^2 = g^1$$

to obtain

$$g^2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Hence, the Jordan form of A is

$$\Lambda = \begin{pmatrix} -4 & 1 \\ 0 & -4 \end{pmatrix}$$

A quick check on MATLAB yields $\Lambda = T^{-1}AT$; where $T = (g^1 \ g^2)$.

Note:

```
>> [t,lam]=eig(a)
```

```
t =
```

```
    0.7071    0.7071
    0.7071    0.7071
```

```
lam =
```

```
   -4     0
     0    -4
```

Whats goin' on!

• Note:

$$J(0, n)^n = 0 \quad J(0, n)^{n-1} \neq 0$$

Hence

$$(J(\lambda, n) - \lambda I)^n = 0 \quad (J(\lambda, n) - \lambda I)^{n-1} \neq 0$$

and (for the special case in this section)

$$(A - \lambda I)^n = 0, \quad (A - \lambda I)^{n-1} \neq 0$$

8.4.2 Block-diagonal matrices

Suppose $A_1, A_2, \dots, A_l$ are l matrices (not necessarily square). By

$$A = \text{blockdiag} (A_1, A_2, \dots, A_l)$$

we mean

$$A = \begin{pmatrix} A_1 & 0 & \dots & 0 \\ 0 & A_2 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & 0 & \dots & A_l \end{pmatrix}$$

Some properties:

$$\alpha A = \text{blockdiag} (\alpha A_1, \alpha A_2, \dots, \alpha A_l)$$

If

$$B = \text{blockdiag} (B_1, B_2 \dots, B_l)$$

then

$$A + B = \text{blockdiag} (A_1 + B_1, A_2 + B_2 \dots, A_l + B_l)$$

$$AB = \text{blockdiag} (A_1 B_1, A_2 B_2, \dots, A_l B_l)$$

If f is an analytic function,

$$f(A) = \text{blockdiag} (f(A_1), f(A_2), \dots, f(A_l))$$

Also,

$$A^* = \text{blockdiag} (A_1^*, A_2^*, \dots, A_l^*)$$

$$A^{-1} = \text{blockdiag} (A_1^{-1}, A_2^{-1}, \dots, A_l^{-1})$$

$$\det A = \det A_1 \det A_2 \dots \det A_l$$

$$\det(sI - A) = \det(sI - A_1) \det(sI - A_2) \dots \det(sI - A_l)$$

$$\text{rank } A = \text{rank } A_1 + \text{rank } A_2 + \dots + \text{rank } A_l$$

$$\text{nullity } A = \text{nullity } A_1 + \text{nullity } A_2 + \dots + \text{nullity } A_l$$

8.4.3 A less special case

Each eigenvalue of A has geometric multiplicity one.

Example 96

$$A = \left(\begin{array}{ccc|cc|c} 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 5 & 1 & 0 \\ 0 & 0 & 0 & 0 & 5 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 7 \end{array} \right)$$

Using properties of properties of blockdiagonal matrices,

$$\begin{aligned} \det(sI - A) &= \det \left(sI - \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix} \right) \det \left(sI - \begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix} \right) \det(s - 7) \\ &= (s - 2)^3(s - 5)^2(s - 7) \end{aligned}$$

So, A has three eigenvalues, 2, 5, 7 and associated with each eigenvalue, there is at most one linearly independent eigenvector; examples are

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

respectively. So, each eigenvalue has geometric multiplicity one.

One can apply procedure of previous section to each eigenvalue to obtain a basis of generalized eigenvectors.

8.4.4 Another less special case

Suppose A has one eigenvalue λ , but the geometric multiplicity of λ is greater than one and less than n .

Example 97

$$A = \left(\begin{array}{ccc|cc|c} 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 2 \end{array} \right)$$

Using properties of properties of blockdiagonal matrices,

$$\begin{aligned}
 \det(sI - A) &= \det \left(sI - \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix} \right) \det \left(sI - \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \right) \det(s - 2) \\
 &= (s - 2)^3 (s - 2)^2 (s - 2) \\
 &= (s - 2)^6
 \end{aligned}$$

So, A has a single eigenvalue equal to 2 and with algebraic multiplicity $n = 6$. A has at most three linearly independent eigenvectors, examples are

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Hence, the single eigenvalue 2 has geometric multiplicity three.

- The *index* of an eigenvalue λ of a matrix A is the smallest integer k such that

$$\text{nullity}((A - \lambda I)^{k+1}) = \text{nullity}((A - \lambda I)^k)$$

In the last example, the index of the eigenvalue 2 is three.

Fact 12 *If q is the geometric multiplicity of an eigenvalue λ , then there is a basis consisting of q chains of generalized eigenvectors.*

$$\begin{aligned}
 &(A - \lambda I)^{k_1-1} g^{k_1}, \quad \dots, \quad g^{k_1} \\
 &\quad \quad \quad \vdots \\
 &(A - \lambda I)^{k_q-1} g^{k_q}, \quad \dots, \quad g^{k_q}
 \end{aligned}$$

where $k_1 \geq k_2 \geq \dots k_q$ and k_1 is the index of λ .

How do you get these generalized eigenvectors? Tricky!

Using these generalized eigenvectors as basis vectors one can readily show that $A = T\Lambda T^{-1}$ where

$$\Lambda = \text{blockdiag} (J(\lambda, k_1), \quad J(\lambda, k_2), \quad \dots, \quad J(\lambda, k_q))$$

8.4.5 General case

Example 98

$$A = \left(\begin{array}{cc|cc|cc|c} 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 5 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 5 \end{array} \right)$$

$$\det(sI - A) = (s - 2)^4(s - 5)^3$$

Both eigenvalues, 2 and 5, have geometric multiplicity two.

Fact 13 *For every square matrix there is a basis of generalized eigenvectors. This basis consists of chains of generalized eigenvectors. For a specific eigenvalue, the number of chains associated with it equals its geometric multiplicity, and the length of the largest chain associated with it equals its index.*

Using these generalized eigenvectors as basis vectors for a similarity transformation T , one can show that

$$A = T\Lambda T^{-1}$$

$$\Lambda = \text{blockdiag} (J_1, J_2, \dots, J_p)$$

where each matrix $J_k, k = 1, 2, \dots, p$ is a Jordan block associated with an eigenvalue of A . The number of Jordan blocks associated with each eigenvalue equals the geometric multiplicity of the eigenvalue. The length of the largest block associated with an eigenvalue equals the index of the eigenvalue.

8.5 LTI systems

Recall that the solution to

$$\dot{x} = Ax, \quad x(0) = x_0$$

is given by

$$x(t) = e^{At}x_0$$

Suppose $\lambda_1, \dots, \lambda_l$ are the eigenvalues of A and n_i is the index of λ_i . We have seen that

$$\begin{aligned} A &= T\Lambda T^{-1} \\ \Lambda &= \text{blockdiag} (J_1, J_2, \dots, J_p), \end{aligned}$$

where each matrix J_k , $k = 1, 2, \dots, p$, is a Jordan block associated with an eigenvalue of A and the length of the largest block associated with a specific eigenvalue equals the index of the eigenvalue. Hence,

$$e^{At} = T \text{ blockdiag} (e^{J_1 t}, e^{J_2 t}, \dots, e^{J_p t}) T^{-1}$$

Recalling the structure of $e^{J(\lambda, n)t}$, we can immediately deduce that e^{At} has the following structure:

$$e^{At} = \sum_{i=1}^l \sum_{j=0}^{n_i-1} t^j e^{\lambda_i t} A_{ij}$$

Hence,

$$x(t) = \sum_{i=1}^l \sum_{j=0}^{n_i-1} t^j e^{\lambda_i t} A_{ij} x_0$$

For the unattached mass,

$$e^{At} = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

8.6 Examples on numerical computations with repeated roots

Example 99 Consider the following polynomial:

$$p(s) = (s - 2)^3$$

Thus $s = 2$ is the only root (multiplicity three). Using MATLAB (long format) to find the roots we obtain:

```
>> format long
>> roots(poly([2 2 2]))
ans =
    2.00000957563204 + 0.00001658568659i
    2.00000957563204 - 0.00001658568659i
    1.99998084873593
```

These results are close, but not exact, although, the polynomial is quite simple. Note

```
>> roots(poly([1 2 3]))

ans =
    3.000000000000000
    2.000000000000000
    1.000000000000000
```

Example 100 Consider

$$p(s) = (s - 1)^{50}$$

Let v be the row vector of length 50 and with each element equal to 1. Then, $\text{poly}(v)$ is the vector whose elements are the coefficients of the above polynomial and $\text{roots}(\text{poly}(v))$, if computed exactly, equals v .

```
>> v2=roots(poly(v))
v2 =
    2.8442
    2.7933 + 0.4054i
    2.7933 - 0.4054i
    2.6286 + 0.7880i
    2.6286 - 0.7880i
    2.3570 + 1.0952i
    2.3570 - 1.0952i
    2.0248 + 1.2884i
    2.0248 - 1.2884i
```

$1.6867 + 1.3662i$
 $1.6867 - 1.3662i$
 $1.3794 + 1.3532i$
 $1.3794 - 1.3532i$
 $1.1148 + 1.2837i$
 $1.1148 - 1.2837i$
 $0.8959 + 1.1853i$
 $0.8959 - 1.1853i$
 $1.1930 + 0.9483i$
 $1.1930 - 0.9483i$
 1.4629
 $0.7268 + 1.0658i$
 $0.7268 - 1.0658i$
 $0.6006 + 0.9337i$
 $0.6006 - 0.9337i$
 $0.7206 + 0.7503i$
 $0.7206 - 0.7503i$
 $0.5070 + 0.8049i$
 $0.5070 - 0.8049i$
 $0.4441 + 0.6888i$
 $0.4441 - 0.6888i$
 $0.4060 + 0.5848i$
 $0.4060 - 0.5848i$
 $0.3852 + 0.4936i$
 $0.3852 - 0.4936i$
 0.6771
 $0.3730 + 0.4165i$
 $0.3730 - 0.4165i$
 $0.3616 + 0.3476i$
 $0.3616 - 0.3476i$
 $0.3536 + 0.2809i$
 $0.3536 - 0.2809i$
 $0.3509 + 0.2192i$
 $0.3509 - 0.2192i$
 $0.3490 + 0.1641i$
 $0.3490 - 0.1641i$
 $0.3446 + 0.1105i$
 $0.3446 - 0.1105i$
 $0.3412 + 0.0554i$
 $0.3412 - 0.0554i$
 0.3402

Example 101 The following matrix has charpoly $p(s) = (s - 1)^{10}$.

```
>> a
a =
    0     1     0     0     0     0     0     0     0     0
    0     0     1     0     0     0     0     0     0     0
    0     0     0     1     0     0     0     0     0     0
    0     0     0     0     1     0     0     0     0     0
    0     0     0     0     0     1     0     0     0     0
    0     0     0     0     0     0     1     0     0     0
    0     0     0     0     0     0     0     1     0     0
    0     0     0     0     0     0     0     0     1     0
    0     0     0     0     0     0     0     0     0     1
   -1    10   -45    120   -210    252   -210    120   -45    10
```

However

```
>> eig(a)
ans =
    1.0528
    1.0421 + 0.0315i
    1.0421 - 0.0315i
    1.0150 + 0.0498i
    1.0150 - 0.0498i
    0.9831 + 0.0484i
    0.9831 - 0.0484i
    0.9586 + 0.0293i
    0.9586 - 0.0293i
    0.9495

>> [t,d]=eig(a); rank(t)
ans = 10
```

The last calculation implies that a is diagonalizable. However, a has only one linearly independent eigenvector. We can see this as follows.

```
>> rank(a - eye(10))
ans = 9

>> null(a-eye(10))
ans =
   -0.3162
   -0.3162
   -0.3162
   -0.3162
```

```
-0.3162  
-0.3162  
-0.3162  
-0.3162  
-0.3162  
-0.3162
```

We can see that 1 is the only eigenvalue as follows.

```
>> b=(a-eye(10));  
  
>> rank(null(b))  
ans = 1  
  
>> rank(null(b^2))  
ans = 2  
  
>> rank(null(b^3))  
ans = 3  
  
>> rank(null(b^4))  
ans = 4  
  
>> rank(null(b^5))  
ans = 5  
  
>> rank(null(b^6))  
ans = 6  
  
>> rank(null(b^7))  
ans = 7  
  
>> rank(null(b^8))  
ans = 8  
  
>> rank(null(b^9))  
ans = 9  
  
>> rank(null(b^10))  
ans = 10
```

Chapter 9

Input-output systems

In this chapter, we look at the behavior of input-output systems.



9.1 Continuous-time systems

In general, a state space description of an LTI (linear time-invariant) continuous-time system with inputs and outputs is given by

$$\begin{array}{c} y \\ \leftarrow \end{array} \boxed{\begin{array}{l} \dot{x} = Ax + Bu \\ y = Cx + Du \end{array}} \begin{array}{c} u \\ \leftarrow \end{array}$$

where A, B, C and D are matrices of dimensions $n \times n$, $n \times m$, $p \times n$ and $p \times m$, respectively. Thus, the state space description is completely specified by the matrices A, B, C, D .

9.1.1 State response and the convolution integral

The solution to

$$\dot{x}(t) = Ax(t) + Bu(t) \quad \text{with} \quad x(t_0) = x_0 \quad (9.1)$$

is unique and is given by

$$\boxed{x(t) = e^{A(t-t_0)}x_0 + \int_{t_0}^t e^{A(t-\tau)}Bu(\tau)d\tau} \quad (9.2)$$

for all t .

PROOF. Consider any time t . Considering any time τ in the interval bounded by t_0 and t , we have

$$\dot{x}(\tau) = Ax(\tau) + Bu(\tau)$$

Pre-multiply both sides by $e^{-A\tau}$ and rearrange to get:

$$e^{-A\tau}\dot{x}(\tau) - e^{-A\tau}Ax(\tau) = e^{-A\tau}Bu(\tau)$$

Hence

$$\frac{d}{d\tau}(e^{-A\tau}x(\tau)) = e^{-A\tau}Bu(\tau)$$

Integrate with respect to τ from t_0 to t and use $x(t_0) = x_0$:

$$e^{-At}x(t) - e^{-At_0}x_0 = \int_{t_0}^t e^{-A\tau}Bu(\tau)d\tau$$

Now premultiply by e^{At} and use $e^{At}e^{-At} = I$ and $e^{At}e^{-At_0} = e^{A(t-t_0)}$ to obtain

$$x(t) - e^{A(t-t_0)}x_0 = \int_{t_0}^t e^{A(t-\tau)}Bu(\tau)d\tau$$

and, hence, the desired result. ■

- Note that *the solution depends linearly on the pair $(x_0, u(\cdot))$.*
- Suppose that for some input function $u(\cdot)$, the function $x^p(\cdot)$ is a particular solution to (9.1) with $x^p(t_0) = x_0^p$. Then all other solutions are given by

$$x(t) = e^{A(t-t_0)}(x_0 - x_0^p) + x^p(t) \quad (9.3)$$

(Why?)

Impulse response

Suppose $t_0 = 0$ and

$$u(t) = \delta(t)v$$

where $\delta(\cdot)$ is the **Dirac delta function** or the **unit impulse function**. (Don't confuse with the δ used in linearization.) Recall that if $[t_0, t]$ is a time interval containing zero and g is any continuous function then,

$$\int_{t_0}^t g(t)\delta(t)dt = g(0)$$

Hence, the solution to (9.1) is given by

$$x(t) = e^{At}x_0 + e^{At}Bv$$

From this we see that the zero initial state response (that is, $x_0 = 0$) to the above impulse input is given by

$$x(t) = e^{At}Bv$$

This is the same as the system response to zero input and initial state, $x_0 = Bv$.

Example 102

$$m\ddot{y} + c\dot{y} + ky = u$$

Response to exponentials

Suppose $t_0 = 0$ and

$$u(t) = e^{\lambda t}v$$

where v is a constant m -vector and λ is not an eigenvalue of A . Then the solution to (9.1) is

$$x(t) = \underbrace{e^{At}(x_0 - x_0^p)}_{\text{transient part}} + \underbrace{e^{\lambda t}x_0^p}_{\text{steady state part}}$$

where

$$x_0^p = (\lambda I - A)^{-1}Bv$$

We need only prove the result for $x_0 = x_0^p$. So considering

$$x(t) = e^{\lambda t}(\lambda I - A)^{-1}Bv$$

we have

$$\dot{x}(t) = \lambda e^{\lambda t}(\lambda I - A)^{-1}Bv = \lambda x(t)$$

and

$$(\lambda I - A)x(t) = e^{\lambda t}Bv = Bu(t)$$

Hence,

$$\lambda x(t) = Ax(t) + Bu(t)$$

and

$$\dot{x}(t) = \lambda x(t) = Ax(t) + Bu(t)$$

■

- If A is asymptotically stable, the transient part goes to zero as t goes to infinity.

9.1.2 Output response

The output response of the system

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

subject to initial condition $x(t_0) = x_0$ is unique and is given by

$$\boxed{y(t) = Ce^{A(t-t_0)}x_0 + \int_{t_0}^t G(t-\tau)u(\tau)d\tau} \quad (9.4)$$

where the impulse response matrix G is defined by

$$\boxed{G(t) = Ce^{At}B + \delta(t)D}$$

where $\delta(\cdot)$ is the Dirac delta function or the unit impulse function. (Don't confuse with the δ used in linearization.)

To see this, recall that if zero is in the interval bounded by t_0 and t and g is any continuous function then,

$$\int_{t_0}^t \delta(t)g(t)dt = g(0)$$

Recalling that

$$x(t) = e^{A(t-t_0)}x_0 + \int_{t_0}^t e^{A(t-\tau)}Bu(\tau)d\tau$$

we see that

$$\begin{aligned} y(t) &= Cx(t) + Du(t) \\ &= Ce^{A(t-t_0)}x_0 + C \int_{t_0}^t e^{A(t-\tau)}Bu(\tau)d\tau + Du(t) \\ &= Ce^{A(t-t_0)}x_0 + \int_{t_0}^t Ce^{A(t-\tau)}Bu(\tau) + \delta(t-\tau)Du(\tau)d\tau \\ &= Ce^{A(t-t_0)}x_0 + \int_{t_0}^t G(t-\tau)u(\tau) d\tau \end{aligned}$$

Zero initial conditions response: $x(0) = 0$.

$$\boxed{y(t) = \int_0^t G(t-\tau)u(\tau)d\tau} \quad (9.5)$$

Sometimes this is represented by

$$y \longleftarrow \boxed{G} \longleftarrow u$$

This defines a linear map from the space of input functions to the space of output functions. The above integral is called a **convolution integral**; it is the convolution of the functions G and u .

Example 103

$$\begin{aligned} \dot{x} &= -x + u \\ y &= x \end{aligned}$$

Here

$$\begin{aligned} G(t) &= Ce^{At}B + \delta(t)D \\ &= e^{-t} \end{aligned}$$

So, for zero initial state,

$$y(t) = \int_0^t e^{-(t-\tau)}u(\tau) d\tau$$

Example 104 Unattached mass

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= u \\ y &= x_1\end{aligned}$$

Here

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 0 \end{pmatrix} \quad D = 0$$

Hence

$$e^{At} = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

and

$$\begin{aligned}G(t) &= Ce^{At}B \\ &= t\end{aligned}$$

So,

$$y(t) = \int_0^t (t-\tau)u(\tau) d\tau$$

Remark 1 Description (9.5) is a very general representation of a linear input-output system. It can also be used for infinite-dimensional systems such as distributed parameter systems and systems with delays.

Example 105 Consider the system with input delay described by

$$\begin{aligned}\dot{x}(t) &= -x(t) + u(t-1) \\ y(t) &= x(t).\end{aligned}$$

Since

$$y(t) = \int_0^t e^{-(t-\hat{\tau})}u(\hat{\tau}-1) d\hat{\tau}$$

(why?) we have (upon changing variable of integration)

$$y(t) = \int_{-1}^{t-1} e^{-(t-1-\tau)}u(\tau) d\tau$$

Considering only inputs for which

$$u(t) = 0 \quad \text{for } t < 0$$

and defining

$$G(t) = \begin{cases} 0 & \text{for } t < 1 \\ e^{-(t-1)} & \text{for } t \geq 1 \end{cases}$$

the input-output response of this system is described by (9.5).

9.1.3 Impulse response

Consider first a SISO (single-input single-output) system with impulse response function g and subject to zero initial conditions, $x(0) = 0$. Suppose such a system is subject to a unit impulse input, that is, $u(t) = \delta(t)$. Then, using (9.4), its output is given by

$$y(t) = \int_0^t g(t-\tau)\delta(\tau) d\tau = g(t).$$

Thus, as its name suggests, *the impulse response function g is the output response of the system to zero initial conditions and a unit impulse input.*

Consider now a MIMO (multi-input multi-output) system with impulse response function G and subject to zero initial conditions. Suppose this system is subject to an input of the form

$$u(t) = \delta(t)v$$

where v is constant m -vector. Then, from (9.4),

$$y(t) = \int_0^t G(t-\tau)v\delta(\tau) d\tau = G(t)v.$$

By considering $v = e^j$, where each component of e^j is zero except for its j -th component which equals one, we see that G_{ij} is the zero initial state response of y_i when $u_j = \delta$ and all other inputs are zero, that is,

$$\left. \begin{array}{l} u_1(t) = 0 \\ \vdots \\ u_j(t) = \delta(t) \\ \vdots \\ u_m(t) = 0 \end{array} \right\} \implies \left\{ \begin{array}{l} y_1(t) = G_{1j}(t) \\ \vdots \\ y_i(t) = G_{ij}(t) \\ \vdots \\ y_p(t) = G_{pj}(t) \end{array} \right.$$

Transfer function and impulse response. Consider the impulse response matrix given by

$$G(t) = Ce^{At}B + \delta(t)D.$$

Using the following Laplace transform results:

$$\mathcal{L}(e^{At})(s) = (sI - A)^{-1}, \quad \mathcal{L}(\delta) = 1$$

we obtain that the Laplace transform of G is given by

$$\mathcal{L}(G) = C(sI - A)^{-1}B + D$$

which we recognize as the transfer function $\hat{G}$ of the system described by (A, B, C, D) ; hence we have the following conclusion.

The transfer function of an LTI system is the Laplace transform of its impulse response matrix, that is,

$$\boxed{\hat{G} = \mathcal{L}(G)}$$

9.1.4 Response to exponentials and sinusoids

Suppose $t_0 = 0$ and

$$u(t) = e^{\lambda t} v$$

where v is a constant m vector and λ is not an eigenvalue of A . Recall that

$$x(t) = e^{At}(x_0 - x_0^p) + e^{\lambda t} x_0^p$$

where $x_0^p = (\lambda I - A)^{-1} B v$. Since $y(t) = Cx(t) + Du(t)$, we obtain that

$$y(t) = \underbrace{C e^{At}(x_0 - x_0^p)}_{\text{transient part}} + \underbrace{e^{\lambda t} \hat{G}(\lambda) v}_{\text{steady state part}}$$

where

$$\boxed{\hat{G}(\lambda) := C(\lambda I - A)^{-1} B + D}$$

If A is asymptotically stable, then the transient part goes to zero as t goes to infinity. Note that $\hat{G}(\lambda)$ is the system transfer function evaluated at λ .

General LTI input-output system. Consider a general LTI system described by

$$y(t) = \int_0^t G(t-\tau) u(\tau) d\tau$$

Then, the transfer function $\hat{G}$ of this system is simply the Laplace transform of the impulse response G , that is,

$$\hat{G}(s) = \int_0^\infty e^{-st} G(t) dt.$$

If we assume that the impulse response G converges to zero exponentially and λ is not a pole of $\hat{G}$, then it can be shown that the steady-state response of this system to an input

$$u(t) = e^{\lambda t} v$$

is given by the output

$$\boxed{y(t) = \hat{G}(\lambda) e^{\lambda t} v}$$

Response to sinusoids. Suppose that $\hat{G}$ is the transfer function of a system with a *real* impulse response that is,

$$\hat{G}(s) = \int_0^\infty e^{-st} G(t) dt$$

where $G(t)$ is real.

Using the property that $e^{-\bar{s}t} = \overline{e^{-st}}$, the realness of $G(t)$ implies that

$$e^{-\bar{s}t} G(t) = \overline{e^{-st} G(t)} = \overline{e^{-st} G(t)};$$

hence

$$\hat{G}(\bar{s}) = \int_0^\infty e^{-\bar{s}t} G(t) dt = \int_0^\infty \overline{e^{-st} G(t)} dt = \overline{\int_0^\infty e^{-st} G(t) dt} = \overline{\hat{G}(s)},$$

that is,

$$\hat{G}(\bar{s}) = \overline{\hat{G}(s)}. \quad (9.6)$$

We now show that the steady state response of this system to

$$u(t) = e^{\alpha t} \sin(\omega t) v,$$

where α and ω are real numbers with v a real vector, is given by

$$\boxed{y(t) = \Im \left(\hat{G}(\lambda) e^{\lambda t} v \right)}$$

where $\lambda = \alpha + j\omega$.

To demonstrate the above result, we first note that

$$e^{\alpha t} \sin(\omega t) = \Im(e^{\lambda t}) = (e^{\lambda t} - e^{\bar{\lambda} t})/(2j)$$

So, the corresponding steady state response is given by

$$y(t) = [\hat{G}(\lambda) e^{\lambda t} v - \hat{G}(\bar{\lambda}) e^{\bar{\lambda} t} v]/2j.$$

Since $\overline{\hat{G}(\lambda)} = \hat{G}(\bar{\lambda})$, we obtain that

$$y(t) = [\hat{G}(\lambda) e^{\lambda t} v - \overline{\hat{G}(\lambda) e^{\lambda t} v}]/(2j) = \Im \left(\hat{G}(\lambda) e^{\lambda t} v \right).$$

SISO (single-input single-output) systems). When $\hat{G}(s)$ is scalar, let $\phi(\lambda)$ be the argument of the complex number $\hat{G}(\lambda)$, that is

$$\hat{G}(\lambda) = |\hat{G}(\lambda)| e^{j\phi(\lambda)} \quad (9.7)$$

where $\phi(\lambda)$ is real. Noting that

$$\hat{G}(\lambda) e^{\lambda t} = |\hat{G}(\lambda)| e^{j\phi(\lambda)} e^{\alpha t} e^{j\omega t} = |\hat{G}(\lambda)| e^{\alpha t} e^{j(\omega t + \phi(\lambda))},$$

it now follows that

$$\Im(\hat{G}(\lambda) e^{\lambda t}) = |\hat{G}(\lambda)| e^{\alpha t} \sin(\omega t + \phi(\lambda)).$$

Hence, the steady state response of the system to input

$$u(t) = e^{\alpha t} \sin(\omega t)$$

is

$$\boxed{y(t) = |\hat{G}(\lambda)| e^{\alpha t} \sin(\omega t + \phi(\lambda))} \quad (9.8)$$

where $s = \alpha + j\omega$. Thus the output has the same form as the input; the non-negative real scalar $|\hat{G}(\lambda)|$ can be regarded as the **system gain** at λ and $\phi(\lambda)$ is the **system phase shift** at λ .

We also see that the steady state response to a purely sinusoidal input

$$u(t) = \sin(\omega t)$$

is given by

$$\boxed{y(t) = |\hat{G}(j\omega)| \sin(\omega t + \phi(j\omega))} \quad (9.9)$$

The significance of zeroes

Recall that a complex number η is a **zero** of a scalar-input scalar-output transfer function $\hat{G}$ if

$$\hat{G}(\eta) = 0. \quad (9.10)$$

The above response results reveal the following results.

If η is a zero of $\hat{G}$ then, the steady-state response of the system to the input $u(t) = e^{\eta t}$ is zero.

If $\eta = \alpha + j\omega$ is a zero of $\hat{G}$ then, the steady-state response of the system to the input $u(t) = e^{\alpha t} \sin(\omega t)$ is zero.

Example 106 (Tuned vibration absorber) Consider a mechanical system subject to a sinusoidal disturbance input:

$$M\ddot{q} + C\dot{q} + Kq = w(t)$$

where $w(t) = A \sin(\omega t)$.

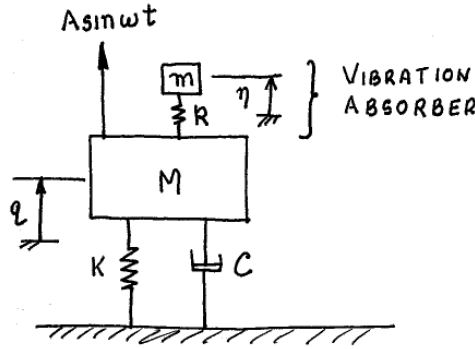


Figure 9.1: Tuned vibration absorber

Attaching a vibration absorber to the system results in

$$M\ddot{q} + C\dot{q} + Kq - k(\eta - q) = w \quad (9.11)$$

$$m\ddot{\eta} + k(\eta - q) = 0. \quad (9.12)$$

Taking the Laplace transform of the above equations yields:

$$(Ms^2 + Cs + K + k)\hat{q} - k\hat{\eta} = \hat{w} \quad (9.13)$$

$$(ms^2 + k)\hat{\eta} - k\hat{q} = 0. \quad (9.14)$$

We see that $\hat{q} = \hat{G}\hat{w}$ where

$$\hat{G} = \frac{ms^2 + k}{\Delta(s)}$$

and

$$\Delta(s) = (Ms^2 + Cs + K + k)(ms^2 + k) - k^2.$$

This transfer function has zeros at $\pm j\sqrt{k/m}$. if we choose

$$k/m = \omega^2$$

then the steady state response of q to w will be zero.

9.2 Discretization and zero order hold

Here we consider the **discretization** of continuous-time systems using a fixed **sampling time**. So, consider a continuous-time system described by

$$\dot{x}_c(t) = A_c x_c(t) + B_c u_c(t)$$

and let $T > 0$ be the sampling time.

Suppose u_d is a discrete-time signal. We will consider the input u_c to the continuous system to be given by a **zero order hold** described by

$$u_c(t) = u_d(k) \quad \text{for} \quad kT \leq t < (k+1)T$$

Thus, the input to the continuous-time system is constant over each sampling interval $[kT, (k+1)T)$. Once the sampling interval is specified, the input to the continuous-time system can be completely characterized by the discrete-time input u_d .

Suppose we sample the state at multiples of the sampling time and let

$$x_d(k) := x_c(kT)$$

Then,

$$\begin{aligned} x_d(k+1) &= x_c((k+1)T) \\ &= e^{A_c T} x_c(kT) + \int_{kT}^{(k+1)T} e^{A_c((k+1)T-t)} B_c u_c(t) dt \\ &= e^{A_c T} x_d(k) + \int_{kT}^{(k+1)T} e^{A_c((k+1)T-t)} B_c u_d(k) dt \\ &= e^{A_c T} x_d(k) + \int_{kT}^{(k+1)T} e^{A_c((k+1)T-t)} dt B_c u_d(k) \end{aligned}$$

Changing the variable of integration from t to $\tau := (k+1)T - t$ yields

$$\int_{kT}^{(k+1)T} e^{A_c((k+1)T-t)} dt = - \int_T^0 e^{A_c\tau} d\tau = \int_0^T e^{A_c\tau} d\tau$$

Hence,

$$x_d(k+1) = A_d x_d(k) + B_d u_d(k)$$

where

$$\boxed{\begin{aligned} A_d &= e^{A_c T} \\ B_d &= \int_0^T e^{A_c\tau} d\tau B_c \end{aligned}} \quad (9.15)$$

Recall that

$$e^{A_c t} = I + A_c t + \frac{1}{2!} A_c^2 t^2 + \frac{1}{3!} A_c^3 t^3 + \dots$$

Thus,

$$\int_0^T e^{A_c\tau} d\tau = TI + \frac{T^2}{2!} A_c + \frac{T^3}{3!} A_c^2 T^3 + \dots$$

and we have the following power series for A_d and B_d :

$$\begin{aligned} A_d &= I + TA_c + \frac{T^2}{2!} A_c^2 + \frac{T^3}{3!} A_c^3 + \dots \\ B_d &= TB_c + \frac{T^2}{2!} A_c B_c + \frac{T^3}{3!} A_c^2 B_c + \dots \end{aligned}$$

If A_c is invertible, then

$$\int_0^T e^{A_c\tau} d\tau = (e^{A_c T} - I)A_c^{-1} = (A_d - I)A_c^{-1}$$

and

$$B_d = (A_d - I)A_c^{-1}B_c$$

If we approximate A_d and B_d by considering only terms up to first order in T in their power series, we obtain

$$A_d \approx I + TA_c \quad \text{and} \quad B_d \approx TB_c.$$

Example 107 (Discretization of forced undamped oscillator)

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1 + u \end{aligned}$$

$$A_c = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad B_c = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Here,

$$e^{A_c t} = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}$$

Hence,

$$A_d = e^{A_c T} = \begin{pmatrix} \cos T & \sin T \\ -\sin T & \cos T \end{pmatrix} \quad B_d = \int_0^T e^{A_c \tau} d\tau B_c = \begin{pmatrix} 1 - \cos T \\ \sin T \end{pmatrix}$$

Consider $T = 2\pi$. Then,

$$A_d = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad B_d = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

So,

$$\begin{aligned} x_{1d}(k+1) &= x_{1d}(k) \\ x_{2d}(k+1) &= x_{2d}(k) \end{aligned}$$

Where did the input go?

What about $T = \pi/2$?

$$A_d = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad B_d = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

>> help c2d

C2D Conversion of state space models from continuous to discrete time.
[Phi, Gamma] = C2D(A,B,T) converts the continuous-time system:

$$\dot{x} = Ax + Bu$$

to the discrete-time state-space system:

$$x[n+1] = \text{Phi} * x[n] + \text{Gamma} * u[n]$$

assuming a zero-order hold on the inputs and sample time T.

See also: C2DM, and D2C.

```
>> a=[0 1 ; -1 0] ; b=[0; 1] ; T= pi/2;
>> [phi, gamma]= c2d(a,b,T)
```



```

phi =
    -0.0000    1.0000
    -1.0000    0.0000

```

```

gamma =
    1.0000
    1.0000

```

9.2.1 Input-output discretization

Consider a continuous-time system:

$$\begin{aligned}\dot{x}_c(t) &= A_c x_c(t) + B_c u_c(t) \\ y_c(t) &= C_c x_c(t) + D_c u_c(t)\end{aligned}$$

Suppose $T > 0$ is the *sampling time*.

- *Zero order hold:*

$$u_c(t) = u_d(k) \quad \text{for } kT \leq t < (k+1)T$$

Letting

$$x_d(k) := x_c(kT) \quad y_d(k) := y_c(kT)$$

we obtain

$$\begin{aligned}x_d(k+1) &= A_d x_d(k) + B_d u_d(k) \\ y_d(k) &= C_d x_d(k) + D_d u_d(k)\end{aligned}$$

where

$A_d = e^{A_c T} \quad B_d = \int_0^T e^{A_c \tau} d\tau B_c \quad C_d = C_c \quad D_d = D_c$

9.3 Response of discrete-time systems*

A general linear time-invariant discrete-time system is described by

$$\begin{aligned}x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k)\end{aligned}$$

where $k \in \mathbb{R}$ is ‘time’, $x(k) \in \mathbb{R}^n$ in the state, $u(k) \in \mathbb{R}^m$ the input, and $y(k) \in \mathbb{R}^p$ is the output.

9.3.1 State response and the convolution sum

The solution to

$$x(k+1) = Ax(k) + Bu(k) \quad x(k_0) = x_0 \quad (9.16)$$

is unique and is given by

$$\boxed{x(k) = A^{(k-k_0)}x_0 + \sum_{j=k_0}^{k-1} A^{(k-1-j)}Bu(j)} \quad (9.17)$$

PROOF. Use induction.

9.3.2 Input-output response and the pulse response matrix

The solution to

$$\begin{aligned}x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k)\end{aligned}$$

subject to initial condition

$$x(k_0) = x_0$$

is unique and is given by

$$\boxed{y(k) = CA^{(k-k_0)}x_0 + \sum_{j=k_0}^k G(k-j)u(j)}$$

where the pulse response matrix G is defined by

$$\boxed{\begin{aligned}G(0) &= D \\ G(k) &= CA^{k-1}B \quad \text{for } k = 1, 2, \dots\end{aligned}}$$

Clearly, the above holds for $k = k_0$. For $k > k_0$, recall that

$$x(k) = A^{(k-k_0)}x_0 + \sum_{j=k_0}^{k-1} A^{(k-1-j)}Bu(j)$$

Hence,

$$\begin{aligned}
y(k) &= CA^{(k-k_0)}x_0 + \sum_{j=k_0}^{k-1} CA^{(k-1-j)}Bu(j) + Du(k) \\
&= CA^{(k-k_0)}x_0 + \sum_{j=k_0}^{k-1} G(k-j)u(j) + G(0)u(k) \\
&= CA^{(k-k_0)}x_0 + \sum_{j=k_0}^k G(k-j)u(j)
\end{aligned}$$

Zero initial conditions response: $x(0) = 0$.

$$y(k) = \sum_{j=0}^k G(k-j)u(j)$$

Sometimes this is represented by

$$y \longleftarrow \boxed{G} \longleftarrow u$$

Pulse response: Suppose $x(0) = 0$ and

$$u(0) = p(k)v \quad \text{for } k = 0, 1, \dots$$

where v is constant m vector and p is the **unit pulse function** defined by

$$\begin{aligned}
p(0) &= 1 \\
p(k) &= 0 \quad \text{for } k = 1, 2, \dots
\end{aligned}$$

Then,

$$y(k) = G(k)v \quad \text{for } k = 0, 1, \dots$$

From this we can see that G_{ij} is the zero initial condition response of output component y_i when input component u_j is a unit pulse and all other inputs are zero, that is,

$$\left. \begin{aligned} u_1(k) &= 0 \\ &\vdots \\ u_j(k) &= p(k) \\ &\vdots \\ u_m(k) &= 0 \end{aligned} \right\} \implies \begin{cases} y_1(k) &= G_{1j}(k) \\ &\vdots \\ y_i(k) &= G_{ij}(k) \\ &\vdots \\ y_p(k) &= G_{pj}(k) \end{cases}$$

- So, for a SISO (single-input single-output) system, G is the output response due a unit pulse input and zero initial conditions.

Exercises

Exercise 108 Obtain (by hand) the response of each of the following systems due to a unit impulse input and the zero initial conditions. For each case, determine whether the response contains all the system modes.

a)

$$\begin{aligned}\dot{x}_1 &= -5x_1 + 2x_2 + u \\ \dot{x}_2 &= -12x_1 + 5x_2 + u\end{aligned}$$

b)

$$\begin{aligned}\dot{x}_1 &= -5x_1 + 2x_2 + u \\ \dot{x}_2 &= -12x_1 + 5x_2 + 2u\end{aligned}$$

c)

$$\begin{aligned}\dot{x}_1 &= -5x_1 + 2x_2 + u \\ \dot{x}_2 &= -12x_1 + 5x_2 + 3u\end{aligned}$$

Exercise 109 Consider the system with disturbance input w and performance output z described by

$$\begin{aligned}\dot{x}_1 &= -x_1 + x_2 + w \\ \dot{x}_2 &= -x_1 + 4x_2 + 2w \\ z &= x_1.\end{aligned}$$

Using an appropriate Lyapunov equation, determine

$$\int_0^\infty \|z(t)\|^2 dt$$

for each of the following situations.

(a)

$$w = 0 \quad \text{and} \quad x(0) = (1, 0).$$

(b)

$$w(t) = \delta(t) \quad \text{and} \quad x(0) = 0.$$

Chapter 10

Stability and boundedness

10.1 Continuous-time systems

The basic results on stability of linear time-invariant systems can be summarized fairly simply. However, we want to be able to make stability determinations of nonlinear systems based on linearization. Before we do this we need to discuss stability concepts for nonlinear systems.

Consider any system described by

$$\dot{x} = f(x) \quad (10.1)$$

where $x(t)$ is an n -vector and t is a scalar. By a **solution** of (10.1) we mean any continuous n -vector valued function $x(\cdot)$ which is defined on the interval $[0, \infty)$ and satisfies $\dot{x}(t) = f(x(t))$ for all $t \geq 0$.

10.1.1 Boundedness of solutions

An important property of most physical systems is that their behavior is bounded. Boundedness is precisely defined as follows.

Boundedness. *A solution $x(\cdot)$ is bounded if there exists $\beta \geq 0$ such that*

$$\|x(t)\| \leq \beta \quad \text{for all } t \geq 0.$$

A solution is unbounded if it is not bounded.

It should be clear from the above definitions that *a solution $x(\cdot)$ is bounded if and only if each component $x_i(\cdot)$ of $x(\cdot)$ is bounded*. Also, a solution is unbounded if and only if at least one of its components is unbounded. So, $x(t) = [e^{-t} \ e^{-2t}]^T$ is bounded while $x(t) = [e^{-t} \ e^t]^T$ is unbounded.

Example 108 All solutions of

$$\dot{x} = -x$$

are bounded.

Example 109 All solutions of

$$\dot{x} = 0$$

are bounded.

Example 110 All solutions (except the zero solution) of

$$\dot{x} = x$$

are unbounded.

Example 111 Here we claim that all solutions of the scalar nonlinear system

$$\dot{x} = x(1 - x^2)$$

are bounded.

Example 112 Some solutions of the

$$\dot{x} = -x(1 - x^2)$$

are bounded and some solutions are unbounded.

Example 113 Undamped oscillator.

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1\end{aligned}$$

All solutions are bounded

Example 114 Simple pendulum. The *undamped* simple pendulum has some bounded and unbounded solutions. All solutions of the *damped* simple pendulum are bounded.

Boundedness and LTI systems. Consider now a general LTI system

$$\dot{x} = Ax \quad (10.2)$$

If A is non-defective, every solution of this system has the form

$$x(t) = \sum_{i=1}^n e^{\lambda_i t} v^i \quad (10.3)$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A , and the constant vectors v^i are corresponding eigenvectors. If A is defective, it has a defective eigenvalue λ_i and it has some solutions containing terms of the form

$$te^{\lambda_i t} v^i$$

Hence we can make the following conclusions.

All solutions of (10.2) are bounded if and only if for each eigenvalue λ_i of A :

(b1) $\Re(\lambda_i) \leq 0$ and

(b2) $\Re(\lambda_i) < 0$ if λ_i is defective.

If there is an eigenvalue λ_i of A such that either

(u1) $\Re(\lambda_i) > 0$ or

(u2) $\Re(\lambda_i) = 0$ and λ_i is defective.

then the system has some unbounded solutions.

Example 115 Unattached mass

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= 0 \end{aligned}$$

Here

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

has a single eigenvalue 0. This eigenvalue has algebraic multiplicity 2 but geometric multiplicity 1; hence some of the solutions of the system are unbounded. One example is

$$x(t) = \begin{pmatrix} t \\ 1 \end{pmatrix}$$

Example 116

$$\begin{aligned}\dot{x}_1 &= 0 \\ \dot{x}_2 &= 0\end{aligned}$$

Here

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

has a single eigenvalue 0. This eigenvalue has algebraic multiplicity 2 and geometric multiplicity 2. Hence all the solutions of the system are bounded. Actually every state is an equilibrium state and every solution is constant.

Example 117 (Resonance) Consider a simple linear oscillator subject to a sinusoidal input of amplitude W :

$$\ddot{q} + q = W \sin(\omega t + \phi)$$

Resonance occurs when $\omega = 1$. To see this, let

$$x_1 := q, \quad x_2 := \dot{q}, \quad x_3 := W \sin(\omega t + \phi), \quad x_4 := \omega W \cos(\omega t + \phi)$$

to yield

$$\dot{x} = Ax$$

where

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -\omega^2 & 0 \end{pmatrix}$$

If $\omega = 1$ then, A has eigenvalues j and $-j$. These eigenvalues have algebraic multiplicity two but geometric multiplicity one; hence the system has unbounded solutions.

10.1.2 Stability of an equilibrium state

A nonlinear system may have several equilibrium states and the qualitative behavior of the system may be different around different equilibrium states. In these cases, we cannot talk about the stability of the system; we need to talk about stability of the equilibrium states. A single nonlinear system may have stable and unstable equilibrium states.

So, suppose x^e is an equilibrium state of a system described by $\dot{x} = f(x)$. We are interested in the behavior of the system near x^e . If the state of the system starts at x^e it stays there. So, suppose now the state is perturbed slightly from x^e ; does the state remain close to x^e or does it move away from x^e ? Roughly speaking, we say that an equilibrium state is stable if it has the property that *solutions which start sufficiently close to the equilibrium state remain close to the equilibrium state*.

To precisely define stability, we use $\|x - y\|$ to measure the **distance of a state x from another state y** . Also, if y is any n -vector and r is any non-negative scalar, the set

$$\{x : \|x - y\| < r\}$$



Figure 10.1: Distance between two vectors

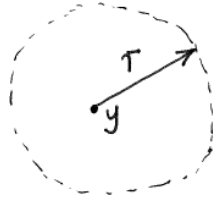


Figure 10.2: Open ball of radius r centered at y

is called the open ball of radius r centered at y .

Stability. An equilibrium state x^e is **stable** if for each $\epsilon > 0$ there is a $\delta > 0$ such that, whenever $\|x(0) - x^e\| < \delta$, we have $\|x(t) - x^e\| < \epsilon$ for all $t \geq 0$.

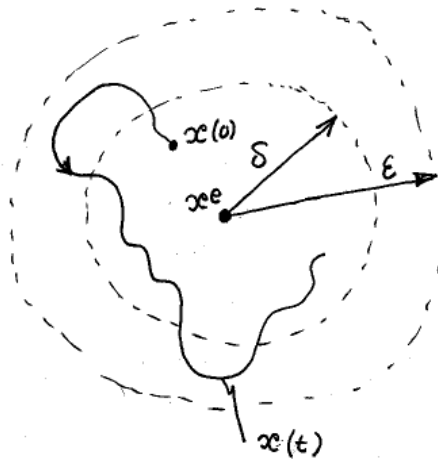


Figure 10.3: Stability of an equilibrium state

Instability. An equilibrium state is said to be **unstable** if it is not stable.

Example 118

$$\dot{x} = -x$$

The origin is stable. (Choose $\delta = \epsilon$)

Example 119

$$\dot{x} = 0$$

All states are stable equilibrium states. (Choose $\delta = \epsilon$)

Example 120

$$\dot{x} = x$$

The origin is unstable. Choose any $\epsilon > 0$. An appropriate δ cannot be found.

Example 121 The origin is an unstable equilibrium state for

$$\dot{x} = x(1 - x^2)$$

Choose any $0 < \epsilon < 1$. An appropriate δ cannot be found. However all solutions of this system are bounded.

Example 122 The origin is an stable equilibrium state for

$$\dot{x} = -x(1 - x^2)$$

However some solutions of this system are unbounded.

Example 123 Undamped oscillator

The origin is stable.

Example 124 Simple pendulum

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\sin x_1\end{aligned}$$

$(0, 0)$ is stable whereas $(\pi, 0)$ is unstable.

Example 125 Van der Pol oscillator

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= (1 - x_1^2)x_2 - x_1\end{aligned}$$

All solutions are bounded, but, $(0, 0)$ is unstable.

Figure 10.4: Van der Pol oscillator

Example 126 Reverse Van der Pol oscillator

$$\begin{aligned}\dot{x}_1 &= -x_2 \\ \dot{x}_2 &= -(1 - x_1^2)x_2 + x_1\end{aligned}$$

The zero state is stable. Note that, although zero is stable, the system has unbounded solutions.

Figure 10.5: Reverse Van der Pol oscillator

LTI systems. It can be shown that every equilibrium state of a LTI system (10.2) is stable if and only if all eigenvalues λ_i of A satisfy conditions (b1) and (b2) above. Hence every equilibrium state of a LTI system is stable if and only if all solutions are bounded.

It can also be shown that every equilibrium state is unstable if and only if there is an eigenvalue λ_i of A which satisfies condition (u1) or (u2) above. Hence every equilibrium state of a LTI system is unstable if and only if there are unbounded solutions.

Thus boundedness of all solutions and stability are equivalent for LTI systems. This is not true for nonlinear systems: the Van der Pol oscillator has a single equilibrium state and it is unstable; however, all solutions are bounded. The reverse Van der Pol oscillator has a single equilibrium state and it is stable; however, this system has some unbounded solutions.

10.1.3 Asymptotic stability

Global asymptotic stability

If x^e is a desirable equilibrium state for a system, then in addition to the system being stable about x^e , we would like the system to always settle down at x^e . This gives rise to the following definition.

Global asymptotic stability (GAS). An equilibrium state x^e is globally asymptotically stable (GAS) if

- (a) It is stable
- (b) Every solution $x(\cdot)$ converges to x^e , that is

$$\lim_{t \rightarrow \infty} x(t) = x^e$$

If x^e is a globally asymptotically stable equilibrium state, then there can be no other equilibrium states. In this case we say *the system $\dot{x} = f(x)$ is globally asymptotically stable*. Note that *if a system is GAS, then all its solutions must be bounded*.

Example 127

$$\dot{x} = -x$$

The origin is GAS.

Example 128

$$\dot{x} = -x^3$$

The origin is GAS.

Asymptotic stability

For nonlinear systems, it is sometimes too much to require that *all* solutions converge to a desired equilibrium state. Sometimes only solutions which start in a *neighborhood* of the equilibrium state converge to the equilibrium state. Hence, we have the following local version of asymptotic stability.

Asymptotic stability (AS). *An equilibrium state x^e is asymptotically stable (AS) if*

(a) *It is stable*

(b) *There exists $R > 0$ such that whenever $\|x(0) - x^e\| < R$ one has*

$$\lim_{t \rightarrow \infty} x(t) = x^e \quad (10.4)$$

The **region of attraction** of an equilibrium state x^e which is AS is the set of initial states which result in (10.4), that is it is the set of initial states which are attracted to x^e . Thus, the region of attraction of a globally asymptotically equilibrium state is the whole state space.

Example 129 For the system,

$$\dot{x} = -x(1 - x^2)$$

The origin is AS but not GAS. (Choose $R = 1$.)

Example 130 Damped simple pendulum

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\sin x_1 - x_2 \end{aligned}$$

The origin is AS but not GAS. Note that, although the origin is AS, the system has unbounded solutions.

Example 131 Reverse Van der Pol oscillator

$$\begin{aligned} \dot{x}_1 &= -x_2 \\ \dot{x}_2 &= -(1 - x_1^2)x_2 + x_1 \end{aligned}$$

The zero state is AS but not GAS. Note that, although the origin is AS, the system has unbounded solutions.

LTI systems. For LTI systems, it should be clear from the general form of the solution that the zero state is AS if and only if all the eigenvalues λ_i of A have negative real parts, that is,

$$\Re(\lambda_i) < 0$$

Also AS is equivalent to GAS.

Example 132 The system

$$\begin{aligned}\dot{x}_1 &= -x_1 + x_2 \\ \dot{x}_2 &= -x_2\end{aligned}$$

is GAS.

Example 133 The system

$$\begin{aligned}\dot{x}_1 &= -x_1 + x_2 \\ \dot{x}_2 &= -x_1 - x_2\end{aligned}$$

is GAS.

10.2 Discrete-time systems

$$x(k+1) = f(x(k)) \quad (10.5)$$

where $x(k)$ is an n -vector and k is an integer. By a solution of (10.5) we mean a sequence $x(\cdot) = (x(0), x(1), x(2), \dots)$ which satisfies $x(k+1) = f(x(k))$ for all $k \geq 0$.

DEFN. (Boundedness of solutions) A solution $x(\cdot)$ is **bounded** if there exists $\beta \geq 0$ such that

$$\|x(k)\| \leq \beta \quad \text{for all } k \geq 0.$$

A solution is *unbounded* if it is not bounded.

Example 134

$$x(k+1) = 0$$

All solutions are bounded.

Example 135

$$x(k+1) = x(k)$$

All solutions are bounded.

Example 136

$$x(k+1) = -2x(k)$$

All solutions (except the zero solution) are unbounded.

- All solutions of the LTI system

$$x(k+1) = Ax(k) \quad (10.6)$$

are bounded if and only if for each eigenvalue λ of A :

(b1) $|\lambda| \leq 1$ and

(b2) if $|\lambda| = 1$ then λ is nondefective.

If there is an eigenvalue λ of A such that either

(u1) $|\lambda| > 1$ or

(u2) $|\lambda| = 1$ and λ is defective.

then the system has unbounded solutions.

Example 137 Discrete unattached mass. Here

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

has a single eigenvalue 1. This eigenvalue has algebraic multiplicity 2 but geometric multiplicity 1; hence some of the solutions of the system $x(k+1) = Ax(k)$ are unbounded. One example is

$$x(k) = \begin{pmatrix} k \\ 1 \end{pmatrix}$$

10.2.1 Stability of an equilibrium state

Suppose x^e is an equilibrium state of system (10.1), that is,

$$f(x^e) = x^e$$

DEFN. (Stability) An equilibrium state x^e is **stable** if for each $\epsilon > 0$ there exists $\delta > 0$ such that whenever $\|x(0) - x^e\| < \delta$, one has $\|x(k) - x^e\| < \epsilon$ for all $k \geq 0$.
 x^e is *unstable* if it is not stable.

• Every equilibrium state of a LTI system (10.6) is stable if and only if all eigenvalues λ of A satisfy conditions (b1) and (b2) above. Hence every equilibrium state of a LTI system is stable if and only if all solutions are bounded.

Every equilibrium state is unstable if and only if there is an eigenvalue λ of A which satisfies condition (u1) or (u2) above. Hence every equilibrium state of a LTI system is unstable if and only if there are unbounded solutions.

10.2.2 Asymptotic stability

Global asymptotic stability

DEFN. (Global asymptotic stability (GAS)) An equilibrium state x^e is *globally asymptotically stable (GAS)* if

- (a) It is stable
- (b) Every solution $x(\cdot)$ converges to x^e , that is,

$$\lim_{k \rightarrow \infty} x(k) = x^e.$$

• If x^e is a globally asymptotically stable equilibrium state, then there are no other equilibrium state. In this case we say *the system (10.5) is globally asymptotically stable*.

Example 138

$$x(k+1) = \frac{1}{2}x(k)$$

GAS

Asymptotic stability

DEFN. (Asymptotic stability (AS)) An equilibrium state x^e is asymptotically stable (AS) if

- (a) It is stable

(b) There exists $R > 0$ such that whenever $\|x(0) - x^e\| < R$ one has

$$\lim_{k \rightarrow \infty} x(k) = x^e$$

- For LTI systems, the zero state is AS if and only if it is GAS.

10.3 A nice table

The following table summarizes the relationship between the stability properties of a LTI system and the eigenproperties of its A -matrix. In the table, unless otherwise stated, a property involving λ must hold for all eigenvalues λ of A .

Stability properties	CT eigenproperties	DT eigenproperties
Global asymptotic stability and boundedness	$\Re(\lambda) < 0$	$ \lambda < 1$
Stability and boundedness	$\Re(\lambda) \leq 0$ $\Re(\lambda) < 0$ if λ is defective	$ \lambda \leq 1$ $ \lambda < 1$ if λ is defective
Instability and some unbounded solutions	A has an eigenvalue λ s.t. $\Re(\lambda) > 0$ or $\Re(\lambda) = 0$ and λ is defective	A has an eigenvalue λ s.t. $ \lambda > 1$ or $ \lambda = 1$ and λ is defective

10.4 Linearization and stability

Consider a nonlinear time-invariant system described by

$$\dot{x} = f(x)$$

where $x(t)$ is an n -vector at time t . Suppose x^e is an equilibrium state for this system, that is, $f(x^e) = 0$, and consider the linearization of this system about x^e :

$$\delta\dot{x} = A\delta x \quad \text{where} \quad A = \frac{\partial f}{\partial x}(x^e).$$

The following results can be demonstrated using nonlinear Lyapunov stability theory.

Asymptotic stability. *If all the eigenvalues of the A matrix of the linearized system have negative real parts, then the nonlinear system is asymptotically stable about x^e .*

Instability. *If at least one eigenvalue of the A matrix of the linearized system has a positive real part, then the nonlinear system is unstable about x^e .*

Undetermined. *Suppose all the eigenvalues of the A matrix of the linearized system have non-positive real parts and at least one eigenvalue of A has zero real part. Then, based on the linearized system alone, one cannot predict the stability properties of the nonlinear system about x^e .*

Note that the first statement above is equivalent to the following statement. If the linearized system is asymptotically stable, then the nonlinear system is asymptotically stable about x^e .

Example 139

$$\dot{x} = x - x^3$$

Example 140 (Damped simple pendulum.) Physically, the system

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\sin x_1 - x_2\end{aligned}$$

has two distinct equilibrium states: $(0, 0)$ and $(\pi, 0)$. The A matrix for the linearization of this system about $(0, 0)$ is

$$A = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$$

Since all the eigenvalues of this matrix have negative real parts, the nonlinear system is asymptotically stable about $(0, 0)$. The A matrix corresponding to linearization about $(\pi, 0)$ is

$$A = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$$

Since this matrix has an eigenvalue with a positive real part, the nonlinear system is unstable about $(\pi, 0)$.

The following example illustrates the fact that if the eigenvalues of the A matrix have non-positive real parts and there is at least one eigenvalue with zero real part, then, one cannot make any conclusions on the stability of the nonlinear system based on the linearization.

Example 141 Consider the scalar nonlinear system:

$$\dot{x} = ax^3$$

This origin is GAS if $a < 0$, unstable if $a > 0$ and stable if $a = 0$. However, the linearization of this system about zero, given by

$$\delta\dot{x} = 0$$

is independent of a and is stable.

The following example illustrates that instability of the linearized system does not imply instability of the nonlinear system.

Example 142 Using nonlinear techniques, one can show that the following system is GAS about the origin.

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1^3 - x_2^3\end{aligned}$$

However, the linearization of this system, given by

$$\begin{aligned}\delta\dot{x}_1 &= \delta x_2 \\ \delta\dot{x}_2 &= 0,\end{aligned}$$

is unstable about the origin.

10.5 Exercises

Exercise 110 Determine (by hand) whether each of the following systems is asymptotically stable, stable, or unstable.

$$\begin{aligned}\dot{x}_1 &= -x_1 + 2001x_2 \\ \dot{x}_2 &= -x_1\end{aligned}$$

$$\begin{aligned}\dot{x}_1 &= -x_1 \\ \dot{x}_2 &= x_2\end{aligned}$$

$$\begin{aligned}\dot{x}_1 &= jx_1 + x_2 \\ \dot{x}_2 &= jx_2\end{aligned}$$

$$\begin{aligned}x_1(k+1) &= -2x_1(k) \\ x_2(k+1) &= 0.5x_2(k)\end{aligned}$$

Exercise 111 Determine (by hand) the stability properties of a linear continuous-time system with

$$A = \begin{pmatrix} \frac{1}{2} & 1 & -\frac{1}{2} & 0 \\ -1 & \frac{1}{2} & 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 & -\frac{1}{2} & 1 \\ 0 & \frac{1}{2} & -1 & -\frac{1}{2} \end{pmatrix}$$

Using the eig command on MATLAB, what would your stability conclusion be?

Exercise 112 Determine the stability properties of the following system about the zero solution.

$$\begin{aligned}2\ddot{y}_1 + \ddot{y}_2 &= y_1 + 2y_2 \\ \ddot{y}_1 + 2\ddot{y}_2 &= 2y_1 + y_2\end{aligned}$$

Exercise 113 If possible, use linearization to determine the stability properties of each of the following systems about the zero equilibrium state.

(i)

$$\begin{aligned}\dot{x}_1 &= (1 + x_1^2)x_2 \\ \dot{x}_2 &= -x_1^3\end{aligned}$$

(ii)

$$\begin{aligned}\dot{x}_1 &= \sin x_2 \\ \dot{x}_2 &= (\cos x_1)x_3 \\ \dot{x}_3 &= e^{x_1}x_2\end{aligned}$$

Exercise 114 If possible, use linearization to determine the stability properties of each of the following systems about the zero equilibrium state.

(a)

$$\begin{aligned}\dot{x}_1 &= (1 + x_1^2)x_2 \\ \dot{x}_2 &= \sin x_1\end{aligned}$$

(b)

$$\begin{aligned}\dot{x}_1 &= (1 + x_1^2)x_2 \\ \dot{x}_2 &= -\sin x_1\end{aligned}$$

Exercise 115 If possible, use linearization to determine the stability properties of each of the following systems about the zero solution.

(a)

$$\begin{aligned}\dot{x}_1 &= \sin x_2 + \cos x_1 - 1 \\ \dot{x}_2 &= -(1 + \sin x_1)x_1\end{aligned}$$

(b)

$$\begin{aligned}(2 + \cos y_2)\ddot{y}_1 - 3 \sin y_2 &= 0 \\ \ddot{y}_2 - (1 + \dot{y}_1^2)y_1 &= 0\end{aligned}$$

Exercise 116 If possible, use linearization to determine the stability properties of each of the following systems about the zero solution.

(a)

$$\begin{aligned}\dot{x}_1 &= -11 \sin x_1 + 20(1 + x_1^2)x_2 \\ \dot{x}_2 &= -6x_1 + x_1^3 + 11(1 - \cos x_1)x_2\end{aligned}$$

(b)

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1^3 - x_2^3\end{aligned}$$

(c)

$$\frac{d^4 y}{dt^4} - \sin y = 0$$

Exercise 117 If possible, use linearization to determine the stability properties of the following system about the zero state.

$$\begin{aligned}\dot{x}_1 &= -x_1 + x_2^3 + 2 \sin x_3 \\ (\cos x_3) \dot{x}_2 &= 3x_1 - 4e^{x_2} + x_3 + x_2 x_3 + 4 \\ (1 + x_1^2) \dot{x}_3 &= -2x_3\end{aligned}$$

Exercise 118 If possible, use linearization to determine the stability properties of each of the following systems about the zero solution.

(a)

$$\begin{aligned}\dot{x}_1 &= x_2^2 x_1 + (\cos x_1) x_2 \\ \dot{x}_2 &= (1 + \sin x_2) x_1 - x_1^2 x_2\end{aligned}$$

(b)

$$\ddot{y} + \dot{y}^3 + y^5 = 0$$

(c)

$$\frac{d^3 y}{dt^3} - (\cos y) \dot{y} = 0$$

Exercise 119 (a) Linearize the following nonlinear system about its zero equilibrium state.

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \sin(\dot{x}_2) &= -x_1^3 - x_2^3\end{aligned}$$

(b) What are the stability properties of the linearized system about zero?

(c) What are the stability properties of the nonlinear system about zero?

Exercise 120 Using linearization determine (if possible) the stability properties of each of the following systems about their corresponding specified equilibrium solution q^e . If not possible, provide a reason.

(a)

$$\ddot{q} + (\dot{q} - 1)|\dot{q} - 1| + 2 \sin q = 0$$

and $q^e = \pi/6$.

(b)

$$\begin{aligned}\dot{q}_1 &= e^{q_1} q_2 - q_1^3 \\ \dot{q}_2 &= -q_1 \cos q_2\end{aligned}$$

and $q^e = [0 \ 0]'$.

(c)

$$\begin{aligned}\ddot{q}_1 &= q_2 \\ \ddot{q}_2 &= \sin q_1\end{aligned}$$

and $q^e = [0 \ 0]'$.

Exercise 121 Determine the stability properties of the following system about the zero solution.

$$\begin{aligned}x_1(k+1) &= x_1(k)^3 + \sin(x_2(k)) \\ x_2(k+1) &= -\frac{1}{2} \cos(x_2(k))x_1(k) + x_2(k)^3\end{aligned}$$

Exercise 122 Circular restricted three body problem (a) Obtain (by any means possible) all equilibrium solutions of the system:

$$\ddot{x} - 2\dot{y} - x = -\frac{(1-\mu)(x+\mu)}{d^3} - \frac{\mu(x-(1-\mu))}{r^3} \quad (10.7)$$

$$\ddot{y} + 2\dot{x} - y = -\frac{(1-\mu)y}{d^3} - \frac{\mu y}{r^3} \quad (10.8)$$

$$\ddot{z} = -\frac{(1-\mu)z}{d^3} - \frac{\mu z}{r^3} \quad (10.9)$$

where d and r are the magnitudes of the vectors

$$(x+\mu, y, z) \quad \text{and} \quad (x-(1-\mu), y, z),$$

respectively. Consider $\mu = 3 \times 10^{-6}$

(b) Determine the stability properties of each equilibrium. (MATLAB might be useful here.)

Exercise 123 Stability properties of the two pendulum cart system. Using MATLAB, determine the stability properties of the linearizations $L1 - L8$. What can you say about the stability properties of the nonlinear system about the corresponding equilibrium states?

Exercise 124 Passive stabililization of the two pendulum cart system. Here attempt passive stabilization of the two pendulum cart system about the $E1$ equilibrium configuration. If we attach a linear spring of spring constant k and a linear damper which damping coefficient c between the cart and the wall, the motion of the system is described by

$$\begin{aligned}(m_0 + m_1 + m_2)\ddot{y} - m_1 l_1 \cos \theta_1 \ddot{\theta}_1 - m_2 l_2 \cos \theta_2 \ddot{\theta}_2 &+ m_1 l_1 \sin \theta_1 \dot{\theta}_1^2 + m_2 l_2 \sin \theta_2 \dot{\theta}_2^2 = -c\dot{y} - ky \\ -m_1 l_1 \cos \theta_1 \ddot{y} + m_1 l_1^2 \ddot{\theta}_1 &+ m_1 l_1 g \sin \theta_1 = 0 \\ -m_2 l_2 \cos \theta_2 \ddot{y} + m_2 l_2^2 \ddot{\theta}_2 &+ m_2 l_2 g \sin \theta_2 = 0\end{aligned}$$

Figure 10.6: Passive stabilization of double pendulum on cart

For parameter sets $P1 - P4$, determine if possible, values of c and k which stabilize the system about $E1$. Simulate the nonlinear system with your chosen values of k and c . For those parameter sets for which the system is not stabilizable, simulate with some chosen positive values of c and k .

Exercise 125 *Two link manipulator.* Using MATLAB, determine the stability properties of the linearizations corresponding to the four equilibrium states found earlier. What can you say about the stability properties of the nonlinear system about these equilibrium states?

Chapter 11

More linear algebra

11.1 The usual scalar product on $\mathbb{R}^n$ and $\mathbb{C}^n$

Consider any pair x, y of real n -vectors. The scalar product (inner product, dot product) of x and y is the real number given by

$$\langle y, x \rangle = y_1 x_1 + \dots + y_n x_n.$$

Regarding x and y as real column matrices, we have

$$\langle y, x \rangle = y^T x.$$

Suppose x and y are complex n -vectors. If we use the above definition as the definition of inner product, we won't have the property that $\langle x, x \rangle \geq 0$. To see this, note that $x = j$ results in $x^2 = -1$. So we define the scalar product (inner product, dot product) of x and y to be the complex number given by

$$\boxed{\langle y, x \rangle := \bar{y}_1 x_1 + \dots + \bar{y}_n x_n}$$

When x and y are real this definition yields the usual scalar product on $\mathbb{R}^n$. Regarding x and y as column matrices, we have

$$\langle y, x \rangle = y' x$$

where y' is the complex conjugate transpose of y .

We can obtain the usual Euclidean norm from the usual scalar product by noting that

$$\begin{aligned} \langle x, x \rangle &= \bar{x}_1 x_1 + \dots + \bar{x}_n x_n \\ &= |x_1|^2 + |x_2|^2 + \dots + |x_n|^2 \\ &= \|x\|^2. \end{aligned}$$

Hence

$$\|x\| = \langle x, x \rangle^{\frac{1}{2}}.$$

Properties of $\langle \cdot, \cdot \rangle$. The usual scalar product on $\mathbb{R}^n$ or $\mathbb{C}^n$ has the following properties.

(i) For every pair of vectors x and y ,

$$\langle x, y \rangle = \overline{\langle y, x \rangle}$$

(ii) For every vector y , every pair of vectors x^1, x^2 , and every pair of scalars α_1, α_2 ,

$$\langle y, \alpha_1 x^1 + \alpha_2 x^2 \rangle = \alpha_1 \langle y, x^1 \rangle + \alpha_2 \langle y, x^2 \rangle$$

(iii) For every vector x ,

$$\langle x, x \rangle \geq 0$$

and

$$\langle x, x \rangle = 0 \quad \text{if and only if} \quad x = 0$$

For real vectors, property (i) reduces to $\langle x, y \rangle = \langle y, x \rangle$. By utilizing properties (i) and (ii), we obtain that for every vector x , every pair of vectors y^1, y^2 , and every pair of scalars α_1, α_2 ,

$$\langle \alpha_1 y^1 + \alpha_2 y^2, x \rangle = \overline{\alpha_1} \langle y^1, x \rangle + \overline{\alpha_2} \langle y^2, x \rangle$$

Any scalar valued function on a real (or complex) vector space which has the above three properties is defined to be an inner product. A vector space equipped with an inner product is called an **inner product space**.

As an example consider $\mathbb{R}^{n \times m}$, the space of real $n \times m$ matrices. An inner product on this space is given by

$$\text{trace}(A^T B) = \sum_{i=1}^n \sum_{j=1}^m a_{ij} b_{ij}$$

As another example, suppose $\mathcal{V}$ is the vector space of all real valued continuous functions defined on the interval $[0, 1]$. For any two functions f and g in $\mathcal{V}$, let

$$\langle f, g \rangle = \int_0^1 f(t)g(t) dt.$$

One may readily show that properties (i)-(iii) above hold.

11.2 Orthogonality

Suppose $\mathcal{V}$ is a real (or complex) vector space which is equipped with an inner product $\langle \cdot, \cdot \rangle$.

A vector y is said to be orthogonal to a vector x if

$$\langle y, x \rangle = 0$$

For example, in $\mathbb{R}^2$ the vector $(1, 1)$ is orthogonal to vector $(1, -1)$.

If y is orthogonal to x , then (using property (ii) above) x is orthogonal to y ; so, we usually just say x and y are orthogonal.

Figure 11.1: Orthogonality

11.2.1 Orthonormal basis

Suppose $\mathcal{V}$ is finite dimensional and

$$b^1, b^2, \dots, b^n$$

is a basis for $\mathcal{V}$. Then this is an **orthonormal basis** if for every i, j ,

$$\begin{aligned} \langle b^i, b^i \rangle &= 1 \\ \langle b^i, b^j \rangle &= 0 \quad \text{when } i \neq j \end{aligned}$$

that is each basis vector has magnitude 1 and is orthogonal to all the other basis vectors.

All finite dimensional vector spaces have an orthonormal basis.

Unitary coordinate transformations

Suppose $b^1, b^2, \dots, b^n$ is an orthonormal basis for $\mathbb{C}^n$ and consider the transformation matrix

$$T = \begin{pmatrix} b^1 & b^2 & \dots & b^n \end{pmatrix}$$

Then

$$\begin{aligned} T'T &= \begin{pmatrix} b^{1'} \\ b^{2'} \\ \vdots \\ b^{n'} \end{pmatrix} \begin{pmatrix} b^1 & b^2 & \dots & b^n \end{pmatrix} \\ &= \begin{pmatrix} b^{1'}b^1 & b^{1'}b^2 & \dots & b^{1'}b^n \\ b^{2'}b^1 & b^{2'}b^2 & \dots & b^{2'}b^n \\ \vdots & \vdots & \vdots & \\ b^{n'}b^1 & b^{n'}b^2 & \dots & b^{n'}b^n \end{pmatrix} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}, \end{aligned}$$

that is, $T'T = I$.

A square complex matrix T is said to be unitary if

$$T'T = I.$$

If T is unitary, then

$$T^{-1} = T'.$$

Example 143 The following matrix is unitary for any θ .

$$T = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Unitary coordinate transformations. *An important property of unitary coordinate transformation is that it preserves inner product and Euclidean norm.* To see this, consider a unitary coordinate transformation

$$x = T\xi$$

with T unitary. Then for any two n vectors ξ^1 and ξ^2 ,

$$\begin{aligned} \langle T\xi^1, T\xi^2 \rangle &= (T\xi^1)' T\xi^2 \\ &= \xi^{1'} T' T\xi^2 \\ &= \xi^{1'} \xi^2 \\ &= \langle \xi^1, \xi^2 \rangle \end{aligned}$$

hence, $\langle T\xi^1, T\xi^2 \rangle = \langle \xi^1, \xi^2 \rangle$; that is, unitary transformations preserve the scalar product and, hence, the Euclidean norm.

11.3 Hermitian matrices

In this section, the square ($n \times n$) complex matrix P is hermitian, that is

$$\boxed{P' = P}$$

If P is real, then hermitian is equivalent to symmetric, that is, $P^T = P$.

- For any complex n -vector x , the scalar $x'Px$ is real:

$$\begin{aligned} \overline{x'Px} &= (x'Px)' \\ &= x'P'x \\ &= x'Px \end{aligned}$$

Since $\overline{x'Px} = x'Px$, the scalar $x'Px$ is real. ■

- Every eigenvalue λ of a hermitian matrix P is real. If λ is an eigenvalue of P then, there is a nonzero vector v such that

$$Pv = \lambda v$$

Hence

$$v'Pv = \lambda v'v = \lambda ||v||^2$$

Since $v'Pv$ is real and $||v||^2$ is nonzero real, λ must be real. ■

• Suppose v_1 and v_2 are eigenvectors corresponding to different eigenvalues of P . Then v_1 and v_2 are orthogonal:

$$Pv_1 = \lambda_1 v_1$$

implies

$$v_2'Pv_1 = \lambda_1 v_2'v_1$$

Similarly,

$$v_1'Pv_2 = \lambda_2 v_1'v_2$$

Taking the complex conjugate transpose of both sides of this last expression and using $P' = P$ and $\overline{\lambda_2} = \lambda_2$ yields

$$v_2'Pv_1 = \lambda_2 v_2'v_1$$

Equating the two expressions for $v_2'Pv_1$ yields

$$\lambda_1 v_2'v_1 = \lambda_2 v_2'v_1$$

Since $\lambda_1 \neq \lambda_2$, we must have $v_2'v_1 = 0$, that is, v_2 is orthogonal to v_1 . ■

Actually, the following more general result is true.

Fact 14 *Every hermitian matrix has an orthonormal basis of eigenvectors.*

The above fact implies that every hermitian matrix is non-defective and diagonalizable. Recall that a non-defective matrix P can be decomposed as $P = T\Lambda T^{-1}$ where T is any matrix whose columns form an eigenvector basis for P and Λ is a diagonal matrix whose diagonal elements are the eigenvalues of P and the number of times an eigenvalue appears equals its algebraic multiplicity.

It now follows that every hermitian matrix P can be expressed as

$$\boxed{P = T\Lambda T'} \quad \text{where } T \text{ is a unitary matrix and } \Lambda \text{ is a real diagonal matrix whose diagonal elements are the eigenvalues of } P \text{ and the number of times an eigenvalue appears equals its algebraic multiplicity.}$$

where T is a unitary matrix and Λ is a real diagonal matrix whose diagonal elements are the eigenvalues of P and the number of times an eigenvalue appears equals its algebraic multiplicity.

11.4 Positive and negative (semi)definite matrices

11.4.1 Quadratic forms

For any square $n \times n$ matrix P we can define an associated quadratic form:

$$x'Px = \sum_{i=1}^n \sum_{j=1}^n p_{ij}x'_i x_j.$$

Quadratic forms arise naturally in mechanics. For example, suppose one considers a rigid body and chooses an orthonormal coordinate system with origin at the body mass center. If ω is a real 3-vector consisting of the components of the angular velocity vector of the body and I is the 3×3 symmetric inertia matrix of the body, then the rotational kinetic energy of the body is given by $\frac{1}{2}\omega^T I \omega$.

If P is hermitian ($P' = P$) then, as we have already shown, the scalar $x'Px$ is real for all $x \in \mathbb{C}^n$. The following result provides useful upper and lower bounds on $x'Px$.

Lemma 1 If P is a hermitian $n \times n$ matrix, then for all $x \in \mathbb{C}^n$

$$\boxed{\lambda_{\min}(P)||x||^2 \leq x'Px \leq \lambda_{\max}(P)||x||^2}$$

where $\lambda_{\min}(P)$ and $\lambda_{\max}(P)$ denote the minimum and maximum eigenvalues of P respectively.

PROOF. Letting $\xi = T'x$, we have

$$\begin{aligned} x'Px &= x'T\Lambda T'x \\ &= \xi'\Lambda\xi \\ &= \lambda_1\xi'_1\xi_1 + \dots + \lambda_n\xi'_n\xi_n \\ &= \lambda_1|\xi_1|^2 + \dots + \lambda_n|\xi_n|^2 \end{aligned}$$

Since, $\lambda_i \leq \lambda_{\max}(P)$ for $i = 1, \dots, n$ we have $\lambda_i|\xi_i|^2 \leq \lambda_{\max}(P)|\xi_i|^2$; hence

$$x'Px \leq \lambda_{\max}(P) (|\xi_1|^2 + \dots + |\xi_n|^2) = \lambda_{\max}(P)||\xi||^2;$$

hence $x'Px \leq \lambda_{\max}(P)||\xi||^2$. Also $||\xi|| = ||x||$; hence,

$$x'Px \leq \lambda_{\max}(P)||x||^2$$

In an analogous fashion one can show that

$$x'Px \geq \lambda_{\min}(P)||x||^2$$

■

11.4.2 Definite matrices

DEFN. A hermitian matrix P is positive definite (pd) if

$$x'Px > 0 \quad \text{for all nonzero } x \text{ in } \mathbb{C}^n$$

We denote this by $P > 0$.

The matrix P is negative definite (nd) if $-P$ is positive definite; we denote this by $P < 0$.

Example 144 For

$$P = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$$

we have (note the completion of squares trick)

$$\begin{aligned} x'Px &= x'_1x_1 - x'_1x_2 - x'_2x_1 + 2x_2^*x_2 \\ &= (x_1 - x_2)'(x_1 - x_2) + x'_2x_2 \\ &= |x_1 - x_2|^2 + |x_2|^2 \end{aligned}$$

Clearly, $x'Px \geq 0$ for all x . If $x'Px = 0$, then $x_1 - x_2 = 0$ and $x_2 = 0$; hence $x = 0$. So, $P > 0$.

The following result is useful in checking positive-definiteness of a matrix.

Fact 15 *The following statements are equivalent for any hermitian matrix P .*

- (a) P is positive definite.
- (b) All the eigenvalues of P are positive.
- (c) All the leading principal minors of P are positive; that is,

$$p_{11} > 0$$

$$\det \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} > 0$$

$$\vdots$$

$$\det(P) > 0$$

Example 145 Consider

$$P = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$$

Since $p_{11} = 1 > 0$ and $\det(P) = 1 > 0$, we must have $P > 0$.

```
>> eig(P)
ans =
    0.3820
    2.6180
```

Note the positive eigenvalues.

11.4.3 Semi-definite matrices

DEFN. A hermitian matrix P is positive semi-definite (psd) if

$$x'Px \geq 0 \quad \text{for all } x \text{ in } \mathbb{C}^n$$

We denote this by $P \geq 0$

The matrix P is negative semi-definite (nsd) if $-P$ is positive semi-definite; we denote this by $P \leq 0$

Fact 16 The following statements are equivalent for any hermitian matrix P .

- (a) P is positive semi-definite.
- (b) All the eigenvalues of P are non-negative.
- (c) All the principal minors of P are non-negative.

Example 146 This example illustrates that non-negativity of only the leading principal minors of P is not sufficient for $P \geq 0$. One needs non-negativity of all the principal minors.

$$P = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$$

We have $p_{11} = 0$ and $\det(P) = 0$. However,

$$x'Px = -|x_2|^2$$

hence, P is not psd. Actually, P is nsd.

Fact 17 Consider any $m \times n$ complex matrix M and let $P = M'M$. Then

- (a) P is hermitian and $P \geq 0$
- (b) $P > 0$ if and only if $\text{rank } M = n$.

Example 147

$$P = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Since

$$P = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix}$$

and

$$\text{rank} \begin{pmatrix} 1 & 1 \end{pmatrix} = 1$$

$P \geq 0$ but P is not pd.

Exercise 126 (Optional)

Suppose P is hermitian and T is invertible. Show that $P > 0$ if and only if $T'PT > 0$.

Exercise 127 (Optional)

Suppose P and Q are two hermitian matrices with $P > 0$. Show that $P + \epsilon Q > 0$ for all real ϵ sufficiently small; that is, there exists $\bar{\epsilon} > 0$ such that whenever $|\epsilon| < \bar{\epsilon}$, one has $P + \epsilon Q > 0$.

11.5 Singular value decomposition

In this section we introduce a matrix decomposition which is very useful for *reliable* numerical computations in linear algebra. First, some notation.

The **diagonal elements** of an $m \times n$ matrix A are

$$A_{11}, A_{22}, \dots, A_{pp} \quad \text{where } p = \min\{m, n\}$$

We say that A is **diagonal** if all its nondiagonal elements are zero, that is, $A_{ij} = 0$ for $i \neq j$. Examples are:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Theorem 4 (Singular value decomposition) *If A is a complex $m \times n$ matrix, then there are unitary matrices $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$ such that*

$$\boxed{A = U\Sigma V'}$$

where Σ is a diagonal matrix of the same dimension as A and with real nonnegative diagonal elements:

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_p \geq 0 \quad \text{where } p = \min\{m, n\}$$

If A is real, then U and V are real.

So, if A is square ($m = n$) then,

$$\Sigma = \begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n \end{pmatrix}.$$

When A is “fat” ($m < n$), we have

$$\Sigma = \begin{pmatrix} \sigma_1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & & \ddots & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \sigma_m & 0 & \cdots & 0 \end{pmatrix}.$$

Finally, a “tall” matrix ($m > n$) yields

$$\Sigma = \begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}.$$

The scalar σ_i is called the i -th **singular value** of A . Let σ_r be the smallest nonzero singular value of A . Then r is less than or equal to both m and n . Moreover, Σ can be expressed as

$$\Sigma = \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & 0 \end{pmatrix}$$

where Σ_{11} is the square diagonal invertible matrix given by

$$\Sigma_{11} = \begin{pmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & & 0 \\ 0 & 0 & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_r \end{pmatrix}$$

If

$$U \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & 0 \end{pmatrix} V'$$

is a singular value decomposition of A , then

$$A'A = V \begin{pmatrix} \Sigma_{11}^2 & 0 \\ 0 & 0 \end{pmatrix} V'$$

Hence, the nonzero singular values of A are uniquely given by

$$\sigma_i = \sqrt{\lambda_i}$$

where $\lambda_1, \dots, \lambda_r$ are the nonzero eigenvalues of the hermitian positive semidefinite matrix $A'A$.

Example 148 The matrix

$$A = \begin{pmatrix} 1.2 & -0.64 & -0.48 \\ 1.6 & 0.48 & 0.36 \end{pmatrix}$$

has the following singular value decomposition

$$A = \underbrace{\begin{pmatrix} 0.6 & 0.8 \\ 0.8 & -0.6 \end{pmatrix}}_U \underbrace{\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & -0.8 & 0.6 \\ 0 & -0.6 & -0.8 \end{pmatrix}'}_{V'}$$

Useful facts

- The **maximum (or largest) singular value** of a matrix satisfies the properties of a norm. Thus, if we consider the set of $m \times n$ matrices A and define

$$||A|| = \sigma_{\max}(A)$$

where $\sigma_{\max}(A)$ denotes the largest singular of A , then the following properties hold:

- (a) $\|A\| \geq 0$ and $\|A\| = 0$ if and only if $A = 0$
- (b) $\|\lambda A\| = |\lambda| \|A\|$
- (c) $\|A + B\| \leq \|A\| + \|B\|$.

In addition to the above usual norm properties, this norm also satisfies

$$\|A\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \max_{\|x\|=1} \frac{\|Ax\|}{\|x\|}$$

where the above vector norms are the usual euclidean vector norms. Thus,

$$\|Ax\| \leq \|A\| \|x\|$$

for all x .

- Using the above norm, we can define the **distance between two matrices** A and B of the same dimensions as $\|A - B\|$.
- *The rank of A equals the number of its nonzero singular values.*
- Suppose A has rank r and

$$U = \begin{pmatrix} u^1 & u^2 & \dots & u^m \end{pmatrix}, \quad V = \begin{pmatrix} v^1 & v^2 & \dots & v^n \end{pmatrix}$$

Then the vectors

$$u^1, u^2, \dots, u^r$$

form an orthonormal basis for the range of A and the vectors

$$v^{r+1}, v^{r+2}, \dots, v^n$$

form an orthonormal basis for the null space of A

SVD and numerical computation of rank. The rank of a (non full-rank) matrix can be changed by arbitrarily small changes in the elements of the matrix. So, in computing the rank of a matrix, it is useful to know how close the matrix is to a matrix of lower rank. The SVD allows us determine how close a given matrix is to a matrix of lower rank.

Suppose k is a nonnegative integer strictly less than the rank r of A . Then in a certain sense (which can be made precise), a matrix of rank k which is “closest” to A is given by

$$A_k = U \Sigma_k V'$$

where Σ_k is the diagonal matrix with diagonal elements

$$\sigma_1, \sigma_2, \dots, \sigma_k, \underbrace{0, \dots, 0}_{r-k \text{ zeros}}$$

that is, the first k diagonal elements of Σ_k are the same as those of Σ and the remaining diagonal elements are zero. Also, the “distance” between A_k and A is σ_{k+1} . So, the *smallest nonzero singular value* σ_r of A is a measure of the distance between A and the closest matrix of rank $r-1$.

MATLAB

```
>> help svd
```

SVD Singular value decomposition.
[U,S,V] = SVD(X) produces a diagonal matrix S, of the same dimension as X and with nonnegative diagonal elements in decreasing order, and unitary matrices U and V so that $X = U*S*V'$.

By itself, SVD(X) returns a vector containing diag(S).

[U,S,V] = SVD(X,0) produces the "economy size" decomposition. If X is m-by-n with $m > n$, then only the first n columns of U are computed and S is n-by-n.

Example 149

A =

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

$$[u \ s \ v] = \text{svd}(A)$$

u =

$$\begin{pmatrix} 0.5774 & -0.0000 & -0.8165 & -0.0000 \\ -0.0000 & -1.0000 & -0.0000 & -0.0000 \\ 0.5774 & -0.0000 & 0.4082 & -0.7071 \\ 0.5774 & -0.0000 & 0.4082 & 0.7071 \end{pmatrix}$$

s =

$$\begin{pmatrix} 3.0000 & 0 & 0 \\ 0 & 1.4142 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

v =

$$\begin{pmatrix} 0.5774 & -0.7071 & 0.4082 \\ 0.5774 & 0.7071 & 0.4082 \\ 0.5774 & 0.0000 & -0.8165 \end{pmatrix}$$

So A has rank 2 and a basis for its range is given by

$$\begin{pmatrix} 0.5774 \\ 0 \\ 0.5774 \\ 0.5774 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \end{pmatrix}.$$

A basis for the null space of A is given by

$$\begin{pmatrix} 0.4082 \\ 0.4082 \\ -0.8165 \end{pmatrix}$$

Chapter 12

Lyapunov theory

Here we will look at another way to ascertain the stability properties of a system. It involves Lyapunov theory which is named after its founder, Aleksandr Mikhailovich Lyapunov (1857-1918). The usefulness of the results in this section go beyond the stability of LTI systems. The main result (for continuous-time systems) of this section is contained in Theorem 5.

12.1 Asymptotic stability results

The scalar system

$$\dot{x} = ax$$

is asymptotically stable if and only if $a + a' < 0$. Consider now a general linear time-invariant system described by

$$\dot{x} = Ax \tag{12.1}$$

The generalization of $a + a' < 0$ to this system is $A + A' < 0$. We will see shortly that this condition is sufficient for asymptotic stability; however, as the following example illustrates, it is not necessary.

Example 150 The continuous time system with

$$A = \begin{pmatrix} -1 & 3 \\ 0 & -1 \end{pmatrix}$$

is asymptotically stable. However, the matrix

$$A + A' = \begin{pmatrix} -2 & 3 \\ 3 & -2 \end{pmatrix}$$

is not negative definite.

Since stability is invariant under a similarity transformation T , that is, the stability of A and $T^{-1}AT$ are equivalent for any nonsingular T , we should consider the more general condition

$$T^{-1}AT + T'A'T^{-'} < 0$$

where $T^{-'} := (T^{-1})'$. Pre- and post-multiplying the above inequality by $T^{-'}$ and T^{-1} , respectively, and introducing the hermitian matrix $P := T^{-'}T^{-1}$, yields

$$\boxed{\begin{array}{rcl} P & > & 0 \\ PA + A'P & < & 0 \end{array}} \quad (12.2)$$

We now show that the existence of a hermitian matrix P satisfying these conditions guarantees asymptotic stability.

Lemma 2 *Suppose there is a hermitian matrix P satisfying (12.2). Then system (12.1) is asymptotically stable*

Example 151 Consider

$$A = \begin{pmatrix} -1 & 3 \\ 0 & -1 \end{pmatrix}$$

and

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$$

Here $P > 0$ and $PA + A'P = -Q$ where

$$Q = \begin{pmatrix} 2 & -3 \\ -3 & 8 \end{pmatrix}$$

Since $Q > 0$ we have $PA + A'P < 0$. Hence $\dot{x} = Ax$ is GAS.

PROOF OF LEMMA 2. Suppose there exists a hermitian matrix P which satisfies inequalities (12.2). Consider any eigenvalue λ of A . Let v be an eigenvector corresponding to λ , that is, $Av = \lambda v$ and $v \neq 0$. Since the matrix $PA + A'P$ is negative definite and v is non-zero, we must have

$$v'PAv + v'A'Pv < 0$$

that is

$$2\Re(v'PAv) < 0.$$

Since $v'PAv = \lambda v'Pv$ and $v'Pv$ is real, we obtain

$$2\Re(\lambda)v'Pv < 0.$$

Since $P > 0$, we must have $v'Pv > 0$; hence $\Re(\lambda) < 0$. Since the above holds for every eigenvalue of A , system (12.1) is asymptotically stable. ■

Remark 2 A hermitian matrix P which satisfies inequalities (12.2) will be referred to as a Lyapunov matrix for (12.1) or A .

Remark 3 Conditions (12.2) are referred to as linear matrix inequalities (LMI's). In recent years, efficient numerical algorithms have been developed to solve LMI's or determine that a solution does not exist.

Lyapunov functions. To obtain an alternative interpretation of inequalities (12.2), consider the quadratic function V of the state defined by

$$V(x) := x'Px$$

Since the matrix P is positive definite, this function has a strict global minimum at zero. Using the second inequality in (12.2), it follows that along any non-zero solution $x(\cdot)$ of (12.1),

$$\begin{aligned} \frac{dV(x(t))}{dt} &= \dot{x}'Px + x'P\dot{x} \\ &= (Ax)'Px + x'PAx \\ &= x'(A'P + PA)x \\ &< 0, \end{aligned}$$

that is, $V(x(\cdot))$ is strictly decreasing along any non-zero solution. Intuitively, one then expects $V(x(t))$ to approach the minimum value of zero as t goes to infinity. Since the minimum of V occurs at the origin of the state space, one then expects $x(t)$ to approach zero as t goes to infinity. The function V is called a **Lyapunov function** for the system. The concept of a Lyapunov function readily generalizes to nonlinear systems and has proved a very useful tool in the analysis of the stability properties of nonlinear systems.

Remark 4 Note that a hermitian matrix L is positive(negative)-definite if and only if, for any non-singular matrix T , the matrix $T'LT$ is positive(negative)-definite. Hence, a matrix inequality is equivalent to an inequality obtained by pre- and post-multiplying the original inequality by T' and T where T is any non-singular matrix. Consider the inequalities in (12.2). Defining the hermitian matrix $S := P^{-1}$ and pre- and post-multiplying these inequalities by S yields the equivalent inequalities:

$$\begin{aligned} S &> 0 \\ AS + SA' &< 0 \end{aligned} \tag{12.3}$$

Hence the above lemma can be stated with S replacing P and the preceding inequalities replacing (12.2).

12.1.1 The Lyapunov equation

So far we have shown that if a LTI system has a Lyapunov matrix, then it is AS. Is the converse true? That is, does every AS LTI system have a Lyapunov matrix? And if this is true how does one find a Lyapunov matrix? To answer this question note that satisfaction of the second inequality in (12.2) is equivalent to

$$\boxed{PA + A'P + Q = 0} \tag{12.4}$$

where Q is a hermitian positive definite matrix. This linear matrix equation is known as the **Lyapunov equation**. So one approach to looking for Lyapunov matrices could be to choose a

pd hermitian Q and determine whether the Lyapunov equation has a pd hermitian solution for P .

The following lemma tells us that every AS LTI system has a Lyapunov matrix and a Lyapunov matrix can be obtained by solving the Lyapunov equation with any pd hermitian Q .

Lemma 3 *Suppose system (12.1) is globally asymptotically stable. Then for every matrix Q , the Lyapunov equation (12.4) has a unique solution for P . This solution satisfies*

$$\boxed{P = \int_0^\infty e^{A't} Q e^{At} dt} \quad (12.5)$$

and, if Q is positive definite hermitian then, this solution is also positive-definite hermitian.

PROOF. Suppose system (12.1) is asymptotically stable and consider any matrix Q . Let

$$P = \int_0^\infty e^{A't} Q e^{At} dt$$

This integral exists because each element of e^{At} is exponentially decreasing.

To show that P satisfies the Lyapunov equation (12.4), use the following properties of e^{At} ,

$$e^{At} A = \frac{de^{At}}{dt} \quad A' e^{A't} = \frac{de^{A't}}{dt}$$

to obtain

$$\begin{aligned} PA + A'P &= \int_0^\infty \left(e^{A't} Q e^{At} A + A' e^{A't} Q e^{At} \right) dt \\ &= \int_0^\infty \left(e^{A't} Q \frac{de^{At}}{dt} + \frac{de^{A't}}{dt} Q e^{At} \right) dt \\ &= \int_0^\infty \frac{d(e^{A't} Q e^{At})}{dt} dt \\ &= \lim_{t_1 \rightarrow \infty} \int_0^{t_1} \frac{d(e^{A't} Q e^{At})}{dt} dt \\ &= \lim_{t_1 \rightarrow \infty} \left(e^{A't_1} Q e^{At_1} - Q \right) \\ &= -Q \end{aligned}$$

Suppose Q is pd hermitian. Then it should be clear that P is pd hermitian.

Uniqueness of solution. We now show that if (12.1) is asymptotically stable and the Lyapunov equation (12.4) has a solution then, the solution is unique. To see this, suppose P_1 and P_2 are two solutions to (12.4). Then,

$$(P_2 - P_1)A + A'(P_2 - P_1) = 0$$

Hence,

$$e^{A't}(P_2 - P_1)Ae^{At} + e^{A't}A'(P_2 - P_1)e^{At} = 0$$

that is,

$$\frac{d(e^{A't}(P_2 - P_1)e^{At})}{dt} = 0$$

This implies that, for all t ,

$$e^{A't}(P_2 - P_1)e^{At} = e^{A'0}(P_2 - P_1)e^{A0} = P_2 - P_1.$$

Since $\dot{x} = Ax$ is asymptotically stable, $\lim_{t \rightarrow \infty} e^{At} = 0$. Hence

$$P_2 - P_1 = \lim_{t \rightarrow \infty} e^{A't}(P_2 - P_1)e^{At} = 0.$$

From this it follows that $P_2 = P_1$. ■

Using the above two lemmas, we can now state the main result of this section.

Theorem 5 *The following statements are equivalent.*

- (a) *The system $\dot{x} = Ax$ is asymptotically stable.*
- (b) *There exist positive definite hermitian matrices P and Q satisfying the Lyapunov equation*

$$\boxed{PA + A'P + Q = 0} \tag{12.6}$$

- (c) *For every positive definite hermitian matrix Q , the Lyapunov equation (12.6) has a unique solution for P and this solution is hermitian positive-definite.*

PROOF. The first lemma yields (b) $\implies$ (a). The second lemma says that (a) $\implies$ (c). Hence, (b) $\implies$ (c).

To see that (c) $\implies$ (b), pick any positive definite hermitian Q . So, (b) is equivalent to (c). Also, (c) $\implies$ (a); hence (a) and (c) are equivalent. ■

Example 152 Suppose we wish to know the values of c for which the following system is GAS.

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1 + cx_2 \end{aligned}$$

Here

$$A = \begin{pmatrix} 0 & 1 \\ -1 & c \end{pmatrix}$$

Choosing

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and letting

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{pmatrix}$$

(note we have taken $p_{21} = p_{12}$ because we are looking for symmetric solutions) the Lyapunov equation results in

$$\begin{aligned} -2p_{12} + 1 &= 0 \\ p_{11} + cp_{12} - p_{22} &= 0 \\ 2p_{12} + 2cp_{22} + 1 &= 0 \end{aligned}$$

We consider three cases:

Case i) $c = 0$. No solution; hence no GAS.

Case ii) $c = -1$. A single solution; this solution is pd; hence GAS.

Case iii) $c = 1$. A single solution; this solution is not pd; hence no GAS.

Remark 5 One can state the above theorem replacing P with S and replacing (12.4) with

$$AS + SA' + Q = 0 \tag{12.7}$$

MATLAB.

```
>> help lyap
```

```
LYAP    Lyapunov equation.
```

```
X = LYAP(A,C) solves the special form of the Lyapunov matrix  
equation:
```

$$A*X + X*A' = -C$$

```
X = LYAP(A,B,C) solves the general form of the Lyapunov matrix  
equation:
```

$$A*X + X*B = -C$$

```
See also DLYAP.
```

Note. In MATLAB, A' is the complex conjugate transpose of A (that is A'). Hence, to solve (12.4) one must use $P = \text{lyap}(A', Q)$. Note the prime on A .

```
>> q=eye(2);
```

```

>> a=[0 1; -1 -1];

>> p=lyap(a', q)
p =
    1.5000    0.5000
    0.5000    1.0000

>> p*a + a'*p
ans =
   -1.0000   -0.0000
         0   -1.0000

>> det(p)
ans =
    1.2500

>> eig(p)
ans =
    1.8090
    0.6910

```

12.1.2 Lyapunov functions and performance

Consider a system with performance output z described by

$$\dot{x} = Ax \quad (12.8)$$

$$z = Cx \quad (12.9)$$

where A is asymptotically stable. Suppose that we are concerned with the “size” of $z(\cdot)$. One measure of the size of $z(\cdot)$ is the **RMS value** of $z(\cdot)$ given by

$$\|z(\cdot)\|_2 := \left(\int_0^\infty \|z(t)\|^2 dt \right)^{\frac{1}{2}} \quad (12.10)$$

Consider any initial state x_0 and let

$$J(x_0) := \int_0^\infty \|z(t)\|^2 dt \quad (12.11)$$

where $z(\cdot)$ is the output due to the initial condition $x(0) = x_0$. Sometimes this is called a **cost function**. We now show that

$$J(x_0) = x_0' P x_0 \quad (12.12)$$

where P solves the Lyapunov equation

$$PA + A'P + C'C = 0. \quad (12.13)$$

Since

$$\|z\|^2 = z'z = (Cx)'Cx = x'(C'C)x$$

and $x(t) = e^{At}x_0$, we have

$$\begin{aligned} J(x_0) &= \int_0^\infty x_0' e^{A't} C' C e^{At} x_0 dt \\ &= x_0' \int_0^\infty e^{A't} C' C e^{At} dt x_0 \end{aligned}$$

Hence the desired result.

12.2 Stability results

For stability, the following lemma is the analog of Lemma 12.2.

Lemma 4 *Suppose there is a hermitian matrix P satisfying*

$$P > 0 \tag{12.14a}$$

$$PA + A'P \leq 0 \tag{12.14b}$$

Then system (12.1) is stable.

We do not have the exact analog of Lemma 3. However the following can be shown.

Lemma 5 *Suppose system (12.1) is stable. Then there exist a positive definite hermitian matrix P and a positive semi-definite hermitian matrix Q satisfying the Lyapunov equation (12.4).*

The above lemma does *not* state that for every positive semi-definite hermitian Q the Lyapunov equation has a solution for P . Also, when the Lyapunov equation has a solution, it is not unique. This is illustrated in the following example.

Example 153

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

With

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

we obtain

$$PA + A'P = 0$$

In this example the Lyapunov equation has a pd solution with $Q = 0$; this solution is not unique; any matrix of the form

$$P = \begin{pmatrix} p & 0 \\ 0 & p \end{pmatrix}$$

(where p is arbitrary) is also a solution.

If we consider the psd matrix

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

the Lyapunov equation has no solution.

Combining the above two lemmas results in the following theorem.

Theorem 6 *The following statements are equivalent.*

- (a) *The system $\dot{x} = Ax$ is stable.*
- (b) *There exist a positive definite hermitian matrix P and a positive semi-definite matrix Q satisfying the Lyapunov equation (12.4).*

12.3 Mechanical systems

Recall the simple structure; it is an example of a general class of systems which arise in structures/mechanical/aerospace applications. Using Lyapunov techniques, we provide some simple natural conditions which guarantee stability and asymptotic stability of these systems.

Example 154 (A simple structure) Recall the simple structure (with no inputs) described by

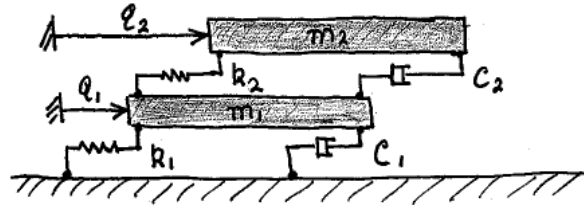


Figure 12.1: A simple structure

$$\begin{aligned} m_1 \ddot{q}_1 + (c_1 + c_2) \dot{q}_1 - c_2 \dot{q}_2 + (k_1 + k_2) q_1 - k_2 q_2 &= 0 \\ m_2 \ddot{q}_2 - c_2 \dot{q}_1 + c_2 \dot{q}_2 - k_2 q_1 + k_2 q_2 &= 0 \end{aligned}$$

Letting

$$q = \begin{pmatrix} q_1 \\ q_2 \end{pmatrix}$$

this system can be described by the following second order vector differential equation:

$$M \ddot{q} + C \dot{q} + K q = 0$$

where the symmetric matrices M, C and K are given by

$$M = \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix} \quad C = \begin{pmatrix} c_1 + c_2 & -c_2 \\ -c_2 & c_2 \end{pmatrix} \quad K = \begin{pmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{pmatrix}$$

Since $m_1, m_2 > 0$, we have $M > 0$. If $k_1, k_2 > 0$ then $K > 0$ (why?) and if $c_1, c_2 > 0$ then $C > 0$.

Note also that

$$\text{kinetic energy} = \frac{1}{2}m_1\dot{q}_1^2 + \frac{1}{2}m_2\dot{q}_2^2 = \frac{1}{2}\dot{q}^T M \dot{q}$$

$$\text{potential energy} = \frac{1}{2}k_1q_1^2 + \frac{1}{2}k_2(q_2 - q_1)^2 = \frac{1}{2}q^T K q$$

Consider now a general linear mechanical system described by

$$\boxed{M\ddot{q} + C\dot{q} + Kq = 0}$$

where the real N -vector $q(t)$ is a vector of generalized coordinates which describes the configuration of the system. The real matrix M is the system **inertia matrix** or **mass matrix** and satisfies

$$\boxed{M^T = M > 0}$$

We call K and C the ‘**stiffness**’ matrix and ‘**damping**’ matrix respectively, associated with the system.

The kinetic energy of the system is given by

$$\frac{1}{2}\dot{q}^T M \dot{q}$$

The potential energy of the system is defined to be

$$\frac{1}{2}q^T K q$$

With $x = (q^T \dot{q}^T)^T$, this system has a state space description of the form $\dot{x} = Ax$ with

$$A = \begin{pmatrix} 0 & I \\ -M^{-1}K & -M^{-1}C \end{pmatrix}$$

A first Lyapunov matrix. Suppose

$$\boxed{K^T = K > 0 \quad C^T = C \geq 0}$$

Consider the following candidate Lyapunov matrix

$$P = \frac{1}{2} \begin{pmatrix} K & 0 \\ 0 & M \end{pmatrix}$$

Then $P' = P > 0$ and

$$PA + A'P + Q = 0$$

where

$$Q = \begin{pmatrix} 0 & 0 \\ 0 & C \end{pmatrix}$$

It should be clear that

$$PA + A'P \leq 0$$

if and only if

$$C \geq 0$$

Hence,

$K^T = K > 0 \text{ and } C^T = C \geq 0 \text{ imply stability.}$
--

Note that

$$\begin{aligned} V(x) &= x^T P x = \frac{1}{2} \dot{q}^T M \dot{q} + \frac{1}{2} q^T K q = \text{total energy} \\ \dot{V}(x) &= -\dot{q}^T C \dot{q} = \text{power of the damping forces} \end{aligned}$$

This yields a physically appealing interpretation of the above results. The total energy has a global minimum at the origin and the energy does not increase along solutions.

A second Lyapunov matrix. If C is positive-definite, we expect the system damping to dissipate all the energy in the system and result in GAS. However, with the Lyapunov results presented so far, the above Lyapunov matrix will only guarantee stability and not GAS. Here we modify the above Lyapunov matrix to achieve GAS when $C > 0$. So suppose

$$K^T = K > 0 \quad \text{and} \quad C^T = C > 0.$$

Consider the following candidate Lyapunov matrix:

$$P = \frac{1}{2} \begin{pmatrix} K + \epsilon C & \epsilon M \\ \epsilon M & M \end{pmatrix} = \frac{1}{2} \begin{pmatrix} K & 0 \\ 0 & M \end{pmatrix} + \frac{\epsilon}{2} \begin{pmatrix} C & M \\ M & 0 \end{pmatrix}$$

For sufficiently small $\epsilon > 0$, P is positive definite.

Now,

$$PA + A'P + Q = 0$$

where

$$Q = \begin{pmatrix} \epsilon K & 0 \\ 0 & C - \epsilon M \end{pmatrix}$$

For sufficiently small $\epsilon > 0$, the matrix $C - \epsilon M$ is pd and, hence, Q is pd. So,

$K^T = K > 0 \text{ and } C^T = C > 0 \text{ imply asymptotic stability.}$
--

Thus, we can conclude without computing eigenvalues that, a general linear mechanical system with positive definite symmetric mass, stiffness and damping matrices is globally asymptotically stable.

12.3.1 A simple stabilization procedure*

Remember Beavis and Butthead:

$$\begin{aligned} m\ddot{q}_1 &= k(q_2 - q_1) \\ m\ddot{q}_2 &= -k(q_2 - q_1) \end{aligned}$$

They are described by

$$M\ddot{q} + C\dot{q} + Kq = 0$$

with

$$M = \begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix} \quad C = 0 \quad K = \begin{pmatrix} k & -k \\ -k & k \end{pmatrix}$$

This system is unstable!

Note that K is not positive definite; it is positive semi-definite. Suppose we could ‘modify’ this system by adding restraining springs and damping so that C and K become

$$C_s = c_s I = \begin{pmatrix} c_s & 0 \\ 0 & c_s \end{pmatrix}$$

$$K_s = K + k_s I = \begin{pmatrix} k + k_s & -k \\ -k & k + k_s \end{pmatrix}$$

respectively, where $k_s, c_s > 0$.

Then, $K_s^T = K_s > 0$ and $C_s^T = C_s > 0$; hence the modified system is asymptotically stable.

A general procedure. First we need the following general result.

Fact 18 *Consider any two hermitian matrices P and Q with $P > 0$. Then all the eigenvalues of $P^{-1}Q$ are real and the hermitian matrix*

$$Q + \sigma P$$

is positive definite for any σ satisfying

$$\sigma > -\lambda_{\min}(P^{-1}Q)$$

Consider a system described by

$$M\ddot{q} + C\dot{q} + Kq = 0$$

where $q(t) \in \mathbb{R}^N$,

$$M^T = M > 0$$

and

$$C^T = C \quad K^T = K$$

Suppose one can modify C and K to

$$C_s = C + c_s W \quad K_s = K + k_s W$$

respectively, where W is a fixed matrix with $W^T = W > 0$ and k_s and c_s can be selected. If we choose c_s and k_s to satisfy

$$c_s > -\lambda_{\min}(W^{-1}C) \quad k_s > -\lambda_{\min}(W^{-1}K)$$

then using above fact we obtain,

$$C_s^T = C_s > 0 \quad K_s^T = K_s > 0$$

and hence the modified system

$$M\ddot{q} + C_s\dot{q} + K_sq = 0$$

is asymptotically stable.

12.4 DT Lyapunov theory

The results in this section are for LTI discrete-time systems described by

$$x(k+1) = Ax(k) \quad (12.15)$$

The results for discrete-time systems are analogous to those for continuous-time systems. However, the Lyapunov equation for discrete-time systems is different than that for continuous-time systems; it is given by

$$\boxed{A'PA - P + Q = 0} \quad (12.16)$$

12.4.1 Asymptotic stability results

Lemma 6 Suppose there is a hermitian matrix P satisfying

$$\begin{aligned} P &> 0 \\ A'PA - P &< 0 \end{aligned}$$

Then system (12.15) is asymptotically stable.

Lemma 7 Suppose system (12.15) is asymptotically stable. Then for every matrix Q , the discrete Lyapunov equation (12.16) has a unique solution for P . This solution satisfies

$$\boxed{P = \sum_{k=0}^{\infty} (A')^k Q A^k}$$

and, if Q is positive definite hermitian then, this solution is also positive-definite hermitian.

PROOF. Let

$$P = \sum_{k=0}^{\infty} (A')^k Q A^k$$

and proceed in a similar fashion to the continuous-time case. ■

Theorem 7 The following statements are equivalent.

- (a) The system $x(k+1) = Ax(k)$ is asymptotically stable.
- (b) There exist positive definite hermitian matrices P and Q satisfying the discrete Lyapunov equation (12.16).
- (c) For every positive definite hermitian matrix Q , the discrete Lyapunov equation (12.16) has a unique solution for P and this solution is hermitian positive-definite.

12.4.2 Stability results

Theorem 8 The following statements are equivalent.

- (a) The system $x(k+1) = Ax(k)$ is stable.
- (b) There exist a positive definite hermitian matrix P and a positive semi-definite matrix Q satisfying the discrete Lyapunov equation (12.16).

Exercises

Exercise 128 Determine (by hand) whether each one of the following matrices is pd, psd, nd, nsd, or none of the above.

$$\begin{pmatrix} 1 & j & 0 \\ -j & 2 & 1 \\ 0 & 1 & 4 \end{pmatrix} \quad \begin{pmatrix} 1 & j \\ -j & 1 \end{pmatrix} \quad \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} \quad \begin{pmatrix} -1 & 1 \\ 1 & -2 \end{pmatrix}$$

Check your answers using the MATLAB command eig.

Exercise 129 Suppose P is a hermitian matrix. Prove that the matrix

$$P + \sigma I$$

is positive definite for any σ which satisfies $\sigma > -\lambda_{\min}(P)$.

Exercise 130 Using the Lyapunov equation determine (by hand) whether or not the system $\dot{x} = Ax$ is asymptotically stable for each one of the following A matrices.

$$\begin{pmatrix} -1 & 2 \\ 0 & -1 \end{pmatrix} \quad \begin{pmatrix} -1 & 2 \\ 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

Check your answers using the MATLAB command lyap.

Exercise 131 Using Lyapunov theory, determine whether or not the following system is asymptotically stable.

$$\begin{aligned} \dot{x}_1 &= x_1 + 2x_2 \\ \dot{x}_2 &= 2x_1 + x_2 \end{aligned}$$

Exercise 132 Consider a system with disturbance input w and performance output z described by

$$\begin{aligned} \dot{x}_1 &= -3x_1 + x_2 + w \\ \dot{x}_2 &= x_1 - 3x_2 + w \\ y &= x_1 + x_2 \end{aligned}$$

Without performing any integration, determine

$$\int_0^\infty |z(t)|^2 dt$$

for each of the following situations.

- (a) $w = 0$ and $x(0) = (1, -1)$.
- (b) $w = \delta(t)$ and $x(0) = (0, 0)$
- (c) $w(t) = 2\delta(t)$ and $x(0) = (2, 0)$.

Exercise 133 Consider the Lyapunov equation

$$PA + A'P + 2\alpha P + Q = 0$$

where A is a square matrix, Q is some positive-definite hermitian matrix and α is some positive scalar. *Show* that this equation has a unique solution for P and this solution is positive-definite hermitian if and only if for every eigenvalue λ of A ,

$$\Re(\lambda) < -\alpha$$

Exercise 134 Can you obtain a matrix P which guarantees stability of the continuous time system with the following A matrix.

$$A = \begin{pmatrix} -1 & 2 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

Exercise 135 Prove that any system of the form

$$M\dot{x} = -Cx$$

with $x(t) \in \mathbb{C}^n$ and

$$M' = M > 0 \quad C + C' > 0$$

is asymptotically stable.

Exercise 136 Prove that if

$$x(k+1) = Ax(k)$$

is asymptotically stable, then for any matrix Q , the solution of

$$A'PA - P + Q = 0$$

is given by

$$P = \sum_{k=0}^{\infty} (A')^k Q A^k$$

Exercise 137 A matrix A is *skew hermitian* if

$$A' = -A$$

Show that if A is skew hermitian, then e^{At} is unitary and every solution of

$$\dot{x} = Ax$$

satisfies $\|x(t)\| \equiv \|x(0)\|$.

Exercise 138 Consider the system with performance output z described by

$$\begin{aligned} \dot{x}_1 &= -x_1 + x_2 \\ \dot{x}_2 &= -x_1 + 4x_2 \\ z &= x_1. \end{aligned}$$

Using an appropriate Lyapunov equation, determine $\int_0^{\infty} \|z(t)\|^2 dt$ for $x(0) = (1, 0)$.

Chapter 13

Observability

13.1 Observability

There are many situations where one needs knowledge of a variable associated with the behavior of a system but cannot measure that variable but can measure other variables. This gives rise to the problem of estimating the unknown variable from the measured variables. We consider here the problem of determining the state of a system based on measurements of its inputs and outputs. This introduces the important concept of **observability**. Roughly speaking, a system is observable if its state can be determined from its inputs and outputs.

In this section, we provide a simple characterization of observability for LTI systems. In a later chapter on observers, we explicitly show how to obtain an estimate of a system state from its inputs and outputs. Consider the LTI system

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}\tag{13.1}$$

with initial condition

$$x(0) = x_0$$

where the n -vector $x(t)$ is the state, the m -vector $u(t)$ is the input and the p -vector $y(t)$ is a vector of measured outputs; we call y the **measured output (vector)**.

Consider any time interval $[0, T]$ with $T > 0$. The basic observability problem is as follows. Suppose we have knowledge of the input $u(t)$ and the output $y(t)$ over the interval $[0, T]$; can we uniquely determine the initial state x_0 ? If we can do this for all input histories and initial states x_0 we say that the system is **observable**. So, for observability, we require that for each input history, different initial states produce different output histories, or, equivalently, if two output histories are identical, then the corresponding initial states must be the same. A formal definition of observability follows the next example. Note that knowledge of the initial state and the input history over the interval $[0, T]$ allows one to compute the state $x(t)$ for $0 \leq t \leq T$. This follows from the relationship

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu(\tau) d\tau.$$

Example 155 Two unattached masses

$$\begin{aligned}\dot{x}_1 &= 0 \\ \dot{x}_2 &= 0 \\ y &= x_1\end{aligned}$$

Clearly, this system is unobservable.

DEFN. System (13.1) is **observable** over an interval $[0, T]$ if the following holds for each input history $u(\cdot)$. If $y^a(\cdot)$ and $y^b(\cdot)$ are any two output histories of system (13.1) due to initial states x_0^a and x_0^b , respectively, and $y^a(t) = y^b(t)$ for $0 \leq t \leq T$; then $x_0^a = x_0^b$.

Thus, observability means that different initial states produce different output histories. If two different initial states produce the same output history then they cannot be distinguished by looking only at the system input and output

13.2 A simple characterization of observability

To describe the main results associated with observability of LTI systems, we introduce the **observability matrix** Q_o associated with system (13.1). This is the following $pn \times n$ matrix

$$Q_o = \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{pmatrix}$$

The following result is the main observability result for continuous-time LTI systems.

Theorem 9 (Main observability result) For any $T > 0$, system (13.1) is observable over the interval $[0, T]$ if and only if its observability matrix has maximum rank, that is,

$$\text{rank} \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{pmatrix} = n$$

where n is the number of state variables.

An immediate consequence of the above theorem is that observability does not depend on the interval $[0, T]$. So, from now on we drop reference to the interval. Since observability only depends on the matrix pair (C, A) , we say that (C, A) is *observable* if system (13.1) is observable.

SO systems. For scalar output ($p = 1$) systems, Q_o is a square $n \times n$ matrix; hence it has rank n if and only if its determinant is nonzero. So, the above theorem has the following corollary.

Corollary 1 *A scalar output system of the form (13.1) is observable if and only if its observability matrix Q_o has non-zero determinant.*

Example 156 (Two unattached masses.) Recall example 155. Here $n = 2$,

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

Hence,

$$Q_o = \begin{pmatrix} C \\ CA \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Since $\text{rank } Q_o = 1 < n$ or $\det(Q_o) = 0$, this system is unobservable.

Example 157 (The unattached mass)

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= 0 \end{aligned}$$

We consider two cases:

(a) (Velocity measurement.) Suppose $y = x_2$. Then

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 0 & 1 \end{pmatrix}$$

So,

$$Q_o = \begin{pmatrix} C \\ CA \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Since $\text{rank } Q_o = 1 < n$ or $\det(Q_o) = 0$, we have unobservability.

(b) (Position measurement.) Suppose $y = x_1$. Then

$$Q_o = \begin{pmatrix} C \\ CA \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Since $\text{rank } Q_o = 2 = n$ or $\det(Q_o) = 1$, we have observability.

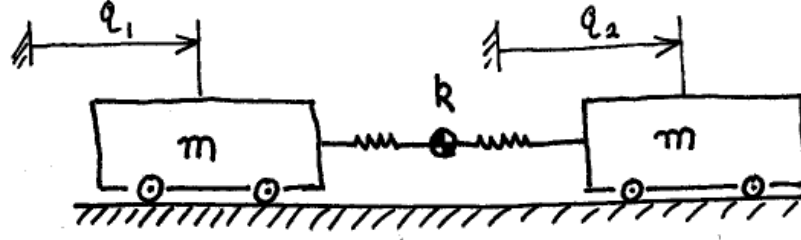


Figure 13.1: Example 158

Example 158 (Beavis and Butthead: mass center measurement)

$$\begin{aligned} m\ddot{q}_1 &= k(q_2 - q_1) \\ m\ddot{q}_2 &= -k(q_2 - q_1) \\ y &= q_1 + q_2 \end{aligned}$$

Actually, y is not the location of the mass center of the system, however there is a one-to-one correspondence between y and the mass center location. Intuitively, one can see why this system is not observable. The system can oscillate with the mass center fixed, that is, a non-zero state motion results in zero output history.

Let us now apply the observeability test. With

$$x_1 = q_1 \quad x_2 = q_2 \quad x_3 = \dot{q}_1 \quad x_4 = \dot{q}_2$$

we have

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -k/m & k/m & 0 & 0 \\ k/m & -k/m & 0 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 1 & 0 & 0 \end{pmatrix}$$

Since,

$$\begin{aligned} CA &= \begin{pmatrix} 0 & 0 & 1 & 1 \end{pmatrix} \\ CA^2 &= (CA)A = \begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix} \\ CA^3 &= (CA^2)A = \begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

the observability matrix for this system is

$$Q_o = \begin{pmatrix} C \\ CA \\ CA^2 \\ CA^3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Since $\text{rank } Q_o = 2 < n$, this system is not observable.

Example 159 (Beavis and Butthead: observable) Here we take the system from the last example and consider the measured output to be the position of the first mass.

$$\begin{aligned} m\ddot{q}_1 &= k(q_2 - q_1) \\ m\ddot{q}_2 &= -k(q_2 - q_1) \\ y &= q_1 \end{aligned}$$

With

$$x_1 = q_1 \quad x_2 = q_2 \quad x_3 = \dot{q}_1 \quad x_4 = \dot{q}_2$$

we have

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -k/m & k/m & 0 & 0 \\ k/m & -k/m & 0 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix}$$

One can argue as follows that this system is observable: If $y = x_1$ is known over an interval then so is $x_3 = \dot{x}_1$ and $\dot{x}_3$. But, $\dot{x}_3 = -k/mx_1 + k/mx_2$. Since $k \neq 0$, x_2 can now be determined. Finally x_4 can be obtained from $x_4 = \dot{x}_2$.

Let us now apply the observability test to verify this.

$$CA = \begin{pmatrix} 0 & 0 & 1 & 0 \end{pmatrix}$$

$$CA^2 = (CA)A = \begin{pmatrix} -k/m & k/m & 0 & 0 \end{pmatrix}$$

$$CA^3 = (CA^2)A = \begin{pmatrix} 0 & 0 & -k/m & k/m \end{pmatrix}$$

So,

$$Q_o = \begin{pmatrix} C \\ CA \\ CA^2 \\ CA^3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -k/m & k/m & 0 & 0 \\ 0 & 0 & -k/m & k/m \end{pmatrix}.$$

Since $\text{rank } Q_o = 4 = n$; this system is observable.

A useful fact for checking observability. In checking observability, the following fact is useful. Suppose there exists n^* such that

$$\mathcal{N} \left(\begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n^*-1} \\ CA^{n^*} \end{pmatrix} \right) = \mathcal{N} \left(\begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n^*-1} \end{pmatrix} \right)$$

or, equivalently,

$$\text{rank} \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n^*-1} \\ CA^{n^*} \end{pmatrix} = \text{rank} \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n^*-1} \end{pmatrix}$$

Then

$$\mathcal{N}(Q_o) = \mathcal{N} \left(\begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n^*-1} \end{pmatrix} \right)$$

and

$$\text{rank}(Q_o) = \text{rank} \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n^*-1} \end{pmatrix}$$

Example 160 (Beavis and Butthead: mass center measurement revisited)

$$\begin{aligned} m\ddot{q}_1 &= k(q_2 - q_1) \\ m\ddot{q}_2 &= -k(q_2 - q_1) \\ y &= q_1 + q_2 \end{aligned}$$

With

$$x_1 = q_1 \quad x_2 = q_2 \quad x_3 = \dot{q}_1 \quad x_4 = \dot{q}_2$$

we have

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -k/m & k/m & 0 & 0 \\ k/m & -k/m & 0 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 1 & 0 & 0 \end{pmatrix}$$

Hence,

$$CA = \begin{pmatrix} 0 & 0 & 1 & 1 \end{pmatrix}$$

$$CA^2 = \begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix}$$

So, using useful fact (b) above,

$$\text{rank}(Q_o) = \text{rank} \begin{pmatrix} C \\ CA \end{pmatrix} = \text{rank} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = 2$$

Since $\text{rank}(Q_o) < n$, this system is not observable.

Another useful fact for checking observability*. Suppose $V^1, \dots, V^k$ is a sequence of matrices which satisfy

$$\mathcal{N}(V^1) = \mathcal{N}(C)$$

$$\mathcal{N} \left(\begin{pmatrix} V^1 \\ \vdots \\ V^{j-1} \\ V^j \end{pmatrix} \right) = \mathcal{N} \left(\begin{pmatrix} V^1 \\ \vdots \\ V^{j-1} \\ V^{j-1}A \end{pmatrix} \right) \quad \text{for } j = 2, \dots, k$$

Then

$$\mathcal{N} \left(\begin{pmatrix} C \\ \vdots \\ CA^k \\ CA^{k+1} \end{pmatrix} \right) = \mathcal{N} \left(\begin{pmatrix} V^1 \\ \vdots \\ V^k \\ V^k A \end{pmatrix} \right)$$

and

$$\text{rank} \begin{pmatrix} C \\ \vdots \\ CA^k \\ CA^{k+1} \end{pmatrix} = \text{rank} \begin{pmatrix} V^1 \\ \vdots \\ V^k \\ V^k A \end{pmatrix}$$

Example 161 (Beavis and Buttthead: observable*)

$$\begin{aligned} m\ddot{q}_1 &= k(q_2 - q_1) \\ m\ddot{q}_2 &= -k(q_2 - q_1) \\ y &= q_1 \end{aligned}$$

With

$$x_1 = q_1 \quad x_2 = q_2 \quad x_3 = \dot{q}_1 \quad x_4 = \dot{q}_2$$

we have

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -k/m & k/m & 0 & 0 \\ k/m & -k/m & 0 & 0 \end{pmatrix} \quad C = (1 \ 0 \ 0 \ 0)$$

Hence,

$$CA = (0 \ 0 \ 1 \ 0)$$

$$CA^2 = (-k/m \ k/m \ 0 \ 0)$$

So,

$$\text{rank} \begin{pmatrix} C \\ CA \\ CA^2 \end{pmatrix} = \text{rank} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

and

$$\text{rank}(Q_o) = \text{rank} \begin{pmatrix} C \\ CA \\ CA^2 \\ CA^3 \end{pmatrix} = \text{rank} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = 4$$

Since $\text{rank } Q_o = n$; this system is observable.

13.3 Development of the main result

To develop the main result presented in the last section, we first demonstrate the following result:

System (13.1) is observable over an interval $[0, T]$ if and only if the uncontrolled system ($u(t) \equiv 0$)

$$\begin{aligned}\dot{x} &= Ax \\ y &= Cx\end{aligned}\tag{13.2}$$

has the property that the zero state is the only initial state which results in an output history which is identically zero; that is, it has the following property:

$$y(t) = 0 \quad \text{for } 0 \leq t \leq T \quad \text{implies} \quad x(0) = 0$$

To see that the above property is necessary for observability, simply note that the zero initial state produces a zero output history; hence if some nonzero initial state produces a zero output history, then there are two different initial states producing the same output history and the system is not observable.

To see that the above property is sufficient, consider any input history $u(\cdot)$ and suppose $y^a(\cdot)$ and $y^b(\cdot)$ are two output histories due to initial states x_0^a and x_0^b , respectively, and $y^a(\cdot) = y^b(\cdot)$. Let $x^a(\cdot)$ and $x^b(\cdot)$ be the state histories corresponding the x_0^a and x_0^b , respectively, that is,

$$\begin{aligned}\dot{x}^a &= Ax^a + Bu, & x^a(0) &= x_0^a \\ y^a &= Cx^a + Du\end{aligned}$$

and

$$\begin{aligned}\dot{x}^b &= Ax^b + Bu, & x^b(0) &= x_0^b \\ y^b &= Cx^b + Du\end{aligned}$$

Then, letting

$$\begin{aligned}x(t) &:= x^b(t) - x^a(t) \\ y(t) &:= y^b(t) - y^a(t)\end{aligned}$$

we obtain

$$\begin{aligned}\dot{x} &= Ax, & x(0) &= x_0^b - x_0^a \\ y &= Cx\end{aligned}$$

Since $y(t) \equiv 0$ we must have $x_0^b - x_0^a = 0$; hence $x_0^b = x_0^a$. So, system (13.1) is observable.

Example 162 (The unattached mass)

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= 0\end{aligned}$$

If $y = x_2$, the system is unobservable: consider nonzero initial state $x_0 = [1 \ 0]^T$; the corresponding output history satisfies $y(t) \equiv 0$. If $y = x_1$, the system is observable. To see this note that $x = [y, \dot{y}]^T$. If $y(t) \equiv 0$ over a nonzero interval, then $\dot{y}(0) = 0$ and hence, $x(0) = 0$.

We have just shown that to investigate the observability of system (13.1), we need only determine the initial states which result in an identically zero output for the system with zero input, that is system (13.2). So, let us look at the consequences of a zero output for the above system. So, suppose x_0 is an initial state which, for some $T > 0$, results in $y(t) = 0$ for $0 \leq t \leq T$, that is,

$$Cx(t) = 0 \quad \text{for } 0 \leq t \leq T \quad (13.3)$$

where

$$\dot{x}(t) = Ax(t), \quad x(0) = x_0.$$

Differentiating the equation in (13.3) with respect to t results in

$$0 = C\dot{x}(t) = CAx(t) = 0 \quad \text{for } 0 \leq t \leq T.$$

Repeated differentiation of (13.3) yields:

$$CA^k x(t) = 0 \quad \text{for } 0 \leq t \leq T \quad \text{and } k = 0, 1, 2, \dots$$

Letting $t = 0$ and using the initial condition $x(0) = x_0$ results in

$$CA^k x_0 = 0 \quad \text{for } k = 0, 1, 2, \dots \quad (13.4)$$

So, we have shown that if an initial state x_0 results in a zero output over some nonzero interval then, (13.4) holds.

Let us now demonstrate the converse, that is, if (13.4) holds, then the resulting output is zero over any nonzero interval. So, suppose (13.4) holds for some initial state x_0 . Recall that $x(t) = e^{At}x_0$; hence

$$y(t) = Ce^{At}x_0$$

Since e^{At} can be expressed as $e^{At} = \sum_{k=0}^{\infty} \beta_k(t)A^k$, we have

$$y(t) = Ce^{At}x_0 = \sum_{k=0}^{\infty} \beta_k(t)CA^k x_0 = 0.$$

So, we now conclude that an initial state x_0 results in a zero output over some nonzero interval if and only if (13.4) holds. Since, for each $k = 0, 1, 2, \dots$, the matrix A^k can be

expressed as a linear combination of $I, A, A^2, \dots, A^{n-1}$, it follows that (13.4) holds if and only if

$$CA^k x_0 = 0 \quad \text{for } k = 0, 1, 2, n-1. \quad (13.5)$$

Thus an initial state x_0 results in a zero output over some nonzero interval if and only if (13.5) holds. Recalling the definition of the **observability matrix** Q_o associated with system (13.1) we obtain the following result:

An initial state x_0 results in zero output over some nonzero interval if and only if $Q_o x_0 = 0$

Hence, system (13.1) is observable if and only if the only vector satisfying $Q_o x_0 = 0$ is the zero vector. This immediately leads to the following result on observability. Recall that the nullity of a matrix is the dimension of its null space.

For any $T > 0$, system (13.1) is observable over the interval $[0, T]$ if and only if the nullity of its observability matrix is zero.

Since the rank and nullity of Q_o must sum to n which is the maximum of Q_o , we now obtain main result, that is Theorem 9.

13.4 The unobservable subspace

Example 163 At first sight, the system

$$\begin{aligned} \dot{x}_1 &= x_1 \\ \dot{x}_2 &= x_2 \\ y &= x_1 + x_2 \end{aligned}$$

might look observable. However, it is not. To see this, introduce new state variables

$$x_o := x_1 + x_2 \quad \text{and} \quad x_{uo} := x_1 - x_2$$

Then

$$\begin{aligned} \dot{x}_o &= x_o \\ \dot{x}_u &= x_u \\ y &= x_o \end{aligned}$$

Hence, $x_o(0) = 0$ implies $x_o(t) = 0$ and, hence, $y(t) = 0$ for all t . In other words if the initial state $x(0)$ is in the set

$$\{x \in \mathbb{R}^2 : x_1 + x_2 = 0\}$$

The resulting output history is zero.

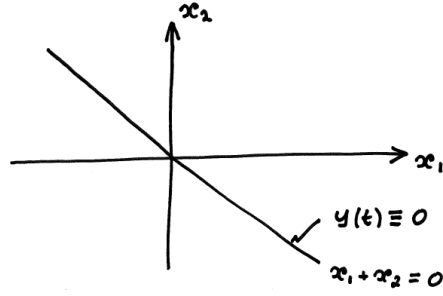


Figure 13.2: Example 163

Recall that observability matrix Q_o associated with (C, A) is defined to be the following $pn \times n$ matrix

$$Q_o = \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{pmatrix}$$

Recall also that an initial state x_0 results in a zero output if and only if $Q_o x_0 = 0$, that is x_0 is in the null space of Q_o . So, we define the **unobservable subspace** $\mathcal{X}_{uo}$ associated with system (13.1) or (C, A) to be the null space of the observability matrix, that is,

$$\boxed{\mathcal{X}_{uo} = \mathcal{N}(Q_o)}$$

and we have the following result.

Lemma 8 (Unobservable subspace lemma) *For any nonzero time interval $[0, T]$, the following statements are equivalent for system (13.2).*

- (i) $y(t) = 0$ for $0 \leq t \leq T$.
- (ii) $Q_o x(0) = 0$, or, equivalently, $x(0)$ is in the unobservable subspace $\mathcal{X}_{uo}$.

Example 164 Recalling the last example we have $n = 2$ and

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 1 \end{pmatrix}$$

Hence,

$$Q_o = \begin{pmatrix} C \\ CA \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Clearly,

$$\mathcal{X}_{uo} = \mathcal{N}(Q_o) = \mathcal{N} \left(\begin{pmatrix} 1 & 1 \end{pmatrix} \right) = \mathcal{N}(C)$$

Hence

$$\mathcal{N}(Q_o) = \{x \in \mathbb{R}^2 : x_1 + x_2 = 0\}$$

13.5 Unobservable eigenvalues and modes

Suppose we are interested in the observability of the the input-output system

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}$$

Since observability is independent of u , we only need to consider the corresponding zero input system:

$$\begin{aligned}\dot{x} &= Ax \\ y &= Cx\end{aligned}\tag{13.6}$$

Recall that a mode of this system is a special solution of the form $x(t) = e^{\lambda t}v$. Such a solution results when $x(0) = v$ and v is an eigenvector of A with eigenvalue λ . We say that $e^{\lambda t}v$ is an **unobservable mode** of the above system if, the above system has a solution of the form $x(t) = e^{\lambda t}v$ and the corresponding output is zero, that is,

$$\boxed{x(t) \equiv e^{\lambda t}v \quad \text{and} \quad y(t) \equiv 0}$$

The eigenvalue λ is called an **unobservable eigenvalue** of the system. We also refer to λ as an unobservable eigenvalue of (C, A) .

Clearly, if a system has an unobservable mode then, the system is not observable. In the next section, we show that if an LTI system is not observable, then it must have an unobservable mode. Thus:

An LTI system is observable if and only if it has no unobservable modes.

This is illustrated in the following example.

Example 165

$$\begin{aligned}
\dot{x}_1 &= \lambda_1 x_1 + b_1 u \\
\dot{x}_2 &= \lambda_2 x_2 + b_2 u \\
y &= c_1 x_1 + c_2 x_2 + d u
\end{aligned}$$

Since

$$Q_o = \begin{pmatrix} C \\ CA \end{pmatrix} = \begin{pmatrix} c_1 & c_2 \\ \lambda_1 c_1 & \lambda_2 c_2 \end{pmatrix}$$

This system is observable if and only if

$$c_1 \neq 0, c_2 \neq 0, \lambda_1 \neq \lambda_2$$

Note that if $c_1 = 0$ then considering $u(t) \equiv 0$ this system has a solution

$$x(t) = e^{\lambda_1 t} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

and the corresponding output satisfies

$$y(t) \equiv 0$$

13.5.1 PBH Test for unobservable eigenvalues

The next lemma provides a useful characterization of unobservable eigenvalues.

Lemma 9 *A complex number λ is an unobservable eigenvalue of the pair (C, A) if and only if the matrix*

$$\begin{pmatrix} A - \lambda I \\ C \end{pmatrix} \tag{13.7}$$

does not have maximum rank (that is n , where n is the number of columns in A .)

PROOF. Recall that λ is an unobservable eigenvalue of (C, A) if the system $\dot{x} = Ax$ has a non-zero solution of the form $x(t) = e^{\lambda t}v$ which satisfies $Cx(t) \equiv 0$. We have already seen that the existence of a nonzero solution of the form $e^{\lambda t}v$ is equivalent to v being an eigenvector of A with eigenvalue λ , that is, $v \neq 0$ and $Av = \lambda v$; this is equivalent to

$$(A - \lambda I)v = 0$$

and $v \neq 0$. Since $x(t) = e^{\lambda t}v$ and $e^{\lambda t} \neq 0$, the statement that $Cx(t) \equiv 0$ is equivalent to

$$Cv = 0.$$

So, we have shown that λ is an unobservable eigenvalue of (C, A) if and only if there is a nonzero vector satisfying

$$\begin{pmatrix} A - \lambda I \\ C \end{pmatrix} v = 0.$$

The existence of a nonzero v satisfying the above equality is equivalent to saying that the columns of the matrix in the equality are linearly dependent, that is, the rank of the matrix is less than the number of its columns. Since the number of columns in the above matrix is less than or equal to the number of its rows, the maximum rank of this matrix equals the number of its columns. Hence, λ being an unobservable eigenvalue of (C, A) is equivalent to the matrix in (13.7) not having maximum rank. ■

Example 166 Consider a system with

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad C = (1 \quad 1) \quad D = 0.$$

Here, the observability matrix is

$$Q_o = \begin{pmatrix} C \\ CA \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Since Q_o has zero determinant, this system is not observable. Hence, the system has some unobservable modes. The eigenvalues of A are 1 and -1 . To determine the unobservable eigenvalues, we carry out the PBH test:

$$\begin{pmatrix} A - \lambda I \\ C \end{pmatrix} = \begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \\ 1 & 1 \end{pmatrix}$$

For $\lambda = -1$, the above matrix is rank deficient; hence -1 is an unobservable eigenvalue. For $\lambda = 1$, the PBH matrix has maximum rank. Hence, this eigenvalue is not unobservable.

Note that the transfer function for this system is given by

$$\hat{G}(s) = \frac{s+1}{s^2-1} = \frac{1}{s-1}.$$

So, *the unobservable eigenvalue is not a pole of the transfer function*. This transfer function can be realized with $A = B = C = 1$ and $D = 0$.

Since we have already stated that an LTI system is observable if and only if it has no unobservable eigenvalues, we obtain the following corollary to Lemma 9. This new result provides another useful test for observability.

Corollary 1 (PBH observability test.) *The pair (C, A) is observable if and only if the matrix*

$$\begin{pmatrix} A - \lambda I \\ C \end{pmatrix}$$

has maximum rank for every eigenvalue λ of A .

Example 167 (Unobservable modes)

```

A
    -2     2     1     2
    -1     1     1     2
    -3     2     2     2
    -1     0    -1     1

B =

     2
     0
     0
     2

C
     1     0     0    -1

Qo= obsv(A, C)
     1     0     0    -1
    -1     2     2     1
    -7     4     4     7
    -9    -2    -2     9

rank(Qo)
     2                                %System is unobservable

eig(A)
    1.0000 + 2.0000i
    1.0000 - 2.0000i
   -1.0000
    1.0000

I=eye(4);

rank([A-(-1)*I; C])                                %PBH Time
     3

rank([A-(1)*I; C])
     3

rank([A-(1+2i)*I; C])
     4

```

```
rank([A-(1-2i)*I; C])
4
```

So the eigenvalues -1 and $+1$ are unobservable. The eigenvalues $1 + i2$ and $1 - i2$ are not unobservable.

```
[num den] = ss2tf(A,B,C,0)

num =  1.0e-013 *
      0   -0.0400    0.0933    0.0044   -0.1954

den =
      1.0000   -2.0000    4.0000    2.0000   -5.0000
```

%Actually, the transfer function is ZERO! Why?

Here we observe that unobservable eigenvalues are invariant under a state transformation. To see this consider any state transformation of the form $x = T\tilde{x}$; then the new PBH observability matrix is given by

$$\begin{pmatrix} \tilde{A} - \lambda I \\ \tilde{C} \end{pmatrix} = \begin{pmatrix} T^{-1}AT - \lambda I \\ CT \end{pmatrix} = \begin{pmatrix} T^{-1} & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} A - \lambda I \\ C \end{pmatrix} T$$

Since T is invertible, we see that the new and the old PBH matrices have the same rank.

13.5.2 Existence of unobservable modes

Here we will show that, if a system is unobservable then, it must have an unobservable mode. So, consider a linear time-invariant input-output system described

$$\dot{x} = Ax + Bu \quad (13.8a)$$

$$y = Cx + Du \quad (13.8b)$$

where $x(t)$ is an n -vector, $u(t)$ is an m -vector, and $y(t)$ is a p -vector. Suppose (C, A) is not observable. In the following section, we show that there is a state transformation $x = T\tilde{x}$ so that the corresponding transformed system has the following special structure:

$\begin{aligned} \dot{x}_o &= A_{oo}x_o + B_o u \\ \dot{x}_u &= A_{uo}x_o + A_{uu}x_u + B_u u \\ y &= C_o x_o + D u \end{aligned}$
--

with

$$\begin{pmatrix} x_o \\ x_u \end{pmatrix} = \tilde{x}.$$

Also, the pair (C_o, A_{oo}) is observable. Roughly speaking, this transformation splits the state into an observable part x_o and an unobservable part x_u ; the subsystem

$$\begin{aligned}\dot{x}_o &= A_{oo}x_o + B_o u \\ y &= C_o x_o + D u\end{aligned}$$

is observable; one can obtain x_o from knowledge of the histories of u and y .

Notice that, in the transformed system description, there is no connection from the state x_u to the output y . We also note that, when $u(t) \equiv 0$ and $x_o(0) = 0$ we must have $x_o(t) \equiv 0$. Also,

$$\begin{aligned}\dot{x}_u &= A_{uu}x_u \\ y &= 0\end{aligned}$$

From this it should be clear that, if λ is any eigenvalue of A_{uu} then the transformed system (with zero input) has a solution which satisfies

$$\tilde{x}(t) = e^{\lambda t} \tilde{v} \quad \text{and} \quad y(t) \equiv 0$$

for some nonzero $\tilde{v}$, that is λ is an unobservable eigenvalue of the transformed system.

Hence, the system

$$\dot{x} = Ax, \quad y = Cx$$

has a nonzero solution x which satisfies

$$x(t) \equiv e^{\lambda t} v \quad \text{and} \quad y(t) \equiv 0$$

Since the matrices A and

$$\tilde{A} = \begin{pmatrix} A_{oo} & 0 \\ A_{uo} & A_{uu} \end{pmatrix}$$

are similar, they have the same characteristic polynomial. Moreover the characteristic polynomial of $\tilde{A}$ is given by

$$\det(sI - \tilde{A}) = \det(sI - A_{oo}) \det(sI - A_{uu})$$

Hence the characteristic polynomial of A is the product of the characteristic polynomials of A_{oo} and A_{uu} . Thus, the eigenvalues of A are the union of those of A_{oo} and A_{uu} . Hence, the eigenvalues of A_{uu} are the **unobservable eigenvalues** of (C, A) . The corresponding modes are the **unobservable modes**.

Transfer function considerations. Consider the reduced order observable system:

$$\begin{aligned}\dot{x}_o &= A_{oo}x_o + B_o u \\ y &= C_o x_o + D u\end{aligned}$$

One can readily show that the transfer function of this system is exactly the same as that of the original system, that is,

$$C(sI - A)^{-1}B + D = C_o(sI - A_o)^{-1}B_o + D$$

13.5.3 A nice transformation

Consider a LTI input-output system described by

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}$$

where $x(t)$ is an n -vector, $u(t)$ is an m -vector, and $y(t)$ is a p -vector. Suppose (C, A) is not observable; then, its unobservable subspace $\mathcal{X}_{uo}$ is nontrivial. The next lemma states that $\mathcal{X}_{uo}$ is an invariant subspace of A .

Lemma 10 *For any $n \times n$ matrix A and any $p \times n$ matrix, C , the subspace*

$$\mathcal{X}_{uo} = \mathcal{N} \left(\begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{pmatrix} \right)$$

is an invariant subspace for A .

PROOF. Consider any element x of $\mathcal{X}_{uo}$. It satisfies

$$CA^k x = 0 \quad \text{for } k = 0, \dots, n-1$$

Since

$$A^n = -\alpha_0 I - \alpha_1 A - \dots - \alpha_{n-1} A^{n-1}$$

we obtain

$$CA^n x = 0$$

from which it follows that

$$CA^k (Ax) = 0 \quad \text{for } k = 0, \dots, n-1$$

hence Ax is contained in $\mathcal{X}_{uo}$. Since this holds for every element x of $\mathcal{X}_{uo}$, it follows that $\mathcal{X}_{uo}$ is A -invariant. ■

Transformation. Suppose n_u is the dimension of the unobservable subspace $\mathcal{X}_{uo}$, that is, n_u is the nullity of the observability matrix, and let

$$n_o := n - n_u.$$

Choose any basis

$$t^{n_o+1}, \dots, t^n$$

for the unobservable subspace, that is, the null space of the observability matrix Q_o . Extend it to a basis

$$t^1, \dots, t^{n_o}, t^{n_o+1}, \dots, t^n$$

for the whole state space $\mathbb{R}^n$ and let

$$T := \begin{pmatrix} t^1 & t^2 & \dots & t^n \end{pmatrix}$$

(One could do this by choosing $T = V$ where USV^* is a singular value decomposition of Q_o). Introduce the state transformation

$$x = T\tilde{x}$$

If we let

$$\begin{pmatrix} x_o \\ x_u \end{pmatrix} = \tilde{x}$$

where x_o is an n_o -vector and x_u is an n_u -vector, then

$$x \text{ is in } \mathcal{X}_{uo} \quad \text{if and only if} \quad x_o = 0$$

The transformed system description is given by

$$\begin{aligned} \dot{\tilde{x}} &= \tilde{A}\tilde{x} + \tilde{B}u \\ y &= \tilde{C}\tilde{x} + Du \end{aligned}$$

where

$$\tilde{A} = T^{-1}AT \quad \tilde{B} = T^{-1}B \quad \tilde{C} = CT$$

Since $\mathcal{X}_{uo}$ is an invariant subspace of A , the matrix $\tilde{A}$ has the following structure:

$$\tilde{A} = \begin{pmatrix} A_{oo} & 0 \\ A_{uo} & A_{uu} \end{pmatrix}$$

where A_{oo} is $n_o \times n_o$. Since the vectors $t^{n_o+1}, \dots, t^n$ are in the null space of C , the matrix $\tilde{C}$ has the following structure

$$\tilde{C} = \begin{pmatrix} C_o & 0 \end{pmatrix}$$

where C_o is $p \times n_o$. If we partition $\tilde{B}$ as

$$\tilde{B} = \begin{pmatrix} B_o \\ B_u \end{pmatrix}$$

where B_o is $n_o \times m$. then the transformed system is described by

$\dot{x}_o$	$= A_{oo}x_o$	$+B_o u$
$\dot{x}_u$	$= A_{uo}x_o + A_{uu}x_u$	$+B_u u$
y	$= C_o x_o$	$+Du$

- The pair (C_o, A_{oo}) is observable.

Example 168 (Unobservable modes)

```

A
    -2     2     1     2
    -1     1     1     2
    -3     2     2     2
    -1     0    -1     1

C
     1     0     0    -1

Qo= obsv(A, C)
     1     0     0    -1
    -1     2     2     1
    -7     4     4     7
    -9    -2    -2     9

rank(Qo)
     2                                %System is unobservable

eig(A)
    1.0000 + 2.0000i
    1.0000 - 2.0000i
   -1.0000
    1.0000

I=eye(4);

rank([A-(-1)*I; C])                    %PBH Time
     3

rank([A-(1)*I; C])
     3

rank([A-(1+2i)*I; C])
     4

rank([A-(1-2i)*I; C])
     4

```

So the eigenvalues -1 and $+1$ are unobservable. The eigenvalues $1 + i2$ and $1 - i2$ are not unobservable.

```
rref(Qo)
```

```

1      0      0      -1
0      1      1      0
0      0      0      0
0      0      0      0

N=[1 0 0 1; 0 1 -1 0]',           %Basis for unobservable subspace
1      0
0      1
0     -1
1      0

Qo*N
0      0           %Just checking!
0      0
0      0
0      0

rref([N I])
1      0      0      0      0      1
0      1      0      0     -1      0
0      0      1      0      0     -1
0      0      0      1      1      0

T=[E(:,1:2) N ]           % A nice transformation
1      0      1      0
0      1      0      1
0      0      0     -1
0      0      1      0

Anew = inv(T)*A*T
-1      2      0      0
-4      3      0      0
-1      0      0      1
3      -2      1      0

Cnew = C*T
1      0      0      0

Aoo= Anew(1:2,1:2)
-1      2
-4      3

Auu= Anew(3:4,3:4)
0      1

```

```

1      0

eig(Auu)
-1.0000                                %Unobservable eigenvalues
1.0000

eig(Aoo)
1.0000 + 2.0000i                        %The remaining eigenvalues
1.0000 - 2.0000i

B
2
0
0
2

[num den] = ss2tf(A,B,C,0) num = 1.0e-013 *
      0   -0.0400    0.0933    0.0044   -0.1954

den =
1.0000   -2.0000    4.0000    2.0000   -5.0000

%Actually, the transfer function is ZERO! Why?

```

13.6 Observability grammians

For each $T > 0$, we define the finite time observability grammian associated with (C, A) :

$$W_o(T) = \int_0^T e^{A^*t} C^* C e^{At} dt$$

To obtain an interpretation of $W_o(T)$, consider the output y due to zero input and any initial state x_0 , that is,

$$\begin{aligned}\dot{x} &= Ax & x(0) &= x_0 \\ y &= Cx\end{aligned}$$

Then $y(t) = C e^{At} x_0$, and

$$\begin{aligned}\int_0^T \|y(t)\|^2 dt &= \int_0^T \|C e^{At} x_0\|^2 dt = \int_0^T (C e^{At} x_0)^* (C e^{At} x_0) dt \\ &= \int_0^T x_0^* e^{A^*t} C^* C e^{At} x_0 dt = x_0^* \left(\int_0^T e^{A^*t} C^* C e^{At} dt \right) x_0 \\ &= x_0^* W_o(T) x_0;\end{aligned}$$

thus

$$\int_0^T \|y(t)\|^2 dt = x_0^* W_o(T) x_0.$$

Remark 6 Note that $W_o(T)$ is the solution at $t = T$ to the following initial value problem:

$$\dot{W}_o = W_o A + A^* W_o + C^* C \quad \text{and} \quad W_o(0) = 0$$

PROOF. With $s = t - \tau$, we have $\tau = t - s$ and

$$W_c(t) = \int_0^t e^{A\tau} C^* C e^{A^* \tau} d\tau = \int_0^t e^{A(t-s)} C^* C e^{A^*(t-s)} ds$$

Hence, $W_c(0) = 0$ and

$$\dot{W}_c = A W_c + W_c A^* + C^* C$$

■

Lemma 11 For each $T > 0$, the matrices $W_o(T)$ and Q_o have the same null space.

PROOF. We need to show that $\mathcal{N}(Q_o) = \mathcal{N}(W_o(T))$. Since

$$Q_o = \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{(n-1)} \end{pmatrix}$$

it follows that x is in $\mathcal{N}(Q_o)$ if and only if

$$CA^k x = 0 \quad \text{for } k = 0, 1, \dots, n-1 \quad (13.9)$$

We first show that $\mathcal{N}(Q_o) \subset \mathcal{N}(W_o(T))$: Consider any x in $\mathcal{N}(Q_o)$. Then (13.9) holds. Recalling the application of the Cayley-Hamilton theorem to functions of a matrix, it follows that for $0 \leq t \leq T$, there are scalars $\beta_0(t), \dots, \beta_{n-1}(t)$ such that

$$e^{At} = \sum_{i=0}^{n-1} \beta_i(t) A^i$$

Hence,

$$C e^{At} x = \sum_{i=0}^{n-1} \beta_i(t) C A^i x = 0$$

and

$$\begin{aligned} W_o(T)x &= \int_0^T e^{A^* t} C^* C e^{At} dt x \\ &= \int_0^T e^{A^* t} C^* C e^{At} x dt \\ &= 0 \end{aligned}$$

that is, x is in $\mathcal{N}(W_o(T))$. Since the above holds for any $x \in \mathcal{N}(Q_o)$, we must have $\mathcal{N}(Q_o) \subset \mathcal{N}(W_o(T))$.

We now show that $\mathcal{N}(W_o(T)) \subset \mathcal{N}(Q_o)$ and hence $\mathcal{N}(Q_o) = \mathcal{N}(W_o(T))$. To this end, consider any x in $\mathcal{N}(W_o(T))$. Then,

$$\begin{aligned} 0 &= x^* W_o(T) x \\ &= x^* \left(\int_0^T e^{A^* t} C^* C e^{At} dt \right) x \\ &= \int_0^T x^* e^{A^* t} C^* C e^{At} x dt \\ &= \int_0^T (C e^{At} x)^* (C e^{At} x) dt \\ &= \int_0^T \|C e^{At} x\|^2 dt. \end{aligned}$$

Since the last integrand is non-negative for all t and the integral is zero, the integrand must be zero for all t , that is,

$$C e^{At} x \equiv 0.$$

If a function is zero for all t , all its derivatives must be zero for all t ; hence

$$C A^k e^{At} x \equiv 0 \quad \text{for } k = 0, 1, \dots, n-1.$$

Considering $t = 0$ yields

$$C A^k x = 0 \quad \text{for } k = 0, 1, \dots, n-1,$$

that is, x is in $\mathcal{N}(Q_o)$. Since the above holds for any x in $\mathcal{N}(W_o(T))$, we must have $\mathcal{N}(W_o(T)) \subset \mathcal{N}(Q_o)$. ■

Example 169 (The unattached mass)

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= 0 \end{aligned}$$

Here

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad e^{At} = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

(a) (Position measurement.) $y = x_1$. Here

$$C = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

Hence,

$$W_o(T) = \int_0^T e^{A^* t} C^* C e^{At} dt = \begin{pmatrix} T & T^2/2 \\ T^2/2 & T^3/3 \end{pmatrix}$$

and $W_o(T)$ and Q_o have the same null space; that is, $\{0\}$.

(b) (Velocity measurement.) $y = x_2$. Here

$$C = \begin{pmatrix} 0 & 1 \end{pmatrix}$$

Hence,

$$W_o(T) = \int_0^T e^{A^*t} C^* C e^{At} dt = \begin{pmatrix} 0 & 0 \\ 0 & T \end{pmatrix}$$

and $W_o(T)$ and Q_o have the same null space.

The above lemma has the following consequence.

The pair (C, A) is observable if and only if for every $T > 0$ the corresponding observability grammian is invertible.

Computation of initial state. One reason for introducing the observability grammian is as follows: Suppose (C, A) is observable; then from the above lemma, $W_o(T)$ is invertible. The output for

$$\begin{aligned} \dot{x} &= Ax + Bu & x(0) &= x_0 \\ y &= Cx + Du \end{aligned}$$

is given by

$$y(t) = Ce^{At}x_0 + \int_0^t Ce^{A(t-\tau)}Bu(\tau) d\tau + Du(t)$$

Letting

$$\tilde{y}(t) = y(t) - \int_0^t Ce^{A(t-\tau)}Bu(\tau) d\tau - Du(t)$$

we have $\tilde{y}(t) = Ce^{At}x_0$ and

$$\begin{aligned} \int_0^T e^{A^*t} C^* \tilde{y}(t) dt &= \int_0^T e^{A^*t} C^* C e^{At} x_0 dt \\ &= W_o(T) x_0. \end{aligned}$$

Since $W_o(T)$ is invertible, we obtain the following explicit expression for the initial state in terms of the input and output histories:

$$\boxed{x_0 = W_o(T)^{-1} \int_0^T e^{A^*t} C^* \tilde{y}(t) dt}$$

The above expression explicitly shows us that if a system is observable, we can obtain the initial state from measurements of the input and output over any time interval. However, this formula is not usually used to obtain the state. One usually uses asymptotic observers or state estimators to asymptotically obtain the state; these are covered in a later chapter.

13.6.1 Infinite time observability grammian

Suppose A is asymptotically stable, that is, all its eigenvalues have negative real parts, and consider the linear matrix equation

$$W_o A + A^* W_o + C^* C = 0$$

This is a Lyapunov equation of the form $PA + A^*P + Q = 0$ with $Q = C^*C$. Since A is asymptotically stable the above equation has a unique solution for W_o and this solution is given by

$$W_o = \int_0^\infty e^{A^*t} C^* C e^{At} dt$$

We call this matrix the (infinite-time) **observability grammian** associated with (C, A) . One may show that W_o and the observability matrix Q_o have the same null-space. Since W_o is a symmetric positive definite matrix, it follows that the null space of W_o is $\{0\}$ if and only if W_o is positive definite. Hence, we have the following result.

Lemma 12 *Suppose all the eigenvalues of A have negative real parts. Then (C, A) is observable if and only if W_o is positive definite.*

13.7 Discrete time

Consider the discrete-time system described by

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned} \quad (13.10)$$

where $k \in \mathbb{Z}$ is time, $x(k) \in \mathbb{R}^n$ is the state, $u(k) \in \mathbb{R}^m$ is the input, and $y(k) \in \mathbb{R}^p$ is the *measured output*.

Consider any fixed time $N > 0$. The basic observability problem is as follows. Suppose we have knowledge of the input sequence $\{u(0), u(1), \dots, u(N)\}$ and output sequence $\{y(0), y(1), \dots, y(N)\}$ over the interval $[0, N]$. Can we uniquely determine the initial state

$$x_0 = x(0)$$

If we can do this for input histories and all initial states we say the system is observable.

DEFN. System (13.10) is observable over the interval $[0, N]$ if the following holds for each input history $u(\cdot)$. Suppose $y^a(\cdot)$ and $y^b(\cdot)$ are any two output histories of system (13.1) due to initial states x_0^a and x_0^b , respectively, and $y^a(k) = y^b(k)$ for $0 \leq k \leq N$; then $x_0^a = x_0^b$.

• Our first observation is that system (13.10) is observable over $[0, N]$ if and only if the system

$$\begin{aligned} x(k+1) &= Ax(k) \\ y(k) &= Cx(k) \end{aligned} \quad (13.11)$$

has the property that the zero state is the only initial state resulting in an output history which is identically zero; that is, it has the the following property:

$$y(k) = 0 \quad \text{for } 0 \leq k \leq N \quad \implies \quad x(0) = 0$$

13.7.1 Main observability result

Theorem 10 (Main observability theorem) For each $\boxed{N \geq n-1}$, system (13.10) is observable over $[0, N]$ if and only if

$$\text{rank} \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{pmatrix} = n$$

PROOF. The proof can be obtained by showing that, for system (13.11) with $x_0 = x(0)$,

$$\begin{pmatrix} y(0) \\ y(1) \\ \vdots \\ y(n-1) \end{pmatrix} = \begin{pmatrix} Cx_0 \\ CAx_0 \\ \vdots \\ CA^{n-1}x_0 \end{pmatrix} = Q_0x_0$$

Also use Cayley Hamilton. ■

13.8 Exercises

Exercise 139 Determine whether or not each of the following systems are observable.

$$\begin{array}{llll}
 \dot{x}_1 & = & -x_1 & \dot{x}_1 & = & -x_1 & \dot{x}_1 & = & x_1 & \dot{x}_1 & = & x_2 \\
 \dot{x}_2 & = & x_2 + u & \dot{x}_2 & = & x_2 + u & \dot{x}_2 & = & x_2 + u & \dot{x}_2 & = & 4x_1 + u \\
 y & = & x_1 + x_2 & y & = & x_2 & y & = & x_1 + x_2 & y & = & -2x_1 + x_2
 \end{array}$$

Exercise 140 Consider the system with input u , output y , and state variables x_1, x_2 described by

$$\begin{aligned}
 \dot{x}_1 &= -3x_1 + 2x_2 + u \\
 \dot{x}_2 &= -4x_1 + 3x_2 + 2u \\
 y &= -2x_1 + x_2
 \end{aligned}$$

- (a) Is this system **observable**?
- (b) If the system is unobservable, determine the **unobservable eigenvalues**.

Exercise 141 Determine (by hand) whether or not the following system is observable.

$$\begin{aligned}
 \dot{x}_1 &= 5x_1 - x_2 - 2x_3 \\
 \dot{x}_2 &= x_1 + 3x_2 - 2x_3 \\
 \dot{x}_3 &= -x_1 - x_2 + 4x_3 \\
 y_1 &= x_1 + x_2 \\
 y_2 &= x_2 + x_3
 \end{aligned}$$

If the system is unobservable, compute the unobservable eigenvalues.

Exercise 142 Determine the unobservable eigenvalues for each of the systems of Exercise 139.

Exercise 143 For each system in Exercise 139 which is not observable, obtain a reduced order system which is observable and has the same transfer function as the original system.

Exercise 144 (Damped linear oscillator.) Recall

$$\begin{aligned}
 \dot{x}_1 &= x_2 \\
 \dot{x}_2 &= -(k/m)x_1 - (d/m)x_2
 \end{aligned}$$

Show the following:

- (a) (Position measurement.) If $y = x_1$, we have observability.
- (b) (Velocity measurement.) If $y = x_2$, we have observability if and only if $k \neq 0$.
- (c) (Acceleration measurement.) If $y = \dot{x}_2$, we have observability if and only if $k \neq 0$.

Exercise 145 (BB in laundromat) Obtain a state space representation of the following system.

$$m\ddot{q}_1 - m\Omega^2 q_1 + k(q_1 - q_2) = 0$$

$$m\ddot{q}_2 - m\Omega^2 q_2 - k(q_1 - q_2) = 0$$

$$y = q_1$$

Determine whether or not your state space representation is observable.

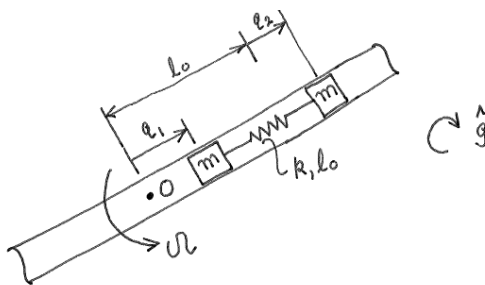


Figure 13.3: Beavis and Butthead in the laundromat

Exercise 146 (BB in laundromat: mass center observations.) Obtain a state space representation of the following system.

$$m\ddot{q}_1 - m\Omega^2 q_1 + k(q_1 - q_2) = 0$$

$$m\ddot{q}_2 - m\Omega^2 q_2 - k(q_1 - q_2) = 0$$

$$y = \frac{1}{2}(q_1 + q_2)$$

(a) Obtain a basis for its unobservable subspace.

(b) Determine the unobservable eigenvalues. Consider $\omega := \sqrt{k/2m} > \Omega$.

Exercise 147 (BB in laundromat: internal observations.) Obtain a state representation of the following system.

$$m\ddot{q}_1 - m\Omega^2 q_1 + k(q_1 - q_2) = 0$$

$$m\ddot{q}_2 - m\Omega^2 q_2 - k(q_1 - q_2) = 0$$

$$y = q_2 - q_1$$

(a) Obtain a basis for its unobservable subspace.

(b) By using a nice transformation, obtain a reduced order observable system which has the same transfer function as the original system.

(c) Determine the unobservable eigenvalues. Consider $\omega := \sqrt{k/2m} > \Omega$.

Exercise 148 Consider a system described by

$$\begin{aligned}\dot{x}_1 &= \lambda_1 x_1 + b_1 u \\ \dot{x}_2 &= \lambda_2 x_2 + b_2 u \\ &\vdots \\ \dot{x}_n &= \lambda_n x_n + b_n u \\ y &= c_1 x_1 + c_2 x_2 + \cdots + c_n x_n\end{aligned}$$

where all quantities are scalar. Obtain conditions on the numbers $\lambda_1, \dots, \lambda_n$ and $c_1, \dots, c_n$ which are necessary and sufficient for the observability of this system.

Exercise 149 Compute the observability grammian for the damped linear oscillator with position measurement and $d, k > 0$.

Exercise 150 Carry out the following for linearizations L1-L8 of the two pendulum cart system.

- (a) Determine which linearizations are observable?
- (b) Compute the singular values of the observability matrix.
- (b) Determine the unobservable eigenvalues for the unobservable linearizations.

Chapter 14

Controllability

14.1 Controllability

Consider a system in which $x(t)$ is the system **state** at time t and $u(t)$ is the **control input** at time t . Consider now any non-trivial time interval $[0, T]$; by non-trivial we mean that $T > 0$. We say that the system is controllable over the interval if it can be ‘driven’ from any one state to any other state over the time interval by appropriate choice of control input over the interval. A formal definition follows the next two examples.

Example 170 (Two unattached masses)

$$\begin{aligned}\dot{x}_1 &= u \\ \dot{x}_2 &= 0\end{aligned}$$

Since the behavior of x_2 is completely unaffected by the control input u , this system is clearly uncontrollable over any time interval.

Example 171 (Integrator) Consider the scalar system described by

$$\dot{x} = u$$

Consider any initial state x_0 and any desired final state x_f . For any any $T > 0$, consider the constant input:

$$u(t) = (x_f - x_0)/T \quad \text{for} \quad 0 \leq t \leq T$$

Then the state trajectory $x(\cdot)$ of this system with $x(0) = x_0$ satisfies $x(T) = x_f$. We consider this system to be controllable over any time interval $[0, T]$.

DEFN.[Controllability] *A system is **controllable** over a non-trivial time interval $[0, T]$ if, for every pair of states x_0 and x_f , there is a control function $u(\cdot)$ defined on the interval such that the solution $x(\cdot)$ of the system with $x(0) = x_0$ satisfies $x(T) = x_f$.*

Note that the definition says nothing about keeping the state at x_f after T .

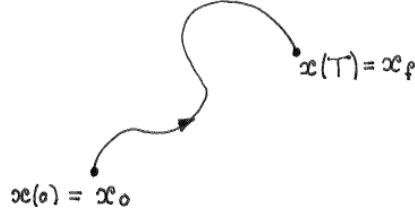


Figure 14.1: Controllability

14.2 Main controllability result

The concept of controllability also applies to linear and nonlinear systems. However, we are concerned here with linear time-invariant systems described by

$$\dot{x} = Ax + Bu \quad (14.1)$$

where the n -vector $x(t)$ is the system state at time t and the m -vector $u(t)$ is the control input at time t . In this section, we introduce a simple algebraic condition which is necessary and sufficient for the controllability of a linear time invariant system over *any* time interval. This condition involves the **controllability matrix** Q_c associated with the above system or the pair (A, B) . This matrix is defined to be the following $n \times nm$ partitioned matrix:

$$Q_c = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix} \quad (14.2)$$

Theorem 11 (Main controllability result) *For any $T > 0$, system (14.1) is controllable over the interval $[0, T]$ if and only if its controllability has maximum rank, that is,*

$$\text{rank} \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix} = n$$

PROOF. A proof is given at the end of Section 14.3.

An immediate consequence of the above theorem is that controllability does not depend on the interval $[0, T]$. So, from now on we drop reference to the interval. Since controllability only depends on the matrix pair (A, B) , we say that this pair is controllable if system (14.1) is controllable.

For scalar input ($m = 1$) systems, the controllability matrix is a square matrix; hence it has maximum rank if and only if its determinant is nonzero. So, the above theorem has the following corollary.

Corollary 2 *A scalar input system of the form (14.1) is controllable if and only if its controllability matrix has non-zero determinant.*

Example 172 (Two unattached masses.) This is a scalar input system with

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Hence,

$$Q_c = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Clearly Q_c has zero determinant, hence this system is not controllable.

Example 173

$$\begin{aligned}\dot{x}_1 &= x_3 + u_1 \\ \dot{x}_2 &= x_3 + u_1 \\ \dot{x}_3 &= u_2\end{aligned}$$

Since there is a control input entering every state equation, this system looks controllable. Is it?

Here $n = 3$ and

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

So,

$$Q_c = \begin{bmatrix} B & AB & A^2B \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

and

$$\text{rank } [Q_c] = \text{rank} \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = 2$$

Since $\text{rank } Q_c = 2$, we have $\text{rank } Q_c \neq n$; hence this system is not controllable.

MATLAB

```
>> help ctrb
```

```
CTRB    Form controllability matrix.
         CTRB(A,B) returns the controllability matrix
         Co = [B AB A^2B ...]
```

```
>> A=[0 0 1; 0 0 1; 0 0 0];
>> B=[1 0; 1 0; 0 1];
```

```
>> ctrb(A,B)
```

```
ans =
     1     0     0     1     0     0
     1     0     0     1     0     0
     0     1     0     0     0     0
```

Useful fact for computing $\text{rank } Q_c$. In computing $\text{rank } Q_c$, the following fact is useful.
Suppose there exists a positive integer k such that

$$\text{rank} \begin{bmatrix} B & AB & \dots & A^k B \end{bmatrix} = \text{rank} \begin{bmatrix} B & AB & \dots & A^{k-1} B \end{bmatrix}$$

Then

$$\text{rank } Q_c = \text{rank} \begin{bmatrix} B & AB & \dots & A^{k-1} B \end{bmatrix}$$

The next example illustrates the above fact.

Example 174 (Beavis and Butthead: self-controlled) Consider the situation in which the system generates a pair of control forces which are equal in magnitude and opposite in direction.

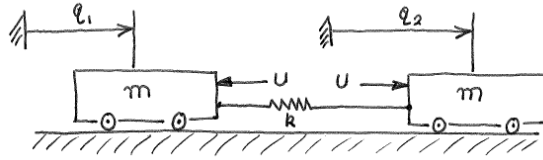


Figure 14.2: B&B: self-controlled

The motion of this system can be described by

$$\begin{aligned} m\ddot{q}_1 &= k(q_2 - q_1) - u \\ m\ddot{q}_2 &= -k(q_2 - q_1) + u \end{aligned}$$

Based on simple mechanical considerations, we can see that this system is not controllable.

Now let's do the controllability test. With

$$x_1 = q_1 \quad x_2 = q_2 \quad x_3 = \dot{q}_1 \quad x_4 = \dot{q}_2$$

we have

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -k/m & k/m & 0 & 0 \\ k/m & -k/m & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ -1/m \\ 1/m \end{bmatrix}$$

Here

$$\begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 0 & -1/m \\ 0 & 1/m \\ -1/m & 0 \\ 1/m & 0 \end{bmatrix}$$

and

$$\begin{bmatrix} B & AB & A^2 B \end{bmatrix} = \begin{bmatrix} 0 & -1/m & 0 \\ 0 & 1/m & 0 \\ -1/m & 0 & 2k/m^2 \\ 1/m & 0 & -2k/m^2 \end{bmatrix}$$

Clearly,

$$\text{rank} \begin{bmatrix} B & AB & A^2B \end{bmatrix} = \text{rank} \begin{bmatrix} B & AB \end{bmatrix}$$

Hence

$$\text{rank } Q_c = \text{rank} \begin{bmatrix} B & AB \end{bmatrix} = 2$$

MATLAB time.

```
>> a = [0      0      1      0
        0      0      0      1
       -1      1      0      0
        1     -1      0      0];
```

```
>> b = [0; 0; -1; 1];
```

```
>> rank(ctrb(a,b))
ans = 2
```

```
>> [U S V] =svd(ctrb(a,b))
```

U =

```
      0    -0.7071    -0.7071    -0.0000
      0     0.7071    -0.7071     0.0000
 -0.7071   -0.0000   -0.0000     0.7071
  0.7071     0.0000     0.0000     0.7071
```

S =

```
  3.1623         0         0         0
      0    3.1623         0         0
      0         0         0         0
      0         0         0         0
```

V =

```
  0.4472         0   -0.8944         0
      0    0.4472         0    0.8944
 -0.8944         0   -0.4472         0
      0   -0.8944         0    0.4472
```

The columns of U form an orthonormal basis for $\mathbb{R}^4$. Since Q_c has two nonzero singular values, the rank of Q_c is two and the first two columns of U form a basis for $\mathcal{R}(Q_c)$.

Example 175 (Beavis and Butthead with external control) Here we consider the system of the previous example and replace the pair of internal forces with a single external control force u . This system is described by

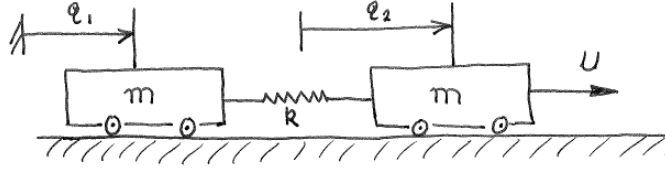


Figure 14.3: B&B: external control

$$\begin{aligned} m\ddot{q}_1 &= k(q_2 - q_1) \\ m\ddot{q}_2 &= -k(q_2 - q_1) + u \end{aligned}$$

With

$$x_1 = q_1 \quad x_2 = q_2 \quad x_3 = \dot{q}_1 \quad x_4 = \dot{q}_2$$

we have

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -k/m & k/m & 0 & 0 \\ k/m & -k/m & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1/m \end{bmatrix}$$

Hence,

$$Q_c = [B \ AB \ A^2B \ A^3B] = \begin{bmatrix} 0 & 0 & 0 & k/m^2 \\ 0 & 1/m & 0 & -k/m^2 \\ 0 & 0 & k/m^2 & 0 \\ 1/m & 0 & -k/m^2 & 0 \end{bmatrix}$$

Clearly, $\text{rank } Q_c = 4 = n$; hence this system is controllable.

Exercise 151 Suppose A is $n \times n$, B is $n \times m$, T is $n \times n$ and nonsingular and let

$$\tilde{A} = T^{-1}AT \quad \tilde{B} = T^{-1}B$$

Then (A, B) is controllable if and only if $(\tilde{A}, \tilde{B})$ is controllable.

14.3 The controllable subspace

Let us now consider the situation when the system $\dot{x} = Ax + Bu$ is not controllable.

Example 176 At first sight the system

$$\begin{aligned} \dot{x}_1 &= x_1 + u \\ \dot{x}_2 &= x_2 + u \end{aligned}$$

might look controllable. However, it is not. To see this, consider the following change of state variables:

$$x_c = x_1 + x_2 \quad \text{and} \quad x_u = x_1 - x_2$$

These variables satisfy

$$\begin{aligned}\dot{x}_c &= x_c + 2u \\ \dot{x}_u &= x_u\end{aligned}$$

Hence, $x_u(0) = 0$ implies $x_u(t) = 0$ for all t . In other words if the initial $x(0)$ is in the set

$$\{(x_1, x_2) \in \mathbb{R}^2 : x_1 - x_2 = 0\}$$

then, regardless of the control input, the state will always stay in this set; it cannot leave this set. So, the system is not controllable.

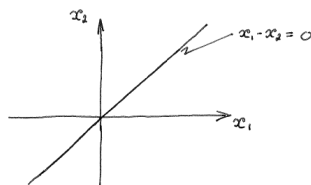


Figure 14.4: Controllable subspace in Example 176

Recall that the controllability matrix associated with the pair (A, B) is defined to be the following $n \times nm$ matrix

$$Q_c = [B \quad AB \quad \dots \quad A^{n-1}B]$$

The controllable subspace is defined as the range of the controllability matrix, that is,

$$\mathcal{R}(Q_c)$$

So, according to the Main Controllability Theorem (Theorem 11), the system $\dot{x} = Ax + Bu$ is controllable if and only if its controllable subspace is the whole state space. It should be clear that a vector x is in the controllable subspace associated with (A, B) if and only if it can be expressed as

$$x = \sum_{i=0}^{n-1} A^i B \nu^i$$

where $\nu^0, \dots, \nu^{n-1}$ are m -vectors.

The following lemma states that when the state of system (14.1) starts at zero, then regardless of the control input, the state always stays in the controllable subspace; that is, it cannot leave this subspace.

Lemma 13 *If $x(0) = 0$ then, for every continuous control input function $u(\cdot)$, the solution $x(\cdot)$ of (14.1) lies in the controllable subspace, that is, $x(t)$ is in the range of Q_c for all $t \geq 0$.*

PROOF. For any time $t > 0$ we have

$$x(t) = \int_0^t e^{A(t-\tau)} B u(\tau) d\tau$$

Recalling the application of the Cayley-Hamilton theorem to functions of a matrix, it follows that for $0 \leq \tau \leq t$, there are scalars $\beta_0(\tau), \dots, \beta_{n-1}(\tau)$ such that

$$e^{A(t-\tau)} = \sum_{i=0}^{n-1} \beta_i(\tau) A^i$$

Hence,

$$e^{A(t-\tau)} B u(\tau) = \sum_{i=0}^{n-1} A^i B \nu_i(\tau) \quad \nu_i(\tau) := \beta_i(\tau) u(\tau)$$

From this it should be clear that for each τ , the n vector $e^{A(t-\tau)} B u(\tau)$ is in $\mathcal{R}(Q_c)$, the range of Q_c . From this one obtains that $x(t) = \int_0^t e^{A(t-\tau)} B u(\tau) d\tau$ is in $\mathcal{R}(Q_c)$. ■

Example 177 For the system of example 176, we have

$$Q_c = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Clearly,

$$\mathcal{R}(Q_c) = \mathcal{R} \left[\begin{bmatrix} 1 \\ 1 \end{bmatrix} \right] = \mathcal{R} [B]$$

If a vector x is in this controllable subspace, then $x = c \begin{bmatrix} 1 & 1 \end{bmatrix}^T$ for some scalar c . This is equivalent to the requirement that $x_1 - x_2 = 0$.

Actually, if the state starts at any point in $\mathcal{R}(Q_c)$ then, regardless of the control input, the state always stays in $\mathcal{R}(Q_c)$; see next exercise. In other words, *the control input cannot drive the state out of the controllable subspace*.

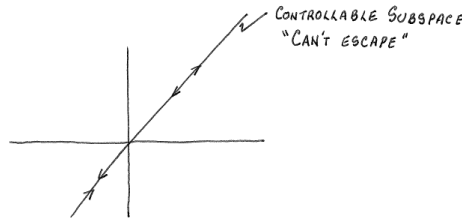


Figure 14.5: Invariance of the controllable subspace

Exercise 152 Prove the following statement. If $x(0)$ is in $\mathcal{R}(Q_c)$, then for every continuous control input function $u(\cdot)$, the solution of (14.1) satisfies $x(t) \in \mathcal{R}(Q_c)$ for all $t \geq 0$.

The next result states that over any time interval, system (14.1) can be driven between any two states in the controllable subspace $\mathcal{R}(Q_c)$ by appropriate choice of control input.

Lemma 14 [Controllable subspace lemma] *Consider any pair of states x_0, x_f in the controllable subspace $\mathcal{R}(Q_c)$. Then for each $T > 0$, there is a continuous control function $u(\cdot) : [0, T] \rightarrow \mathbb{R}^m$ such that the solution $x(\cdot)$ of (14.1) with $x(0) = x_0$ satisfies $x(T) = x_f$*

PROOF. A proof is contained in section 14.4.

Example 178

$$\begin{aligned}\dot{x}_1 &= -x_1 + x_2 + u \\ \dot{x}_2 &= -x_1 + x_2 + u\end{aligned}$$

Here,

$$A = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Hence

$$Q_c = [B \quad AB] = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

and

$$\mathcal{R}(Q_c) = \mathcal{R} \left[\begin{bmatrix} 1 \\ 1 \end{bmatrix} \right]$$

To illustrate the controllable subspace lemma, introduce new states

$$x_c = x_1 \quad x_u = -x_1 + x_2$$

Then

$$x = x_c \begin{bmatrix} 1 \\ 1 \end{bmatrix} + x_u \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and

$$\begin{aligned}\dot{x}_c &= x_u + u \\ \dot{x}_u &= 0\end{aligned}$$

It should be clear that x is in the controllable subspace if and only if $x_u = 0$. In this case x is completely specified by the scalar x_c and

$$\dot{x}_c = u$$

Recalling example 171, we see that given any x_{c0}, x_{cf} and any $T > 0$, one can always find a control input which ‘drives’ x_c from x_{c0} to x_{cf} over the interval $[0, T]$. Hence, the state x can be driven between any two points in the controllable subspace.

Proof of main controllability result. First note that $\text{rank } Q_c = n$ if and only if $\mathcal{R}(Q_c) = \mathbb{R}^n$, that is, the controllability matrix has rank n if and only if the controllable subspace is the whole state space. If $\text{rank } Q_c = n$, then Lemma 14 implies controllability. If $\text{rank } Q_c < n$, then the controllable subspace is not the whole state space and from Lemma 13 any state which is not in the controllable subspace cannot be reached from the origin. ■

14.4 Proof of the controllable subspace lemma

14.4.1 Finite time controllability grammian

Before proving the controllable subspace lemma, we first establish a preliminary lemma. For each $T > 0$, we define the finite time controllability grammian associated with (A, B) :

$$W_c(T) := \int_0^T e^{At} B B' e^{A't} dt \quad (14.3)$$

Note that this matrix is hermitian and positive semi-definite.

Remark 7 Note that $W_c(T)$ is the solution at $t = T$ of the following initial value problem:

$$\dot{W}_c = A W_c + W_c A' + B B' \quad \text{and} \quad W_c(0) = 0$$

PROOF.

$$W_c(t) = \int_0^t e^{A\tau} B B' e^{A'\tau} d\tau = \int_0^t e^{A(t-s)} B B' e^{A'(t-s)} ds$$

Hence, $W_c(0) = 0$ and

$$\dot{W}_c = A W_c + W_c A' + B B'$$

■

Lemma 15 For each $T > 0$,

$$\mathcal{R}(W_c(T)) = \mathcal{R}(Q_c)$$

PROOF. We will show that $\mathcal{N}(Q'_c) = \mathcal{N}(W_c(T))$. From this it will follow that $\mathcal{R}(Q_c) = \mathcal{R}(W_c(T))' = \mathcal{R}(W_c(T))$.

Since

$$Q'_c = \begin{bmatrix} B' \\ B' A' \\ \vdots \\ B' A^{*(n-1)} \end{bmatrix}$$

it follows that x is in $\mathcal{N}(Q'_c)$ if and only if

$$B' A^{*k} x = 0 \quad \text{for } k = 0, 1, \dots, n-1 \quad (14.4)$$

We first show that $\mathcal{N}(Q'_c) \subset \mathcal{N}(W_c(T))$: Consider any $x \in \mathcal{N}(Q'_c)$. Then (14.4) holds. Recalling the application of the Cayley-Hamilton theorem to functions of a matrix, it follows that for $0 \leq t \leq T$, there are scalars $\beta_0(t), \dots, \beta_{n-1}(t)$ such that

$$e^{A't} = \sum_{i=0}^{n-1} \beta_i(t) A^{*i}$$

Hence,

$$B'e^{A't}x = \sum_{i=0}^{n-1} \beta_i(t) B'A^{*i}x = 0$$

and

$$\begin{aligned} W_c(T)x &= \int_0^T e^{At} B B' e^{A't} dt x \\ &= \int_0^T e^{At} B B' e^{A't} x dt \\ &= 0 \end{aligned}$$

that is, $x \in \mathcal{N}(W_c(T))$. Since the above holds for any $x \in \mathcal{N}(Q'_c)$, we must have $\mathcal{N}(Q'_c) \subset \mathcal{N}(W_c(T))$.

We now show that $\mathcal{N}(W_c(T)) \subset \mathcal{N}(Q'_c)$ and hence $\mathcal{N}(Q'_c) = \mathcal{N}(W_c(T))$: Consider any $x \in \mathcal{N}(W_c(T))$. Then,

$$\begin{aligned} 0 &= x' W_c(T) x \\ &= x' \left[\int_0^T e^{At} B B' e^{A't} dt \right] x \\ &= \int_0^T x' e^{At} B B' e^{A't} x dt \\ &= \int_0^T \left[B' e^{A't} x \right]' \left[B' e^{A't} x \right] dt \\ &= \int_0^T \|B' e^{A't} x\|^2 dt \end{aligned}$$

Since the integrand is non-negative for all t and the integral is zero, the integrand must be zero for all t , that is,

$$B' e^{A't} x \equiv 0$$

If a function is zero for all t , all its derivatives must be zero for all t ; hence

$$B' A^{*k} e^{A't} x \equiv 0 \quad \text{for } k = 0, 1, \dots, n-1$$

Considering $t = 0$ yields

$$B' A^{*k} x = 0 \quad \text{for } k = 0, 1, \dots, n-1$$

that is $x \in \mathcal{N}(Q'_c)$. Since the above holds for any $x \in \mathcal{N}(W_c(T))$, we must have $\mathcal{N}(W_c(T)) \subset \mathcal{N}(Q'_c)$. ■

From our main controllability result and the above lemma, we see that (A, B) is controllable if and only if, for any $T > 0$, its finite time controllability grammian $W_c(T)$ has full rank. Since $W_c(T)$ is square, it has full rank if and only if it is invertible. Moreover, since $W_c(T)$ is symmetric and positive semidefinite, it has full rank if and only if it is positive definite. This yields the following result.

Corollary 2 *The system $\dot{x} = Ax + Bu$ is controllable if and only if its controllability gramian over any non-trivial interval is positive definite.*

14.4.2 Proof of the controllable subspace lemma

Consider any $T > 0$ and any pair of states x_0, x_f in $\mathcal{R}(Q_c)$. The solution of (14.1) at time T is:

$$x(T) = e^{AT}x_0 + \int_0^T e^{A(T-\tau)}Bu(\tau) d\tau$$

Hence, a control input drives the state from x_0 to x_f if and only if

$$\int_0^T e^{A(T-\tau)}Bu(\tau) d\tau = z := x_f - e^{AT}x_0$$

We first show that

$$z \in \mathcal{R}(Q_c)$$

Since $x_0 \in \mathcal{R}(Q_c)$,

$$x_0 = \sum_{i=0}^{n-1} A^i B \nu^i$$

hence

$$e^{AT}x_0 = \sum_{i=0}^{n-1} e^{AT} A^i B \nu^i$$

By application of Cayley Hamilton to functions of a matrix

$$e^{AT} A^i = \sum_{k=0}^{n-1} \beta_{ik} A^k$$

so,

$$e^{AT}x_0 = \sum_{i=0}^{n-1} \sum_{k=0}^{n-1} \beta_{ik} A^k B \nu^i = \sum_{k=0}^{n-1} A^k B \tilde{\nu}^k$$

where $\tilde{\nu}^k := \sum_{i=0}^{n-1} \beta_{ik} \nu^i$. From this it should be clear that $e^{AT}x_0$ is in $\mathcal{R}(Q_c)$. Since x_f is also in $\mathcal{R}(Q_c)$ we have $z \in \mathcal{R}(Q_c)$.

Since z is in $\mathcal{R}(Q_c)$, it follows from the previous lemma that z is also in $\mathcal{R}(W_c(T))$. Hence there exists an n -vector y such that

$$z = W_c(T)y$$

Considering the control input

$$u(\tau) = B'e^{A'(T-\tau)}y$$

we obtain

$$\begin{aligned}
\int_0^T e^{A(T-\tau)} B u(\tau) d\tau &= \int_0^T e^{A(T-\tau)} B B' e^{A'(T-\tau)} y d\tau \\
&= \int_0^T e^{A(T-\tau)} B B' e^{A'(T-\tau)} d\tau y \\
&= \int_0^T e^{At} B B' e^{A't} dt y \\
&= W_c(T) y \\
&= z
\end{aligned}$$

hence, this control input drives the state from x_0 to x_f . ■

Remark 8 Suppose (A, B) is controllable. Then $\text{rank } W_c(T) = \text{rank } Q_c = n$. Hence $W_c(T)$ is invertible and the control which drives x_0 to x_f over the interval $[0, T]$ is given by

$$u(t) = B' e^{A'(T-t)} \tilde{x} \quad \tilde{x} = W_c(T)^{-1} (x_f - e^{AT} x_0)$$

14.5 Uncontrollable modes

14.5.1 Eigenvectors revisited: left eigenvectors

Suppose that λ is an eigenvalue of a square matrix A . Then an eigenvector corresponding to λ is any nonzero vector satisfying $Av = \lambda v$; sometimes v is called a right eigenvector. We say that a nonzero vector w is a left eigenvector of A corresponding to λ if it satisfies

$$w' A = \lambda w' \tag{14.5}$$

To see that left eigenvectors exist, note that the above equation is equivalent to $A'w = \bar{\lambda}w$, that is, $\bar{\lambda}$ is an eigenvalue of A' and w is an eigenvector of A' corresponding to $\bar{\lambda}$. We now recall that λ is an eigenvalue of A if and only if $\bar{\lambda}$ is an eigenvalue of A' . Thus, we can say that λ is an eigenvalue of A if and only if it satisfies (14.5) for some non-zero vector w . Moreover, we can also say that every eigenvalue λ of A has left eigenvectors and these vectors are simply the eigenvectors of A' corresponding to $\bar{\lambda}$.

Consider now any two eigenvectors λ_1 and λ_2 of A with $\lambda_1 \neq \lambda_2$. Suppose that v_1 is an eigenvector for λ_1 and w_2 is a left eigenvector for λ_2 . Then

$$w_2' v_1 = 0.$$

To see this, we first use the eigenvector definitions to obtain

$$Av_1 = \lambda_1 v_1 \quad \text{and} \quad w_2' A = \lambda_2 w_2'.$$

Pre-multiplying the first equation by w_2' we obtain that $w_2' Av_1 = \lambda_1 w_2' v_1$. Post-multiplying the second equation by v_1 we obtain that $w_2' Av_1 = \lambda_2 w_2' v_1$. This implies that $\lambda_1 w_2' v_1 = \lambda_2 w_2' v_1$, that is $(\lambda_1 - \lambda_2) w_2' v_1 = 0$. Since $\lambda_1 - \lambda_2 \neq 0$, we must have $w_2' v_1 = 0$.

System significance of left eigenvectors. Consider now a system described by

$$\dot{x} = Ax \quad (14.6)$$

Recall that λ is an eigenvalue of A if and only if the above system has a solution of the form $x(t) = e^{\lambda t}v$ where v is a nonzero constant vector. When this occurs, the vector v can be any eigenvector of A corresponding to λ and the above special solution is described as a mode of the system. We now demonstrate the following result:

A scalar λ is an eigenvalue of A if and only if there is a nonzero vector w such that every solution $x(\cdot)$ of system (14.6) satisfies

$$\boxed{w'x(t) = e^{\lambda t}w'x(0)} \quad (14.7)$$

When this occurs, the vector w can be any left eigenvector of A corresponding to λ .

In particular, if $x(0) = w/\|w\|^2$ then, $w'x(t) = e^{\lambda t}$. If $w'x(0) = 0$ then $w'x(t) = 0$ for all t .

To gain further insight into the above result, introduce the scalar variable $\xi(t) = w'x(t)$. Then $\xi(0) = w'x(0)$ and (14.7) is equivalent to

$$\xi(t) = e^{\lambda t}\xi(0),$$

that is, the behavior of ξ is governed by

$$\dot{\xi} = \lambda\xi.$$

To demonstrate the above result, we first suppose that λ is an eigenvalue of A . Then there is a nonzero vector w such that (14.5) holds; also w is a left eigenvector of A corresponding to λ . Consider now any solution $x(\cdot)$ of system (14.6). Since it satisfies $\dot{x} = Ax$, it follows from (14.5) that

$$\frac{d(w'x)}{dt} = w'\dot{x} = w'Ax = \lambda(w'x)$$

It now follows that (14.7) holds for all t .

To demonstrate the converse, suppose that there is a nonzero vector w so that (14.7) holds for every solution $x(\cdot)$. This implies that

$$w'Ax(t) = w'\dot{x}(t) = \frac{d(w'x)}{dt} = \lambda e^{\lambda t}w'x(0) = \lambda w'x(t);$$

hence,

$$w'(A - \lambda I)x(t) = 0.$$

Since the above holds for any $x(t)$, we must have $w'(A - \lambda I) = 0$. Thus, $w'A = \lambda w'$. This implies that λ is an eigenvalue of A with left eigenvector w .

14.5.2 Uncontrollable eigenvalues and modes

Consider now a system with input u described by

$$\dot{x} = Ax + Bu \quad (14.8)$$

We say that λ is an **uncontrollable eigenvalue** of this system or λ is an **uncontrollable eigenvalue** of (A, B) if there is a nonzero vector w such that for every input $u(\cdot)$, every solution $x(\cdot)$ of the system satisfies

$$\boxed{w'x(t) = e^{\lambda t}w'x(0)} \quad (14.9)$$

To gain further insight into the above concept, introduce the scalar variable $\xi(t) = w'x(t)$. Then, as we have shown above, (14.9) is equivalent to the statement that the behavior of ξ is governed by

$$\dot{\xi} = \lambda\xi.$$

Thus the behavior of ξ is completely unaffected by the input u . So, the mode $e^{\lambda t}$ is completely unaffected by the input u . We say that this is an **uncontrollable mode**.

When system (14.8) has an uncontrollable mode, then the system is not controllable. To see this choose any initial state $x(0) = x_0$ such that $w'x_0 \neq 0$. Then, regardless of the control input, we have $w'x(t) = e^{\lambda t}w'x_0 \neq 0$ for all t . Hence, regardless of the control input, $x(t) \neq 0$ for all t . This means that over any time interval $[0, T]$ we cannot obtain a control history which drives the system from x_0 to 0. Hence the system is not controllable. In the next section, we show that if a system is not controllable then, it must have some uncontrollable modes.

14.5.3 Existence of uncontrollable modes

In this section, we show that if a LTI system is not controllable, then it has uncontrollable modes. Consider a linear time-invariant input-output system described

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned} \quad (14.10)$$

where $x(t)$ is an n -vector, $u(t)$ is an m -vector, and $y(t)$ is a p -vector. Suppose (A, B) is not controllable; then, its controllable subspace $\mathcal{X}_c$ is not equal to the whole state space. In the following section, we show that there is a state transformation $x = T\tilde{x}$ so that the corresponding transformed system is described by

$$\begin{aligned} \dot{x}_c &= A_{cc}x_c + A_{cu}x_u + B_cu \\ \dot{x}_u &= A_{uu}x_u \\ y &= C_cx_c + C_u x_u + Du \end{aligned} \quad (14.11)$$

with

$$\begin{bmatrix} x_c \\ x_u \end{bmatrix} = \tilde{x}.$$

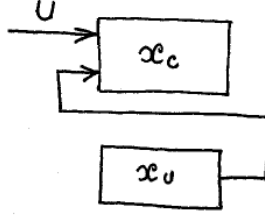


Figure 14.6: Controllable/uncontrollable decomposition

Also, the pair (A_{cc}, B_c) is controllable.

Clearly, the state x_u of the subsystem

$$\dot{x}_u = A_{uu}x_u \quad (14.12)$$

is completely unaffected by the input u . Also, if $x_u(0) = 0$ then $x_u(t) = 0$ for all t and the input-output system in (14.11) can be described by the controllable subsystem:

$$\begin{aligned} \dot{x}_c &= A_{cc}x_c + B_c u \\ y &= C_c x_c + D u \end{aligned} \quad (14.13)$$

Hence, when the initial state $x(0)$ of the input-output system (14.10) is zero, its behavior can be described by the lower order system (14.13).

Since the matrices A and

$$\tilde{A} = \begin{bmatrix} A_{cc} & A_{cu} \\ 0 & A_{uu} \end{bmatrix}$$

are similar, they have the same characteristic polynomial. In particular, the characteristic polynomial of $\tilde{A}$ is given by

$$\det(sI - \tilde{A}) = \det(sI - A_{cc}) \det(sI - A_{uu})$$

Hence the characteristic polynomial of A is the product of the characteristic polynomials of A_{cc} and A_{uu} . Thus, the eigenvalues of A are the union of those of A_{cc} and A_{uu} . We now make the following claim:

The eigenvalues of A_{uu} are the uncontrollable eigenvalues of A .

PROOF. To prove this claim, let λ be any eigenvalue of A_{uu} . We will show that λ is an uncontrollable eigenvalue of (A, B) . Since λ is an eigenvalue of A_{uu} there is a nonzero vector w_u such that for every solution $x_u(\cdot)$ of (14.12), we have

$$w'_u x_u(t) = e^{\lambda t} w'_u x_u(0).$$

Let $\tilde{w}' = [0 \quad w'_u]$. Then every solution $\tilde{x}$ of (14.11) satisfies

$$\tilde{w}' \tilde{x}(t) = e^{\lambda t} \tilde{w}' \tilde{x}(0)$$

With $w = T\tilde{w}$, we now obtain that every solution of (14.10) satisfies $w'x(t) = e^{\lambda t} w'x(0)$. Thus, λ is an uncontrollable eigenvalue of A .

We now prove the converse, that is, if λ is an uncontrollable eigenvalue of (A, B) , then λ must be an eigenvalue of A_{uu} . So, suppose that λ is an uncontrollable eigenvalue of (A, B) . Then (A, B) is not controllable and there is a state transformation $x = T[x'_c \ x'_u]'$ such that the system can be represented as in (14.11) with a nontrivial x_u and with (A_{cc}, B_c) controllable. Suppose, on the contrary, that λ is not an eigenvalue of A_{uu} . Then λ must be an eigenvalue of A_{cc} . Let w be such that (14.9) holds and let $[w'_u \ w'_c]' = T^{-1}w$. Then we must have

$$w'_c x_c(t) + w'_u x_u(t) = e^{\lambda t} [w'_c x_c(0) + w'_u x_u(0)]$$

for every input u and every solution $\tilde{x}(\cdot)$ of (14.11). Considering those solutions with $x_u(0) = 0$, we obtain that $x_u(t) = 0$ for all t and

$$\dot{x}_c = A_c x_c + B_c u. \quad (14.14)$$

Thus $w'_c x_c(t) = e^{\lambda t} w'_c x_c(0)$ for every input and every solution $x_c(\cdot)$ of (14.14); hence λ is an uncontrollable eigenvalue of (A_c, B_c) . This implies that (A_c, B_c) is not controllable and we have a contradiction. So, λ must be an eigenvalue of A_{uu} . ■

Transfer function considerations. Consider the reduced order controllable subsystem:

$$\begin{aligned} \dot{x}_c &= A_{cc} x_c + B_c u \\ y &= C_c x_c + D u \end{aligned}$$

One can readily show that the transfer function of this system is exactly the same as that of the original system, that is

$$C(sI - A)^{-1}B + D = C_c(sI - A_c)^{-1}B_c + D.$$

An immediate consequence of this result is that a minimal state space realization of a transfer function must be controllable.

14.5.4 A nice transformation

We first show that the controllability subspace $\mathcal{X}_c$ is an invariant subspace for A , that is, if x is in $\mathcal{X}_c$, then Ax is in $\mathcal{X}_c$.

Lemma 16 *For any pair of matrices $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times m}$, the controllable subspace*

$$\mathcal{X}_c = \mathcal{R} \left(\begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix} \right)$$

is an invariant subspace for A .

PROOF. Consider any element x of $\mathcal{X}_c$. It can be expressed as

$$x = B\nu_1 + AB\nu_2 + \cdots + A^{n-1}B\nu_n$$

Hence

$$Ax = AB\nu_1 + A^2B\nu_2 + \cdots + A^nB\nu_n$$

Since

$$A^n = -\alpha_0 I - \alpha_1 A - \dots - \alpha_{n-1} A^{n-1}$$

we obtain

$$Ax = B(-\alpha_0 \nu_n) + AB(\nu_1 - \alpha_1 \nu_n) + \dots + A^{n-1}B(\nu_{n-1} - \alpha_{n-1} \nu_n)$$

hence Ax is contained in $\mathcal{X}_c$. Since this holds for every element x of $\mathcal{X}_c$, it follows that $\mathcal{X}_c$ is A -invariant. ■

A nice transformation. Choose any basis

$$t^1, \dots, t^{n_c}$$

for the controllable subspace $\mathcal{X}_c$ of (A, B) . Extend it to a basis

$$t^1, \dots, t^{n_c}, t^{n_c+1}, \dots, t^n$$

for $\mathbb{R}^n$ and let

$$T := \begin{bmatrix} t^1 & t^2 & \dots & t^n \end{bmatrix}$$

One could do this by choosing $T = U$ where USV' is a singular value decomposition of the controllability matrix, that is

$$USV' = \begin{bmatrix} B & AB & \dots & A^{n-1}B \end{bmatrix}$$

Introduce the state transformation

$$x = T\tilde{x}$$

Let

$$\begin{bmatrix} x_c \\ x_u \end{bmatrix} = \tilde{x}$$

where x_c is a n_c -vector and x_u is a $(n-n_c)$ -vector. Then

$$x \text{ is in } \mathcal{X}_c \quad \text{if and only if} \quad x_u = 0$$

The transformed system description is given by

$$\begin{aligned} \dot{\tilde{x}} &= \tilde{A}\tilde{x} + \tilde{B}u \\ y &= \tilde{C}\tilde{x} + Du \end{aligned}$$

where

$$\tilde{A} = T^{-1}AT \quad \tilde{B} = T^{-1}B \quad \tilde{C} = CT$$

Since $\mathcal{X}_c$ is an invariant subspace of A , the matrix $\tilde{A}$ has the following structure:

$$\tilde{A} = \begin{bmatrix} A_{cc} & A_{cu} \\ 0 & A_{uu} \end{bmatrix}$$

where A_{cc} is $n_c \times n_c$. Since the columns of B are contained in the $\mathcal{X}_c$, the matrix $\tilde{B}$ has the following structure

$$\tilde{B} = \begin{bmatrix} B_c \\ 0 \end{bmatrix}$$

where B_c is $n_c \times m$. If we partition $\tilde{C}$ as

$$\tilde{C} = \begin{bmatrix} C_c & C_u \end{bmatrix}$$

where C_c is $p \times n_c$, then the transformed system is described by

$$\boxed{\begin{aligned} \dot{x}_c &= A_{cc}x_c + A_{cu}x_u + B_c u \\ \dot{x}_u &= A_{uu}x_u \end{aligned}}$$

$$y = C_c x_c + C_u x_u + Du$$

- The pair (A_{cc}, B_c) is controllable.

Example 179 Consider

$$\begin{aligned} \dot{x}_1 &= x_2 - u \\ \dot{x}_2 &= x_1 + u \\ y &= x_2 \end{aligned}$$

Here,

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix} \quad D = 0$$

and the system transfer function is given by

$$\hat{G}(s) = \frac{1}{s+1}$$

Since the controllability matrix

$$Q_c = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

has rank less than two, this system is not controllable. The controllable subspace is spanned by the vector

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix} = B$$

Introducing the state transformation

$$x = T\tilde{x} \quad \tilde{x} = \begin{bmatrix} x_c \\ x_u \end{bmatrix} \quad T = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

we obtain an equivalent system description with

$$\tilde{A} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \quad \tilde{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \tilde{C} = \begin{bmatrix} 1 & 1 \end{bmatrix} \quad \tilde{D} = 0$$

that is

$$\begin{aligned}\dot{x}_c &= -x_c + u \\ \dot{x}_u &= x_u \\ y &= x_c + x_u\end{aligned}$$

The transfer function of the reduced order controllable system

$$\begin{aligned}\dot{x}_c &= -x_c + u \\ y &= x_c\end{aligned}$$

is clearly the same as that of the original system.

14.6 PBH test

The next lemma provides a useful characterization of uncontrollable eigenvalues.

Lemma 17 *A complex number λ is an uncontrollable eigenvalue of the pair (A, B) if and only if*

$$\text{rank} [A - \lambda I \quad B] < n$$

where n is the number of rows in A .

PROOF. When the above rank condition holds, there exists a non-zero n -vector w such that

$$w' [A - \lambda I \quad B] = 0,$$

that is

$$w'A = \lambda w' \quad \text{and} \quad w'B = 0.$$

So, if $x(\cdot)$ is any solution of the system $\dot{x} = Ax + Bu$ corresponding to any input $u(\cdot)$, we obtain that

$$\frac{d(w'x)}{dt} = w'Ax + w'Bu = \lambda(w'x).$$

Hence,

$$w'x(t) = e^{\lambda t} w'x(0). \quad (14.15)$$

So λ is an uncontrollable eigenvalue.

To prove the converse, suppose now that λ is an uncontrollable eigenvalue of (A, B) . Then, (14.15) holds for any solution $x(\cdot)$ of the system $\dot{x} = Ax + Bu$ corresponding to any input $u(\cdot)$. Since (14.15) implies that

$$w'\dot{x}(t) = \lambda e^{\lambda t} w'x(0) = \lambda w'x(t),$$

it follows that

$$w'Ax(t) + w'Bu(t) = \lambda w'x(t);$$

hence

$$w' \begin{bmatrix} A - \lambda I & B \end{bmatrix} \begin{bmatrix} x(t) \\ u(t) \end{bmatrix} = 0$$

Since $x(t)$ and $u(t)$ can be arbitrary chosen, we must have

$$w' [A - \lambda I \quad B] = 0,$$

Since w is non zero, it follows that the rank of $[A - \lambda I \quad B]$ is less than n . ■

ALTERNATE PROOF. We first show that when the above rank condition holds, the pair (A, B) is not controllable. When this rank condition holds, there exists a non-zero n vector x such that

$$x' [A - \lambda I \quad B] = 0$$

hence,

$$x'A = \lambda x', \quad x'B = 0$$

It now follows that

$$x'AB = \lambda x'B = 0$$

and, by induction, one can readily prove that

$$x'A^k B = 0 \quad \text{for } k = 0, \dots, n-1$$

Hence

$$x'Q_c = 0$$

where Q_c is the controllability matrix associated with (A, B) . This last condition implies that Q_c has rank less than n , hence, (A, B) is not controllable.

So, if either (A, B) has an uncontrollable eigenvalue or the above rank condition holds then, (A, B) is not controllable and there exists a nonsingular matrix T such that the matrices $\tilde{A} = T^{-1}AT$, $\tilde{B} = T^{-1}B$ have the following structure;

$$\tilde{A} = \begin{bmatrix} A_{cc} & A_{cu} \\ 0 & A_{uu} \end{bmatrix} \quad \tilde{B} = \begin{bmatrix} B_c \\ 0 \end{bmatrix}$$

with (A_{cc}, B_c) controllable. Noting that

$$\begin{bmatrix} \tilde{A} - \lambda I & \tilde{B} \end{bmatrix} = T^{-1} [A - \lambda I \quad B] \begin{bmatrix} T & 0 \\ 0 & I \end{bmatrix}$$

it follows that

$$\text{rank} \begin{bmatrix} \tilde{A} - \lambda I & \tilde{B} \end{bmatrix} = \text{rank} [A - \lambda I \quad B]$$

Since (A_{cc}, B_c) is controllable,

$$\text{rank} \begin{bmatrix} A_{cc} - \lambda I & B_c \end{bmatrix} = n_c$$

where n_c is the number of rows of A_c ; hence

$$\begin{aligned} \text{rank} \begin{bmatrix} \tilde{A} - \lambda I & \tilde{B} \end{bmatrix} &= \text{rank} \begin{bmatrix} A_{cc} - \lambda I & A_{cu} & B_c \\ 0 & A_{uu} - \lambda I & 0 \end{bmatrix} \\ &= \text{rank} \begin{bmatrix} A_{cc} - \lambda I & B_c & A_{cu} \\ 0 & 0 & A_{uu} - \lambda I \end{bmatrix} \\ &= \text{rank} \begin{bmatrix} A_{cc} - \lambda I & B_c & 0 \\ 0 & 0 & A_{uu} - \lambda I \end{bmatrix} \\ &= n_c + \text{rank} [A_{uu} - \lambda I] \end{aligned}$$

It now follows that

$$\text{rank} [A - \lambda I \quad B] < n$$

if and only if

$$\text{rank} [A_{uu} - \lambda I] < n_u$$

where $n_u = n - n_c$ is the number of rows of A_{uu} . This last condition is equivalent to λ being an uncontrollable eigenvalue of (A, B) . ■

Corollary 3 (PBH controllability test.) *The pair (A, B) is controllable if and only if*

$$\text{rank} [A - \lambda I \quad B] = n \quad \text{for all } \lambda \in \mathbb{C}$$

Example 180 Two unattached masses. There,

$$[A - \lambda I \quad B] = \begin{bmatrix} -\lambda & 0 & 0 \\ 0 & -\lambda & 1 \end{bmatrix}$$

The above rank test fails for $\lambda = 0$.

Example 181 Recall example 179. There

$$[A - \lambda I \quad B] = \begin{bmatrix} -\lambda & 1 & -1 \\ 1 & -\lambda & 1 \end{bmatrix}$$

The above rank test fails for $\lambda = 1$

Exercise 153 If α is any complex number, prove the following result. The pair $(A + \alpha I, B)$ is controllable if and only if (A, B) is controllable.

Example 182 (Beavis and Butthead: self-controlled) Recall the system of Example 174.

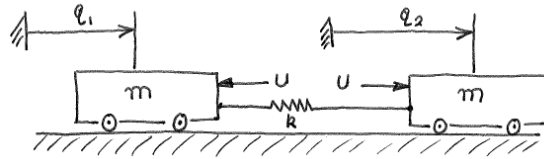


Figure 14.7: B&B: self-controlled

Here

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -k/m & k/m & 0 & 0 \\ k/m & -k/m & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ -1/m \\ 1/m \end{bmatrix}$$

So, the PBH controllability matrix for the zero eigenvalue is

$$[A \quad B] = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -k/m & k/m & 0 & 0 & -1/m \\ k/m & -k/m & 0 & 0 & 1/m \end{bmatrix}$$

Since the last two rows of this matrix are not linearly independent, this matrix does not have the maximum rank of 4; hence zero is an uncontrollable eigenvalue. This eigenvalue corresponds to the rigid body mode of this system.

14.7 Other characterizations of controllability

14.7.1 Infinite time controllability grammian

Suppose A is asymptotically stable, that is all its eigenvalues have negative real parts, and consider the linear matrix equation

$$\boxed{AW_c + W_cA' + BB' = 0}$$

This is a Lyapunov equation of the form $AS + SA' + Q = 0$ with $Q = BB'$. Since A is asymptotically stable the above equation has a unique solution for W_c and this solution is given by

$$W_c = \int_0^\infty e^{At} BB' e^{A't} dt.$$

We call this matrix the (infinite-time) controllability grammian associated with (A, B) . For asymptotically stable systems, this matrix provides another controllability test.

Lemma 18 *Suppose all the eigenvalues of A have negative real parts. Then (A, B) is controllable if and only if $W_c > 0$*

PROOF. When A is asymptotically stable, one can readily generalize Lemma 15 to include the case $T = \infty$; hence

$$\mathcal{R}(W_c) = \mathcal{R}(Q_c)$$

Since $W_c > 0$ if and only if it has rank n , the lemma now follows from the main controllability theorem. ■

MATLAB

```
>> help gram
```

```
GRAM    Controllability and observability gramians.  
GRAM(A,B) returns the controllability gramian:
```

$$G_c = \text{integral} \{ \exp(tA) BB' \exp(tA') \} dt$$

```
GRAM(A',C') returns the observability gramian:
```

$$G_o = \text{integral} \{ \exp(tA') C' C \exp(tA) \} dt$$

```
See also DGRAM, CTRB and OBSV.
```

14.8 Controllability, observability, and duality

Recall that for any matrix M , we have $\text{rank } M' = \text{rank } M$. It now follows from the main controllability theorem that a pair (A, B) is controllable if and only if the rank of Q'_c is n , that is,

$$\text{rank} \begin{bmatrix} B' \\ B'A' \\ \vdots \\ B'A^{*(n-1)} \end{bmatrix} = n$$

Note that the above matrix is the observability matrix associated with the pair (B', A') . Recalling the main observability theorem, we obtain the following result:

- (A, B) is controllable if and only if (B', A') is observable.

The above statement is equivalent to

- (C, A) is observable if and only if (A', C') is controllable.

This is what is meant by the **duality** between controllability and observability.

14.9 Discrete time

With one exception, all the results we have developed for continuous time also hold in an analogous fashion for discrete time. In discrete-time we need to consider time intervals $[0, N]$ where N is bigger. Consider the system described by

$$x(k+1) = Ax(k) + Bu(k) \quad (14.16)$$

where $k \in \mathbb{Z}$ is time, $x(k) \in \mathbb{R}^n$ is the state, and $u(k) \in \mathbb{R}^m$ is the *control input*.

Consider any fixed time $N > 0$. We say that this system is controllable if it can be ‘driven’ from any state to any other state over the time interval $[0, N]$ by appropriate choice of control input.

DEFN. System (14.16) is **controllable over the interval $[0, N]$** if for every pair of states $x_0, x_f \in \mathbb{R}^n$, there is a control function $u(\cdot) : \{0, 1, \dots, N\} \rightarrow \mathbb{R}^m$ such that the solution $x(\cdot)$ of

$$x(k+1) = Ax(k) + Bu(k) \quad x(0) = x_0 \quad (14.17)$$

satisfies $x(N) = x_f$.

14.9.1 Main controllability theorem

Theorem 12 (Main controllability theorem.) For each $\boxed{N \geq n}$, system (14.16) is controllable over $[0, N]$ if and only if

$$\boxed{\text{rank} \begin{bmatrix} B & AB & \dots & A^{n-1}B \end{bmatrix} = n}$$

Chapter 15

State feedback and stabilizability

In this chapter we look at the problem of changing the dynamics of a system using state feedback. In particular, we look at the problem of stabilization via state feedback.

15.1 Stabilizability and state feedback

Consider a linear time-invariant system with control input described by

$$\dot{x} = Ax + Bu. \quad (15.1)$$

For now, we consider linear static (memoryless) state feedback controllers, that is, at each instant t of time the current control input $u(t)$ depends linearly on the current state $x(t)$, that is, we consider the control input to be given by

$$u(t) = Kx(t) \quad (15.2)$$

where K is a constant $m \times n$ matrix, sometimes called a **state feedback gain matrix**. When system (15.1) is subject to such a controller, its behavior is governed by

$$\dot{x} = (A + BK)x \quad (15.3)$$

We call this the **closed loop system** resulting from controller (15.2).

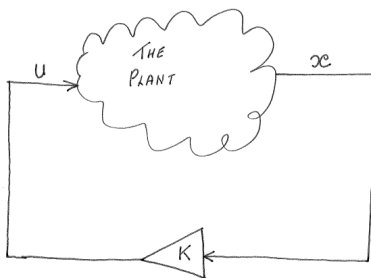


Figure 15.1: State feedback

Suppose the **open loop system**, that is, the system with zero feedback, or, $\dot{x} = Ax$ is unstable or at most marginally stable. A natural question is the following. Can we choose the gain matrix K so that $A + BK$ is asymptotically stable? If yes, we say that system (15.1) is stabilizable.

DEFN.[Stabilizability] *System (15.1) is stabilizable if there exists a matrix K such that $A + BK$ is asymptotically stable.*

We sometimes say that the pair (A, B) is stabilizable if system (15.1) is stabilizable.

The big question is: under what conditions is a given pair (A, B) stabilizable? We shall see in Section 15.4.2 that controllability is a sufficient condition for stabilizability; that is, if a system is controllable it is stabilizable. However, controllability is not necessary for stabilizability, that is, it is possible for a system to be stabilizable but not controllable. To see this, consider a system with A asymptotically stable and B equal to zero; see also the next two examples.

Example 183 (*Not stabilizable and not controllable*)

$$\begin{aligned}\dot{x}_1 &= x_1 \\ \dot{x}_2 &= u\end{aligned}$$

For any gain matrix K , the matrix

$$A + BK = \begin{bmatrix} 1 & 0 \\ k_1 & k_2 \end{bmatrix}$$

has eigenvalues 1 and k_2 . Hence, $A + BK$ is unstable for all K . So, this system is not stabilizable.

Example 184 (*Stabilizable but not controllable.*)

$$\begin{aligned}\dot{x}_1 &= -x_1 \\ \dot{x}_2 &= u\end{aligned}$$

For any gain matrix K , the matrix

$$A + BK = \begin{bmatrix} -1 & 0 \\ k_1 & k_2 \end{bmatrix}$$

has eigenvalues -1 and k_2 . Hence, $A + BK$ is asymptotically stable provided $k_2 < 0$. So, this system is stabilizable. However, it is not controllable.

15.2 Eigenvalue placement (pole placement) by state feedback

Recall the stabilizability problem for the system

$$\dot{x} = Ax + Bu$$

and suppose A and B are real matrices. Here we consider a more general question: *Can we arbitrarily assign the eigenvalues of $A + BK$ by choice of the gain matrix K ?* The main result of this section is that the answer is yes if and only if the pair (A, B) is controllable. We consider first single input systems.

15.2.1 Single input systems

Here we consider scalar input ($m = 1$) systems. We first demonstrate that the characteristic polynomial of $A + BK$ depends in an affine linear fashion on the gain matrix K . To achieve this, we need to use the following result whose proof is given in the appendix at the end of this chapter.

Fact 19 *Suppose M and N are any two matrices of dimensions $n \times m$ and $m \times n$, respectively. Then,*

$$\det(I + MN) = \det(I + NM).$$

In particular, if $m = 1$, we have

$$\det(I + MN) = 1 + NM.$$

Whenever s is not an eigenvalue of A , we have $sI - A - BK = (sI - A)(I - (sI - A)^{-1}BK)$. Thus, using Fact 19, we obtain that

$$\begin{aligned} \det(sI - A - BK) &= \det(sI - A) \det(I - (sI - A)^{-1}BK) \\ &= \det(sI - A) \det(I - K(sI - A)^{-1}B) \\ &= \det(sI - A)(1 - K(sI - A)^{-1}B), \end{aligned}$$

that is, the characteristic polynomial of $A + BK$ satisfies

$$\det(sI - A - BK) = \det(sI - A)(1 - K(sI - A)^{-1}B). \quad (15.4)$$

From this expression, we can make the following statement:

For a single input system, the characteristic polynomial of $A + BK$ depends in an affine linear fashion on the gain matrix K .

We now show that we can arbitrarily assign the eigenvalues of $A + BK$ by choice of the gain matrix K if (A, B) is controllable. Consider any bunch of complex numbers $\lambda_1, \dots, \lambda_n$ and suppose we wish to choose a gain matrix K such that the eigenvalues of $A + BK$ are precisely these complex numbers. Let

$$\hat{d}(s) = \prod_{i=1}^n (s - \lambda_i)$$

that is, $\hat{d}$ is the unique monic polynomial whose roots are the desired closed loop eigenvalues, $\lambda_1, \dots, \lambda_n$. If the eigenvalues are chosen with the property that if λ is in the bunch then so

is $\bar{\lambda}$, the coefficients of the polynomial $\hat{d}$ will be real. We need to show that there is a gain matrix K so that $\det(sI - A - BK) = \hat{d}(s)$. Let d be the characteristic polynomial of A , that is, $d(s) = \det(sI - A)$. Then, recalling (15.4) we need to show that there exists a gain matrix K such that

$$\hat{d}(s) = d(s)(1 - K(sI - A)^{-1}B)$$

that is,

$$d(s)K(sI - A)^{-1}B = d(s) - \hat{d}(s). \quad (15.5)$$

Let

$$\begin{aligned} d(s) &= a_0 + a_1s + \cdots + a_{n-1}s^{n-1} + s^n \\ \hat{d}(s) &= \hat{a}_0 + \hat{a}_1s + \cdots + \hat{a}_{n-1}s^{n-1} + s^n. \end{aligned}$$

Recall the power series expansion for $(sI - A)^{-1}$:

$$(sI - A)^{-1} = \frac{1}{s}I + \frac{1}{s^2}A + \cdots + \frac{1}{s^k}A^{k-1} + \cdots$$

Substituting this expression into (15.5) and equating the coefficients of like powers of s we obtain that (15.5) holds if and only if

$$\begin{aligned} s^0 : \quad a_1KB + a_2KAB + \cdots + a_{n-1}KA^{n-2}B + KA^{n-1}B &= a_0 - \hat{a}_0 \\ s^1 : \quad a_2KB + \cdots + a_{n-1}KA^{n-3}B + KA^{n-2}B &= a_1 - \hat{a}_1 \\ &\vdots \\ s^{n-2} : \quad a_{n-1}KB + KAB &= a_{n-2} - \hat{a}_{n-2} \\ s^{n-1} : \quad KB &= a_{n-1} - \hat{a}_{n-1} \end{aligned}$$

that is,

$$\begin{bmatrix} KB & KAB & \cdots & KA^{n-1}B \end{bmatrix} \Upsilon = \begin{bmatrix} a_0 - \hat{a}_0 & a_1 - \hat{a}_1 & \cdots & a_{n-1} - \hat{a}_{n-1} \end{bmatrix} \quad (15.6)$$

where Υ is the invertible matrix given by

$$\Upsilon = \begin{bmatrix} a_1 & a_2 & \cdots & a_{n-2} & a_{n-1} & 1 \\ a_2 & a_3 & \cdots & a_{n-1} & 1 & 0 \\ a_3 & a_4 & \cdots & 1 & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots & \vdots \\ a_{n-1} & 1 & \cdots & 0 & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix}. \quad (15.7)$$

Let Q_c be the controllability matrix associated with (A, B) , that is,

$$Q_c = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix}.$$

Then condition (15.6) on K reduces to

$$KQ_c\Upsilon = \begin{bmatrix} a_0 - \hat{a}_0 & a_1 - \hat{a}_1 & \cdots & a_{n-1} - \hat{a}_{n-1} \end{bmatrix}.$$

Because the pair (A, B) is controllable, Q_c is invertible. This readily implies that the gain matrix K is uniquely given by

$$K = \begin{bmatrix} a_0 - \hat{a}_0 & a_1 - \hat{a}_1 & \cdots & a_{n-1} - \hat{a}_{n-1} \end{bmatrix} \Upsilon^{-1} Q_c^{-1}. \quad (15.8)$$

We have just demonstrated the following result.

Theorem 13 (Eigenvalue (pole) placement theorem: SI case) *Suppose the matrix pair (A, B) is a single input controllable pair with state dimension n and $\lambda_1, \dots, \lambda_n$ is any bunch of n complex numbers. Then there exists a unique matrix K such that $\lambda_1, \dots, \lambda_n$ are the eigenvalues (multiplicities included) of $A + BK$.*

If A and B are real and the eigenvalues are chosen with the property that if λ is in the bunch then so is $\bar{\lambda}$, the matrix K is real.

MATLAB

```
>> help acker
```

```
ACKER Pole placement gain selection using Ackermann's formula.
K = ACKER(A,B,P) calculates the feedback gain matrix K such that
the single input system
```

$$\dot{x} = Ax + Bu$$

```
with a feedback law of u = -Kx has closed loop poles at the
values specified in vector P, i.e., P = eig(A-B*K).
```

```
See also PLACE.
```

```
Note: This algorithm uses Ackermann's formula. This method
is NOT numerically reliable and starts to break down rapidly
for problems of order greater than 10, or for weakly controllable
systems. A warning message is printed if the nonzero closed loop
poles are greater than 10\% from the desired locations specified
in P.
```

Example 185 Consider

$$A = \begin{bmatrix} 3 & -2 \\ 4 & -3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Here the controllability matrix

$$Q_c = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 3 & -1 \end{bmatrix}$$

is full rank; hence (A, B) is controllable. Considering any state feedback gain matrix

$$K = \begin{bmatrix} k_1 & k_2 \end{bmatrix}$$

, we have

$$A + BK = \begin{bmatrix} 3 + 2k_1 & -2 + 2k_2 \\ 4 + 3k_1 & -3 + 3k_2 \end{bmatrix}$$

and

$$\det(sI - A - BK) = s^2 - (2k_1 + 3k_2)s - 1 + k_2.$$

Note that the coefficients of the characteristic polynomial of $A + BK$ depend in an affine linear fashion on K . Suppose we desire the eigenvalues of $A + BK$ to be -2 and -3 . Then the desired characteristic polynomial $\hat{d}$ of $A + BK$ is given by

$$\hat{d}(s) = (s + 2)(s + 3) = s^2 + 5s + 6.$$

By equating coefficients of like powers of s in $\det(sI - A - BK)$ and $\hat{d}(s)$, we obtain

$$\begin{aligned} -1 + k_2 &= 6 \\ -2k_1 - 3k_2 &= 5 \end{aligned}$$

Solving these equations yields (uniquely) $k_1 = -13$ and $k_2 = 7$. Hence the gain matrix is given by

$$K = \begin{bmatrix} -13 & 7 \end{bmatrix}.$$

Checking our answer in Matlab, we obtain

```
K = acker(A,B,[-2 -3])
K =
    -13     7
```

Example 186 (Beavis and Butthead with external control) Considering $m = k =$

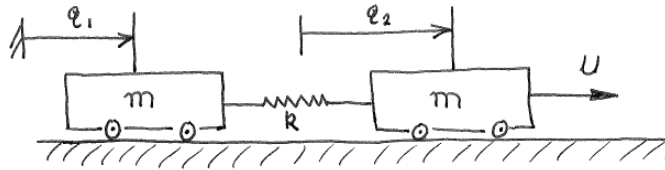


Figure 15.2: B&B: external control

1, a state space description of this system is given by

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

We have already shown that this system is controllable.

Here,

$$\det(sI - A) = s^4 + 2s^2$$

Let

$$\hat{d}(s) = s^4 + \hat{a}_3 s^3 + \hat{a}_2 s^2 + \hat{a}_1 s + \hat{a}_0$$

be the desired characteristic polynomial of $A + BK$.

Since this is a scalar input system, the coefficients of the characteristic polynomial of $A + BK$ depend affinely on K and we can use the following solution approach to find K .

Let

$$K = \begin{bmatrix} k_1 & k_2 & k_3 & k_4 \end{bmatrix}$$

Then

$$\begin{aligned} \det(sI - A - BK) &= \det \begin{bmatrix} s & 0 & -1 & 0 \\ 0 & s & 0 & -1 \\ 1 & -1 & s & 0 \\ -1 - k_1 & 1 - k_2 & -k_3 & s - k_4 \end{bmatrix} \\ &= s^4 - k_4 s^3 + (2 - k_2) s^2 - (k_3 + k_4) s - (k_1 + k_2). \end{aligned}$$

Note the affine dependence of the coefficients of the above polynomial on K . Letting the above polynomial equal the desired polynomial $\hat{d}$ and equating like coefficients results in the following four linear equations in the four unknowns components of K :

$$\begin{aligned} -k_4 &= \hat{a}_3 \\ 2 - k_2 &= \hat{a}_2 \\ -k_3 - k_4 &= \hat{a}_1 \\ -k_1 - k_2 &= \hat{a}_0 \end{aligned}$$

Solving for $k_1, \dots, k_4$ uniquely yields

$$\begin{aligned} k_1 &= -2 - \hat{a}_0 + \hat{a}_2 \\ k_2 &= 2 - \hat{a}_2 \\ k_3 &= -\hat{a}_1 + \hat{a}_3 \\ k_4 &= -\hat{a}_3 \end{aligned}$$

- To illustrate MATLAB, consider desired closed loop eigenvalues: $-1, -2 - 3, -4$.

```
>> poles=[-1 -2 -3 -4];
```

```
>> poly(poles)
```

```
ans =
```

```
1    10    35    50    24
```

Hence

$$\hat{a}_0 = 24 \quad \hat{a}_1 = 50 \quad \hat{a}_2 = 35 \quad \hat{a}_3 = 10$$

and the above expressions for the gain matrix yield

$$k_1 = 9 \quad k_2 = -33 \quad k_3 = -40 \quad k_4 = -10$$

Using MATLAB pole placement commands, we obtain:

```
>> k= place(a,b,poles)
place: ndigits= 18

k =
   -9.0000   33.0000   40.0000   10.0000

>> k= acker(a,b,poles)

k =
   -9    33    40    10
```

Remembering that MATLAB yields $-K$, these results agree with our ‘hand’ calculated results. Lets check to see if we get the desired closed loop eigenvalues.

```
>> eig(a-b*k)

ans =
   -1.0000
   -2.0000
   -3.0000
   -4.0000
```

YEES!

15.2.2 Multi-input systems

The following theorem (which we do not prove here) states that the eigenvalue placement result also holds in the general multi-input case.

Theorem 14 (Eigenvalue (pole) placement theorem) *Suppose the matrix pair (A, B) is controllable with state dimension n and $\lambda_1, \dots, \lambda_n$ is any bunch of n complex numbers. Then there exists a matrix K such that $\lambda_1, \dots, \lambda_n$ are the eigenvalues (multiplicities included) of $A + BK$.*

If A and B are real and the eigenvalues are chosen with the property that if λ is in the bunch then so is $\bar{\lambda}$, the matrix K is real.

It follows from the above result that

$$\boxed{\text{controllability} \quad \Rightarrow \quad \text{stabilizability}}$$

MATLAB

```
>> help place
```

```
PLACE  K = place(A,B,P)  computes the state feedback matrix K such that
the eigenvalues of A-B*K are those specified in vector P.
The complex eigenvalues in the vector P must appear in consecutive
complex conjugate pairs. No eigenvalue may be placed with
multiplicity greater than the number of inputs.
```

```
The displayed "ndigits" is an estimate of how well the
eigenvalues were placed. The value seems to give an estimate
of how many decimal digits in the eigenvalues of A-B*K match
the specified numbers given in the array P.
```

```
A warning message is printed if the nonzero closed loop poles
are greater than 10% from the desired locations specified in P.
```

```
See also: LQR and RLOCUS.
```

15.3 Uncontrollable eigenvalues (you can't touch them)

In the previous section, we saw that, if the pair (A, B) is controllable, then one could arbitrarily place the eigenvalues of $A + BK$ by appropriate choice of the gain matrix K ; hence controllability is a sufficient condition for arbitrary eigenvalue placement. In this section, we show that controllability is also necessary for arbitrary eigenvalue placement. In particular, we show that if λ is an uncontrollable eigenvalue of (A, B) , then λ is an eigenvalue of $A + BK$ for every K . Recall Examples 183 and 184 in which the matrix A had an eigenvalue λ with the property that for every gain matrix K , it is an eigenvalue of $A + BK$. We now demonstrate the following result:

If λ is an uncontrollable eigenvalue of (A, B) , then λ is an eigenvalue of $A + BK$ for every gain matrix K .

To see this, recall that λ is an uncontrollable eigenvalue of (A, B) if and only if

$$\text{rank} \begin{bmatrix} A - \lambda I & B \end{bmatrix} < n \quad (15.9)$$

where A is $n \times n$. Since the above PBH matrix has n rows, when it has rank less than n , the rows are linearly dependent; hence, there exists a nonzero n -vector w such that

$$w' \begin{bmatrix} A - \lambda I & B \end{bmatrix} = 0.$$

that is

$$w'A = \lambda w' \quad \text{and} \quad w'B = 0$$

So, for any gain matrix K , we have

$$w'(A + BK) = \lambda w'$$

that is, λ is an eigenvalue of $A + BK$ with left eigenvector w . So, regardless of the feedback gain matrix, λ is always an eigenvalue of $A + BK$ with left eigenvector w . We cannot alter this eigenvalue by feedback. In particular, if λ has a non-negative real part, then the pair (A, B) is not stabilizable.

Another way to see the above result is to consider the nice transformation introduced earlier, namely,

$$x = T\tilde{x} \quad \text{and} \quad \tilde{x} = \begin{bmatrix} x_c \\ x_u \end{bmatrix}$$

which results in the equivalent system description,

$$\dot{\tilde{x}} = \tilde{A}\tilde{x} + \tilde{B}u$$

where

$$\tilde{A} = T^{-1}AT = \begin{bmatrix} A_{cc} & A_{cu} \\ 0 & A_{uu} \end{bmatrix} \quad \text{and} \quad \tilde{B} = T^{-1}B = \begin{bmatrix} B_c \\ 0 \end{bmatrix}$$

and the pair (A_{cc}, B_c) is controllable. If $u = Kx$, then $u = \tilde{K}\tilde{x}$ where $\tilde{K} = KT$. Letting $\tilde{K} = \begin{bmatrix} K_c & K_u \end{bmatrix}$, we obtain

$$\tilde{A} + \tilde{B}\tilde{K} = \begin{bmatrix} A_{cc} + B_c K_c & A_{cu} + B_c K_u \\ 0 & A_{uu} \end{bmatrix}$$

Since

$$\tilde{A} + \tilde{B}\tilde{K} = T^{-1}[A + BK]T$$

the eigenvalues of $A + BK$ are exactly the same as those of $\tilde{A} + \tilde{B}\tilde{K}$ which are equal to the union of the eigenvalues of $A_{cc} + B_c K_c$ and A_{uu} . From this we see that, regardless of K , the eigenvalues of A_{uu} are always eigenvalues of $A + BK$. Now recall that the eigenvalues of A_{uu} are the uncontrollable eigenvalues of (A, B) .

Furthermore, since the pair (A_{cc}, B_c) is controllable, the eigenvalues of $A_{cc} + B_c K_c$ can be arbitrarily placed by appropriate choice of K_c . Hence, except for the eigenvalues of A_{uu} , all eigenvalues of $A + BK$ can be arbitrarily placed. In particular, if all the eigenvalues of A_{uu} have negative real parts, then K can be chosen so that all the eigenvalues of $A + BK$ have negative real parts. So, we have the following conclusion.

A pair (A, B) is stabilizable if and only if all the uncontrollable eigenvalues of (A, B) have negative real parts.

Recalling the PBH test for uncontrollable eigenvalues, we now have the following PBH result on stabilizability.

Theorem 15 (PBH stabilizability theorem) *A pair (A, B) is stabilizable if and only if*

$$\text{rank} \begin{bmatrix} A - \lambda I & B \end{bmatrix} = n$$

for every eigenvalue λ of A with nonnegative real part.

Example 187

A =

$$\begin{bmatrix} 0 & -1 & 0 \\ 1 & -1 & -1 \\ -1 & 0 & 0 \end{bmatrix}$$

B =

$$\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

rank(ctrb(A,B))

2

%Uncontrollable

eig(A)

```

0 + 1.0000i
0 - 1.0000i
-1.0000

rank([ A-i*eye(3) B])
3                               %Stabilizable

rank([ A+eye(3) B])
2                               %-1 is an uncontrollable eigenvalue

```

Considering $K = \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix}$ we obtain that

$$\det(sI - A - BK) = s^3 + (1 - k_1 + k_2 - k_3)s^2 + (1 - 2k_1)s + (1 - k_1 - k_2 + k_3).$$

As expected, we can write this as

$$\det(sI - A - BK) = (s + 1) \left[s^2 + (-k_1 + k_2 - k_3)s + (1 - k_1 - k_2 + k_3) \right]$$

Suppose that the desired eigenvalues of $A + BK$ are $-1 - 2, -3$. Then

$$\hat{d}(s) = (s + 1)(s + 2)(s + 3) = s^3 + 6s^2 + 11s + 6.$$

Equating coefficients, we obtain

$$\begin{aligned} 1 - k_1 - k_2 + k_3 &= 6 \\ 1 - 2k_1 &= 11 \\ 1 - k_1 + k_2 - k_3 &= 6 \end{aligned}$$

Solving yields

$$\begin{aligned} k_1 &= -5 \\ k_2 &= k_3 \\ k_3 &\text{ is arbitrary} \end{aligned}$$

We choose

$$K = \begin{bmatrix} -5 & 0 & 0 \end{bmatrix}.$$

```

eig(A+B*K)
-3.0000
-2.0000
-1.0000
                                %Just checking!

```

Remark 9 If system (15.1) is not stabilizable via a linear static state controller of the form $u = Kx$ then, it is not stabilizable with any controller. This can be seen as follows. The fact that the system is not stabilizable via a linear controller implies that the system has an uncontrollable eigenvalue λ with non-negative real part. Hence there exists a nonzero vector w such that, regardless of the input u , we have

$$w'x(t) = e^{\lambda t}w'x(0)$$

for all t . Considering any initial state $x(0)$ for which $w'x(0) \neq 0$ (for example, let $x(0)$ be the real part of w), we see that, regardless of the input u , the resulting solution $x(t)$ does not go to zero as t goes to infinity.

15.4 Controllable canonical form

Consider a scalar input system of the form $\dot{x} = Ax + Bu$ where A and B have the following structure:

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-2} & -a_{n-1} \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ \vdots \\ 0 \\ 1 \end{bmatrix}.$$

We say that the pair (A, B) is in **controllable canonical form**. We saw this structure when we constructed realizations for scalar transfer functions. Note that $\dot{x} = Ax + Bu$ looks like:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ &\vdots \\ \dot{x}_{n-1} &= x_n \\ \dot{x}_n &= -a_0x_1 + a_1x_2 + \dots + a_{n-1}x_n + u \end{aligned}$$

Also,

$$\det(sI - A) = a_0 + a_1s + \dots + a_{n-1}s^{n-1} + s^n$$

Example 188 (*Controlled unattached mass*)

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= u \end{aligned}$$

Here,

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The following result is easily shown using the PBH test for controllability.

Fact 20 *The pair $(A + BK, B)$ is controllable if and only if (A, B) is controllable.*

Using this result allows to obtain the next result.

Fact 21 *If the pair (A, B) is in controllable canonical form, then it is controllable.*

To prove the above result, let $K = [a_0 \ a_1 \ \cdots \ a_{n-1}]$. Then

$$A + BK = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}$$

The controllability matrix Q_c for the pair $(A + BK, B)$ is simply given by

$$Q_c = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 1 & 0 \\ \vdots & \vdots & \ddots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{bmatrix}$$

This matrix has rank n ; hence $(A + BK, B)$ is controllable. It now follows that (A, B) is controllable.

15.4.1 Transformation to controllable canonical form*

Here we show that any controllable single-input system

$$\dot{x} = Ax + Bu$$

can be transformed via a state transformation to a system which is in controllable canonical form.

State transformations. Suppose T is a nonsingular matrix and consider the state transformation

$$x = T\tilde{x}$$

Then the evolution of $\tilde{x}$ is governed by

$$\dot{\tilde{x}} = \tilde{A}\tilde{x} + \tilde{B}u$$

where $\tilde{A} = T^{-1}AT$ and $\tilde{B} = T^{-1}B$.

Note that *controllability is invariant under a state transformation*, that is, (A, B) is controllable if and only if $(T^{-1}AT, T^{-1}B)$ is controllable. This can readily be seen by noting the controllability matrix $\tilde{Q}_c$ for the pair $(T^{-1}AT, T^{-1}B)$ is given by $\tilde{Q}_c = T^{-1}Q_c$ where Q_c is the controllability matrix for (A, B) ; hence $\tilde{Q}_c$ and Q_c have the same rank.

Transformation to controllable canonical form. Suppose (A, B) is controllable and $m = 1$. Then the following algorithm yields a state transformation which transforms (A, B) into controllable canonical form.

Algorithm. Let

$$a_0 + \dots + a_{n-1}s^{n-1} + s^n = \det(sI - A)$$

Recursively define the following sequence of n -vectors:

$$\boxed{\begin{array}{lcl} t^n & = & B \\ t^j & = & At^{j+1} + a_j B, \quad j = n-1, \dots, 1 \end{array}}$$

and let

$$T = [t^1 \quad \dots \quad t^n] \quad \blacksquare$$

Fact 22 *If (A, B) is controllable then, the matrix T is generated by the above algorithm is nonsingular and the pair $(T^{-1}AT, T^{-1}B)$ is in controllable canonical form.*

PROOF. We first need to prove that T is nonsingular. By applying the algorithm, we obtain

$$\begin{aligned} t^n &= B \\ t^{n-1} &= AB + a_{n-1}B \\ t^{n-2} &= A(AB + a_{n-1}B) + a_{n-2}B = A^2B + a_{n-1}AB + a_{n-2}B \end{aligned}$$

One may readily show by induction that for $j = n-1, n-2, \dots, 1$, one has

$$t^j = A^{n-j}B + a_{n-1}A^{n-j-1}B + \dots + a_{j+1}AB + a_jB \quad (15.10)$$

From this it should be clear that for any $j = n, n-1, \dots, 1$,

$$\text{rank} [t^n \quad t^{n-1} \quad \dots \quad t^j] = \text{rank} [B \quad AB \quad \dots \quad A^{n-j}B]$$

Hence

$$\text{rank } T = \text{rank} [t^n \quad t^{n-1} \quad \dots \quad t^1] = \text{rank} [B \quad AB \quad \dots \quad A^n B] = \text{rank } Q_c$$

Since (A, B) is controllable, Q_c has rank n . Hence T has rank n and is invertible.

We now show that $T^{-1}AT$ and $T^{-1}B$ have the required structure. It follows from (15.10) that

$$t^1 = A^{n-1}B + a_{n-1}A^{n-2}B + \dots + a_2AB + a_1B$$

Using the Cayley Hamilton Theorem, we obtain

$$At^1 = [A^n + a_{n-1}A^{n-1} + \dots + a_2A^2 + a_1A]B = -a_0B,$$

that is $At^1 = -a_0t^n$. Also from the algorithm, we have

$$At^j = t^{j-1} - a_{j-1}t^n \quad \text{for } j = 2, 3, \dots, n$$

Recalling a useful fact on similarity transformations, it follows that $T^{-1}AT$ has the required structure. Since $B = t^n$, it follows that $T^{-1}B$ also has the required structure. $\blacksquare$

We can now state the following result.

Theorem 16 *A single input system (A, B) is controllable if and only if there is a nonsingular matrix T such that $(T^{-1}AT, T^{-1}B)$ is in controllable canonical form.*

Example 189 *(Beavis and Butthead with external control.)* Consider $m = k = 1$. Then,

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Hence,

$$\det(sI - A) = s^4 + 2s^2$$

and

$$\begin{aligned} t^4 &= B \\ t^3 &= At^4 + a_3B = At^4 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \\ t^2 &= At^3 + a_2B = At^3 + 2B = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \\ t^1 &= At^2 + a_1B = At^2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \end{aligned}$$

So, the transformation matrix is given by:

$$T = \begin{bmatrix} t^1 & t^2 & t^3 & t^4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

```
>> inv(t)
```

```
ans =
```

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

```
>> inv(t)*a*t
```

```
ans =
```

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{array}{cccc} 0 & 0 & 0 & 1 \\ 0 & 0 & -2 & 0 \end{array}$$

15.4.2 Eigenvalue placement by state feedback*

Recall that if (A, B) is controllable then, we arbitrarily assign the eigenvalues of $A + BK$ by choice of the gain matrix K ? We can demonstrate this result for single input systems using the controllable canonical form. Consider any bunch of complex numbers $\lambda_1, \dots, \lambda_n$ with the property that if λ is in the bunch then so is $\bar{\lambda}$ and suppose we wish to choose a gain matrix K such that the eigenvalues of $A + BK$ are precisely these complex numbers.

Let

$$\hat{d}(s) = s^n + \hat{a}_{n-1}s^{n-1} + \dots + \hat{a}_0 = \prod_{i=1}^n (s - \lambda_i)$$

that is, the real numbers, $\hat{a}_0, \dots, \hat{a}_{n-1}$, are the coefficients of the unique monic polynomial $\hat{d}$ whose roots are the desired closed loop eigenvalues, $\lambda_1, \dots, \lambda_n$. If (A, B) is controllable and $m = 1$, then there exists a nonsingular matrix T such that the pair $(\tilde{A}, \tilde{B})$ is in controllable canonical form, where $\tilde{A} = T^{-1}AT$ and $\tilde{B} = T^{-1}B$, that is,

$$\tilde{A} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-2} & -a_{n-1} \end{bmatrix}, \quad \tilde{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

where

$$s^n + a_{n-1}s^{n-1} + \dots + a_0 = \det(sI - A).$$

Considering the $1 \times n$ real gain matrix

$$\tilde{K} = [a_0 - \hat{a}_0 \quad a_1 - \hat{a}_1 \quad \dots \quad a_{n-1} - \hat{a}_{n-1}]$$

we obtain

$$\tilde{A} + \tilde{B}\tilde{K} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ -\hat{a}_0 & -\hat{a}_1 & -\hat{a}_2 & \dots & -\hat{a}_{n-2} & -\hat{a}_{n-1} \end{bmatrix}.$$

Clearly, the matrix $\tilde{A} + \tilde{B}\tilde{K}$ has the desired characteristic polynomial $\hat{d}$. Let

$$K = \tilde{K}T^{-1}.$$

Then

$$\begin{aligned} A + BK &= T\tilde{A}T^{-1} + T\tilde{B}\tilde{K}T^{-1} \\ &= T[\tilde{A} + \tilde{B}\tilde{K}]T^{-1} \end{aligned}$$

and hence

$$\det(sI - A - BK) = \det(sI - \tilde{A} - \tilde{B}\tilde{K}) = \hat{d}(s)$$

in other words, the eigenvalues of $A + BK$ are as desired.

Remark 10 For scalar input systems, the coefficients of the characteristic polynomial of $A + BK$ depend affinely on $\tilde{K}$. Since $K = \tilde{K}T^{-1}$, it follows that the coefficients of the characteristic polynomial of $A + BK$ depend affinely on K .

Example 190 (*Beavis and Buttthead with external control*) Here

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

We have already shown that this system is controllable and it can be transformed to controllable canonical form with the matrix

$$T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Also,

$$\det(sI - A) = s^4 + 2s^2$$

Let

$$\hat{p}(s) = s^4 + \hat{a}_3s^3 + \hat{a}_2s^2 + \hat{a}_1s + \hat{a}_0$$

be the desired characteristic polynomial of $A + BK$. Then

$$\tilde{K} = \begin{bmatrix} -\hat{a}_0 & -\hat{a}_1 & 2-\hat{a}_2 & -\hat{a}_3 \end{bmatrix}$$

and

$$\begin{aligned} K &= \tilde{K}T^{-1} = \begin{bmatrix} -\hat{a}_0 & -\hat{a}_1 & 2-\hat{a}_2 & -\hat{a}_3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -2-\hat{a}_0+\hat{a}_2 & 2-\hat{a}_2 & -\hat{a}_1+\hat{a}_3 & -\hat{a}_3 \end{bmatrix} \end{aligned}$$

This is the same as the result we previously obtained for this example.

15.5 Discrete-time systems

Basically, everything we have said in this chapter so far also holds for discrete time systems, except that in talking about asymptotic stability of discrete-time systems, we need the eigenvalues to have magnitude strictly less than one.

15.5.1 Dead beat controllers

One cannot drive the state of a LTI continuous-time system to zero in a finite time using a static linear state-feedback controller. However, this is possible for a controllable discrete-time system. To see this, consider

$$x(k+1) = Ax(k) + Bu(k)$$

where $x(k)$ is an n -vector and $u(k)$ is an m -vector and suppose that the pair (A, B) is controllable. Then there exists a gain matrix K so that all the eigenvalues of the corresponding closed loop system matrix $A_{cl} := A + BK$ are all zero. Hence the characteristic polynomial of A_{cl} is $p(s) = s^n$. From the Cayley-Hamilton Theorem, we obtain that

$$A_{cl}^n = 0.$$

Since the solutions $x(\cdot)$ of the closed loop system

$$x(k+1) = A_{cl}x(k)$$

satisfy $x(k) = A_{cl}^k x(0)$ for all $k \geq 0$, we see that

$$x(k) = 0 \quad \text{for} \quad k \geq n.$$

Thus, all solutions of the closed loop system go to zero in at most n steps.

One could use the above result to obtain a controller which drives the state of a continuous-time system to zero. Simply, pick a sampling time T and discretize the continuous-time system. Now design a deadbeat controller for the discrete-time system. Using a zero order hold, apply this controller to the continuous-time system. Note that the resulting controller is not a static controller for the continuous-time system.

15.6 Stabilization of nonlinear systems

Consider a nonlinear system described by

$$\dot{x} = F(x, u)$$

and suppose we wish to stabilize this system about some controlled equilibrium state x^e . Let u^e be a constant input that achieves the desired controlled equilibrium state x^e , that is, $F(x^e, u^e) = 0$. Let

$$\delta\dot{x} = A\delta x + B\delta u \quad (15.11)$$

be the linearization of the nonlinear system about (x^e, u^e) . Thus,

$$A = \frac{\partial F}{\partial x}(x^e, u^e) \quad \text{and} \quad B = \frac{\partial F}{\partial u}(x^e, u^e).$$

15.6.1 Continuous-time controllers

If the pair (A, B) is stabilizable, the following procedure yields a controller which stabilizes the nonlinear system about x^e .

Choose a gain matrix K such that all eigenvalues of $A + BK$ have negative real part and let

$$\boxed{u = u^e + K(x - x^e)} \quad (15.12)$$

We now show that the above controller results in the closed loop nonlinear system being asymptotically about x^e . To this end, note that the closed loop system is described by

$$\dot{x} = f(x) = F(x, Kx + u^e - Kx^e)$$

Note that

$$f(x^e) = F(x^e, u^e) = 0;$$

hence x^e is an equilibrium state for the closed loop system. Linearizing the closed loop system about x^e , we obtain

$$\delta\dot{x} = (A + BK)\delta x$$

Since all the eigenvalues of $A + BK$ have negative real parts, the closed loop nonlinear system is asymptotically stable about x^e .

15.6.2 Discrete-time controllers

Linearize the continuous-time nonlinear system to obtain the LTI continuous-time system (15.11).

Choose a sample time $T > 0$ and discretize (15.11) to obtain

$$\delta x_d(k+1) = A_d \delta x_d(k) + B_d \delta u_d(k) \quad (15.13)$$

If the pair (A_d, B_d) is stabilizable, choose matrix K so that all the eigenvalues of $A_d + B_d K$ have magnitude less than one.

Let

$$u_d(k) = K(x_d(k) - x^e) + u^e = K[x(kT) - x^e] + u^e. \quad (15.14)$$

The continuous control input u is obtained from u_d via a zero order hold.

Exercises

Exercise 154 Consider the system with input u , output y , and state variables x_1, x_2, x_3, x_4 described by

$$\begin{aligned} \dot{x}_1 &= -x_1 \\ \dot{x}_2 &= x_1 - 2x_2 \\ \dot{x}_3 &= x_1 - 3x_3 + u \\ \dot{x}_4 &= x_1 + x_2 + x_3 + x_4 \\ y &= x_3 \end{aligned}$$

- (a) Is this system **controllable**?
- (b) Is this system **stabilizable using state feedback**? Justify your answers.

Exercise 155 (*BB in laundromat: external excitation.*) Obtain a state space representation of the following system.

$$\begin{aligned} m\ddot{\phi}_1 - m\Omega^2\phi_1 + \frac{k}{2}(\phi_1 - \phi_2) &= 0 \\ m\ddot{\phi}_2 - m\Omega^2\phi_2 - \frac{k}{2}(\phi_1 - \phi_2) &= u \end{aligned}$$

Show that this system is controllable.

Considering

$$\Omega = 1, m = 1, k = 2$$

obtain a linear state feedback controller which results in a closed loop system with eigenvalues

$$-1, -2, -3, -4$$

Use the following methods.

- (a) Method based on computing the transformation matrix which transforms to controllable canonical form.
- (b) Obtaining an expression for the closed loop characteristic polynomial in terms of the components of the gain matrix.
- (c) **acker**
- (d) **place**

Numerically simulate the closed loop system with the same initial conditions you used in to simulate the open-loop system.

Exercise 156 Consider the system described by

$$\begin{aligned}\dot{x}_1 &= x_1^3 + \sin x_2 - 1 + u \\ \dot{x}_2 &= -e^{x_1} + (\cos x_2)u\end{aligned}$$

Obtain a **state feedback controller** which results in a closed loop system which is asymptotically stable about the zero state.

Exercise 157 *Stabilization of 2-link manipulator with linear state feedback.*

$$[m_1 l c_1^2 + m_2 l_1^2 + I_1] \ddot{q}_1 + [m_2 l_1 l c_2 \cos(q_1 - q_2)] \ddot{q}_2 + m_2 l_1 l c_2 \sin(q_1 - q_2) \dot{q}_2^2 - [m_1 l c_1 + m_2 l_1] g \sin(q_1) = u$$

$$[m_2 l_1 l c_2 \cos(q_1 - q_2)] \ddot{q}_1 + [m_2 l c_2^2 + I_2] \ddot{q}_2 - m_2 l_1 l c_2 \sin(q_1 - q_2) \dot{q}_1^2 - m_2 g l c_2 \sin(q_2) = 0$$

Linearize this system about

$$q_1 = q_2 = u = 0$$

and obtain a state space description.

For rest of exercise, use MATLAB and the following data: Is the linearization control-

m_1	l_1	$l c_1$	I_1	m_2	l_2	$l c_2$	I_2	$m_{payload}$
kg	m	m	$kg.m^2$	kg	m	m	$kg.m^2$	kg
10	1	0.5	10/12	5	1	0.5	5/12	0

lable? Design a state feedback controller which stabilizes this linearized system . Apply the controller to the nonlinear system and simulate the nonlinear closed loop system for various initial conditions.

Exercise 158 Consider the system described by

$$\begin{aligned}\dot{x}_1 &= x_2 + u \\ \dot{x}_2 &= x_1 + u\end{aligned}$$

- Is this system stabilizable ?
- Does there exist a linear state feedback controller which results in closed loop eigenvalues $-1, -2$?
- Does there exist a linear state feedback controller which results in closed loop eigenvalues $-2, -3$?

In parts (b) and (c): If no controller exists, explain why; if one does exist, give an example of one.

Exercise 159 Consider the system described by

$$\begin{aligned}\dot{x}_1 &= -x_2 + u \\ \dot{x}_2 &= -x_1 - u\end{aligned}$$

where all quantities are scalars.

- (a) Is this system stabilizable via state feedback?
- (b) Does there exist a linear state feedback controller which results in closed loop eigenvalues $-1, -4$?
- (c) Does there exist a linear state feedback controller which results in closed loop eigenvalues $-2, -4$?

In parts (b) and (c): If no controller exists, explain why; if one does exist, give an example of one.

Exercise 160 Consider the system

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1 + u\end{aligned}$$

- (a) Design a continuous-time linear state feedback controller which stabilizes this system about the zero state. Illustrate the effectiveness of your controller with a simulation.
- (b) For various sampling times, discretize the controller found in part (a) and simulate the corresponding closed loop. Is the closed loop system stable for any sampling time?
- (c) Design a stabilizing discrete-time controller based on a discretization of the system. Consider sampling times $T = 0.1, 1, 2\pi, 10$. Illustrate the effectiveness of your controller with simulations of the continuous-time system subject to the discrete-time controller.
- (d) Design a dead-beat controller based on a discretization of the system. Consider sampling times $T = 0.1, 1, 10$. Illustrate the effectiveness of your controller with simulations.

Exercise 161 (*BB in laundromat: external excitation.*) Obtain a state space representation of the following system.

$$\begin{aligned}m\ddot{\phi}_1 - m\Omega^2\phi_1 + \frac{k}{2}(\phi_1 - \phi_2) &= 0 \\ m\ddot{\phi}_2 - m\Omega^2\phi_2 - \frac{k}{2}(\phi_1 - \phi_2) &= u\end{aligned}$$

- (a) Show that this system is controllable.
- (b) Considering

$$\Omega = 1, m = 1, k = 2$$

obtain a linear state feedback controller which results in a closed loop system with eigenvalues

$$-1, -2, -3, -4$$

Use the following methods.

- (i) Obtain (by hand) an expression for the closed loop characteristic polynomial in terms of the components of the gain matrix.
- (ii) `acker`
- (iii) `place`

Illustrate the effectiveness of your controllers with numerical simulations.

Exercise 162 (Stabilization of cart pendulum system via state feedback.) Carry out the following for parameter sets P2 and P4 and equilibriums $E1$ and $E2$. Illustrate the effectiveness of your controllers with numerical simulations.

Using eigenvalue placement techniques, obtain a state feedback controller which stabilizes the nonlinear system about the equilibrium.

What is the largest value of δ (in degrees) for which your controller guarantees convergence of the closed loop system to the equilibrium for initial condition

$$(y, \theta_1, \theta_2, \dot{y}, \dot{\theta}_1, \dot{\theta}_2)(0) = (0, \theta_1^e - \delta, \theta_2^e + \delta, 0, 0, 0)$$

where θ_1^e and θ_2^e are the equilibrium values of θ_1 and θ_2 .

Exercise 163 (Discrete-time stabilization of cart pendulum system via state feedback.)

Carry out the following for parameter set P4 and equilibriums $E1$ and $E2$. Illustrate the effectiveness of your controllers with numerical simulations.

Using eigenvalue placement techniques, obtain a discrete-time state feedback controller which stabilizes the nonlinear system about the equilibrium.

What is the largest value of δ (in degrees) for which your controller guarantees convergence of the closed loop system to the equilibrium for initial condition

$$(y, \theta_1, \theta_2, \dot{y}, \dot{\theta}_1, \dot{\theta}_2)(0) = (0, \theta_1^e - \delta, \theta_2^e + \delta, 0, 0, 0)$$

where θ_1^e and θ_2^e are the equilibrium values of θ_1 and θ_2 .

15.7 Appendix

Proof of Fact 19. We need to prove that for any two matrices M and N of dimensions $m \times n$ and $n \times m$, respectively, we have

$$\det(I + MN) = \det(I + NM).$$

Consider the following matrix equation:

$$\begin{bmatrix} I & M \\ -N & I \end{bmatrix} \begin{bmatrix} I & O \\ N & I \end{bmatrix} = \begin{bmatrix} I + MN & M \\ O & I \end{bmatrix} \quad (15.15)$$

Since the matrix on the right of the equality is block diagonal, its determinant is $\det(I) \det(I + MN) = \det(I + MN)$. Since the determinant of the product of two matrices is the product of the determinants of the two matrices, we obtain that the product of the determinants of

the two matrices on the left of the equation must be $\det(I + MN)$. Reversing the order of the two matrices on the left of (15.15), we obtain

$$\begin{bmatrix} I & O \\ N & I \end{bmatrix} \begin{bmatrix} I & M \\ -N & I \end{bmatrix} = \begin{bmatrix} I & M \\ O & I + NM \end{bmatrix}$$

From this we can conclude that the product of the determinants of the two matrices on the left of equation (15.15) is $\det(I + NM)$. Hence we obtain the desired result that $\det(I + MN) = \det(I + NM)$.

Chapter 16

Observers, state estimators and detectability

16.1 Observers, state estimators and detectability

Consider a system (plant) with state $x(t)$, input $u(t)$ and output $y(t)$. Suppose that at each instant of time t we can measure the plant output $y(t)$ and the plant input $u(t)$ and we wish to estimate the plant state $x(t)$. In this section, we consider LTI systems and demonstrate how to obtain an estimate $\hat{x}(t)$ of the plant state $x(t)$ with the property that as $t \rightarrow \infty$ the state estimation error $\hat{x}(t) - x(t)$ goes to zero. We do this by constructing an **observer** or **state estimator**.

Observer or state estimator. An observer or state estimator for a plant is a system whose inputs consist of the plant input and plant output while its output is an estimate of the plant state; see Figure 16.1.

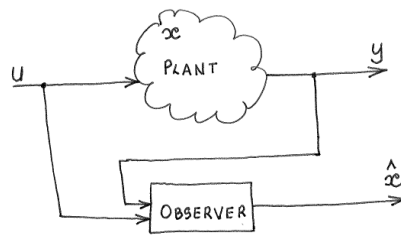


Figure 16.1: Observer

Consider an LTI plant.

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}\tag{16.1}$$

We consider here observers which have the following structure.

$$\boxed{\begin{aligned}\dot{\hat{x}} &= A\hat{x} + Bu + L(\hat{y} - y) \\ \hat{y} &= C\hat{x} + Du\end{aligned}} \quad \hat{x}(0) = \hat{x}_0\tag{16.2}$$

where the n -vector $\hat{x}(t)$ is the **estimated state**; the matrix L is called the **observer gain matrix** and is yet to be determined. The initial state $\hat{x}_0$ for the observer is arbitrary. One can regard $\hat{x}_0$ as an initial guess of the initial plant state x_0 . Note that the observer has a very simple structure: it consists of a copy of the plant plus the “correction term,” $L(\hat{y} - y)$. If we rewrite the observer description as

$$\dot{\hat{x}} = (A + LC)\hat{x} + (B + LD)u - Ly,$$

it should be clear that we can regard the observer as a linear system whose inputs are the plant input u and plant output y . We can regard the estimated state $\hat{x}$ as the observer output.

To discuss the behavior of the above estimator, we introduce the **state estimation error**

$$\tilde{x} := \hat{x} - x$$

Using (16.1) and (16.2) one can readily show that the evolution of the estimation error is simply governed by

$$\boxed{\dot{\tilde{x}} = (A + LC)\tilde{x}}$$

Hence, if one can choose L such that all the eigenvalues of the matrix $A + LC$ have negative real parts then, we obtain the desired result that

$$\lim_{t \rightarrow \infty} \tilde{x}(t) = 0.$$

In other words, the state estimation error goes to zero as t goes to infinity. When this happens, we call the observer an **asymptotic observer**. This leads to the next definition.

DEFN. *System (16.1) or the pair (C, A) is **detectable** if there exists a matrix L such that $A + LC$ is asymptotically stable.*

Note that the matrix $A + LC$ is asymptotically stable if and only if the matrix

$$(A + LC)' = A' + C'L'$$

is asymptotically stable. This follows from the fact that the eigenvalues of $(A + LC)'$ are simply the complex conjugate of the eigenvalues of $A + LC$; note that if all matrices are real, $A + LC$ and $(A + LC)'$ have the same eigenvalues. Thus, the problem of choosing a matrix L so that $A + LC$ is asymptotically stable is equivalent to the stabilizability problem of choosing K so that $A' + C'K$ is asymptotically stable. If one solves the stabilizability problem for K , then $L = K'$ solves the original problem.

From the above observations, we obtain the following duality between stabilizability and detectability.

- (C, A) is detectable if and only if (A', C') is stabilizable.
- (A, B) is stabilizable if and only if (B', A') is detectable.
- Also, it follows that

$$\boxed{\text{observability} \quad \implies \quad \text{detectability}}$$

16.2 Eigenvalue placement for estimation error dynamics

Suppose (C, A) is a real observable pair. Then (A', C') is a real controllable pair. Now consider any bunch of complex numbers

$$\{\lambda_1, \lambda_2, \dots, \lambda_n\}$$

with the property that if λ is in the bunch then so is $\bar{\lambda}$. Since (A', C') is real and controllable, there is a real gain matrix K such that $\{\lambda_1, \lambda_2, \dots, \lambda_n\}$ are precisely the eigenvalues of $A' + C'K$. Letting $L = K'$ and recalling that a real matrix and its transpose have the same eigenvalues, we obtain that the eigenvalues of $A + LC$ are precisely $\{\lambda_1, \lambda_2, \dots, \lambda_n\}$. We have just demonstrated the following result.

If (C, A) is observable, one can arbitrarily place the eigenvalues of $A + LC$ by appropriate choice of the observer gain matrix L .

Example 191 (*The unattached mass with position measurement and an input.*)

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= u \\ y &= x_1\end{aligned}$$

Here

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad C = [1 \quad 0]$$

Since this system is observable, it is detectable. If

$$L = \begin{bmatrix} l_1 \\ l_2 \end{bmatrix}$$

we have

$$A + LC = \begin{bmatrix} l_1 & 1 \\ l_2 & 0 \end{bmatrix}$$

Hence,

$$\det(sI - A - LC) = s^2 - l_1s - l_2$$

and $A + LC$ is asymptotically stable if

$$l_1 < 0 \quad l_2 < 0$$

An asymptotic observer is then given by

$$\begin{aligned}\dot{\hat{x}}_1 &= \hat{x}_2 + l_1(\hat{x}_1 - y) \\ \dot{\hat{x}}_2 &= u + l_2(\hat{x}_1 - y)\end{aligned}$$

and the estimation error $\tilde{x} = \hat{x} - x$ satisfies

$$\begin{aligned}\dot{\tilde{x}}_1 &= l_1\tilde{x}_1 + \tilde{x}_2 \\ \dot{\tilde{x}}_2 &= l_2\tilde{x}_1\end{aligned}$$

16.3 Unobservable modes (you can't see them)

Suppose system (16.1) is unobservable. Then, as we have already seen, this system will have at least one unobservable eigenvalue λ . This means that, when $u(t) \equiv 0$, the system has a nonzero solution of the form $x^u(t) = e^{\lambda t}v$ which produces zero output, $y(t) \equiv 0$. Thus one cannot distinguish this state motion from the zero motion, $x(t) \equiv 0$. If $\Re(\lambda) < 0$ then $x^u(t)$ asymptotically goes to zero. However, if $\Re(\lambda) \geq 0$ then, $x^u(t)$ does not asymptotically go to zero. Hence, even using nonlinear observers, there is no way that one can asymptotically estimate the state of this system. For linear observers, we have the following result.

If λ an unobservable eigenvalue of (C, A) , then λ is an eigenvalue of $A + LC$ for every observer gain matrix L .

To see this, recall that λ is an unobservable eigenvalue of (C, A) if and only if

$$\text{rank} \begin{bmatrix} A - \lambda I \\ C \end{bmatrix} < n$$

where A is $n \times n$. Since the above PBH matrix has n columns, this means that there is a nonzero n -vector v such that

$$\begin{bmatrix} A - \lambda I \\ C \end{bmatrix} v = 0$$

or, equivalently,

$$\begin{aligned} Av &= \lambda v \\ Cv &= 0 \end{aligned}$$

Hence, for any matrix L

$$(A + LC)v = \lambda v$$

Since $v \neq 0$, it follows that λ is an eigenvalue of $A + LC$. ■

The next result states that a necessary and sufficient condition for detectability is that all unobservable eigenvalues are asymptotically stable.

Theorem 17 (PBH detectability theorem) *The pair (C, A) is detectable if and only if*

$$\text{rank} \begin{bmatrix} A - \lambda I \\ C \end{bmatrix} = n$$

for every eigenvalue λ of A with nonnegative real part.

PROOF. Use duality and the corresponding result for stabilizability.

Example 192 (The unattached mass with velocity measurement.)

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= 0 \\ y &= x_2 \end{aligned}$$

Here

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

Hence

$$\begin{bmatrix} A - \lambda I \\ C \end{bmatrix} = \begin{bmatrix} -\lambda & 1 \\ 0 & -\lambda \\ 0 & 1 \end{bmatrix}$$

The above matrix has rank $1 < n$ for $\lambda = 0$; hence $\lambda = 0$ is an unobservable eigenvalue and this system is not detectable.

Note that for any observer gain matrix

$$L = \begin{bmatrix} l_1 \\ l_2 \end{bmatrix}$$

we have

$$A + LC = \begin{bmatrix} 0 & 1 + l_1 \\ 0 & l_2 \end{bmatrix}$$

Hence,

$$\det(sI - A - LC) = s(s - l_2)$$

So, regardless of choice of L , the matrix $A + LC$ always has an eigenvalue at 0.

16.4 Observable canonical form*

Suppose A and C have the following structure:

$$A = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 \end{bmatrix}.$$

Then we say that the pair (C, A) is in **observable canonical form**. Note that (C, A) is in observable canonical form if and only if (A', C') is in controllable canonical form. If (C, A) is in observable canonical form, then $\dot{x} = Ax$ and $y = Cx$ look like

$$\begin{aligned} \dot{x}_1 &= -a_0 x_n \\ \dot{x}_2 &= -a_1 x_n + x_1 \\ &\vdots \\ \dot{x}_n &= -a_{n-1} x_n + x_{n-1} \\ y &= x_n \end{aligned}$$

Also,

$$\det(sI - A) = s^n + a_{n-1}s^{n-1} + \cdots + a_0.$$

If we reverse the ordering of the states, then

$$\begin{aligned}\dot{x}_1 &= -a_{n-1}x_1 + x_2 \\ &\vdots \\ \dot{x}_{n-1} &= -a_1x_1 + x_n \\ \dot{x}_n &= -a_0x_1 \\ y &= x_1\end{aligned}$$

and the corresponding A and C matrices have the following structure:

$$A = \begin{bmatrix} -a_{n-1} & 1 & 0 & \cdots & 0 \\ -a_{n-2} & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ -a_1 & 1 & 0 & \cdots & 1 \\ -a_0 & 0 & 0 & \cdots & 0 \end{bmatrix} \quad \text{and} \quad C = [1 \ 0 \ \cdots \ 0 \ 0]$$

16.5 Discrete-time systems

The results for discrete-time systems are basically the same as those for continuous-time systems. Consider a discrete-time system described by

$$\begin{aligned}x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k)\end{aligned} \tag{16.3}$$

Observer or state estimator. Recalling the observer structure in the continuous-time case, we consider here observers which have the following structure.

$$\boxed{\begin{aligned}\hat{x}(k+1) &= A\hat{x}(k) + Bu(k) + L(\hat{y}(k) - y(k)) \\ \hat{y}(k) &= C\hat{x}(k) + Du(k)\end{aligned}} \quad \hat{x}(0) = \hat{x}_0 \tag{16.4}$$

where the n -vector $\hat{x}(k)$ is the **estimated state**; the matrix L is called the **observer gain matrix**.

Introducing the **state estimation error**

$$\tilde{x} := \hat{x} - x$$

and using (16.3) and (16.4) one can readily show that the evolution of the estimation error is simply governed by

$$\boxed{\tilde{x}(k+1) = (A + LC)\tilde{x}(k)}$$

Hence, if one can choose L such that all the eigenvalues of the matrix $A + LC$ have magnitude less than one then, we obtain the desired result that

$$\lim_{t \rightarrow \infty} \tilde{x}(k) = 0.$$

Dead beat observer. If (C, A) is observable, choose L so that all the eigenvalues of $A + LC$ are zero. Then, the state estimation error goes to zero in at most n steps.

Exercises

Exercise 164 *BB in laundromat:* Obtain a state space representation of the following system.

$$m\ddot{\phi}_1 - m\Omega^2\phi_1 + \frac{k}{2}(\phi_1 - \phi_2) = 0$$

$$m\ddot{\phi}_2 - m\Omega^2\phi_2 - \frac{k}{2}(\phi_1 - \phi_2) = u$$

$$y = \phi_2$$

Show that this system is observable. Considering

$$\Omega = 1, m = 1, k = 2$$

obtain a state estimator (observer) which results in an estimation error system with eigenvalues

$$-1, -2, -3, -4$$

Use the following methods.

- (a) Obtaining an expression for the characteristic polynomial of the error system in terms of the components of the observer gain matrix.
- (b) `acker`
- (c) `place`

Illustrate the effectiveness of your observer(s) with numerical simulations.

Exercise 165 Consider the system described by

$$\dot{x}_1 = -x_2 + u$$

$$\dot{x}_2 = -x_1 - u$$

$$y = x_1 - x_2$$

where all quantities are scalars.

- (a) Is this system observable?
- (b) Is this system detectable?
- (c) Does there exist an asymptotic state estimator for this system? If an estimator does not exist, explain why; if one does exist, give an example of one.

Chapter 17

Climax: output feedback controllers

Plant. (The object of your control desires.) Consider a system described by

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx\end{aligned}\tag{17.1}$$

where the **state** $x(t)$ is an n -vector, the **control input** $u(t)$ is an m -vector, and the **measured output** $y(t)$ is a p -vector. Initially, we consider $D = 0$ for simplicity of presentation.

17.1 Memoryless (static) output feedback

The simplest type of controller is a memoryless (or static) linear output feedback controller; this is of the form

$$\boxed{u = Ky}$$

where K is a real $m \times p$ matrix, sometimes called a **gain matrix**. This controller results in the following closed loop system:

$$\dot{x} = (A + BKC)x$$

If the **open loop system** $\dot{x} = Ax$ is not asymptotically stable, a natural question is whether one can choose K so that the closed loop system is asymptotically stable. For full state feedback ($C = I$), we have seen that it is possible to do this if (A, B) is controllable or, less restrictively, if (A, B) is stabilizable. If (C, A) is observable or detectable, we might expect to be able to stabilize the plant with static output feedback. This is not the case as the following example illustrates.

Example 193 Unattached mass with position measurement.

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= u \\ y &= x_1\end{aligned}$$

This system is both controllable and observable. All linear static output feedback controllers are given by

$$u = ky$$

which results in

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= kx_1\end{aligned}$$

Such a system is never asymptotically stable.

Note that if the plant has some damping in it, that is,

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -dx_2 + u \\ y &= x_1\end{aligned}$$

where $d > 0$, the closed loop system is given by

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= kx_1 - dx_2\end{aligned}$$

This is asymptotically stable provided $k < 0$.

Example 194 Consider

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_1 + u \\ y &= -2x_1 + x_2\end{aligned}$$

For a SISO system, one could use **root locus** techniques to determine whether or not the system is stabilizable via static output feedback. For a general MIMO system, there are currently no easily verifiable conditions which are both necessary and sufficient for stabilizability via static output feedback. It is a topic of current research. So where do we go now?

17.2 Dynamic output feedback

A dynamic output feedback controller is a dynamic input-output system whose input is the measured output of the plant and whose output is the control input to the plant. Thus, a linear, time-invariant, dynamic, output-feedback controller is described by

$$\boxed{\begin{aligned}\dot{x}_c &= A_c x_c + B_c y \\ u &= C_c x_c + D_c y\end{aligned}} \quad (17.2)$$

where the n_c -vector $x_c(t)$ is called the **controller state** and n_c is the **order of the controller**. We regard a memoryless controller as a controller of order zero.

Using transfer functions, the above controller can be described by

$$\hat{u}(s) = \hat{G}_c(s)\hat{y}(s)$$

where

$$\hat{G}_c(s) = C_c(sI - A_c)^{-1}B_c + D_c$$

Closed loop system. Application of the dynamic controller (17.2) to the plant (17.1) yields the following closed loop system:

$$\begin{aligned}\dot{x} &= (A + BD_cC)x + BC_cx_c \\ \dot{x}_c &= B_cCx + A_cx_c\end{aligned}\tag{17.3}$$

This is an LTI system whose state is

$$\begin{bmatrix} x \\ x_c \end{bmatrix}$$

and whose “A-matrix” is given by

$$\mathcal{A} = \begin{bmatrix} A + BD_cC & BC_c \\ B_cC & A_c \end{bmatrix}$$

So the order of the closed loop system is $n + n_c$.

Example 195 Consider a SISO system and recall that a PI (proportional integral) controller can be described by

$$u(t) = -k_P y(t) - k_I \int_0^t y(\tau) dt$$

Letting

$$x_c(t) = \int_0^t y(\tau) dt$$

it can readily be seen that this controller is a first order dynamic system described by

$$\begin{aligned}\dot{x}_c &= y \\ u &= -k_I x_c - k_P y\end{aligned}$$

with initial condition $x_c(0) = 0$.

Exercise 166 Show that the unattached mass with position feedback can be stabilized with a first order dynamic output feedback controller. What controller parameters place all the eigenvalues of the closed loop system at -1 ?

We now demonstrate the following result:

If a complex number λ is an unobservable eigenvalue of (C, A) then λ is an eigenvalue of $\mathcal{A}$.

To see this, suppose λ is an unobservable eigenvalue of (C, A) . Then

$$\text{rank} \begin{bmatrix} A - \lambda I \\ C \end{bmatrix} < n$$

Since the above matrix has n columns, it must have non-zero nullity; this means there exists a nonzero n -vector v with

$$\begin{bmatrix} A - \lambda I \\ C \end{bmatrix} v = 0$$

or, equivalently,

$$\begin{aligned} Av &= \lambda v \\ Cv &= 0 \end{aligned}$$

Letting $\clubsuit$ be the $n + p$ vector

$$\clubsuit = \begin{bmatrix} v \\ 0 \end{bmatrix}$$

it should be clear that

$$\mathcal{A}\clubsuit = \lambda\clubsuit$$

Since $\clubsuit \neq 0$, it follows that λ is an eigenvalue of $\mathcal{A}$ ■

Exercise 167 Prove the following statement. If λ is an uncontrollable eigenvalue of (A, B) , then λ is an eigenvalue of $\mathcal{A}$.

We can now state the following intuitively appealing result.

Lemma 19 *If either (A, B) is not stabilizable or (C, A) is not detectable, then plant (17.1) is not stabilizable by a LTI dynamic output feedback controller of any order.*

The last lemma states that stabilizability of (A, B) and detectability of (C, A) is necessary for stabilizability via linear dynamic output feedback control. In the next section, we demonstrate that if these conditions are satisfied then closed loop asymptotic stability can be achieved with a controller of order no more than the plant order.

17.3 Observer based controllers

Consider a general LTI plant described by

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned} \tag{17.4}$$

An observer based controller has the following structure:

$$\begin{aligned} \dot{\hat{x}} &= A\hat{x} + Bu + L(C\hat{x} + Du - y) \\ u &= K\hat{x} \end{aligned}$$

(17.5)

Note that this controller is completely specified by specifying the gain matrices K and L ; also this controller can be written as

$$\begin{aligned}\dot{\hat{x}} &= [A + BK + L(C + DK)]\hat{x} - Ly \\ u &= K\hat{x}\end{aligned}$$

This is a dynamic output feedback controller with state $x_c = \hat{x}$ (hence $n_c = n$, that is, the controller has same order as plant) and $A_c = A + BK + L(C + DK)$, $B_c = -L$, $C_c = K$, and $D_c = 0$.

Closed loop system. Combining the plant description (17.1) with the controller description (17.5), the closed loop system can be described by

$$\begin{aligned}\dot{x} &= Ax + BK\hat{x} \\ \dot{\hat{x}} &= -LCx + (A + BK + LC)\hat{x}\end{aligned}$$

Let $\tilde{x}$ be the state estimation error, that is,

$$\tilde{x} = \hat{x} - x. \quad (17.6)$$

Then the closed loop system is described by

$$\begin{aligned}\dot{x} &= (A + BK)x + BK\tilde{x} \\ \dot{\tilde{x}} &= (A + LC)\tilde{x}\end{aligned} \quad (17.7)$$

This is an LTI system with “A matrix”

$$\mathcal{A} = \begin{bmatrix} A + BK & BK \\ 0 & A + LC \end{bmatrix}$$

Noting that

$$\det(sI - \mathcal{A}) = \det(sI - A - BK) \det(sI - A - LC) \quad (17.8)$$

it follows that the set of eigenvalues of the closed loop system are simply the union of those of $A + BK$ and those of $A + LC$. So, if both $A + BK$ and $A + LC$ are asymptotically stable, the closed loop system is asymptotically stable. If (A, B) is stabilizable, one can choose K so that $A + BK$ is asymptotically stable. If (C, A) is detectable, one can choose L so that $A + LC$ is asymptotically stable. Combining these observations with Lemma 19 leads to the following result.

Theorem 18 *The following statements are equivalent.*

- (a) (A, B) is stabilizable and (C, A) is detectable.
- (b) Plant (17.1) is stabilizable via a LTI dynamic output feedback controller.
- (c) Plant (17.1) is stabilizable via a LTI dynamic output feedback controller whose order is less than or equal to that of the plant.

Example 196 The unattached mass with position measurement

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= u \\ y &= x_1\end{aligned}$$

Observer based controllers are given by

$$\begin{aligned}\dot{\hat{x}}_1 &= \hat{x}_2 + l_1(\hat{x}_1 - y) \\ \dot{\hat{x}}_2 &= u + l_2(\hat{x}_1 - y) \\ u &= k_1\hat{x}_1 + k_2\hat{x}_2\end{aligned}$$

which is the same as

$$\begin{aligned}\dot{\hat{x}}_1 &= l_1\hat{x}_1 + \hat{x}_2 - l_1y \\ \dot{\hat{x}}_2 &= (k_1 + l_2)\hat{x}_1 + k_2\hat{x}_2 - l_2y\end{aligned}$$

The closed loop system is described by

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= k_1x_1 + k_2x_2 + k_1\tilde{x}_1 + k_2\tilde{x}_2 \\ \dot{\tilde{x}}_1 &= l_1\tilde{x}_1 + \tilde{x}_2 \\ \dot{\tilde{x}}_2 &= l_2\tilde{x}_1\end{aligned}$$

where $\tilde{x}_i = \hat{x}_i - x_i$, $i = 1, 2$ are the estimation error state variables. The characteristic polynomial of the closed loop system is

$$p(s) = (s^2 - k_2s - k_1)(s^2 - l_1s - l_2)$$

Hence, if

$$k_1, k_2, l_1, l_2 < 0$$

the closed loop system is asymptotically stable.

17.4 Discrete-time systems

Basically, everything said above for continuous-time systems also holds for discrete-time systems. The only differences are the way systems are described are the characterization of stability, stabilizability, and detectability in terms of eigenvalues.

Plant.

$$x(k+1) = Ax(k) + Bu(k) \quad (17.9a)$$

$$y(k) = Cx(k) \quad (17.9b)$$

with state $x(k) \in \mathbb{R}^n$, control input $u(k) \in \mathbb{R}^m$, and measured output $y(k) \in \mathbb{R}^p$.

17.4.1 Memoryless output feedback

A *memoryless (or static) linear output feedback controller* is of the form

$$\boxed{u(k) = Ky(k)}$$

where K is a real $m \times p$ matrix, sometimes called a *gain matrix*. This controller results in the following *closed loop system*:

$$x(k+1) = (A + BKC)x(k)$$

17.4.2 Dynamic output feedback

In general, a *linear dynamic output feedback controller* is described by

$$\boxed{\begin{aligned} x_c(k+1) &= A_c x_c(k) + B_c y(k) \\ u(k) &= C_c x_c(k) + D_c y(k) \end{aligned}} \quad (17.10)$$

where $x_c(k) \in \mathbb{R}^{n_c}$ is called the *controller state*; n_c is the *order of the controller*.

Closed loop system.

$$\begin{aligned} x(k+1) &= (A + BD_cC)x(k) + BC_c x_c(k) \\ x_c(k+1) &= B_c C x(k) + A_c x_c(k) \end{aligned} \quad (17.11)$$

This is an lti system with state $\begin{bmatrix} x \\ x_c \end{bmatrix}$ and “A-matrix”

$$\mathcal{A} = \begin{bmatrix} A + BD_cC & BC_c \\ B_c C & A_c \end{bmatrix}$$

17.4.3 Observer based controllers

Consider

$$x(k+1) = Ax(k) + Bu(k) \quad (17.12a)$$

$$y(k) = Cx(k) + Du(k) \quad (17.12b)$$

An observer based controller has the following structure:

$$\boxed{\begin{aligned} \hat{x}(k+1) &= A\hat{x}(k) + Bu + L[C\hat{x}(k) + Du(k) - y(k)] \\ u(k) &= K\hat{x}(k) \end{aligned}} \quad (17.13)$$

Closed loop system. Let $\tilde{x}$ be the estimation error, that is, $\tilde{x} = \hat{x} - x$. Then the closed loop system is described by

$$\begin{aligned} x(k+1) &= (A + BK)x(k) + BK\tilde{x}(k) \\ \tilde{x}(k+1) &= (A + LC)\tilde{x}(k) \end{aligned} \quad (17.14)$$

This is an lti system with “A matrix”

$$\mathcal{A} = \begin{bmatrix} A + BK & BK \\ 0 & A + LC \end{bmatrix}$$

Dead-beat controllers. Place all eigenvalues of $A + BK$ and $A + LC$ at zero.

Exercises

Exercise 168 Consider the system,

$$\begin{aligned}\dot{x}_1 &= x_2 + u \\ \dot{x}_2 &= x_1 \\ y &= x_1\end{aligned}$$

where all quantities are scalar. Obtain an output feedback controller which results in an asymptotically stable closed loop system.

Exercise 169 Consider the system with input u , output y , and state variables x_1, x_2, x_3, x_4 described by

$$\begin{aligned}\dot{x}_1 &= -x_1 \\ \dot{x}_2 &= x_1 - 2x_2 \\ \dot{x}_3 &= x_1 - 3x_3 + u \\ \dot{x}_4 &= x_1 + x_2 + x_3 + x_4 \\ y &= x_3\end{aligned}$$

(a) Is this system observable? (b) Is this system stabilizable using output feedback? Justify your answers.

Exercise 170 Consider the system,

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_1 + u \\ y &= x_1\end{aligned}$$

where all quantities are scalar. Obtain an output feedback controller which results in an asymptotically stable closed loop system.

Exercise 171 Consider the system

$$\begin{aligned}\dot{x}_1 &= -x_1 + x_3 \\ \dot{x}_2 &= u \\ \dot{x}_3 &= x_2 \\ y &= x_3\end{aligned}$$

with scalar control input u and scalar measured output y .

- (a) Obtain an observer-based output feedback controller which results in an asymptotically stable closed loop system.
- (b) Can all the eigenvalues of the closed loop system be arbitrarily placed?

Exercise 172 Consider the system,

$$\begin{aligned}\dot{x}_1 &= x_2 + u_1 \\ \dot{x}_2 &= u_2 \\ y &= x_1\end{aligned}$$

where all quantities are scalar. Obtain an output feedback controller which results in an asymptotically stable closed loop system.

Exercise 173 *Stabilization of 2-link manipulator with dynamic output feedback.*

$$\begin{aligned}[m_1lc_1^2 + m_2l_1^2 + I_1]\ddot{q}_1 + [m_2l_1lc_2 \cos(q_1 - q_2)]\ddot{q}_2 + m_2l_1lc_2 \sin(q_1 - q_2)\dot{q}_2^2 \\ - [m_1lc_1 + m_2l_1]g \sin(q_1) = u\end{aligned}$$

$$\begin{aligned}[m_2l_1lc_2 \cos(q_1 - q_2)]\ddot{q}_1 + [m_2lc_2^2 + I_2]\ddot{q}_2 - m_2l_1lc_2 \sin(q_1 - q_2)\dot{q}_1^2 - m_2glc_2 \sin(q_2) = 0 \\ y = q_2\end{aligned}$$

Recall the linearization of this system about

$$q_1 = q_2 = u = 0$$

and recall its state space description.

Use parameter values previously given.

Is the linearization observable? Design an observer based dynamic output feedback controller which asymptotically stabilizes this linearized system. Apply the controller to the nonlinear system and simulate the nonlinear closed loop system for various initial conditions. (Use zero initial conditions for the observer.)

Chapter 18

Constant output tracking in the presence of constant disturbances

18.1 Zeros

Consider a SISO system described by

$$\dot{x} = Ax + Bu \quad (18.1a)$$

$$y = Cx + Du \quad (18.1b)$$

where $u(t)$ and $y(t)$ are scalars while $x(t)$ is an n -vector. The transfer function $\hat{g}$ associated with this system is given by

$$\hat{g}(s) = C(sI - A)^{-1}B + D. \quad (18.2)$$

Recall that a scalar λ is a **zero** of $\hat{g}$ if

$$\hat{g}(\lambda) = 0.$$

Every rational scalar transfer function $\hat{g}$ can be written as $\hat{g} = n/d$ where n and d are polynomials with no common zeros. With this representation, the zeros of $\hat{g}$ are the zeros of n while the poles of $\hat{g}$ are the zeros of d . Hence, a scalar cannot be simultaneously a pole and a zero of $\hat{g}$. Also, the eigenvalues of A which are neither uncontrollable nor unobservable are the poles of $\hat{g}$.

18.1.1 A state space characterization of zeros

For each scalar λ we define T_λ to be the $(n+1) \times (n+1)$ matrix given by

$$T_\lambda = \begin{bmatrix} A - \lambda I & B \\ C & D \end{bmatrix}. \quad (18.3)$$

This matrix yields the following state space characterization of zeros.

Theorem 19 *Consider a SISO system described by (18.1) and let T_λ be the matrix defined by (18.3). Then $\det T_\lambda = 0$ if and only if at least one of the following conditions hold.*

(a) λ is an uncontrollable eigenvalue of (A, B) .

(b) λ is an unobservable eigenvalue of (C, A) .

(c) λ is a zero of $\hat{g}$.

PROOF. We first demonstrate that if at least one of conditions (a), (b), or (c) hold, then $\det T_\lambda = 0$. When λ is an uncontrollable eigenvalue of (A, B) , we have

$$\text{rank} \begin{bmatrix} A - \lambda I & B \end{bmatrix} < n.$$

This implies that $\text{rank } T_\lambda < n + 1$; hence $\det T_\lambda = 0$. In a similar fashion, if λ is an unobservable eigenvalue of (C, A) , then

$$\text{rank} \begin{bmatrix} A - \lambda I \\ C \end{bmatrix} < n.$$

From this it also follows that $\det T_\lambda = 0$. Now consider the case in which λ is a zero of $\hat{g}$ but is neither an uncontrollable eigenvalue of (A, B) nor an unobservable eigenvalue of (C, A) . Since λ is a zero of $\hat{g}$ it follows that λ is not a pole of $\hat{g}$. Thus λ is not an eigenvalue of A . Hence $\lambda I - A$ is invertible. Letting $x_0 = (\lambda I - A)^{-1}B$, we obtain that

$$(A - \lambda I)x_0 + B = 0$$

and

$$Cx_0 + D = C(\lambda I - A)^{-1}B + D = \hat{g}(\lambda) = 0.$$

This implies that

$$\begin{bmatrix} A - \lambda I & B \\ C & D \end{bmatrix} \begin{bmatrix} x_0 \\ 1 \end{bmatrix} = 0.$$

Hence, $\det T_\lambda = 0$.

We now show that if $\det T_\lambda = 0$, then at least one of conditions (a), (b), or (c) must hold. To this end suppose that $\det T_\lambda = 0$ and (a) and (b) do not hold. Then we need to show that (c) holds. Since $\det T_\lambda = 0$, there is an n -vector x_0 and a scalar u_0 , such that

$$\begin{bmatrix} A - \lambda I & B \\ C & D \end{bmatrix} \begin{bmatrix} x_0 \\ u_0 \end{bmatrix} = 0. \quad (18.4)$$

and u_0 and x_0 cannot both be zero. We now claim that u_0 cannot be zero. If $u_0 = 0$, then $x_0 \neq 0$ and

$$\begin{bmatrix} A - \lambda I \\ C \end{bmatrix} x_0 = 0.$$

This contradicts the assumption that λ is not an unobservable eigenvalue of (C, A) . Hence u_0 is non-zero and, by scaling x_0 , we can consider $u_0 = 1$. It follows from (18.4) that

$$(A - \lambda I)x_0 + B = 0 \quad (18.5a)$$

$$Cx_0 + D = 0 \quad (18.5b)$$

Hence, $B = -(A - \lambda I)x_0$, that is B is a linear combination of the columns of $A - \lambda I$. This implies that

$$\text{rank} \begin{bmatrix} A - \lambda I & B \end{bmatrix} = \text{rank}(A - \lambda I).$$

Using this conclusion and the fact that λ is not an uncontrollable eigenvalue of (A, B) , we obtain that

$$n = \text{rank} \begin{bmatrix} A - \lambda I & B \end{bmatrix} = \text{rank}(A - \lambda I).$$

Since $\text{rank}(A - \lambda I) = n$, it follows that λ is not an eigenvalue of A . It now follows from (18.5a) that $x_0 = (\lambda I - A)^{-1}B$; hence, using (18.5a), we obtain

$$0 = Cx_0 + D = C(\lambda I - A)^{-1}B + D = \hat{g}(\lambda),$$

that is, λ is a zero of $\hat{g}$. ■

Corollary 4 *Suppose (A, B, C, D) is a state space realization of a SISO transfer function $\hat{g}$ with (A, B) controllable and (C, A) observable. Then a scalar λ is a zero of $\hat{g}$ if and only if*

$$\det \begin{bmatrix} A - \lambda I & B \\ C & D \end{bmatrix} = 0.$$

Example 197 Consider the transfer function

$$\hat{g}(s) = \frac{s^2 - 1}{s^2 + 1}.$$

The zeros of this transfer function are 1 and -1 . A controllable/unobservable realization of this transfer function is given by

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} -2 & 0 \end{bmatrix}, \quad D = 1.$$

For this realization,

$$\det T_\lambda = \det \begin{bmatrix} A - \lambda I & B \\ C & D \end{bmatrix} = \det \begin{bmatrix} -\lambda & 1 & 0 \\ -1 & -\lambda & 1 \\ -2 & 0 & 1 \end{bmatrix} = \lambda^2 - 1.$$

Clearly, the zeros of $\hat{g}$ are given by $\det T_\lambda = 0$.

18.2 Tracking and disturbance rejection with state feedback

Consider a system described by

$$\begin{aligned} \dot{x} &= Ax + B_1w + B_2u \\ z &= Cx + D_1w + D_2u \end{aligned} \tag{18.6}$$



Figure 18.1: The plant to be controlled

where the n -vector $x(t)$ is the state; the m_1 -vector w is an **unknown, constant, disturbance input**; the m_2 -vector $u(t)$ is the control input; and the p -vector $z(t)$ is a **performance output**. Initially, we assume that the state x and the performance output z can be measured.

The control problem we consider here is the following: Suppose r is any **constant reference output**; we wish to design a feedback controller such that for all initial conditions $x(0) = x_0$, we have

$$\lim_{t \rightarrow \infty} z(t) = r$$

and the state $x(t)$ is bounded for all $t \geq 0$.

Assumptions. *Clearly, we need the pair (A, B_2) to be stabilizable. We will also need the following condition:*

$$\boxed{\text{rank} \begin{bmatrix} A & B_2 \\ C & D_2 \end{bmatrix} = n + p} \quad (18.7)$$

that is, the the rank of the above matrix must equal the number of its rows.

For a system with scalar control input u and scalar performance output z , the above rank condition is equivalent to

$$\det \begin{bmatrix} A & B_2 \\ C & D_2 \end{bmatrix} \neq 0.$$

By Theorem 19, this is equivalent to the requirement that 0 is neither an uncontrollable eigenvalue of (A, B_2) , an unobservable eigenvalue of (C, A) , nor a zero of the transfer function $T_{zu}(s) = C(sI - A)^{-1}B_2 + D_2$.

We now claim that condition (18.7) is necessary to achieve our control objectives. In fact, we demonstrate the following result.

If condition (18.7) does not hold then, there is a reference output r_0 such that one cannot achieve $\lim_{t \rightarrow \infty} z(t) = r_0$ with a bounded state history.

To see this, suppose the above condition does not hold. Then there is a nonzero vector

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

with $v_1 \in \mathbb{R}^n$ and $v_2 \in \mathbb{R}^p$ such that

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}' \begin{bmatrix} A & B_2 \\ C & D_2 \end{bmatrix} = 0,$$

that is,

$$v'_1 A + v'_2 C = 0, \quad v'_1 B_2 + v'_2 D_2 = 0$$

Since (A, B_2) is stabilizable, we must have $v_2 \neq 0$. (Why?) Consider $w = 0$, pre-multiply the first and second equations in the system description by v'_1 and v'_2 , respectively, and add to obtain

$$v'_1 \dot{x} + v'_2 z = 0,$$

that is,

$$\frac{dv'_1 x}{dt} = -v'_2 z$$

Suppose

$$\lim_{t \rightarrow \infty} z(t) = r_0 := -v_2$$

Then,

$$\lim_{t \rightarrow \infty} -v'_2 z(t) = -v'_2(-v_2) = \|v_2\|^2 \neq 0.$$

Hence there exists a time T and a positive number ϵ such that

$$\frac{dv'_1 x}{dt} = -v'_2 z(t) \geq \epsilon \quad \text{for all } t \geq T$$

which implies that, for $t \geq T$,

$$v'_1 x(t) \geq v'_1 x(T) + \epsilon(t - T)$$

This implies that $v'_1 x$ is unbounded, So, x is unbounded. ■

Example 198 Consider

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1 - x_2 + u \\ z &= x_2 \end{aligned}$$

Here

$$\begin{bmatrix} A & B_2 \\ C & D_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & -1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Clearly rank condition (18.7) does not hold. It should be clear from the first differential equation describing this system that, in order to keep $z = x_2$ equal to a nonzero constant, x_1 must be unbounded. Note that the transfer function from u to z , given by

$$\frac{s}{s^2 + s + 1},$$

has a zero at 0.

Since the rank of a matrix is always less than or equal to the number of its columns, it follows that if rank condition (18.7) holds then,

$$p \leq m_2,$$

that is, the number of performance output variables is less than or equal to the number of control input variables.

18.2.1 Nonrobust control

Consider first the question of the existence of steady state values of x and u such that z has a steady state value of r . Letting $x(t) \equiv x^e$, $u(t) \equiv u^e$, and $z(t) \equiv r$ in system description (18.6) results in

$$\begin{bmatrix} A & B_2 \\ C & D_2 \end{bmatrix} \begin{bmatrix} x^e \\ u^e \end{bmatrix} = \begin{bmatrix} -B_1 w \\ -D_1 w + r \end{bmatrix}$$

Since the matrix $\begin{bmatrix} A & B_2 \\ C & D_2 \end{bmatrix}$ has full row rank, there exists a solution (x^e, u^e) to the above equation. Consider now the controller

$$u = u^e + K[x - x^e]$$

Letting

$$\delta x := x - x^e$$

the closed loop system is described by

$$\begin{aligned} \delta \dot{x} &= [A + B_2 K] \delta x \\ z &= C \delta x + r \end{aligned}$$

Choosing K so that $A + B_2 K$ is asymptotically stable results in $\lim_{t \rightarrow \infty} \delta x(t) = 0$ and, hence, $\lim_{t \rightarrow \infty} z(t) = r$.

In addition to requiring knowledge of w , this approach also requires exact knowledge of the matrices A, B_2, C, D_2 . Inexact knowledge of these matrices yields inexact values of the required steady values x_e and u_e . This results in the steady state value of z being offset from the desired value r .

18.2.2 Robust controller

To robustly reject constant disturbances and track constant reference outputs, we need controllers which use integral action. This is what we consider here. Also, we reduce the original disturbance rejection/output tracking problem to a stabilization problem.

Integration of output error. Introduce a new state variable x_I given by

$$\boxed{\dot{x}_I = z - r}$$

where $x_I(0) = x_{I0}$ and x_{I0} is arbitrary. Roughly speaking, this state is the integral of the output tracking error, $z - r$; specifically, it is given by

$$x_I(t) = x_{I0} + \int_0^t (z(\tau) - r) d\tau.$$

Introducing new state, disturbance input and performance output,

$$\tilde{x} := \begin{bmatrix} x \\ x_I \end{bmatrix}, \quad \tilde{w} := \begin{bmatrix} w \\ r \end{bmatrix}, \quad \tilde{z} := z - r,$$

respectively, we obtain the **augmented plant**,

$$\begin{aligned}\dot{\tilde{x}} &= \tilde{A}\tilde{x} + \tilde{B}_1\tilde{w} + \tilde{B}_2u \\ \tilde{z} &= \tilde{C}\tilde{x} + \tilde{D}_1\tilde{w} + \tilde{D}_2u\end{aligned}$$

where

$$\begin{aligned}\tilde{A} &= \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} & \tilde{B}_1 &= \begin{bmatrix} B_1 & 0 \\ D_1 & -I \end{bmatrix} & \tilde{B}_2 &= \begin{bmatrix} B_2 \\ D_2 \end{bmatrix} \\ \tilde{C} &= \begin{bmatrix} C & 0 \end{bmatrix} & \tilde{D}_1 &= \begin{bmatrix} D_1 & -I \end{bmatrix} & \tilde{D}_2 &= D_2\end{aligned}$$

We have now the following result.

Lemma 20 *The pair $(\tilde{A}, \tilde{B}_2)$ is controllable (stabilizable) if and only if*

- (a) *the pair (A, B_2) is controllable (stabilizable), and*
- (b) *rank condition (18.7) holds.*

PROOF. The pair $(\tilde{A}, \tilde{B}_2)$ is controllable (stabilizable) iff

$$\text{rank} \begin{bmatrix} \tilde{A} - \lambda I & \tilde{B}_2 \end{bmatrix} = n + p$$

for all complex numbers λ (for all complex λ with $\Re(\lambda) \geq 0$). Also,

$$\begin{bmatrix} \tilde{A} - \lambda I & \tilde{B}_2 \end{bmatrix} = \begin{bmatrix} A - \lambda I & 0 & B_2 \\ C & -\lambda I & D_2 \end{bmatrix}$$

Considering $\lambda = 0$, we obtain

$$\text{rank} \begin{bmatrix} \tilde{A} - \lambda I & \tilde{B}_2 \end{bmatrix} = \text{rank} \begin{bmatrix} A & 0 & B_2 \\ C & 0 & D_2 \end{bmatrix} = \text{rank} \begin{bmatrix} A & B_2 \\ C & D_2 \end{bmatrix}. \quad (18.8)$$

Considering $\lambda \neq 0$, we obtain

$$\begin{aligned}\text{rank} \begin{bmatrix} \tilde{A} - \lambda I & \tilde{B}_2 \end{bmatrix} &= \text{rank} \begin{bmatrix} A - \lambda I & 0 & B_2 \\ C & -\lambda I & D_2 \end{bmatrix} \\ &= \text{rank} \begin{bmatrix} A - \lambda I & B_2 & 0 \\ 0 & 0 & -\lambda I \end{bmatrix} \\ &= \text{rank} \begin{bmatrix} A - \lambda I & B_2 \end{bmatrix} + p\end{aligned} \quad (18.9)$$

Since the pair (A, B_2) is controllable (stabilizable) iff $\text{rank} \begin{bmatrix} A - \lambda I & B_2 \end{bmatrix} = n$ for all complex numbers λ (for all complex λ with $\Re(\lambda) \geq 0$), the desired result now follows from (18.8) and (18.9). ■

Controller design. Suppose (A, B_2) is stabilizable and rank condition (18.7) holds. Then $(\tilde{A}, \tilde{B}_2)$ is also stabilizable. Choose *any* matrix $\tilde{K}$ such that $\tilde{A} + \tilde{B}_2\tilde{K}$ is asymptotically stable. Let

$$u = \tilde{K}\tilde{x}$$

If we partition $\tilde{K}$ as

$$\tilde{K} = \begin{bmatrix} K_P & K_I \end{bmatrix}$$

where K_P is given by the first n columns of $\tilde{K}$ and K_I is given by the last p columns of $\tilde{K}$, then the controller is given by

$$\begin{array}{lcl} \dot{x}_I & = & z - r \\ u & = & K_P x + K_I x_I \end{array}$$

Note that this controller can be written as

$$u = K_P x + K_I \int (z - r)$$

This is a generalization of the classical PI (Proportional Integral) Controller.

Closed loop system. The closed loop system is given by

$$\begin{array}{lcl} \dot{\tilde{x}} & = & [\tilde{A} + \tilde{B}_2\tilde{K}]\tilde{x} + \tilde{B}_1\tilde{w} \\ \dot{\tilde{z}} & = & [\tilde{C} + \tilde{D}_2\tilde{K}]\tilde{x} + \tilde{D}_1\tilde{w} \end{array}$$

Since the matrix $\tilde{A} + \tilde{B}_2\tilde{K}$ is asymptotically stable and $\tilde{w}$ is constant, all solutions $\tilde{x}$ are bounded for $t \geq 0$. Thus, $x(t)$ is bounded for all $t \geq 0$.

Letting $\eta = \dot{\tilde{x}}$ and noting that $\tilde{w}$ is constant, we obtain that

$$\dot{\eta} = [\tilde{A} + \tilde{B}_2\tilde{K}]\eta.$$

Since the matrix $\tilde{A} + \tilde{B}_2\tilde{K}$ is asymptotically stable, we must have

$$\lim_{t \rightarrow \infty} \dot{\tilde{x}} = \lim_{t \rightarrow \infty} \eta(t) = 0.$$

In particular, we have

$$\lim_{t \rightarrow \infty} \dot{x}_I = 0;$$

hence

$$\lim_{t \rightarrow \infty} z(t) = r.$$

Thus, the proposed controller achieves the desired behavior.

Example 199 Consider the constantly disturbed and controlled harmonic oscillator described by

$$\begin{array}{lcl} \dot{x}_1 & = & x_2 \\ \dot{x}_2 & = & -x_1 + w + u \\ z & = & x_1 \end{array}$$

Here

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad B_1 = B_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D_1 = D_2 = 0$$

The pair (A, B_2) is controllable and

$$\text{rank} \begin{bmatrix} A & B_2 \\ C & D_2 \end{bmatrix} = \text{rank} \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = 3 = n + p$$

Hence

$$\tilde{A} = \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad \text{and} \quad \tilde{B}_2 = \begin{bmatrix} B_2 \\ D_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Letting

$$\tilde{K} = \begin{bmatrix} k_1 & k_2 & k_I \end{bmatrix}$$

we obtain

$$\tilde{A} + \tilde{B}_2 \tilde{K} = \begin{bmatrix} 0 & 1 & 0 \\ -1 + k_1 & k_2 & k_I \\ 1 & 0 & 0 \end{bmatrix}$$

The characteristic polynomial p of $\tilde{A} + \tilde{B}_2 \tilde{K}$ is given by

$$p(s) = s^3 - k_2 s^2 + (1 - k_1)s - k_I$$

Choosing k_1, k_2, k_I so that this polynomial has roots with negative real parts, a controller yielding constant output tracking, rejecting constant disturbances and yielding bounded states is given by

$$\begin{aligned} \dot{x}_I &= x_1 - r \\ u &= k_1 x_1 + k_2 x_2 + k_I x_I \end{aligned}$$

18.3 Measured Output Feedback Control

Consider a system described by

$$\begin{aligned} \dot{x} &= Ax + B_1 w + B_2 u \\ z &= C_1 x + D_{11} w + D_{12} u \\ y &= C_2 x + D_{21} w \end{aligned}$$

where the n -vector $x(t)$ is the state; the m_1 -vector w is an **unknown, constant, disturbance input**; the m_2 -vector $u(t)$ is the control input; and the p_1 -vector $z(t)$ is a **performance output**. We will assume $z(t)$ can be measured and that instead of x we can only measure the **measured output** $y(t)$ which is a p_2 -vector.

Suppose the p_1 -vector r is any **constant reference output** and we wish to design a controller such that for any initial condition $x(0) = x_0$,

$$\lim_{t \rightarrow \infty} z(t) = r$$

and the state $x(t)$ is bounded for all $t \geq 0$. To achieve this objective, we need the following assumptions.

Assumptions.

- (a) The pair (A, B_2) is stabilizable.
- (b) The pair (C_2, A) is detectable.
- (c)

$$\text{rank} \begin{bmatrix} A & B_2 \\ C_1 & D_{12} \end{bmatrix} = n + p_1$$

As before, we introduce a new state variable x_I given by

$$\dot{x}_I = z - r$$

where $x_I(0)$ is arbitrary. Letting

$$\tilde{x} := \begin{bmatrix} x \\ x_I \end{bmatrix}, \quad \tilde{w} := \begin{bmatrix} w \\ r \end{bmatrix}, \quad \tilde{z} := z - r, \quad \tilde{y} := \begin{bmatrix} y \\ x_I \\ r \end{bmatrix}$$

we obtain the augmented plant

$$\begin{aligned} \dot{\tilde{x}} &= \tilde{A}\tilde{x} + \tilde{B}_1\tilde{w} + \tilde{B}_2u \\ \tilde{z} &= \tilde{C}_1\tilde{x} + \tilde{D}_{11}\tilde{w} + \tilde{D}_{12}u \\ \tilde{y} &= \tilde{C}_2\tilde{x} + \tilde{D}_{21}\tilde{w} \end{aligned} \tag{18.10}$$

with

$$\begin{aligned} \tilde{A} &= \begin{bmatrix} A & 0 \\ C_1 & 0 \end{bmatrix} & \tilde{B}_1 &= \begin{bmatrix} B_1 & 0 \\ D_{11} & -I \end{bmatrix} & \tilde{B}_2 &= \begin{bmatrix} B_2 \\ D_{12} \end{bmatrix} \\ \tilde{C}_1 &= \begin{bmatrix} C_1 & 0 \end{bmatrix} & \tilde{D}_{11} &= \begin{bmatrix} D_{11} & -I \end{bmatrix} & \tilde{D}_{12} &= D_{12} \\ \tilde{C}_2 &= \begin{bmatrix} C_2 & 0 \\ 0 & I \\ 0 & 0 \end{bmatrix} & \tilde{D}_{21} &= \begin{bmatrix} D_{21} & 0 \\ 0 & 0 \\ 0 & I \end{bmatrix} \end{aligned}$$

Consider now any output feedback controller of the form

$$\begin{aligned} \dot{x}_c &= A_c x_c + B_c \tilde{y} \\ u &= C_c x_c + D_c \tilde{y} \end{aligned} \tag{18.11}$$

The resulting closed loop system is described by

$$\begin{bmatrix} \dot{\tilde{x}} \\ \dot{x}_c \end{bmatrix} = \underbrace{\begin{bmatrix} \tilde{A} + \tilde{B}_2 D_c \tilde{C}_2 & \tilde{B}_2 C_c \\ B_c \tilde{C}_2 & A_c \end{bmatrix}}_{A_{cl}} \begin{bmatrix} \tilde{x} \\ x_c \end{bmatrix} + \begin{bmatrix} \tilde{B}_1 + \tilde{B}_2 D_c \tilde{D}_{21} \\ D_c \tilde{D}_{21} \end{bmatrix} \tilde{w} \tag{18.12}$$

From the lemma of the previous section, assumptions (a) and (c) imply that the pair $(\tilde{A}, \tilde{B}_2)$ is stabilizable. Also, detectability of the pair (C_2, A) is equivalent to detectability of the pair $(\tilde{C}_2, \tilde{A})$. Hence, one can always choose matrices A_c, B_c, C_c, D_c so that so that the closed loop system matrix A_{cl} is asymptotically stable.

Controller design. Choose any matrices A_c, B_c, C_c, D_c so that the closed loop system matrix A_{cl} in (18.12) is asymptotically stable. Then the controller is given by (18.11).

If we let

$$\begin{aligned} B_c &= [B_{cP} \quad B_{cI} \quad B_{cr}] \\ D_c &= [B_{cP} \quad B_{cI} \quad B_{cr}] \end{aligned}$$

where the above partitions correspond to the partitions of $\tilde{y}$, we can express this controller as

$$\begin{aligned} \dot{x}_I &= z - r \\ \dot{x}_c &= A_c x_c + B_{cI} x_I + B_{cP} y + B_{cr} r \\ u &= C_c x_c + D_{cI} x_I + D_{cP} y + D_{cr} r \end{aligned}$$

Note that A_{cl} does not depend on B_{cr} and D_{cr} ; hence these two matrices can be arbitrarily chosen.

Closed loop system. The closed loop system is described by (18.12). Since the closed loop system matrix A_{cl} is asymptotically stable and $\tilde{w}$ is constant, it follows that $\tilde{x}$ and x_c are bounded. Hence x is bounded.

Letting

$$\eta = \begin{bmatrix} \dot{\tilde{x}} \\ \dot{x}_c \end{bmatrix}$$

and noting that $\tilde{w}$ is constant, we obtain that

$$\dot{\eta} = A_{cl} \eta$$

Since the matrix A_{cl} is asymptotically stable, we must have

$$\lim_{t \rightarrow \infty} \begin{bmatrix} \dot{\tilde{x}} \\ \dot{x}_c \end{bmatrix} = \lim_{t \rightarrow \infty} \eta(t) = 0$$

In particular, we have

$$\lim_{t \rightarrow \infty} \dot{x}_I = 0;$$

hence

$$\lim_{t \rightarrow \infty} z(t) = r$$

18.3.1 Observer based controllers

As we have seen in a previous chapter, one approach to the design of stabilizing output feedback controllers is to combine an asymptotic state estimator with a stabilizing state feedback gain matrix. Since there is no need to estimate the state x_I of the augmented plant (18.10), we design the observer to estimate only x . This yields the following design.

- Choose gain matrix $\tilde{K}$ so that the $\tilde{A} + \tilde{B}_2 \tilde{K}$ is asymptotically stable and let

$$\tilde{K} = [K_P \quad K_I]$$

where K_P is given by the first n columns of $\tilde{K}$ and K_I is given by the last p_1 columns of $\tilde{K}$.

- Choose observer gain matrix L so that $A + LC_2$ is asymptotically stable.
- An observer based controller is given by

$\dot{x}_I = z - r$	integrator
$\dot{x}_c = Ax_c + B_2u + L(C_2x_c - y)$	observer
$u = K_Px_c + K_Ix_I$	control input

where $x_c(0)$ and $x_I(0)$ are arbitrary.

If we let

$$e := x_c - x$$

the closed loop system due to this controller is described by

$$\begin{bmatrix} \dot{\tilde{x}} \\ \dot{e} \end{bmatrix} = \begin{bmatrix} \tilde{A} + \tilde{B}_2\tilde{K} & * \\ 0 & A + LC_2 \end{bmatrix} \begin{bmatrix} \tilde{x} \\ e \end{bmatrix} + \begin{bmatrix} * \\ * \end{bmatrix} \tilde{w}$$

Since the eigenvalues of the closed loop system matrix

$$\begin{bmatrix} \tilde{A} + \tilde{B}_2\tilde{K} & * \\ 0 & A + LC_2 \end{bmatrix}$$

are the union of those of $\tilde{A} + \tilde{B}_2\tilde{K}$ and $A + LC_2$, it follows that this matrix is asymptotically stable. Hence, this controller yields the desired behavior.

Exercises

Exercise 174 Consider a system described by

$$\hat{z}(s) = T_{zw}(s)\hat{w}(s) + T_{zu}(s)\hat{u}(s)$$

with

$$T_{zw}(s) = \frac{1}{s^2 - 1} \quad \text{and} \quad T_{zu}(s) = \frac{s - 2}{s^2 - 1}$$

where $\hat{z}$, $\hat{w}$, and $\hat{u}$ are the Laplace transforms of the performance output z , disturbance input w , and control input u , respectively. Let r be a desired constant output and assume the disturbance input w is an unknown constant. Design an output (z) feedback controller which guarantees that for all r and w ,

$$\lim_{t \rightarrow \infty} z(t) = r$$

and all signals are bounded. Illustrate your results with simulations.

Exercise 175 Consider a system described by

$$\begin{aligned}\dot{x} &= Ax + B_1w + B_2u \\ y &= Cx + D_1w + D_2u\end{aligned}$$

where $x(t)$, $u(t)$, and w , and $y(t) \in \mathbb{R}^p$ are as above. Suppose that we can only measure the control input u and the measured output y . It is desired to obtain an asymptotic observer for x and w ; that is, we wish to design an observer producing estimates $\hat{x}$ and $\hat{w}$ of x and w with the property that for any initial value x_0 of x and any w , one has

$$\lim_{t \rightarrow \infty} [\hat{x}(t) - x(t)] = 0, \quad \lim_{t \rightarrow \infty} [\hat{w}(t) - w] = 0$$

- (a) State the least restrictive conditions on the system which allows one to construct an asymptotic estimator.
- (b) Give a procedure for constructing an asymptotic estimator.
- (c) Illustrate your results with example(s) and simulation results.

Chapter 19

Lyapunov revisited

19.1 Stability, observability, and controllability

Recall the damped linear oscillator described by

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{k}{m}x_1 - \frac{d}{m}x_2\end{aligned}\tag{19.1}$$

with m, d and k positive. This system is globally asymptotically stable. Suppose that we wanted to prove this using the total energy of the system as a Lyapunov function. which is given by

$$V(x) = \frac{k}{2}x_1^2 + \frac{m}{2}x_2^2 = x'Px$$

where

$$P = \begin{bmatrix} \frac{k}{2} & 0 \\ 0 & \frac{m}{2} \end{bmatrix}.$$

Along any solution, the rate of change of the energy is given by

$$\frac{dV}{dt} = -dx_2^2 = -x'C'Cx$$

where

$$C = \begin{bmatrix} 0 & d^{\frac{1}{2}} \end{bmatrix}$$

From this it should be clear that

$$\begin{aligned}P &> 0 \\ PA + A'P + C'C &= 0\end{aligned}$$

The matrix $Q = C'C$ is hermitian positive semi-definite, but it is not positive definite. Hence, using this P matrix, our existing Lyapunov results only guarantee that the system is stable; they do not guarantee asymptotic stability. Noting that the pair (C, A) is observable,

we present a result in this section which allows us to use the above P matrix to guarantee asymptotic stability of (19.1).

Consider now a general LTI system described by

$$\dot{x} = Ax \quad (19.2)$$

where $x(t)$ is an n -vector and A is a constant $n \times n$ matrix. We have the following result.

Lemma 21 *Suppose there is a hermitian matrix P satisfying*

$$\boxed{\begin{array}{rcl} P & > & 0 \\ PA + A'P + C'C & \leq & 0 \end{array}} \quad (19.3)$$

where (C, A) is observable. Then system (19.2) is asymptotically stable.

PROOF. Suppose there exists a hermitian matrix P which satisfies inequalities (19.3). Consider any eigenvalue λ of A ; we will show that $\Re(\lambda) < 0$. Let $v \neq 0$ be an eigenvector corresponding to λ , that is,

$$Av = \lambda v. \quad (19.4)$$

Then pre-multiplication by $v'P$ yields:

$$v'PAv = \lambda v'Pv;$$

and taking the complex conjugate transpose of both sides of this last equation yields

$$v'A'Pv = \bar{\lambda}v'Pv$$

Pre- and post-multiplying the second inequality in (19.3) by v' and v , respectively, yields

$$v'PAv + v'APv + v'C'Cv \leq 0$$

which implies that

$$\lambda v'Pv + \bar{\lambda}v'Pv + \|Cv\|^2 \leq 0,$$

that is,

$$2\Re(\lambda)v'Pv \leq -\|Cv\|^2.$$

Since $P > 0$, we must have $v'Pv > 0$.

We now show that we must also have $\|Cv\| > 0$. Suppose $\|Cv\| = 0$. Then $Cv = 0$ and, recalling (19.4), results in

$$\begin{bmatrix} A - \lambda I \\ C \end{bmatrix} v = 0$$

Since (C, A) is observable, the above matrix has rank n , and hence $v = 0$; this contradicts $v \neq 0$. So, we must have $\|Cv\| > 0$.

Hence,

$$\Re(\lambda) = -\|Cv\|^2 / 2v'Pv < 0$$

Since $\Re(\lambda) < 0$ for every eigenvalue λ of A , system (19.2) is asymptotically stable. ■

Example 200 Returning to the damped linear oscillator, we see that inequalities (19.3) are satisfied by

$$P = \begin{bmatrix} \frac{k}{2} & 0 \\ 0 & \frac{m}{2} \end{bmatrix} \quad C = \begin{bmatrix} 0 & d^{\frac{1}{2}} \end{bmatrix}$$

Since the oscillator is described by (19.2) with

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{d}{m} \end{bmatrix}$$

we have

$$\begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 0 & d^{\frac{1}{2}} \\ \frac{-d^{\frac{1}{2}}k}{m} & -\frac{d^{\frac{3}{2}}}{m} \end{bmatrix}$$

Since this observability matrix is invertible, (C, A) is observable. Using Lemma 21, one can use the above matrix P to guarantee asymptotic stability of the damped oscillator.

Exercise 176 Prove the following result. Suppose there is a hermitian matrix P satisfying

$$\begin{aligned} P &\geq 0 \\ PA + A'P + C'C &\leq 0 \end{aligned}$$

where (C, A) is detectable. Then the system (19.2) is asymptotically stable.

Recalling that controllability of (A, B) is equivalent to observability of (B', A') , and that asymptotic stability of A and A' are equivalent, we obtain the following result

Lemma 22 Suppose there is a hermitian matrix S satisfying

$$\boxed{\begin{aligned} S &> 0 \\ AS + SA' + BB' &\leq 0 \end{aligned}} \quad (19.5)$$

where (A, B) is controllable. Then system (19.2) is asymptotically stable

Exercise 177 Suppose there is a hermitian matrix S satisfying

$$\begin{aligned} S &\geq 0 \\ AS + SA' + BB' &\leq 0 \end{aligned}$$

where (A, B) is stabilizable. Then system (19.2) is asymptotically stable

Suppose system (19.2) is asymptotically stable, Then for any matrix C (of appropriate dimensions) the Lyapunov equation

$$PA + A'P + C'C = 0 \quad (19.6)$$

has the unique solution

$$P = \int_0^\infty e^{A't} C' C e^{At} dt$$

Recall that this is the infinite time **observability gramian** associated with (C, A) , that is, $P = W_o$; it is always hermitian and positive semi-definite. In addition, as we have seen earlier, W_o is positive definite if and only if (C, A) is observable. These observations and Lemma 21 lead to the following result.

Theorem 20 *The following statements are equivalent.*

- (a) *The system $\dot{x} = Ax$ is asymptotically stable.*
- (b) *There exist a positive definite hermitian matrix P and a matrix C with (C, A) observable which satisfy the Lyapunov equation*

$$\boxed{PA + A'P + C'C = 0} \quad (19.7)$$

- (c) *For every matrix C with (C, A) observable, the Lyapunov equation (19.7) has a unique solution for P and this solution is hermitian positive-definite.*

Recall the original Lyapunov equation

$$PA + A'P + Q = 0$$

If Q is hermitian positive semi-definite, it can always be expressed as $Q = C'C$ by appropriate choice of C and one can readily show that (Q, A) is observable iff (C, A) is observable. Also $\text{rank } C = n$ if and only if Q is positive definite. If C has rank n , the pair (C, A) is observable for any A . So, if Q is positive definite hermitian, the original Lyapunov equation is of the form (19.7) with (C, A) observable.

Recalling that controllability of (A, B) is equivalent to observability of (B', A') , and that asymptotic stability of A and A' are equivalent, we can obtain the following result.

Theorem 21 *The following statements are equivalent.*

- (a) *The system $\dot{x} = Ax$ is asymptotically stable.*
- (b) *There exist a positive definite hermitian matrix S and a matrix B with (A, B) controllable which satisfy the Lyapunov equation*

$$\boxed{AS + SA' + BB' = 0} \quad (19.8)$$

- (c) *For every matrix B with (A, B) controllable, the Lyapunov equation (19.8) has a unique solution for S and this solution is hermitian positive-definite.*

Note that if A is asymptotically stable, the solution S to Lyapunov equation (19.8) is the infinite-time controllability grammian associated with (A, B) , that is $S = W_c$. Also,

$$S = \int_0^\infty e^{At} B B' e^{A't} dt$$

19.2 A simple stabilizing controller*

Consider a linear time-invariant system described by

$$\dot{x} = Ax + Bu$$

where $x(t)$ is an n -vector and $u(t)$ is an m -vector. Consider any positive $T > 0$ and let

$$W = W_c(-T) := \int_{-T}^0 e^{At} BB' e^{A't} dt$$

The above matrix can be regarded as the finite time controllability grammian (over the interval $[-T, 0]$) associated with (A, B) . If (A, B) is controllable, W is invertible.

We now show that the controller

$$u = -B'W^{-1}x$$

yields an asymptotically stable closed loop system.

Using the following properties of e^{At} ,

$$Ae^{At} = \frac{de^{At}}{dt}; \quad e^{A't} A' = \frac{de^{A't}}{dt}$$

we obtain

$$\begin{aligned} AW + WA' &= \int_{-T}^0 \left[Ae^{At} BB' e^{A't} + e^{At} BB' e^{A't} A' \right] dt \\ &= \int_{-T}^0 \left[\frac{de^{At}}{dt} BB' e^{A't} + e^{At} BB' \frac{de^{A't}}{dt} \right] dt \\ &= \int_{-T}^0 \frac{d \left[(e^{At} BB' e^{A't}) \right]}{dt} dt \\ &= BB' - e^{-AT} BB' e^{-A'T}, \end{aligned}$$

that is,

$$AW + WA' - BB' = -e^{-AT} BB' e^{-A'T}$$

Considering the closed loop system

$$\dot{x} = (A - BB'W^{-1})x$$

we have

$$\begin{aligned} (A - BB'W^{-1})W + W(A - BB'W^{-1})' &= AW + WA' - 2BB' \\ &= -BB' - e^{-AT} BB' e^{-A'T} \\ &\leq -BB', \end{aligned}$$

that is,

$$(A - BB'W^{-1})W + W(A - BB'W^{-1})' + BB' \leq 0.$$

Controllability of (A, B) implies controllability of $(A - BB'W^{-1}, B)$. This fact, along with $W > 0$ and the previous inequality imply that the closed-loop system is asymptotically stable. ■

So, the above yields another proof of

controllability	$\implies$	stabilizability
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Chapter 20

Performance*

20.1 The L_2 norm

20.1.1 The rms value of a signal

Consider any scalar signal s which is defined for $0 \leq t < \infty$. One measure of the size of the signal is given by the **rms value** of the signal; we denote this by $\|s\|_2$ and define it by

$$\|s\|_2 = \left(\int_0^\infty |s(t)|^2 dt \right)^{\frac{1}{2}}$$

This is also called the L_2 **norm** of the signal. One can readily show that the three required properties of a norm are satisfied. This norm is simply the generalization of the usual Euclidean norm for n -vectors to signals or functions. Clearly, a signal has a finite L_2 norm if and only if it is square integrable.

Example 201 Consider the exponential signal $s(t) = e^{\lambda t}$ where $\lambda = -\alpha + j\omega$ and α and ω are both real with $\alpha > 0$. Since $|e^{\lambda t}| = e^{\alpha t}$, we obtain, for any $T > 0$,

$$\int_0^T |s(t)|^2 dt = \int_0^T e^{-2\alpha t} dt = \frac{1}{-2\alpha} (e^{-2\alpha T} - 1).$$

Thus,

$$\int_0^\infty |s(t)|^2 dt = \lim_{T \rightarrow \infty} \int_0^T |s(t)|^2 dt = \lim_{T \rightarrow \infty} \frac{1}{-2\alpha} (e^{-2\alpha T} - 1) = \frac{1}{2\alpha}.$$

Hence

$$\|s\|_2 = 1/\sqrt{2\alpha} \quad \text{for} \quad s(t) = e^{(-\alpha + j\omega)t} \quad \text{with} \quad \alpha > 0.$$

Consider now a signal s where $s(t)$ is an n -vector for $0 \leq t < \infty$. We define the L_2 **norm** of the signal s by

$$\|s\|_2 = \left(\int_0^\infty \|s(t)\|^2 dt \right)^{\frac{1}{2}}$$

where $\|s(t)\|$ is the usual Euclidean norm of the n -vector $s(t)$ as given by

$$\|s(t)\|^2 = |s_1(t)|^2 + |s_2(t)|^2 + \cdots + |s_n(t)|^2.$$

One can readily show that the three required properties of a norm are satisfied.

20.1.2 The L_2 norm of a time-varying matrix

Consider now a function G defined on $[0, \infty)$ where $G(t)$ is a matrix for $0 \leq t < \infty$. We define the L_2 norm of G by

$$\|G\|_2 = \left(\int_0^\infty \text{trace}[G^*(t)G(t)] dt \right)^{\frac{1}{2}}$$

Suppose $G(t)$ is $m \times n$ and the m -vectors $g_1(t), g_2(t), \dots, g_n(t)$ are the columns of $G(t)$. Then

$$\begin{aligned} \text{trace}(G(t)^*G(t)) &= g_1(t)^*g_1(t) + g_2(t)^*g_2(t) + \cdots + g_n(t)^*g_n(t) \\ &= \|g_1(t)\|^2 + \|g_2(t)\|^2 + \cdots + \|g_n(t)\|^2. \end{aligned}$$

So we conclude that

$$\|G\|_2^2 = \sum_{i=1}^n \int_0^\infty \|g_i(t)\|^2 dt = \sum_{i=1}^n \|g_i\|_2^2 \quad (20.1)$$

where the g_i is the i -th column of G .

20.2 The H_2 norm of an LTI system

Consider an input-output system with disturbance input w and performance output z described by

$$\begin{aligned} \dot{x} &= Ax + Bw \\ z &= Cx \end{aligned} \quad (20.2)$$

and suppose A is asymptotically stable. Recall that zero initial condition input-output behavior of this system can be completely characterized by its impulse response

$$G(t) = Ce^{At}B. \quad (20.3)$$

Consider first a SISO system with impulse response g . When A is asymptotically stable, it follows that g is square integrable and the L_2 norm $\|g\|_2$ of g is finite where

$$\|g\|_2 = \left(\int_0^\infty |g(t)|^2 dt \right)^{\frac{1}{2}}. \quad (20.4)$$

We refer to $\|g\|_2$ as the H_2 norm of the system. Thus the H_2 norm of an asymptotically stable SISO system is the rms value of its impulse response. This norm is a **performance measure** for the system; it can be regarded as a measure of the ability of the system to mitigate the effect of the disturbance w on the performance output z .

Consider now a general linear time-invariant system described by (20.2) with impulse response given by (20.3). When A is asymptotically stable, it follows that each element of G is square integrable, and we define the L_2 norm of the impulse response by

$$\|G\|_2 = \left(\int_0^\infty \text{trace} [G^*(t)G(t)] dt \right)^{\frac{1}{2}}. \quad (20.5)$$

We refer to $\|G\|_2$ as the H_2 norm of the system.

We now obtain a frequency domain interpretation of $\|G\|_2$. Recall that the transfer function of this system is given by $\hat{G}(s) = C(sI - A)^{-1}B$. When A is asymptotically stable, we define the 2-norm of the transfer function by

$$\|\hat{G}\|_2 = \left(\frac{1}{2\pi} \int_{-\infty}^\infty \text{trace} [G(j\omega)^*G(j\omega)] d\omega \right)^{\frac{1}{2}}. \quad (20.6)$$

For a SISO system with transfer function $\hat{g}$, we have

$$\|\hat{g}\|_2^2 = \frac{1}{2\pi} \int_{-\infty}^\infty |\hat{g}(j\omega)|^2 d\omega.$$

Since $\hat{G}$ is the Laplace transform of G , it follows from Parseval's Theorem that

$$\|\hat{G}\|_2 = \|G\|_2, \quad (20.7)$$

that is, the 2-norm of the transfer function equals the L_2 norm of the impulse response.

20.3 Computation of the system H_2 norm

We now show how the H_2 -norm can be computed using Lyapunov equations. Suppose A is asymptotically stable and let W_o be the observability grammian associated with (C, A) , that is, $W_o = \int_0^\infty e^{A^*t}C^*Ce^{At}dt$. Recall that W_o is the unique solution to the Lyapunov equation

$$W_oA + A^*W_o + C^*C = 0.$$

We now obtain that

$$\int_0^\infty G^*(t)G(t)dt = \int_0^\infty B^*e^{A^*t}C^*Ce^{At}Bdt = B^* \int_0^\infty e^{A^*t}C^*Ce^{At}dt B = B^*W_oB,$$

that is,

$$\int_0^\infty G^*(t)G(t)dt = B^*W_oB. \quad (20.8)$$

Using the linearity of the trace operation, it now follows that

$$\|G(\cdot)\|_2^2 = \int_0^\infty \text{trace} (G^*(t)G(t))dt = \text{trace} \left[\int_0^\infty G^*(t)G(t)dt \right] = \text{trace} (B^*W_oB);$$

hence

$$\boxed{\|G\|_2^2 = \text{trace}(B^*W_oB)} . \quad (20.9)$$

We can also compute the H_2 -norm using the controllability grammian. Since

$$\text{trace}[G^*(t)G(t)] = \text{trace}[G(t)G^*(t)] ;$$

it follows that the H_2 -norm is also given by

$$\|G\|_2 = \left(\int_0^\infty \text{trace}[G(t)G^*(t)] dt \right)^{\frac{1}{2}} . \quad (20.10)$$

Let W_c be the controllability grammian associated with (A, B) , that is, $W_o = \int_0^\infty e^{At}BB^*e^{A^*t}dt$. Recall that W_c is the unique solution to the Lyapunov equation

$$AW_c + W_cA^* + BB^* = 0 .$$

Hence,

$$\int_0^\infty G(t)G(t)^* dt = \int_0^\infty Ce^{At}BB^*e^{A^*t}C^* dt = C \int_0^\infty e^{At}BB^*e^{A^*t} dt C^* = CW_cC^* ,$$

that is,

$$\int_0^\infty G(t)G^*(t) dt = CW_cC^* . \quad (20.11)$$

Using the linearity of the trace operation, it now follows that

$$\|G(\cdot)\|_2^2 = \int_0^\infty \text{trace}[G(t)G^*(t)] dt = \text{trace} \left[\int_0^\infty G(t)G^*(t) dt \right] = \text{trace}(CW_cC^*) ;$$

hence

$$\boxed{\|G\|_2^2 = \text{trace}(CW_cC^*)} . \quad (20.12)$$

Example 202 Consider the scalar system described by

$$\begin{aligned} \dot{x} &= -\alpha x + u \\ y &= x \end{aligned}$$

with $\alpha > 0$. This system has transfer function $\hat{g}(s) = \frac{1}{s+\alpha}$. Hence,

$$\begin{aligned} \|\hat{g}\|_2^2 &= \frac{1}{2\pi} \int_{-\infty}^\infty |\hat{g}(j\omega)|^2 d\omega = \frac{1}{2\pi} \int_{-\infty}^\infty \frac{1}{\alpha^2 + \omega^2} d\omega = \frac{1}{2\pi\alpha} \int_{-\infty}^\infty \frac{1}{1 + \eta^2} d\eta \quad (\eta = \omega/\alpha) \\ &= \frac{1}{2\pi\alpha} \left|_{-\infty}^\infty \tan^{-1}(\eta) \right. \\ &= \frac{1}{2\alpha} . \end{aligned}$$

The impulse response for this system is given by $g(t) = e^{-\alpha t}$. Hence

$$\begin{aligned}\|g\|_2^2 &= \int_0^\infty g(t)^2 dt = \int_0^\infty e^{-2\alpha t} dt = \left|_0^\infty \frac{1}{-2\alpha} e^{-2\alpha t}\right. \\ &= \frac{1}{2\alpha}.\end{aligned}$$

This clearly illustrates the fact that $\|\hat{g}\|_2 = \|g\|_2$ and the H_2 norm of this system is $(1/2\alpha)^{\frac{1}{2}}$.

The observability grammian W_o is given by the solution to the Lyapunov equation

$$-W_o\alpha - \alpha W_o + 1 = 0.$$

This yields $W_o = 1/2\alpha$. Since $B = 1$, we readily see that the H_2 norm is given by $\text{trace}(B^*W_oB)^{\frac{1}{2}}$. The controllability grammian W_c is given by the solution to the Lyapunov equation

$$-\alpha W_c - W_c\alpha + 1 = 0.$$

This yields $W_c = 1/2\alpha$. Since $C = 1$, we readily see that the H_2 norm is given by $\text{trace}(CW_cC^*)^{\frac{1}{2}}$.

MATLAB

```
>> help normh2
```

```
NORMH2 Continuous H2 norm.
```

```
[NMH2] = NORMH2(A,B,C,D) or
```

```
[NMH2] = NORMH2(SS_) computes the H2 norm of the given state-space  
realization. If the system not strictly proper, INF is returned.  
Otherwise, the H2 norm is computed as
```

$$\text{NMH2} = \text{trace}[\text{CPC}']^{0.5} = \text{trace}[\text{B}'\text{QB}]^{0.5}$$

```
where P is the controllability grammian, and Q is the observability  
grammian.
```

```
>> help norm2
```

```
nh2 = norm2(sys)
```

```
Computes the H2 norm of a stable and strictly proper LTI  
system
```

$$\text{SYS} = \text{C} \left(\text{sE} - \text{A} \right)^{-1} \text{B}$$

This norm is given by

$$\sqrt{\text{Trace}(C*P*C')}$$

where P solves the Lyapunov equation: $A*P+P*A'+B*B' = 0$.

See also LTISYS.

Exercises

Exercise 178 Consider the system described by

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -2x_1 - 3x_2 + w \\ y &= 4x_1 + 2x_2\end{aligned}$$

Obtain the H_2 norm of this system using the following methods.

- (a) The definition in (20.4).
- (b) Equation (20.9).
- (c) Equation (20.11).
- (d) MATLAB.

Exercise 179 Without carrying out any integration, obtain the H_2 norm of the system whose impulse response g is given by

$$g(t) = e^{-t} + e^{-2t}.$$

Chapter 21

Linear quadratic regulators (LQR)

21.1 The linear quadratic optimal control problem

In this chapter we learn another method of designing stabilizing state feedback controllers. These controllers result from solving the linear quadratic (LQ) optimal control problem are called **linear quadratic regulators (LQR)**. This problem is characterized by a linear system and a quadratic cost. Consider a linear time-invariant system (plant)

$$\boxed{\dot{x} = Ax + Bu} \quad (21.1)$$

with initial condition

$$x(0) = x_0 \quad (21.2)$$

where the real scalar t is the time variable, the n -vector $x(t)$ is the plant state, and the m -vector $u(t)$ is the control input. When we refer to a **control history**, **control signal**, or **open loop control** we will mean a continuous control input function $u(\cdot)$ which is defined for $0 \leq t < \infty$.

Suppose Q and R are hermitian matrices with

$$R > 0 \quad \text{and} \quad Q \geq 0$$

We call them **weighting matrices**. For each initial state x_0 and each control history $u(\cdot)$, we define the corresponding **quadratic cost** by

$$\boxed{J(x_0, u(\cdot)) := \int_0^\infty x(t)' Q x(t) + u(t)' R u(t) dt}$$

where $x(\cdot)$ is the unique solution of (21.1) with initial condition (21.2). We wish to minimize this cost by appropriate choice of the control history. Although the above cost can be infinite, it follows from the assumptions on Q and R that it is always non-negative, that is,

$$0 \leq J(x_0, u(\cdot)) \leq \infty.$$

In addition to minimizing the above cost, we will demand that the control input drives the state to zero, that is, we require that

$$\lim_{t \rightarrow \infty} x(t) = 0.$$

So, for a fixed initial state x_0 , we say that a control history $u(\cdot)$ is **admissible** at x_0 if it results in $\lim_{t \rightarrow \infty} x(t) = 0$. We say that a control history $u^{opt}(\cdot)$ is **optimal** at x_0 if it is admissible at x_0 and

$$J(x_0, u^{opt}(\cdot)) \leq J(x_0, u(\cdot)) \quad (21.3)$$

for every admissible control history $u(\cdot)$. We refer to

$$J^{opt}(x_0) := J(x_0, u^{opt}(\cdot))$$

as the **optimal cost** at x_0 .

In general, the optimal control history and the optimal cost depend on the initial state. Also, for a fixed initial state, the optimal cost is unique whereas the optimal control history need not be. In some cases an optimal control history might not exist. Since the cost is bounded below by zero, there is always an **infimal cost** $J^{inf}(x_0)$. That is, one cannot achieve better than this cost but can come arbitrarily close to it; more precisely, for every admissible control history $u(\cdot)$, one has

$$J^{inf}(x_0) \leq J(x_0, u(\cdot))$$

and for every $\epsilon > 0$ there is an admissible control history $u^\epsilon(\cdot)$ such that

$$J(x_0, u^\epsilon(\cdot)) \leq J^{inf}(x_0) + \epsilon$$

When an optimal control history exists, we must have $J^{inf}(x_0) = J^{opt}(x_0)$. If there exists at least one admissible control history yielding a bounded cost, then $J^{inf}(x_0) < \infty$.

Example 203 Consider the scalar integrator plant described by

$$\dot{x} = u$$

with initial condition $x(0) = 1$. Suppose we wish to drive the state of this plant to zero while minimizing the cost

$$\int_0^\infty u(t)^2 dt.$$

Consider any positive $\delta > 0$ and let

$$u(t) = -\delta e^{-\delta t}.$$

Then, the resulting state history, $x(t) = e^{-\delta t}$, has the desired behavior and the corresponding cost is

$$\int_0^\infty \delta^2 e^{-2\delta t} dt = \lim_{\epsilon \rightarrow \infty} \left[-\frac{\delta}{2} e^{-2\delta t} \right]_0^\epsilon = \frac{\delta}{2}.$$

By choosing $\delta > 0$ arbitrarily small, we can make the cost arbitrarily close to zero. Since the cost is bounded below by zero, it follows that the infimal cost is zero. However, we cannot make the cost zero. If the cost is zero, that is $\int_0^\infty u(t)^2 dt = 0$, then we must have $u(t) \equiv 0$; hence $x(t) \equiv 1$. This control history does not drive the state to zero and, hence, is not admissible. So there is no optimal control history for this problem. Note that in this problem, $(Q, A) = (0, 0)$ has an imaginary axis unobservable eigenvalue.

Optimal feedback controllers. Suppose one has a feedback controller $\mathcal{C}$ with the property that for any initial state x_0 the control histories generated by the controller are admissible. We say that $\mathcal{C}$ is an **optimal feedback controller** if for each initial state x_0 the corresponding control histories generated by $\mathcal{C}$ are optimal at x_0 . We will shortly demonstrate the following fact for linear quadratic optimal control problems:

If there is an optimal control history for every initial state then, for every initial state, the optimal control history is unique and is given by a linear memoryless state feedback controller.

Since we are going to look for feedback controllers which minimize the cost and result in

$$\lim_{t \rightarrow \infty} x(t) = 0$$

we need to make the following assumption for the rest of the chapter:

- The pair (A, B) is stabilizable.

21.2 The algebraic Riccati equation (ARE)

We will see that, when an optimal feedback controller exists, its determination involves the following matrix equation which is known as the Algebraic Riccati Equation (ARE):

$$\boxed{PA + A'P - PBR^{-1}B'P + Q = 0} \quad (21.4)$$

The matrices A, B, Q, R are determined by the LQR problem data. Solving the LQR problem involves solving the ARE for a hermitian matrix P . Since the ARE depends quadratically on P , it is a nonlinear matrix equation and, in general, has more than one solution.

A solution P is said to be a **stabilizing solution** if the matrix $A - BR^{-1}B'P$ is Hurwitz, that is, all its eigenvalues have negative real parts. This is the solution we will be looking for. The significance of this property is that the linear state feedback controller

$$u = -R^{-1}B'Px$$

results in the asymptotically stable closed loop system:

$$\dot{x} = (A - BR^{-1}B'P)x$$

It is shown in Section 21.4 that *if there is an optimal control history for every initial state then, the ARE must have a stabilizing solution*. Hence, if the ARE does not have a stabilizing solution then, there is not an optimal control history for every initial state.

It is also shown in Section 21.4 that *if the pair (Q, A) has no imaginary unobservable eigenvalues then, the ARE must have a stabilizing solution*.

Section 21.4 also shows that *a stabilizing solution is maximal* in the sense that if P is a stabilizing and $\tilde{P}$ is any other solution then, $P \geq \tilde{P}$. Hence there can only be one stabilizing solution.

21.3 Linear state feedback controllers

Suppose

$$u(t) = Kx(t)$$

where K is a constant state feedback gain matrix. The resulting closed loop system is described by

$$\dot{x} = (A + BK)x$$

and for each initial state x_0 the controller results in the cost

$$J(x_0, u(\cdot)) = \hat{J}(x_0, K) := \int_0^\infty x(t)'[K'RK + Q]x(t) dt$$

This feedback controller is an optimal feedback controller if and only if it is a stabilizing controller (that is, $A + BK$ is asymptotically stable) and for each initial state x_0 we have

$$\hat{J}(x_0, K) \leq J(x_0, u(\cdot))$$

for every admissible control history $u(\cdot)$.

If K is stabilizing then, recalling a result from Lyapunov theory,

$$\hat{J}(x_0, K) = x_0'Px_0 \quad (21.5)$$

where P is the unique solution to the following Lyapunov equation

$$P(A + BK) + (A + BK)'P + K'RK + Q = 0 \quad (21.6)$$

Since $A + BK$ is asymptotically stable and $K'RK + Q$ is hermitian positive semi-definite, we have $P' = P \geq 0$.

Example 204 Consider the simple real scalar system

$$\dot{x} = \alpha x + u$$

with cost

$$\int_0^\infty u(t)^2 dt$$

If we do not require $\lim_{t \rightarrow \infty} x(t) = 0$ then regardless of initial state x_0 , this cost is uniquely minimized by

$$u(t) \equiv 0$$

and the resulting controlled system is described by

$$\dot{x} = \alpha x$$

This system is not asymptotically stable for $\alpha \geq 0$. So, in looking for an optimal control we restrict consideration to control inputs that result in $\lim_{t \rightarrow \infty} x(t) = 0$

Now consider any linear controller $u = kx$. This generates a closed loop system

$$\dot{x} = (\alpha + k)x$$

which is asymptotically stable if and only if $\alpha + k < 0$. Utilizing (21.5), (21.6), the cost corresponding to a stabilizing linear controller is Px_0^2 where

$$P = \frac{k^2}{-2(\alpha + k)}$$

If $\alpha = 0$, $P = -k/2$. Since we require asymptotic stability of the closed loop system, this cost can be made arbitrarily small but nonzero. So for $\alpha = 0$, we have an *infimum cost* (that is, zero) but no minimum cost. The infimum cannot be achieved by controls which drive the state to zero. Note that in this case, $(Q, A) = (0, 0)$ has an imaginary axis unobservable eigenvalue.

21.4 Infimal cost*

21.4.1 Reduced cost stabilizing controllers.

Suppose one has a stabilizing gain matrix K and wishes to obtain another stabilizing gain matrix which results in reduced cost. We will show that this is achieved by the gain matrix

$$\tilde{K} = -R^{-1}B^*P \quad (21.7)$$

where P solves the Lyapunov equation (21.6).

Fact 23 *Suppose $A + BK$ is asymptotically stable and $\tilde{K}$ is given by (21.7) and (21.6). Then,*

(i) *the matrix $A + B\tilde{K}$ is asymptotically stable;*

(ii) *for every initial state x_0 ,*

$$\hat{J}(x_0, \tilde{K}) \leq \hat{J}(x_0, K)$$

PROOF. The Appendix contains a proof. ■

21.4.2 Cost reducing algorithm

The above result suggests the following algorithm to iteratively reduce the cost.

Jack Benny algorithm.

(0) Choose any gain matrix K_0 so that $A + BK_0$ is asymptotically stable.

(1) For $k = 0, 1, 2, \dots$, let P_k be the unique solution to the Lyapunov equation

$$P_k(A + BK_k) + (A + BK_k)^*P_k + K_k^*RK_k + Q = 0 \quad (21.8)$$

and let

$$K_{k+1} = -R^{-1}B^*P_k \quad (21.9)$$

Fact 24 Suppose (A, B) is stabilizable, $R' = R > 0$ and $Q' = Q \geq 0$. Consider any sequence of matrices $\{P_k\}$ generated by the Jack Benny algorithm. Then this sequence has a limit $\bar{P}$, that is,

$$\lim_{k \rightarrow \infty} P_k = \bar{P}.$$

Furthermore, $\bar{P}$ has the following properties.

(a)

$$\bar{P}' = \bar{P} \geq 0$$

(b) $P = \bar{P}$ solves

$$\boxed{PA + A^*P - PBR^{-1}B^*P + Q = 0} \quad (21.10)$$

(c) Every eigenvalue of the matrix $A - BR^{-1}B'\bar{P}$ has real part less than or equal to zero.

(d) If (Q, A) has no imaginary axis unobservable eigenvalues, then every eigenvalue of the matrix $A - BR^{-1}B'\bar{P}$ has negative real part.

PROOF. (a) Since P_k solves (21.8) we have $P_k \geq 0$. From consequence (ii) of fact 23, we have that $P_{k+1} \leq P_k$. Hence,

$$0 \leq P_{k+1} \leq P_k$$

From this one can show that there is a matrix $\bar{P}$ such that

$$\bar{P} := \lim_{k \rightarrow \infty} P_k$$

and $\bar{P} \geq 0$.

(b) In the limit, and using (21.9),

$$\bar{K} := \lim_{k \rightarrow \infty} K_k = \lim_{k \rightarrow \infty} K_{k+1} = -R^{-1}B'\bar{P}$$

hence, it follows from (21.8) that

$$\bar{P}(A + B\bar{K}) + (A + B\bar{K})^*\bar{P} + \bar{K}^*R\bar{K} + Q = 0 \quad (21.11)$$

Substituting

$$\bar{K} = -R^{-1}B'\bar{P}$$

results in (21.10).

(c) For each k , $\Re(\lambda) < 0$ for all eigenvalues λ of $A + B\bar{K}$. Since $\bar{K} = \lim_{k \rightarrow \infty} K_k$ and the eigenvalues of a matrix depend continuously on its coefficients, we must have $\Re(\lambda) \leq 0$ for all eigenvalues λ of $A + B\bar{K}$.

(d) Now suppose $A + B\bar{K}$ has an eigenvalue λ with $\Re(\lambda) = 0$. Letting v be an eigenvector 0 to λ and pre- and post-multiplying (21.11) by v' and v yields

$$v'\bar{K}^*R\bar{K}v + v'Qv = 0$$

Hence,

$$\bar{K}v = 0 \quad Qv = 0$$

and

$$\lambda v = (A + B\bar{K})v = Av$$

So,

$$\begin{bmatrix} A - \lambda I \\ Q \end{bmatrix} v = 0$$

that is, λ is an unobservable eigenvalue of (Q, A) . So, if (Q, A) has no imaginary axis unobservable eigenvalues, $A + B\bar{K}$ has no imaginary axis eigenvalues. So all the eigenvalues of $A + B\bar{K}$ have negative real parts. ■

Example 205 Recall first example. Here the algorithm yields:

$$\begin{aligned} P_k &= \frac{K_k^2}{-2(\alpha + K_k)} \\ K_{k+1} &= -P_k \end{aligned}$$

Hence,

$$P_{k+1} = \frac{P_k^2}{2(-\alpha + P_k)}$$

1. ($\alpha < 0$): Here $\bar{P} = \lim_{k \rightarrow \infty} P_k = 0$ and $A - BR^{-1}B'\bar{P} = \alpha < 0$.
2. ($\alpha > 0$): Here $\bar{P} = \lim_{k \rightarrow \infty} P_k = 2\alpha$ and $A - BR^{-1}B'\bar{P} = -\alpha < 0$.
3. ($\alpha = 0$): Here $P_{k+1} = P_k/2$ for all k . Hence, $\bar{P} = \lim_{k \rightarrow \infty} P_k = 0$ and $A - BR^{-1}B'\bar{P} = 0$.

Note that this is the only case in which (Q, A) has an imaginary axis unobservable eigenvalue.

Remark 11 The matrix equation (21.10) is called an **algebraic Riccati equation (ARE)**. Since it depends quadratically on P , it is a nonlinear matrix equation and, in general, has more than one solution.

Remark 12 We say that a hermitian matrix P is a **stabilizing solution** to ARE (21.10) if all the eigenvalues of the matrix $A - BR^{-1}B'P$ have negative real parts. The significance of this is that the controller

$$u = -R^{-1}B'Px$$

results in the asymptotically stable closed loop system:

$$\dot{x} = (A - BR^{-1}B'P)x$$

21.4.3 Infimal cost

The following result provides a lower bound on the infimal cost $J^{inf}(x_0)$ in terms of a specific solution to the ARE.

Fact 25 (Completion of squares lemma.) *Suppose $R' = R > 0$ and P is any hermitian solution to ARE (21.10). Then for every initial condition x_0 and every continuous control input history $u(\cdot)$ which guarantees*

$$\lim_{t \rightarrow \infty} x(t) = 0$$

we have

$$x_0' P x_0 \leq J(x_0, u(\cdot));$$

hence

$$x_0' P x_0 \leq J^{inf}(x_0).$$

PROOF. Suppose P is any hermitian solution P to the ARE (21.10), let x_0 be any initial state, and consider any control history $u(\cdot)$ which results in $\lim_{t \rightarrow \infty} x(t) = 0$. Consider the function $V(x(\cdot))$ given by

$$V(x(t)) = x'(t) P x(t)$$

Its derivative is given by

$$\begin{aligned} \frac{dV(x(t))}{dt} &= 2x'(t) P \dot{x}(t) \\ &= 2x'(t) P A x + 2x'(t) P B u(t) \end{aligned}$$

Utilizing the ARE (21.10) we obtain

$$\begin{aligned} 2x' P A x &= x'(P A + A' P)x \\ &= x' P B R^{-1} B' P x - x' Q x \end{aligned}$$

Hence (omitting the argument t for clarity),

$$\begin{aligned} \frac{dV(x(t))}{dt} &= x' P B R^{-1} B' P x + 2x' P B u - x' Q x \\ &= x' P B R^{-1} B' P x + 2x' P B u + u' R u - x' Q x - u' R u \\ &= (u + R^{-1} B' P x)' R (u + R^{-1} B' P x) - x' Q x - u' R u \end{aligned}$$

Hence

$$x' Q x + u' R u = -\frac{dV(x)}{dt} + (u + R^{-1} B' P x)' R (u + R^{-1} B' P x)$$

Considering any $t \geq 0$ and integrating over the interval $[0, t]$ yields

$$\int_0^t [x' Q x + u' R u] dt = -V(x(t)) + V(x(0)) + \int_0^t (u + R^{-1} B' P x)' R (u + R^{-1} B' P x) dt$$

Recall $V(x) = x'Px$, $x(0) = x_0$, and $\lim_{t \rightarrow \infty} x(t) = 0$; so, considering the limit as $t \rightarrow \infty$ and recalling the definition of the cost, we have

$$\boxed{J(x_0, u(\cdot)) = x_0'Px_0 + \int_0^\infty (u + R^{-1}B'Px)'R(u + R^{-1}B'Px) dt} \quad (21.12)$$

Since $R > 0$, we immediately have

$$J(x_0, u(\cdot)) \geq x_0'Px_0 \quad (21.13)$$

■

The next result states that for any initial state x_0 , the infimal cost is $x_0'\bar{P}x_0$ where $\bar{P}$ is the limit of any sequence $\{P_k\}$ generated by Jack Benny algorithm.

Lemma 23 (Infimal cost lemma.) *Suppose (A, B) is stabilizable, $R' = R > 0$ and $Q' = Q \geq 0$. Consider any sequence of matrices $\{P_k\}_{k=0}^\infty$ generated by the Jack Benny algorithm. Then this sequence has a limit $\bar{P}$, that is,*

$$\lim_{k \rightarrow \infty} P_k = \bar{P}$$

and $\bar{P}$ has the following properties.

(a) *If P is any hermitian solution to ARE (21.10), then*

$$\bar{P} \geq P$$

(b) *For every initial state x_0 ,*

$$\boxed{J^{\text{inf}}(x_0) = x_0'\bar{P}x_0}$$

(c) *If there exists an optimal control history for every initial state x_0 then $\bar{P}$ is a stabilizing solution to ARE (21.10).*

PROOF. Consider any sequence of matrices $\{P_k\}$ generated by the Jack Benny algorithm. Then, from fact (24), this sequence has a limit $\bar{P}$, that is,

$$\lim_{k \rightarrow \infty} P_k = \bar{P}$$

and it satisfies ARE (21.10).

Consider any initial state x_0 and any $k = 1, 2, \dots$. The control history given by

$$u_k(t) = -R^{-1}B'P_{k-1}x(t)$$

results in the cost

$$J(x_0, u_k(\cdot)) = x_0'P_kx_0$$

Hence

$$J^{\text{inf}}(x_0) \leq x_0'P_kx_0$$

Since $\bar{P}$ is a solution to the ARE, it follows from the completion of squares lemma that

$$x_0' \bar{P} x_0 \leq J^{\inf}(x_0)$$

Hence

$$x_0' \bar{P} x_0 \leq J^{\inf}(x_0) \leq x_0' P_k x_0$$

Considering the limit as $k \rightarrow \infty$ we must have

$$x_0' \bar{P} x_0 \leq J^{\inf}(x_0) \leq x_0' \bar{P} x_0;$$

so

$$J^{\inf}(x_0) = x_0' \bar{P} x_0$$

If P is any hermitian solution to ARE (21.10), we must have

$$x_0' P x_0 \leq J(x_0, u_k(\cdot)) = x_0' P_k x_0$$

Taking the limit as $k \rightarrow \infty$, we obtain

$$x_0' P x_0 \leq x_0' \bar{P} x_0$$

Since this is true for all x_0 we have $P \leq \bar{P}$.

Suppose an optimal control history u^{opt} exists for every x_0 . Then

$$J(x_0, u^{opt}) = J^{opt}(x_0) = J^{\inf}(x_0) = x_0' \bar{P} x_0$$

Recalling the proof of the completion of squares lemma, we have

$$J(x_0, u^{opt}(\cdot)) = x_0' \bar{P} x_0 + \int_0^\infty (u^{opt} + R^{-1} B' \bar{P} x)' R (u^{opt} + R^{-1} B' \bar{P} x) dt$$

Hence

$$\int_0^\infty (u^{opt} + R^{-1} B' \bar{P} x)' R (u^{opt} + R^{-1} B' \bar{P} x) dt = 0$$

Since $R > 0$ we must have

$$u^{opt} = -R^{-1} B' \bar{P} x$$

Since $u^{opt}(\cdot)$ is admissible, it results in $\lim_{t \rightarrow \infty} x = 0$. Since the above is true for any initial state x_0 , it follows that the system

$$\dot{x} = (A - B R^{-1} B') x$$

must be asymptotically stable. Hence $\bar{P}$ is a stabilizing solution to ARE. ■

Remark 13 A solution to the ARE which satisfies condition (a) above is called the **maximal solution** to the ARE. Clearly, it must be unique.

21.5 Optimal stabilizing controllers

The solution to the LQR problem is based on the following matrix equation which is known as the Algebraic Riccati Equation:

$$\boxed{PA + A^*P - PBR^{-1}B^*P + Q = 0} \quad (21.14)$$

Since this is a nonlinear equation in P it can have more than one solution. We will be interested in a **stabilizing solution** which is defined as a hermitian matrix P with the property that all the eigenvalues of the matrix $A - BR^{-1}B^*P$ have negative real parts.

Theorem 22 *Suppose $R' = R > 0$ and the ARE (21.14) has a stabilizing solution P . Then for every initial state x_0 , there is a unique optimal control history u^{opt} ; it is given by*

$$\boxed{u^{opt}(t) = -R^{-1}B^*Px(t)}$$

and the optimal cost is

$$J^{opt}(x_0) = x_0'Px_0.$$

PROOF. Let x_0 be any initial state, and consider any control history $u(\cdot)$ which results in $\lim_{t \rightarrow \infty} x(t) = 0$. Define the function V given by

$$V(t) = x'(t)Px(t)$$

The derivative of V is given by

$$\begin{aligned} \frac{dV}{dt} &= 2x'(t)P\dot{x}(t) \\ &= 2x'(t)PAx(t) + 2x'(t)PBu(t) \end{aligned}$$

Utilizing the ARE (21.14) we obtain

$$\begin{aligned} 2x'PAx &= x'(PA + A^*P)x \\ &= x'PBR^{-1}B^*Px - x'Qx \end{aligned}$$

Hence (omitting the argument t for clarity),

$$\begin{aligned} \frac{dV}{dt} &= x'PBR^{-1}B^*Px + 2x'PBu - x'Qx \\ &= x'PBR^{-1}B^*Px + 2x'PBu + u'Ru - x'Qx - u'Ru \\ &= (u + R^{-1}B^*Px)'R(u + R^{-1}B^*Px) - x'Qx - u'Ru \end{aligned}$$

Hence

$$x'Qx + u'Ru = -\frac{dV}{dt} + (u + R^{-1}B^*Px)'R(u + R^{-1}B^*Px)$$

Considering any $t \geq 0$ and integrating over the interval $[0, t]$ yields

$$\int_0^t [x'Qx + u'Ru] dt = -V(t) + V(0) + \int_0^t (u + R^{-1}B'Px)'R(u + R^{-1}B'Px) dt$$

Recall $V(t) = x'(t)Px(t)$, $x(0) = x_0$, and $\lim_{t \rightarrow \infty} x(t) = 0$; so, considering the limit as $t \rightarrow \infty$ and recalling the definition of the cost, we have

$$\boxed{J(x_0, u(\cdot)) = x_0'Px_0 + \int_0^\infty (u + R^{-1}B'Px)'R(u + R^{-1}B'Px) dt} \quad (21.15)$$

Since P is a stabilizing solution to ARE (21.10) the control history given by $u(t) = \hat{u}(t) := -R^{-1}B'Px(t)$ results in $\lim_{t \rightarrow \infty} x(t) = 0$; hence

$$J(x_0, \hat{u}(\cdot)) = x_0'Px_0$$

Consider any other admissible control history $u(\cdot)$. Since $R > 0$, we have

$$J(x_0, u(\cdot)) \geq x_0'Px_0$$

This implies that $\hat{u}$ is optimal at x_0 and the optimal cost is $x_0'Px_0$.

Also, (21.15) implies that if any other control $\tilde{u}$ yields the optimal cost $x_0'Px_0$, then the $\tilde{u}(t) = -R^{-1}B'Px(t)$; hence $\tilde{u}(t) = \hat{u}(t)$. So, $\hat{u}$ is a unique optimal control history. ■

Example 206 Returning to Example 204 we have $A = \alpha$, $B = 1$, $Q = 0$, and $R = 1$. Hence the ARE is

$$2\alpha P - P^2 = 0$$

This has two solutions

$$P = P_1 := 0 \quad P = P_2 := 2\alpha$$

Hence

$$A - BR^{-1}B'P_1 = \alpha \quad A - BR^{-1}B'P_2 = -\alpha$$

We have three cases.

Case 1. ($\alpha < 0$): $P_1 = 0$ is the stabilizing solution and optimal controller is $u = 0$.

Case 2. ($\alpha > 0$): $P_1 = 2\alpha$ is the stabilizing solution and optimal controller is $u = -2\alpha x$.

Case 3. ($\alpha = 0$): No stabilizing solution.

Remark 14 If P is a stabilizing solution to ARE (21.10) then, it can be shown that P must be the maximal solution $\bar{P}$. This follows from

$$x_0'Px_0 = J^{opt}(x_0) = J^{inf}(x_0) = x_0'\bar{P}x_0.$$

Hence there can be at most one stabilizing solution.

21.5.1 Existence of stabilizing solutions to the ARE

Lemma 24 *Suppose (A, B) is stabilizable, $R' = R > 0$ and $Q' = Q$. Then ARE (21.10) has a stabilizing solution if and only if (Q, A) has no imaginary axis unobservable eigenvalues.*

PROOF. From Fact 24 it follows that if (Q, A) has no imaginary unobservable eigenvalues then the ARE has a stabilizing solution.

We now prove the converse by contradiction. Suppose P is a stabilizing solution to ARE and λ is an imaginary unobservable eigenvalue of (Q, A) . Then, there is a nonzero vector v such that

$$Av = \lambda v \quad \text{and} \quad Qv = 0$$

Pre- and postmultiplying the ARE by v' and v yields

$$2\Re(\lambda)v'Pv - v'PBR^{-1}B'Pv = 0$$

Hence, $R^{-1}B'Pv = 0$ and

$$\begin{aligned} (A - BR^{-1}B'P)v &= Av - BR^{-1}B'Pv \\ &= \lambda v \end{aligned}$$

that is, λ is an eigenvalue of the matrix $A - BR^{-1}B'P$. This contradicts the asymptotic stability of this matrix. ■

Lemma 25 *Suppose (A, B) is stabilizable, $R' = R > 0$ and $Q' = Q \geq 0$. If (Q, A) is detectable, then*

- (a) *The ARE has a stabilizing solution.*
- (b) *P is a stabilizing solution if and only if $P \geq 0$*

PROOF. Part (a) follows from previous lemma.

(b) We have already seen that if P is stabilizing then $P \geq 0$. So, all we have to show is that if P solves the ARE and $P \geq 0$ then P must be stabilizing. To prove this we write the ARE as

$$P(A + BK) + (A + BK)^*P + K'RK + Q = 0$$

where $K = -R^{-1}B'P$. Hence,

$$P(A + BK) + (A + BK)^*P + Q \leq 0$$

Since $P \geq 0$ and (Q, A) is detectable, it follows from a previous Lyapunov result that $A + BK$ is asymptotically stable; hence P is stabilizing. ■

21.6 Summary

The main results of this chapter are summarized in the following theorem.

Theorem 23 *Suppose (A, B) is stabilizable, $R' = R > 0$ and $Q' = Q \geq 0$. Then the following statements are equivalent.*

- (a) *For every initial state x_0 , there is an optimal control history.*
- (b) *(Q, A) has no imaginary axis unobservable eigenvalues.*
- (c) *ARE (21.10) has a stabilizing solution.*
- (d) *ARE (21.10) has a unique stabilizing solution P ; also P is positive semi-definite and is the maximal solution.*

Suppose any of conditions (a)-(d) hold. Then for every initial state x_0 , the optimal control history is unique and is generated by the linear state feedback controller

$$\boxed{u(t) = -R^{-1}B'Px(t)}$$

where P is the stabilizing solution to ARE (21.10). Also, the optimal cost is given by

$$\boxed{J^{opt}(x_0) = x_0'Px_0}$$

Example 207 Consider the problem of finding an optimal stabilizing controller in the problem of minimizing the cost,

$$\int_0^\infty x_2^2 + u^2 dt$$

for the system,

$$\begin{aligned}\dot{x}_1 &= \alpha x_1 + x_2 \\ \dot{x}_2 &= u\end{aligned}$$

This is an LQR (linear quadratic regulator) problem with

$$A = \begin{bmatrix} \alpha & 1 \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad Q = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \quad R = 1$$

The pair (Q, A) has a single unobservable eigenvalue α . Hence an optimal stabilizing controller exists if and only if $\alpha \neq 0$.

Considering

$$P = \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix}$$

(we are looking for a symmetric P) the ARE yields the following three scalar equations

$$\begin{aligned} 2\alpha p_{11} - p_{12}^2 &= 0 \\ p_{11} + \alpha p_{12} - p_{12}p_{22} &= 0 \\ 2p_{12} - p_{22}^2 + 1 &= 0 \end{aligned} \quad (21.16)$$

We consider three cases:

Case 1 ($\alpha < 0$.) The first equation in (21.16) yields $p_{11} = p_{12}^2/\alpha$. Since we require $p_{11} \geq 0$ we must have

$$p_{11} = p_{12} = 0$$

The second equation in (21.16) is now automatically satisfied and (requiring $p_{22} \geq 0$) the third equation yields

$$p_{22} = 1$$

Since

$$P = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

is the only positive semi-definite solution to the ARE; it must be the stabilizing one. The fact that it is stabilizing is evident from

$$A_{cl} = A - BR^{-1}B'P = \begin{bmatrix} \alpha & 1 \\ 0 & -1 \end{bmatrix}$$

So, the optimal stabilizing controller for this case is given by

$$u = -x_2$$

Case 2 ($\alpha > 0$.) We obtain three positive semi-definite solutions:

$$P_i = \begin{bmatrix} \beta_i^2/2\alpha & \beta_i \\ \beta_i & (1 + 2\beta_i)^{\frac{1}{2}} \end{bmatrix} \quad \beta_1 = 2\alpha(\alpha + 1), \quad \beta_2 = 2\alpha(\alpha - 1), \quad \beta_3 = 0$$

Since $(P_1)_{11} \geq (P_2)_{11}, (P_3)_{11}$, and the stabilizing solution is maximal, the matrix P_1 must be the stabilizing solution. Let us check that P_1 is indeed a stabilizing solution. The corresponding closed loop matrix,

$$A_{cl} = A - BR^{-1}B'P_1 = \begin{bmatrix} \alpha & 1 \\ -\beta_1 & -(1 + 2\beta_1)^{\frac{1}{2}} \end{bmatrix}$$

has characteristic polynomial

$$q(s) = s^2 + a_1s + a_0$$

with

$$\begin{aligned} a_1 &= (1 + 2\beta_1)^{\frac{1}{2}} - \alpha &= (4\alpha^2 + 4\alpha + 1)^{\frac{1}{2}} - \alpha &> 0 \\ a_0 &= \beta_1 - \alpha(1 + 2\beta_1)^{\frac{1}{2}} &= 2\alpha(\alpha + 1) - \alpha(4\alpha^2 + 4\alpha + 1)^{\frac{1}{2}} &> 0 \end{aligned}$$

Since all coefficients of this second order polynomial are positive, all roots of this polynomial have negative real part; hence A_{cl} is asymptotically stable. So, the optimal stabilizing controller for this case is given by

$$u = -2\alpha(\alpha + 1)x_1 - (4\alpha^2 + 4\alpha + 1)^{\frac{1}{2}}x_2$$

Case 3 ($\alpha = 0$.) We already know that there is no optimal stabilizing controller for this case. However, to see this explicitly note that the only positive semi-definite solution to the ARE is

$$P = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Since

$$A_{cl} = A - BR^{-1}B'P = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix},$$

this solution is not stabilizing; hence there is no stabilizing solution and no optimal stabilizing controller.

21.7 MATLAB

>> help lqr

LQR Linear quadratic regulator design for continuous systems.
 $[K,S,E] = \text{LQR}(A,B,Q,R)$ calculates the optimal feedback gain matrix K such that the feedback law $u = -Kx$ minimizes the cost function:

$$J = \text{Integral} \{x'Qx + u'Ru\} dt$$

subject to the constraint equation:

$$\dot{x} = Ax + Bu$$

Also returned is S , the steady-state solution to the associated algebraic Riccati equation and the closed loop eigenvalues E :

$$0 = SA + A'S - SBR^{-1}B'S + Q \quad E = \text{EIG}(A-B*K)$$

$[K,S,E] = \text{LQR}(A,B,Q,R,N)$ includes the cross-term N that relates u to x in the cost function.

$$J = \text{Integral} \{x'Qx + u'Ru + 2x'Nu\}$$

The controller can be formed with REG.

See also: LQRY, LQR2, and REG.

>> help lqr2

LQR2 Linear-quadratic regulator design for continuous-time systems.
 $[K,S] = \text{LQR2}(A,B,Q,R)$ calculates the optimal feedback gain matrix K such that the feedback law $u = -Kx$ minimizes the cost function

$$J = \text{Integral} \{x'Qx + u'Ru\} dt$$

subject to the constraint equation: $\dot{x} = Ax + Bu$

Also returned is S , the steady-state solution to the associated algebraic Riccati equation:

$$0 = SA + A'S - SBR^{-1}B'S + Q$$

$[K,S] = \text{LQR2}(A,B,Q,R,N)$ includes the cross-term $2x'Nu$ that relates u to x in the cost functional.

The controller can be formed with REG.

LQR2 uses the SCHUR algorithm of [1] and is more numerically reliable than LQR, which uses eigenvector decomposition.

See also: ARE, LQR, and LQE2.

>> help are

ARE Algebraic Riccati Equation solution.
X = ARE(A, B, C) returns the stabilizing solution (if it exists) to the continuous-time Riccati equation:

$$A'X + XA - XBX + C = 0$$

assuming B is symmetric and nonnegative definite and C is symmetric.

See also: RIC.

21.8 Minimum energy controllers

Suppose we are interested in stabilizing a system with the minimum amount of control effort. Then, we might consider the problem of minimizing

$$\int_0^\infty \|u(t)\|^2 dt$$

subject to

$$\dot{x} = Ax + Bu \quad \text{and} \quad x(0) = x_0 \quad \text{and} \quad \lim_{t \rightarrow \infty} x(t) = 0. \quad (21.17)$$

This is an LQR problem with $Q = 0$ and $R = I$. The corresponding ARE is

$$PA + A'P - PBR^{-1}B'P = 0. \quad (21.18)$$

Since $Q = 0$, it follows from our main LQR result that *an optimal stabilizing controller exists if and only if A has no imaginary eigenvalues*.

The optimal controllers for this problem result in a closed loop system having the following interesting property.

If λ is an eigenvalue of $\tilde{A} := A + BK = A - BR^{-1}B'P$, then either λ or $-\bar{\lambda}$ is an eigenvalue of A.

Thus, an optimal controller in this case does not alter the stable eigenvalues of A and each unstable eigenvalue λ of A is replaced with $-\bar{\lambda}$. So, for a problem with real matrices, each unstable λ is replaced with $-\lambda$.

To demonstrate the above property, rewrite ARE as

$$P\tilde{A} + A'P = 0.$$

Suppose λ is an eigenvalue of $\tilde{A}$. Then there is a nonzero vector v such that $\tilde{A}v = \lambda v$. Post-multiplying the ARE by v yields

$$\lambda Pv + A'Pv = 0$$

If $Pv \neq 0$, then Pv is an eigenvector of A' with eigenvalue $-\lambda$. Hence, $-\bar{\lambda}$ is an eigenvalue of A . If $Pv = 0$, then $\tilde{A}v = Av - BB'Pv = Av$; hence $Av = \lambda v$ and λ is an eigenvalue of A .

Example 208 Suppose we wish to obtain a stabilizing state feedback controller which always minimizes $\int_0^\infty u(t)^2 dt$ for the scalar system

$$\dot{x} = ax + u.$$

Here

$$A = a, \quad B = 1, \quad R = 1, \quad Q = 0.$$

So the problem has a solution provided a is not on the imaginary axis. The ARE corresponding to this problem,

$$(a + \bar{a})P - P^2 = 0,$$

has two solutions

$$P = 0 \quad \text{and} \quad P = a + \bar{a}.$$

The two corresponding values of $\tilde{A} := A - BR^{-1}B'P$ are

$$a \quad \text{and} \quad -\bar{a}$$

respectively.

From this we can explicitly see that, when a is on the imaginary axis, there is no stabilizing P .

When a has negative real part, $P = 0$ is stabilizing, the optimal controller is zero and $\tilde{A} = A$.

When a has positive real part, $P = a + \bar{a}$ is stabilizing, the optimal controller is $u = -(a + \bar{a})x$ and $\tilde{A} = -\bar{a}$.

21.9 H_2 optimal controllers*

Consider a system described by

$$\begin{aligned}\dot{x} &= Ax + B_1w + B_2u \\ z &= Cx + Du\end{aligned}$$

where the m_1 -vector $w(t)$ is a **disturbance input**; the m_2 -vector $u(t)$ is a control input; and the p -vector z **performance output**. We assume that

(i) $\text{rank} D = m_2$.

(ii) $D'C = 0$.

With the above assumptions, the energy $\|z\|_2$ associated with the output is given by

$$\|z\|_2^2 = \int_0^\infty \|z(t)\|^2 dt = \int_0^\infty \|Cx(t) + Du(t)\|^2 dt = \int_0^\infty [x(t)'Qx(t) + u(t)'Ru(t)] dt$$

with

$$Q = C'C \quad \text{and} \quad R = D'D.$$

Since D has full column rank, it follows that R is positive definite.

A linear state feedback controller

$$u = Kx$$

results in the closed loop system

$$\dot{x} = (A + B_2K)x + B_1w \tag{21.19a}$$

$$z = (C + DK)x \tag{21.19b}$$

Suppose that $A + B_2K$ is asymptotically stable and recall the definition of the H_2 -norm of a linear system. If G_K is the impulse response matrix from w to z then the H_2 norm of the above system is given by $\|G_K\|_2 = [\text{trace} \int_0^\infty G_K'(t)G_K(t) dt]^{\frac{1}{2}}$. Noting that

$$(C + DK)'(C + DK) = C'C + K'D'DK$$

it follows that

$$\|G_K\|_2^2 = \text{trace } B_1'P_KB_1$$

where P_K is the unique solution to

$$P_K(A + B_2K) + (A + B_2K)'P_K + K'D'DK + C'C = 0.$$

From this one can readily show that if one considers the LQ problem with

$$B = B_2, \quad R = D'D, \quad Q = C'C$$

then an optimal LQ controller minimizes the H_2 norm of the closed loop system.

The observability condition on (Q, A) is equivalent to:

- The pair (C, A) has no imaginary axis unobservable eigenvalues.

So provided this condition holds the controller which minimizes $\|G_K\|_2$ is given by

$$u = -(D'D)^{-1}B_2'P$$

where P is the stabilizing solution to

$$PA + A'P - PB_1'(D'D)^{-1}B_1'P + C'C = 0$$

- If $A + B_2K$ is asymptotically stable, it follows from an identity in Lyapunov revisited that

$$\|G_K\|_2^2 = \text{trace}(C + DK)'X(C + DK)$$

where X is the unique solution to

$$(A + B_2K)X + X(A + B_2K)' + B_1'B_1 = 0$$

The hermitian positive semi-definite matrix X is sometimes referred to as the *state covariance matrix* associated with the above system.

21.10 Exercises

Exercise 180 Consider the system

$$\dot{x} = 2x + u,$$

where x and u are real scalars. Obtain an optimal stabilizing feedback controller which minimizes

$$\int_0^\infty u(t)^2 dt.$$

Exercise 181 Consider the system

$$\begin{aligned}\dot{x}_1 &= -x_1 + x_2 \\ \dot{x}_2 &= x_2 + u,\end{aligned}$$

where x_1, x_2 and u are real scalars. Obtain an optimal stabilizing feedback controller which minimizes

$$\int_0^\infty u(t)^2 dt.$$

Exercise 182 Consider the system

$$\begin{aligned}\dot{x}_1 &= -x_2 + u \\ \dot{x}_2 &= -x_1 - u\end{aligned}$$

where all quantities are scalar. Obtain an optimal stabilizing feedback controller which minimizes

$$\int_0^\infty u(t)^2 dt.$$

Exercise 183 For the system

$$\begin{aligned}\dot{x}_1 &= \alpha x_1 + x_2 \\ \dot{x}_2 &= -x_2 + u\end{aligned}$$

Obtain an optimal feedback controller which minimizes

$$\int_0^\infty u(t)^2 dt$$

You may have to distinguish between different cases of α .

Exercise 184 *2-link manipulator* Recall the two link manipulator with measured output $y = q_2$. Recall the linearization of this system about

$$q_1 = q_2 = u = 0$$

and recall its state space description.

Obtain an LQG controller for this linearized system. Apply the controller to the nonlinear system and simulate the nonlinear closed loop system for various initial conditions. (Use zero initial conditions for the observer.)

21.11 Appendix

21.11.1 Proof of fact 1

Asymptotic Stability

$$\begin{aligned}
P(A + B\tilde{K}) + (A + B\tilde{K})'P &= P(A - BR^{-1}B^*P) + (A - BR^{-1}B^*P)^*P \\
&= PA + A^*P - 2PBR^{-1}B^*P \\
&= -PBK - K^*B^*P - K^*RK - 2PBR^{-1}B^*P - Q \\
&= -PBR^{-1}B^*P - Q - (K - \tilde{K})'R(K - \tilde{K})
\end{aligned}$$

Hence

$$P(A + B\tilde{K}) + (A + B\tilde{K})'P + PBR^{-1}B^*P + Q + (K - \tilde{K})'R(K - \tilde{K}) = 0 \quad (21.20)$$

Consider any eigenvalue λ of $A + B\tilde{K}$ and let $v \neq 0$ be a corresponding eigenvector. Then

$$2\Re(\lambda)v'Pv + v'(K - \tilde{K})'R(K - \tilde{K})v \leq 0$$

First note that if

$$v'(K - \tilde{K})'R(K - \tilde{K})v = 0$$

we must have

$$\tilde{K}v = Kv$$

and hence

$$(A + B\tilde{K})v = (A + BK)v$$

that is, λ is an eigenvalue of $A + BK$. Since K is stabilizing, $\Re(\lambda) < 0$. So suppose $v'(K - \tilde{K})'R(K - \tilde{K})v > 0$. Then

$$2\Re(\lambda)v'Pv < 0$$

Since $v'Pv \geq 0$ we must have $\Re(\lambda) < 0$. Since $\Re(\lambda) < 0$ for every eigenvalue λ of $A + B\tilde{K}$, we have asymptotic stability of $A + B\tilde{K}$.

Reduced cost: The cost associated with $\tilde{K}$ is given by

$$\hat{J}(x_0, \tilde{K}) = x_0'\tilde{P}x_0$$

where $\tilde{P}$ is the unique solution to

$$\tilde{P}(A + B\tilde{K}) + (A + B\tilde{K})^T\tilde{P} + \tilde{K}^TR\tilde{K} + Q = 0$$

Substitution for $\tilde{K}$ yields

$$\tilde{P}(A + B\tilde{K}) + (A + B\tilde{K})'\tilde{P} + PBR^{-1}B^*P + Q = 0 \quad (21.21)$$

Combining (21.21) and (21.20) yields

$$(P - \tilde{P})(A + B\tilde{K}) + (A + B\tilde{K})'(P - \tilde{P}) \leq 0$$

In other words, for some hermitian positive semi-definite matrix ΔQ ,

$$(P - \tilde{P})(A + B\tilde{K}) + (A + B\tilde{K})'(P - \tilde{P}) + \Delta Q = 0$$

Since $A + B\tilde{K}$ is asymptotically stable,

$$P - \tilde{P} = \int_0^\infty e^{(A+B\tilde{K})'t} \Delta Q e^{(A+B\tilde{K})t} dt$$

Hence $P - \tilde{P} \geq 0$, or

$$\tilde{P} \leq P$$

■

Chapter 22

LQG controllers

In the previous chapter, we saw how to obtain a stabilizing state feedback gain matrix K by solving a Riccati equation. Anytime one has a technique for obtaining stabilizing state feedback gain matrices, one also has a technique for obtaining stabilizing observer gain matrices L . This simply follows from the fact that the matrix $A + LC$ is Hurwitz if and only if the same is true for $A' + C'L'$. To see this, suppose one has a technique for obtaining K such that $A + BK$ is Hurwitz. Then one can replace A with A' and B with C' to obtain K such that $A' + C'K$ is Hurwitz; letting $L = K'$, we obtain that $A' + C'L'$ is Hurwitz; hence $A + LC$ is Hurwitz.

In this chapter, we use the results of the last chapter to obtain an observer design method which is based on a Riccati equation which we call the **observer Riccati equation**. We then combine these results with the results of the previous chapter to obtain a methodology for the design of stabilizing output feedback controllers via the solution of two uncoupled Riccati equations.

The observer presented here is the same as the state estimator commonly referred to as a **Kalman filter**. The Kalman filter is developed in a stochastic setting where one considers stochastic disturbances to the plant and stochastic measurement noise.

22.1 LQ observer

- Plant

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}$$

where t is the time variable, the n -vector $x(t)$ is the plant state, the m -vector $u(t)$ is the control input and the p -vector $y(t)$ is the measured output.

- Observer

$$\dot{x}_c = Ax_c + Bu + L(Cx_c + Du - y)$$

- Error dynamics : $\tilde{x} = x_c - x$

$$\dot{\tilde{x}} = (A + LC)\tilde{x}$$

In what follows the dimensions of the weighting matrices V and W are $p \times p$ and $n \times n$, respectively.

Lemma 26 *Suppose (C, A) is detectable, $V' = V > 0$, and $W' = W \geq 0$. Then the following are equivalent.*

- (i) *The pair (A, W) has no imaginary axis uncontrollable eigenvalues.*
- (ii) *The Riccati equation*

$$\boxed{AS + SA' - SC'V^{-1}CS + W = 0} \quad (22.1)$$

has a hermitian solution S with $A - SC'V^{-1}C$ asymptotically stable.

PROOF. By duality. ■

- We will refer to ARE (22.1) as an **observer Riccati equation** associated with the plant.
- We say that S is a **stabilizing solution** to (22.1) if $A - SC'V^{-1}C$ is asymptotically stable. The significance of this is that if one chooses the observer gain L as

$$\boxed{L = -SC'V^{-1}} \quad (22.2)$$

then the error dynamics are described by the asymptotically stable system

$$\dot{\tilde{x}} = (A - SC'V^{-1}C)\tilde{x}$$

22.2 LQG controllers

Plant

$$\boxed{\begin{array}{lcl} \dot{x} & = & Ax + Bu \\ y & = & Cx + Du \end{array}}$$

Basically we use an LQR state feedback controller and supply it with an estimate of the state from an LQ observer.

So suppose the the plant satisfies the following minimal requirements for stabilization via output feedback:

- (i) (A, B) is stabilizable.
- (ii) (C, A) is detectable.

Choose weighting matrices

$$Q \quad R \quad W \quad V$$

(of appropriate dimensions) which satisfy the following conditions:

- (i) $R' = R > 0$ and $Q' = Q \geq 0$, and (Q, A) has no imaginary axis unobservable eigenvalues.
- (ii) $V' = V > 0$ and $W' = W \geq 0$, and (A, W) has no imaginary axis uncontrollable eigenvalues.

Let P and S be the stabilizing solutions to the Riccati equations

$$\begin{array}{lcl} PA + A'P - PBR^{-1}B'P + Q & = & 0 \\ AS + SA' - SC'V^{-1}CS + W & = & 0 \end{array}$$

LQG Controller. The corresponding LQG controller is given by

$$\begin{array}{lcl} \dot{x}_c & = & (A - BR^{-1}B'P)x_c - SC'V^{-1}(Cx_c + Du - y) \\ u & = & -R^{-1}B'Px_c \end{array}$$

Note that this can be written as

$$\begin{array}{lcl} \dot{x}_c & = & (A - BR^{-1}B'P - SC'V^{-1}C + SC'V^{-1}DR^{-1}B'P)x_c + SC'V^{-1}y \\ u & = & -R^{-1}B'Px_c \end{array}$$

Letting $\dot{\tilde{x}} = \dot{x}_c - \dot{x}$, the closed loop system is described by

$$\begin{array}{lcl} \dot{x} & = & (A - BR^{-1}B'P)x - BR^{-1}B'P\tilde{x} \\ \dot{\tilde{x}} & = & (A - SC'V^{-1}C)\tilde{x} \end{array}$$

As we have seen before the eigenvalues of this closed loop system are simply the union of those of the asymptotically matrices $A - BR^{-1}B'P$ and $A - SC'V^{-1}C$. Hence, the closed loop system is asymptotically stable.

22.3 H_2 Optimal controllers*

22.3.1 LQ observers

Consider

$$\begin{array}{lcl} \dot{x} & = & Ax + B_1w + B_2u \\ y & = & Cx + Dw \end{array}$$

where the measurement vector $y(t)$ is a p -vector. The vector w represents both disturbance inputs to the plant and measurement noise. We assume that

- (i) $\text{rank} D = p$
- (ii) $DB_1' = 0$

With an observer

$$\dot{x}_c = Ax_c + B_2u + L(Cx_c - y),$$

the corresponding observer error dynamics are described by

$$\dot{\tilde{x}} = (A + LC)\tilde{x} + (B_1 - LD)w.$$

Note that

$$(B_1 - LD)(B_1 - LD)' = B_1B_1' + LDD'L'$$

Consider any “performance variable”

$$\tilde{z} = M\tilde{x}$$

Suppose $A + LC$ is asymptotically stable and G_L is the impulse response matrix from w to $\tilde{z}$. Then, the H_2 norm of G_L is given by

$$\|G_L\|_2^2 = \int_0^\infty \text{trace} (G'(t)G(t)) dt = MS_LM^*$$

where S_L uniquely solves

$$(A + LC)S_L + S_L(A + LC)^* + LDD^*L' + B_1B_1^* = 0$$

With

$$V = DD^* \quad W = B_1B_1^*$$

the corresponding LQ observer minimizes the H_2 norm of G_L .

The controllability condition on (A, W) is equivalent to:

- The pair (A, B_1) has no imaginary axis uncontrollable eigenvalues.

So provided this condition holds the observer gain which minimizes the H_2 norm of G_L is given by

$$L = -SC^*(DD^*)^{-1}$$

where S is the stabilizing solution to

$$AS + SA' - SC^*(DD')^{-1}CS + B_1B_1^* = 0$$

22.3.2 LQG controllers

Consider

$$\begin{aligned} \dot{x} &= Ax + B_1w + B_2u \\ z &= C_1x + D_{12}u \\ y &= C_2x + D_{21}w \end{aligned}$$

Chapter 23

Systems with inputs and outputs: II

Recall

$$\leftarrow \begin{array}{l} \dot{x} = Ax + Bu \\ y = Cx + Du \end{array} \leftarrow \quad (23.1)$$

23.1 Minimal realizations

Recall that (A, B, C, D) is a realization of a transfer matrix $\hat{G}$ if

$$\hat{G}(s) = C(sI - A)^{-1}B + D.$$

- A realization is minimal if the dimension of its state-space is less than or equal to the state space dimension of any other realization.

SISO systems. Consider a SISO system described by

$$\hat{G}(s) = \frac{n(s)}{d(s)}$$

where n and d are polynomials. If n and d are coprime (i.e., they have no common zeros), then the order of a minimal realization is simply the degree of d .

Equivalent systems. System

$$\begin{array}{l} \dot{x} = Ax + Bu \\ y = Cx + Du \end{array}$$

is equivalent to system

$$\begin{array}{l} \dot{\tilde{x}} = \tilde{A}\tilde{x} + \tilde{B}u \\ y = \tilde{C}\tilde{x} + \tilde{D}u \end{array}$$

if there is a nonsingular matrix T such that

$$\begin{array}{ll} \tilde{A} = T^{-1}AT & \tilde{B} = T^{-1}B \\ \tilde{C} = CT & \tilde{D} = D \end{array}$$

Theorem 24

- (i) (A, B, C, D) is a minimal realization of $\hat{G}(s)$ iff (A, B) is controllable and (C, A) is observable.
- (ii) All minimal realizations are equivalent.

23.2 Exercises

Exercise 185 Obtain a state space realization of the transfer function,

$$\hat{G}(s) = \begin{bmatrix} \frac{1}{s-1} & \frac{1}{s-1} \\ \frac{1}{s+1} & -\frac{1}{s+1} \end{bmatrix}.$$

Is your realization minimal?

Exercise 186 Obtain a state space realization of the transfer function,

$$\hat{G}(s) = \begin{bmatrix} \frac{s}{s-1} & \frac{1}{s+1} \end{bmatrix}.$$

Is your realization minimal?

Problem 2 Obtain a minimal state space realization of the transfer function

$$\hat{G}(s) = \frac{s^2 - 3s + 2}{s^2 + s - 2}.$$

Chapter 24

Mechanical/Aerospace systems

24.1 Description

Here we consider systems described by

$$\boxed{\begin{aligned} M\ddot{q} + D\dot{q} + Kq &= Wu \\ y &= C_1q + C_2\dot{q} \end{aligned}} \quad (24.1)$$

where $q(t) \in \mathbb{R}^N$ is a vector of generalized coordinates which describes the configuration of the system at time t ; $u(t) \in \mathbb{R}^m$ is a vector of control input variables; $y(t) \in \mathbb{R}^p$ is a vector of measured output variables. Sometimes the square matrices M, D, K are referred to as *mass (or inertia), damping, and stiffness matrices*, respectively. Examples include linearized models of robots, flexible space structures, buildings, and anything whose equations of motion are obtained from mechanics. Recall the examples given at beginning of this course.

The purpose of this section is to look at the concepts of stability, controllability, observability, etc. for these systems. In particular, we want to derive tests for the above attributes in terms of the M, D, K matrices. These tests eliminate the need for putting the system into first order form and are usually easier to apply and are sometimes more transparent than the original tests.

- We assume throughout that M is a square invertible matrix.

Example 209 Guess who!!

$$\begin{aligned} m\ddot{q}_1 &= k(q_2 - q_1) - u \\ m\ddot{q}_2 &= -k(q_2 - q_1) + u \\ y &= q_1 + q_2 \end{aligned}$$

Here

$$M = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \quad D = 0 \quad K = \begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \quad W = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad C_1 = \begin{bmatrix} 1 & 1 \end{bmatrix} \quad C_2 = 0$$

State space description. Letting

$$x := \begin{bmatrix} x^1 \\ x^2 \end{bmatrix} := \begin{bmatrix} q \\ \dot{q} \end{bmatrix}$$

system (24.1) is described by

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned}$$

with

$$A = \begin{bmatrix} 0 & I \\ -M^{-1}K & -M^{-1}D \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ M^{-1}W \end{bmatrix} \quad C = \begin{bmatrix} C_1 & C_2 \end{bmatrix} \quad (24.2)$$

Transfer function matrix. Taking the Laplace transform of (24.1) with $q(0) = \dot{q}(0) = 0$ yields

$$\begin{aligned} (s^2M + sD + K)\hat{q} &= W\hat{u} \\ \hat{y} &= (C_1 + sC_2)\hat{q} \end{aligned}$$

Solving for $\hat{y}$ we obtain

$$\hat{y} = (sC_2 + C_1)(s^2M + sD + K)^{-1}W\hat{u}$$

hence the transfer function matrix from $\hat{u}$ to $\hat{y}$ is given by

$$\hat{G}(s) = (sC_2 + C_1)(s^2M + sD + K)^{-1}W$$

Example 210 Guess who!! Recalling example 209 we have

$$(s^2M + sD + K)^{-1} = \begin{bmatrix} ms^2 + k & -k \\ -k & ms^2 + k \end{bmatrix}^{-1} = \frac{1}{\Delta(s)} \begin{bmatrix} ms^2 + k & k \\ k & ms^2 + k \end{bmatrix}$$

where

$$\Delta(s) = ms^2(ms^2 + 2k)$$

Hence,

$$\begin{aligned} \hat{G}(s) &= (sC_2 + C_1)(s^2M + sD + K)^{-1}W \\ &= \frac{1}{\Delta(s)} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} ms^2 + k & k \\ k & ms^2 + k \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \\ &= 0 \end{aligned}$$

What's goin' on!!

24.2 Modes, eigenvalues and eigenvectors

Lemma 27 *A complex number λ is an eigenvalue of A (given by (24.2)) iff*

$$\boxed{\det(\lambda^2 M + \lambda D + K) = 0} \quad (24.3)$$

If λ is an eigenvalue of A , all its eigenvectors are given by

$$v = \begin{bmatrix} q \\ \lambda q \end{bmatrix}$$

where $q \neq 0$ satisfies

$$\boxed{(\lambda^2 M + \lambda D + K)q = 0} \quad (24.4)$$

PROOF. A complex number λ is an eigenvalue of A iff there is a nonzero $2N$ -vector v such that

$$Av = \lambda v \quad (24.5)$$

Letting

$$v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

where v_1 and v_2 are N vectors, and using (24.2), the relationship (24.5) is equivalent to

$$\begin{aligned} v_2 &= \lambda v_1 \\ -M^{-1}Kv_1 - M^{-1}Dv_2 &= \lambda v_2 \end{aligned}$$

These two equations are equivalent to

$$\begin{aligned} v_2 &= \lambda v_1 \\ (\lambda^2 M + \lambda D + K)v_1 &= 0 \end{aligned} \quad (24.6)$$

From the first of the above equations, it should be clear that $v \neq 0$ iff $v_1 \neq 0$. Satisfaction of the second equation in (24.6) by a nonzero v_1 is equivalent to

$$\det(\lambda^2 M + \lambda D + K) = 0$$

■

- We will sometimes refer to the polynomial

$$\boxed{p(s) = \det(s^2 M + sD + K)}$$

as the characteristic polynomial associated with system (24.1) or (M, D, K) . So the eigenvalues of A are precisely the roots of p . The algebraic multiplicity of an eigenvalue λ is its multiplicity as a root of p .

- Sometimes a vector q satisfying (24.4) is called a *mode shape* associated with λ .

- From the above lemma, one can deduce that

$$\text{nullity}(A - \lambda I) = \text{nullity}(\lambda^2 M + \lambda D + K)$$

i.e., the dimension of the eigenspace associated with λ is the dimension of the null space of the matrix $\lambda^2 M + \lambda D + K$. Hence the geometric multiplicity of an eigenvalue λ is the dimension of the null space of $\lambda^2 M + \lambda D + K$.

Example 211 We're back!

$$s^2 M + sD + K = \begin{bmatrix} s^2 m + k & -k \\ -k & s^2 m + k \end{bmatrix}$$

Hence

$$p(s) = \det(s^2 M + sD + K) = ms^2(ms^2 + 2k)$$

The roots of this characteristic polynomial yield three eigenvalues

$$\lambda_1 = 0, \quad \lambda_2 = j\omega, \quad \lambda_3 = -j\omega; \quad \omega := \sqrt{2k/m}$$

The eigenvalue $\lambda_1 = 0$ has algebraic multiplicity 2 but has one linearly independent mode shape, e.g.,

$$q^1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

this yields one linearly independent eigenvector of the A matrix

$$v^1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Hence the geometric multiplicity of this eigenvalue is 1.

Eigenvalues λ_2, λ_3 have the same mode shapes; there is only one linearly independent one:

$$q^2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Linearly independent eigenvectors of the A matrix corresponding to λ_2 and λ_3 are

$$v^2 = \begin{bmatrix} 1 \\ -1 \\ j\omega \\ -j\omega \end{bmatrix} \quad v^3 = \begin{bmatrix} 1 \\ -1 \\ -j\omega \\ j\omega \end{bmatrix}$$

respectively.

24.2.1 Undamped systems

By an “undamped system” we mean a system for which

$$D = 0$$

see last example. Hence, an undamped system is described by

$$\boxed{M\ddot{q} + Kq = 0}$$

and its the corresponding “ A matrix” is

$$A = \begin{bmatrix} 0 & I \\ -M^{-1}K & 0 \end{bmatrix}$$

The characteristic polynomial of an undamped system is given by

$$p(s) = \det(s^2M + K)$$

and modal equation (24.4) becomes

$$(\lambda^2M + K)q = 0$$

This is equivalent to

$$\lambda^2Mq = -Kq$$

or

$$M^{-1}Kq = -\lambda^2q$$

From this it should be clear that λ is an eigenvalue of A iff $-\lambda^2$ is an eigenvalue of $M^{-1}K$; hence

$$\lambda = j\sqrt{\sigma} \quad \text{or} \quad \lambda = -j\sqrt{\sigma}$$

where σ is an eigenvalue of $M^{-1}K$, i.e., for some nonzero vector q

$$M^{-1}Kq = \sigma q$$

or equivalently,

$$(\sigma M - K)q = 0$$

• *Rigid body modes:* If K is singular, then $M^{-1}K$ has an eigenvalue $\sigma = 0$. This yields a defective eigenvalue $\lambda = 0$ of the original A matrix. (By a defective eigenvalue we mean an eigenvalue whose geometric multiplicity is not equal to its algebraic multiplicity.) This corresponds to a *rigid body mode* of the physical system. So, rigid body mode shapes can be obtained from

$$Kq = 0$$

Note that if q is a rigid body mode shape then

$$v = \begin{bmatrix} q \\ 0 \end{bmatrix}$$

is an eigenvector of the A matrix and

$$g = \begin{bmatrix} 0 \\ q \end{bmatrix}$$

is a generalized eigenvector of the A matrix.

Symmetric M and K

In this case, it can be shown that each eigenvalue σ of $M^{-1}K$ is real and nondefective. (By a nondefective eigenvalue we mean an eigenvalue whose geometric multiplicity is the same as its algebraic multiplicity.)

Each nonzero σ gives rise to two nondefective eigenvalues $j\sqrt{\sigma}$, $-j\sqrt{\sigma}$ of the original A matrix.

If $\sigma = 0$ the original A matrix has a defective eigenvalue $\lambda = 0$.

$M > 0$ and $K \geq 0$. In this case all the σ 's are nonnegative; hence

$$\sigma = \omega^2$$

for some nonnegative real number ω and all the ω 's are given by

$$\boxed{\det(\omega^2 M - K) = 0}$$

Each nonzero ω results in two nondefective eigenvalues

$$\lambda = j\omega \quad \lambda = -j\omega$$

of the original A matrix. A nonzero ω is called a *natural frequency* of the system. The mode shapes corresponding to a natural frequency ω are given by

$$\boxed{(\omega^2 M - K)q = 0} \quad (24.7)$$

Example 212

$$\begin{aligned} m\ddot{q}_1 + 2kq_1 - kq_2 &= 0 \\ m\ddot{q}_2 - kq_1 + 2kq_2 &= 0 \end{aligned}$$

This is an undamped system with

$$M = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \quad K = \begin{bmatrix} 2k & -k \\ -k & 2k \end{bmatrix}$$

Both of these matrices are positive definite and symmetric.

The natural frequencies are the zeroes of

$$\begin{aligned} \det(\omega^2 M - K) &= \det \begin{bmatrix} \omega^2 m - 2k & k \\ k & \omega^2 m - 2k \end{bmatrix} \\ &= \omega^4 m^2 - 4km\omega^2 + 3k^2 \\ &= (m\omega^2 - k)(m\omega^2 - 3k) \end{aligned}$$

Hence two distinct natural frequencies

$$\omega_1 = \sqrt{k/m} \quad \omega_2 = \sqrt{3k/m}$$

For $\omega_1 = \sqrt{k/m}$, modal equation (24.7) yields one linearly independent mode shape

$$q^1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\omega_2 = \sqrt{3k/m}$, modal equation (24.7) yields one linearly independent mode shape

$$q^2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

24.3 Controllability and stabilizability

Recall the concept of an uncontrollable eigenvalue of a matrix pair (A, B) .

Lemma 28 *A complex number λ is an uncontrollable eigenvalue of (A, B) (given by (24.2)) iff*

$$\boxed{\text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K & W \end{bmatrix} < N}$$

PROOF.

Recall that λ is a uncontrollable eigenvalue of A iff

$$\text{rank} \begin{bmatrix} A - \lambda I & B \end{bmatrix} < n = 2N$$

Since

$$\begin{aligned} \text{rank} \begin{bmatrix} A - \lambda I & B \end{bmatrix} &= \text{rank} \begin{bmatrix} -\lambda I & I & 0 \\ -M^{-1}K & -\lambda I - M^{-1}D & M^{-1}W \end{bmatrix} \\ &= \text{rank} \begin{bmatrix} 0 & I & 0 \\ -\lambda^2 I - \lambda M^{-1}D - M^{-1}K & -\lambda I - M^{-1}D & M^{-1}W \end{bmatrix} \\ &= \text{rank} \begin{bmatrix} 0 & 0 \\ -\lambda^2 I - \lambda M^{-1}D - M^{-1}K & M^{-1}W \end{bmatrix} + N \\ &= \text{rank} \begin{bmatrix} -\lambda^2 I - \lambda M^{-1}D - M^{-1}K & M^{-1}W \end{bmatrix} + N \\ &= \text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K & W \end{bmatrix} + N \end{aligned}$$

Hence λ is uncontrollable iff

$$\text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K & W \end{bmatrix} < N$$

■

Corollary 3 *The pair (A, B) is controllable iff*

$$\text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K & W \end{bmatrix} = N \quad \text{for all } \lambda \in \mathbb{C}$$

Corollary 4 *The pair (A, B) is stabilizable iff*

$$\text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K & W \end{bmatrix} = N \quad \text{for all } \lambda \in \mathbb{C} \quad \text{with } \Re(\lambda) \geq 0$$

Example 213 Guess!

$$\begin{bmatrix} \lambda^2 M + \lambda D + K & W \end{bmatrix} = \begin{bmatrix} \lambda^2 m + k & -k & -1 \\ -k & \lambda^2 m + k & 1 \end{bmatrix}$$

For $\lambda = 0$

$$\text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K & W \end{bmatrix} = \text{rank} \begin{bmatrix} k & -k & -1 \\ -k & k & 1 \end{bmatrix} = 1 < N$$

Hence, $\lambda = 0$ is uncontrollable. So, this system is not stabilizable.

For $\lambda^2 = -2k/m$

$$\text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K & W \end{bmatrix} = \text{rank} \begin{bmatrix} -k & -k & -1 \\ -k & -k & 1 \end{bmatrix} = 2 = N$$

Hence the eigenvalues $j\omega, -j\omega$ are controllable.

24.4 Observability and detectability

Recall the concept of an unobservable eigenvalue of a matrix pair (C, A) .

Lemma 29 *A complex number λ is an unobservable eigenvalue of (C, A) (given by (24.2)) iff*

$$\boxed{\text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K \\ \lambda C_2 + C_1 \end{bmatrix} < N}$$

PROOF Recall that λ is a unobservable eigenvalue of (C, A) iff

$$\text{rank} \begin{bmatrix} A - \lambda I \\ C \end{bmatrix} < n = 2N$$

Since

$$\begin{aligned} \text{rank} \begin{bmatrix} A - \lambda I \\ C \end{bmatrix} &= \text{rank} \begin{bmatrix} -\lambda I & I \\ -M^{-1}K & -\lambda I - M^{-1}D \\ C_1 & C_2 \end{bmatrix} \\ &= \text{rank} \begin{bmatrix} 0 & I \\ -\lambda^2 I - \lambda M^{-1}D - M^{-1}K & -\lambda I - M^{-1}D \\ C_1 + \lambda C_2 & C_2 \end{bmatrix} \\ &= \text{rank} \begin{bmatrix} 0 \\ -\lambda^2 I - \lambda M^{-1}D - M^{-1}K \\ C_1 + \lambda C_2 \end{bmatrix} + N \\ &= \text{rank} \begin{bmatrix} -\lambda^2 I - \lambda M^{-1}D - M^{-1}K \\ C_1 + \lambda C_2 \end{bmatrix} + N \\ &= \text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K \\ \lambda C_2 + C_1 \end{bmatrix} \end{aligned}$$

Hence λ is an unobservable eigenvalue iff

$$\text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K \\ \lambda C_2 + C_1 \end{bmatrix} < N$$

■

Corollary 5 *The pair (C, A) is observable iff*

$$\text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K \\ \lambda C_2 + C_1 \end{bmatrix} = N \quad \text{for all } \lambda \in \mathbb{C}$$

Corollary 6 *The pair (C, A) is detectable iff*

$$\text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K \\ \lambda C_2 + C_1 \end{bmatrix} = N \quad \text{for all } \lambda \in \mathbb{C} \quad \text{with } \Re(\lambda) \geq 0$$

Example 214 B&B

$$\begin{bmatrix} \lambda^2 M + \lambda D + K \\ \lambda C_2 + C_1 \end{bmatrix} = \begin{bmatrix} \lambda^2 m + k & -k \\ -k & \lambda^2 m + k \\ 1 & 1 \end{bmatrix}$$

For $\lambda = 0$

$$\text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K \\ \lambda C_2 + C_1 \end{bmatrix} = \text{rank} \begin{bmatrix} k & -k \\ -k & k \\ 1 & 1 \end{bmatrix} = 2$$

Hence this eigenvalue is observable.

For $\lambda^2 = -2k/m$

$$\text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K \\ \lambda C_2 + C_1 \end{bmatrix} = \text{rank} \begin{bmatrix} -k & -k \\ -k & -k \\ 1 & 1 \end{bmatrix} = 1 < N$$

Hence the two imaginary eigenvalues are not observable. So, this system is not detectable.

24.5 Lyapunov revisited

Consider

$$\boxed{M\ddot{q} + D\dot{q} + Kq = 0} \quad (24.8)$$

and suppose

$$\boxed{M^T = M > 0 \quad K^T = K > 0 \quad D^T = D \geq 0} \quad (24.9)$$

Using a Lyapunov argument, we have already seen that this system is stable, and if $D > 0$, it is asymptotically stable. However $D > 0$ is not necessary for asymptotic stability. We present in this section a necessary and sufficient condition for asymptotic stability.

• By an *undamped natural frequency* of this system we mean a positive real number ω which satisfies

$$\det(\omega^2 M - K) = 0$$

Lemma 30 *A system described by (24.8) and satisfying (24.9) is asymptotically stable iff*

$$\boxed{\text{rank} \begin{bmatrix} \omega^2 M - K \\ D \end{bmatrix} = N}$$

for every undamped natural frequency ω of (24.8).

PROOF. Sufficiency. We first show that satisfaction of the rank condition guarantees asymptotic stability. Recall that if we let

$$P = \frac{1}{2} \begin{bmatrix} K & 0 \\ 0 & M \end{bmatrix}$$

Then $P^* = P > 0$ and

$$PA + A^*P + Q = 0$$

where

$$Q = \begin{bmatrix} 0 & 0 \\ 0 & D \end{bmatrix}$$

If we show that (Q, A) is observable, then A is asymptotically stable. (Q, A) is observable iff (C, A) is observable where

$$C = [0 \quad D]$$

and (C, A) is observable iff

$$\text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K \\ \lambda D \end{bmatrix} = N \quad \text{for all } \lambda \in \mathbb{C}$$

However,

$$\text{rank} \begin{bmatrix} \lambda^2 M + \lambda D + K \\ \lambda D \end{bmatrix} = \text{rank} \begin{bmatrix} \lambda^2 M + K \\ \lambda D \end{bmatrix}$$

This matrix has rank N when $\lambda^2 M + K$ is nonsingular. When $\lambda^2 M + K$ is singular, then $\lambda^2 = -\omega^2 < 0$ where ω is an undamped natural frequency of the system. In this case $\lambda \neq 0$ and

$$\text{rank} \begin{bmatrix} \lambda^2 M + K \\ \lambda D \end{bmatrix} = \text{rank} \begin{bmatrix} \lambda^2 M + K \\ D \end{bmatrix} = \text{rank} \begin{bmatrix} \omega^2 M - K \\ D \end{bmatrix}$$

Necessity. Now suppose the rank condition fails. Then there is a nonzero N -vector q such that

$$\begin{bmatrix} \omega^2 M - K \\ D \end{bmatrix} q = 0$$

or, equivalently

$$\begin{aligned} (\omega^2 M - K)q &= 0 \\ Dq &= 0 \end{aligned}$$

Hence,

$$(\lambda^2 M + \lambda D + K)q = 0 \quad \lambda = j\omega$$

which means the system has eigenvalue $j\omega$ and, hence, is not asymptotically stable. ■

Example 215

$$\begin{aligned} m\ddot{q}_1 + d\dot{q}_1 - d\dot{q}_2 + 2kq_1 - kq_2 &= 0 \\ m\ddot{q}_2 - d\dot{q}_1 + d\dot{q}_2 - kq_1 + 2kq_2 &= 0 \end{aligned}$$

Here

$$M = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \quad D = \begin{bmatrix} d & -d \\ -d & d \end{bmatrix} \quad K = \begin{bmatrix} 2k & -k \\ -k & 2k \end{bmatrix}$$

The undamped natural frequencies are the zeroes of

$$\begin{aligned} \det(\omega^2 M - K) &= \det \begin{bmatrix} \omega^2 m - 2k & k \\ k & \omega^2 m - 2k \end{bmatrix} \\ &= \omega^4 m^2 - 4km\omega^2 + 3k^2 \\ &= (m\omega^2 - k)(m\omega^2 - 3k) \end{aligned}$$

Hence two undamped natural frequencies

$$\omega_1 = \sqrt{k/m} \quad \omega_2 = \sqrt{3k/m}$$

$$\begin{bmatrix} \omega^2 M - K \\ D \end{bmatrix} = \begin{bmatrix} \omega^2 m - 2k & k \\ k & \omega^2 m - 2k \\ d & -d \\ -d & d \end{bmatrix}$$

For $\omega_2 = \sqrt{3k/m}$

$$\text{rank} \begin{bmatrix} \omega^2 M - K \\ D \end{bmatrix} = \text{rank} \begin{bmatrix} k & k \\ k & k \\ d & -d \\ -d & d \end{bmatrix} = 2 = N$$

For $\omega_1 = \sqrt{k/m}$

$$\text{rank} \begin{bmatrix} \omega^2 M - K \\ D \end{bmatrix} = \text{rank} \begin{bmatrix} -k & k \\ k & -k \\ d & -d \\ -d & d \end{bmatrix} = 1 < N$$

Hence, the system is not asymptotically stable.

Example 216

$$\begin{aligned} m\ddot{q}_1 + d\dot{q}_1 + 2kq_1 - kq_2 &= 0 \\ m\ddot{q}_2 - kq_1 + 2kq_2 &= 0 \end{aligned}$$

Here

$$M = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \quad D = \begin{bmatrix} d & 0 \\ 0 & 0 \end{bmatrix} \quad K = \begin{bmatrix} 2k & -k \\ -k & 2k \end{bmatrix}$$

The undamped natural frequencies are the same as last example:

$$\omega_1 = \sqrt{k/m} \quad \omega_2 = \sqrt{3k/m}$$

$$\begin{bmatrix} \omega^2 M - K \\ D \end{bmatrix} = \begin{bmatrix} \omega^2 m - 2k & k \\ k & \omega^2 m - 2k \\ d & 0 \\ 0 & 0 \end{bmatrix}$$

For $\omega_2 = \sqrt{3k/m}$

$$\text{rank} \begin{bmatrix} \omega^2 M - K \\ D \end{bmatrix} = \text{rank} \begin{bmatrix} k & k \\ k & k \\ d & 0 \\ 0 & 0 \end{bmatrix} = 2 = N$$

For $\omega_1 = \sqrt{k/m}$

$$\text{rank} \begin{bmatrix} \omega^2 M - K \\ D \end{bmatrix} = \text{rank} \begin{bmatrix} -k & k \\ k & -k \\ d & 0 \\ 0 & 0 \end{bmatrix} = 2 = N$$

Hence, the system is asymptotically stable.

Chapter 25

Time-varying systems

25.1 Linear time-varying systems (LTVs)

Here we are concerned with linear time-varying systems (LTVs) systems described by

$$\begin{array}{l} \dot{x}(t) = A(t)x(t) + B(t)u(t) \\ y(t) = C(t)x(t) + D(t)u(t) \end{array} \quad (25.1)$$

where t is the time variable, the state $x(t)$ is an n -vector, the input $u(t)$ is an m -vector while the output $y(t)$ is a p -vector. We suppose that the time-varying matrices $A(t), B(t), C(t), D(t)$ depend in piecewise continuously on t .

25.1.1 Linearization about time-varying trajectories

Quite often, an LTV system arises from linearizing a nonlinear system about a time-varying trajectory. Consider a nonlinear system described by

$$\dot{x} = F(x, u) \quad (25.2a)$$

$$y = H(x, u). \quad (25.2b)$$

Suppose $\bar{u}(\cdot)$ is a specific time-varying input history to the nonlinear system and $\bar{x}(\cdot)$ is any corresponding state trajectory. Then

$$\dot{\bar{x}}(t) = F(\bar{x}(t), \bar{u}(t)).$$

Let $\bar{y}(\cdot)$ be the corresponding output history, that is,

$$\bar{y}(t) = H(\bar{x}(t), \bar{u}(t)).$$

Then, introducing the perturbed variables

$$\delta x = x - \bar{x} \quad \text{and} \quad \delta u = u - \bar{u} \quad \text{and} \quad \delta y = y - \bar{y},$$

the linearization of system (25.2) about $(\bar{x}, \bar{u})$ is defined by

$$\delta \dot{x} = A(t)\delta x + B(t)\delta u \quad (25.3a)$$

$$\delta y = C(t)\delta x + D(t)\delta u \quad (25.3b)$$

where

$$\begin{aligned} A(t) &= \frac{\partial F}{\partial x}(\bar{x}(t), \bar{u}(t)) \quad \text{and} \quad B(t) = \frac{\partial F}{\partial u}(\bar{u}(t), \bar{u}(t)) \\ C(t) &= \frac{\partial H}{\partial x}(\bar{x}(t), \bar{u}(t)) \quad \text{and} \quad D(t) = \frac{\partial H}{\partial u}(\bar{x}(t), \bar{u}(t)). \end{aligned}$$

Note that, although the original nonlinear system is time-invariant, its linearization about a time-varying trajectory can be time-varying.

25.2 The state transition matrix

Consider a linear time-varying system described by

$$\boxed{\dot{x} = A(t)x} \tag{25.4}$$

where t is the time variable, $x(t)$ is an n -vector and the time-varying matrix $A(t)$ depends continuously on t .

Fact 26 *We have existence and uniqueness of solutions, that is, for each initial condition,*

$$x(t_0) = x_0, \tag{25.5}$$

there is a unique solution to the differential equation (25.4) which satisfies the initial condition. Also, this solution is defined for all time.

Fact 27 *Solutions depend linearly on initial state. More specifically, for each pair of times t, t_0 , there is a matrix $\Phi(t, t_0)$ such that for every initial state x_0 , the solution $x(t)$ at time t due to initial condition (25.5) is given by*

$$\boxed{x(t) = \Phi(t, t_0)x_0}$$

The matrix $\Phi(t, t_0)$ is called the **state transition matrix** associated with system (25.4). Note that it is uniquely defined by

$$\Phi(t_0, t_0) = I$$

$$\frac{\partial \Phi}{\partial t}(t, t_0) = A(t)\Phi(t, t_0)$$

LTI systems

$$\dot{x} = Ax$$

$$\Phi(t, t_0) = e^{A(t-t_0)}$$

Scalar systems. Consider any scalar system described by

$$\dot{x} = a(t)x$$

where $x(t)$ and $a(t)$ are scalars. The solution to this system corresponding to initial condition $x(t_0) = x_0$ is given by

$$x(t) = e^{\int_{t_0}^t a(\tau) d\tau} x_0.$$

Hence,

$$\Phi(t, t_0) = e^{\int_{t_0}^t a(\tau) d\tau}$$

Some properties of the state transition matrix

(a)

$$\Phi(t_2, t_1)\Phi(t_1, t_0) = \Phi(t_2, t_0)$$

(b) $\Phi(t, t_0)$ is invertible and

$$\Phi(t_0, t) = \Phi(t, t_0)^{-1}$$

(c)

$$\frac{\partial \Phi}{\partial t_0}(t, t_0) = -\Phi(t, t_0)A(t_0)$$

(d)

$$\det[\Phi(t, t_0)] = \exp \int_{t_0}^t \text{trace}[A(\tau)] d\tau$$

Computation of the state transition matrix. Let X be the solution of

$$\dot{X}(t) = A(t)X(t) \tag{25.6a}$$

$$X(0) = I \tag{25.6b}$$

where I is the $n \times n$ identity matrix. Then $X(t) = \Phi(t, 0)$. Hence, using property (b) above, $X(t_0)^{-1} = \Phi(0, t_0)$. Now use property (a) to obtain

$$\Phi(t, t_0) = X(t)X(t_0)^{-1} \tag{25.7}$$

25.3 Stability

Recall that when A is a constant matrix, system (25.4) is asymptotically stable if and only if all the eigenvalues of A have negative real parts. This leads to the following natural conjecture: system (25.4) is asymptotically stable if the eigenvalues of $A(t)$ have negative real parts for all time t . This conjecture is, in general, false. This is demonstrated by the following example. Actually, in this example, the eigenvalues of $A(t)$ are the same for all t and have negative real parts. However, the system has unbounded solutions and is unstable.

Example 217 (Markus and Yamabe, 1960.) This is an example of an unstable second order system whose time-varying A matrix has constant eigenvalues with negative real parts. Consider

$$A(t) = \begin{bmatrix} -1 + a \cos^2 t & 1 - a \sin t \cos t \\ -1 - a \sin t \cos t & -1 + a \sin^2 t \end{bmatrix}$$

with $1 < a < 2$. Here,

$$\Phi(t, 0) = \begin{bmatrix} e^{(a-1)t} \cos t & e^{-t} \sin t \\ -e^{(a-1)t} \sin t & e^{-t} \cos t \end{bmatrix}$$

Since $a > 1$, the system corresponding to this $A(t)$ matrix has unbounded solutions. However, for all t , the characteristic polynomial of $A(t)$ is given by

$$p(s) = s^2 + (2 - a)s + 2$$

Since $2 - a > 0$, the eigenvalues of $A(t)$ have negative real parts.

Periodic time-varying systems. Suppose $A(\cdot)$ is a periodic function, that is, there exists a time period $T > 0$ such that

$$A(t + T) = A(t)$$

for all t . Consider any t_0 and let

$$\boxed{A_d = \Phi(t_0 + T, t_0)} \quad (25.8)$$

Then the stability of the system is determined by the discrete-time stability of the constant matrix A_d .

To see this, we first note that the periodicity of $A(\cdot)$ implies that, for any t and t_0 ,

$$\Phi(t + T, t_0 + T) = \Phi(t, t_0)$$

Hence,

$$\Phi(t, t_0) = \Phi(t - T, t_0 - T) = \Phi(t - T, t_0) \Phi(t_0, t_0 - T) = \Phi(t - T, t_0) \Phi(t_0 + T, t_0)$$

that is,

$$\Phi(t, t_0) = \Phi(t - T, t_0) A_d \quad (25.9)$$

where A_d is given by (25.8).

Any $t \geq t_0$ can be represented as

$$t = \tau + kT \quad \text{where} \quad t_0 \leq \tau < t_0 + T$$

for some integer k . Using (25.9) it follows by induction that we have

$$\Phi(t, t_0) = \Phi(\tau, t_0) A_d^k \quad (25.10)$$

25.4 Systems with inputs and outputs

$$\dot{x} = A(t)x + B(t)u \quad x(t_0) = x_0 \quad (25.11)$$

$$y = C(t)x + D(t)u \quad (25.12)$$

- The solution to (25.11) is uniquely given by

$$x(t) = \Phi(t, t_0)x_0 + \int_{t_0}^t \Phi(t, \tau)B(\tau)u(\tau) d\tau$$

Impulse response matrix

$$y(t) = C(t)\Phi(t, t_0)x_0 + \int_{t_0}^t C(t)\Phi(t, \tau)B(\tau)u(\tau) dt + D(t)u(t)$$

Hence, for $x(t_0) = 0$,

$$y(t) = \int_{t_0}^t G(t, \tau)u(\tau) d\tau$$

where

$$G(t, \tau) := C(t)\Phi(t, \tau)B(\tau) + D(\tau)\delta(\tau)$$

25.5 DT

$$x(k) = \underbrace{\Phi(k, k_0)}_{\text{state transition matrix}} x_0$$

Fact 28 (a)

$$\Phi(k_0, k_0) = I$$

(b)

$$\Phi(k_2, k_1)\Phi(k_1, k_0) = \Phi(k_2, k_0)$$

(c)

$$\Phi(k+1, k_0) = A(k)\Phi(k, k_0)$$

LTI

$$\Phi(k) = A^k$$

Chapter 26

Appendix A: complex stuff

26.1 Complex numbers

Think about ordered pairs (α, β) of real numbers α, β and suppose we define addition operation and a multiplication operation by

$$(\alpha, \beta) + (\gamma, \delta) = (\alpha + \gamma, \beta + \delta)$$

$$(\alpha, \beta)(\gamma, \delta) = (\alpha\gamma - \beta\delta, \alpha\delta + \beta\gamma)$$

respectively. It can readily be shown that these two operations satisfy the ‘usual’ rules of addition and multiplication for real numbers.

Consider now any two ordered pairs of the form $(\alpha, 0)$ and $(\gamma, 0)$. Using the above definitions of addition and multiplication we obtain

$$(\alpha, 0) + (\gamma, 0) = (\alpha + \gamma, 0)$$

and

$$(\alpha, 0)(\beta, 0) = (\alpha\beta, 0)$$

(check it out) So we can identify each ordered pair of the form $(\alpha, 0)$ with the real number α and the above definitions of addition and multiplication reduce to the usual ones for real numbers.

Now consider $j := (0, 1)$. Using the above definition of multiplication, we obtain

$$j^2 = (0, 1)(0, 1) = (-1, 0)$$

In other words (identifying $(-1, 0)$ with the real number -1),

$$j^2 = -1$$

We also note that for any real number β ,

$$j\beta = (0, 1)(\beta, 0) = (0, \beta)$$

hence,

$$(\alpha, \beta) = (\alpha, 0) + (0, \beta) = \alpha + j\beta$$

If we regard the set $\mathbb{C}$ of complex numbers as the set of ordered pairs of real numbers for which an addition and multiplication are as defined above, we arrive at the following representation of complex numbers.

- A complex number λ can be written as

$$\lambda = \alpha + j\beta$$

where α, β are real numbers and $j^2 = -1$. The real numbers α and β are called the *real part* and *imaginary part* of λ and are denoted by

$$\alpha := \Re(\lambda) \quad \beta := \Im(\lambda)$$

respectively.

Note

$$(\alpha + j\beta) + (\gamma + j\delta) = \alpha + \gamma + j(\beta + \delta)$$

and

$$\begin{aligned} (\alpha + j\beta)(\gamma + j\delta) &= \alpha\gamma + j\alpha\delta + j\beta\gamma + j^2\beta\delta \\ &= (\alpha\gamma - \beta\delta) + j(\alpha\delta + \beta\gamma) \end{aligned}$$

In other words, we recovered our original definitions of complex addition and multiplication.

- If $\lambda = \alpha + j\beta$, its *complex conjugate* $\bar{\lambda}$ is defined by

$$\bar{\lambda} := \alpha - j\beta$$

Some properties:

$$\begin{aligned} \Re(\lambda) &= \frac{1}{2}(\lambda + \bar{\lambda}) \\ \Im(\lambda) &= \frac{1}{2j}(\lambda - \bar{\lambda}) \\ \overline{\lambda + \eta} &= \bar{\lambda} + \bar{\eta} \\ \overline{\lambda\eta} &= \bar{\lambda}\bar{\eta} \end{aligned}$$

- The absolute value of λ :

$$\begin{aligned} |\lambda| &= \sqrt{\bar{\lambda}\lambda} \\ &= \sqrt{\alpha^2 + \beta^2} \end{aligned}$$

- Suppose p is an n -th order polynomial, i.e.,

$$p(s) = a_0 + a_1s + \dots + a_ns^n$$

where each coefficient a_i is a complex number and $a_n \neq 0$. A complex number λ is a *root* or *zero* of p if

$$p(\lambda) = 0$$

Each p has at least one root and at most n distinct roots. Suppose p has l *distinct* roots, $\lambda_1, \lambda_2, \dots, \lambda_l$; then

$$p(s) = a_n \prod_{i=1}^l (s - \lambda_i)^{m_i}$$

The integer m_i is called the *algebraic multiplicity* of λ_i .

Suppose

$$p(s) = a_0 + a_1s + \dots + a_ns^n$$

and define

$$\bar{p}(s) = \bar{a}_0 + \bar{a}_1s + \dots + \bar{a}_ns^n$$

Then λ is a root of p iff $\bar{\lambda}$ is a root of $\bar{p}$. To see this note that $\overline{p(\lambda)} = \bar{p}(\bar{\lambda})$, hence

$$p(\lambda) = 0 \quad \text{iff} \quad \bar{p}(\bar{\lambda}) = 0$$

Real coefficients. If the coefficients a_i of p are real, then the roots occur in complex conjugate pairs, i.e., if λ is a root then so is $\bar{\lambda}$.

26.1.1 Complex functions

The *exponential function*

$$e^\lambda = 1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \dots + \frac{\lambda^n}{n!} + \dots$$

It satisfies:

$$\frac{de^z}{dz} = e^z$$

and

$$e^{(\lambda+\eta)} = e^\lambda e^\eta$$

Can define $\cos(\lambda)$ and $\sin(\lambda)$. They satisfy

$$\cos(\lambda)^2 + \sin(\lambda)^2 = 1$$

Also, for any complex λ

$$e^{j\lambda} = \cos(\lambda) + j\sin(\lambda)$$

Hence, for any real number β ,

$$e^{j\beta} = \cos(\beta) + j \sin(\beta)$$

and

$$|e^{j\beta}| = \sqrt{\cos(\beta)^2 + \sin(\beta)^2} = 1$$

Also,

$$e^{(\alpha+j\beta)} = e^\alpha e^{j\beta} = e^\alpha \cos \beta + j e^\alpha \sin \beta$$

Hence,

$$|e^{(\alpha+j\beta)}| = |e^\alpha| |e^{j\beta}| = e^\alpha$$

26.2 Complex vectors and $\mathbb{C}^n$

Read all the material on real vectors and replace the word real with the word complex and $\mathbb{R}$ with $\mathbb{C}$.

If $x \in \mathbb{C}^n$, the *complex conjugate of x* :

$$\bar{x} := \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_n \end{bmatrix}$$

Some properties:

$$\begin{aligned} \overline{x+y} &= \bar{x} + \bar{y} \\ \overline{\lambda x} &= \bar{\lambda} \bar{x} \end{aligned}$$

$$\begin{aligned} \Re(x) &= \frac{1}{2}(x + \bar{x}) \\ \Im(x) &= \frac{1}{2j}(x - \bar{x}) \end{aligned}$$

26.3 Complex matrices and $\mathbb{C}^{m \times n}$

Read all the material on real matrices and replace the word real with the word complex and $\mathbb{R}$ with $\mathbb{C}$

If $A \in \mathbb{C}^{m \times n}$, the *complex conjugate of A* :

$$\bar{A} := \begin{bmatrix} \bar{a}_{11} & \bar{a}_{12} & \dots & \bar{a}_{1n} \\ \bar{a}_{21} & \bar{a}_{22} & \dots & \bar{a}_{2n} \\ \vdots & \vdots & & \vdots \\ \bar{a}_{m1} & \bar{a}_{m2} & \dots & \bar{a}_{mn} \end{bmatrix}$$

Some properties:

$$\begin{aligned}\overline{A+B} &= \bar{A} + \bar{B} \\ \overline{\lambda A} &= \bar{\lambda} \bar{A} \\ \overline{AB} &= \bar{A} \bar{B}\end{aligned}$$

$$\begin{aligned}\Re(A) &= \frac{1}{2}(A + \bar{A}) \\ \Im(A) &= \frac{1}{2j}(A - \bar{A})\end{aligned}$$

The *complex conjugate transpose* of A :

$$\begin{aligned}A^* &:= \bar{A}^T \\ &= \begin{bmatrix} \bar{a}_{11} & \bar{a}_{21} & \dots & \bar{a}_{m1} \\ \bar{a}_{12} & \bar{a}_{22} & \dots & \bar{a}_{m2} \\ \vdots & \vdots & & \vdots \\ \bar{a}_{1n} & \bar{a}_{2n} & \dots & \bar{a}_{mn} \end{bmatrix}\end{aligned}$$

Some properties:

$$\begin{aligned}(A+B)^* &= A^* + B^* \\ (\lambda A)^* &= \bar{\lambda} A^* \\ (AB)^* &= B^* A^*\end{aligned}$$

DEFN. A is *hermitian* if

$$A^* = A$$

Chapter 27

Appendix B: Norms

27.1 The Euclidean norm

What is the size or magnitude of a vector? Meet **norm**. We have the following definition for any real complex n -vector x .

The Euclidean norm or 2- norm of x is the nonnegative real number given by

$$||x|| := (|x_1|^2 + \dots + |x_n|^2)^{\frac{1}{2}}$$

Note that

$$\begin{aligned} ||x|| &= (\bar{x}_1 x_1 + \dots + \bar{x}_n x_n)^{\frac{1}{2}} \\ &= (x'x)^{\frac{1}{2}} \end{aligned}$$

If x is *real*, these expressions become

$$\begin{aligned} ||x|| &= (x_1^2 + \dots + x_n^2)^{\frac{1}{2}} \\ &= (x^T x)^{\frac{1}{2}} \end{aligned}$$

```
>> norm([3; 4])
```

```
ans = 5
```

```
>> norm([1; j])
```

```
ans = 1.4142
```

Note that in the last example $x^T x = 0$, but $x'x = 2$.

There are other norms, but, the Euclidean one is the most common for n -vectors. All norms are functions from a vector space to the set of real numbers are usually denoted by

$\|\cdot\|$ and must have the following three properties. Of course the Euclidean norm has these properties.

Properties of a norm $\|\cdot\|$

(i) For every vector x ,

$$\|x\| \geq 0$$

and

$$\|x\| = 0 \quad \text{iff} \quad x = 0$$

(ii) (**Triangle Inequality.**) For every pair of vectors x, y ,

$$\|x + y\| \leq \|x\| + \|y\|$$

(iii) For every vector x and every scalar λ .

$$\|\lambda x\| = |\lambda| \|x\|$$

Any real valued function on a vector space which has the above three properties is defined to be a norm. Two other commonly encountered norms on the space of n -vectors are the 1-norm:

$$\|x\|_1 := |x_1| + \dots + |x_n|$$

and the ∞ -norm:

$$\|x\|_\infty := \max_i |x_i|$$

As an example of a norm on the set of $m \times n$ matrices A , consider

$$\|A\| = \text{Tr}(A'A)^{\frac{1}{2}} = \left(\sum_{i=1}^m \sum_{j=1}^n |A_{ij}|^2 \right)^{\frac{1}{2}}$$

where $\text{Tr}(M)$ denotes the trace of a square matrix M , that is the sum of its diagonal elements. This norm is called the **Frobenius norm** and is the matrix analog of the Euclidean norm for n -vectors. We will meet other matrix norms later.

Bibliography

- [1] Skelton, R.E., *Dynamic Systems Control*, Wiley, 1988.
- [2] Chen, C.T., *Linear System Theory and Design*, Harcourt Brace Jovanovich, 1984.